# Learning to generate

Distributions, autoencoders and VAEs, built from one handwritten 3



A 3 turns into a 7. Every frame is decoded from a point on a straight line through the code space of the VAE we train today.

Nipun Batra · IIT Gandhinagar

START WITH A QUESTION

# Who are these people?



**Look closely: skin, hair, glasses, lighting, background. Anything odd?**

Karras, Laine & Aila, *A Style-Based Generator Architecture for GANs* (StyleGAN), 2019 · CC BY-NC 4.0

START WITH A QUESTION

# Who are these people?



**No one. Each face is a random vector of 512 numbers pushed through a trained network.**

Karras, Laine & Aila, *A Style-Based Generator Architecture for GANs* (StyleGAN), 2019 · CC BY-NC 4.0

YOUR TURN TO BE THE JUDGE

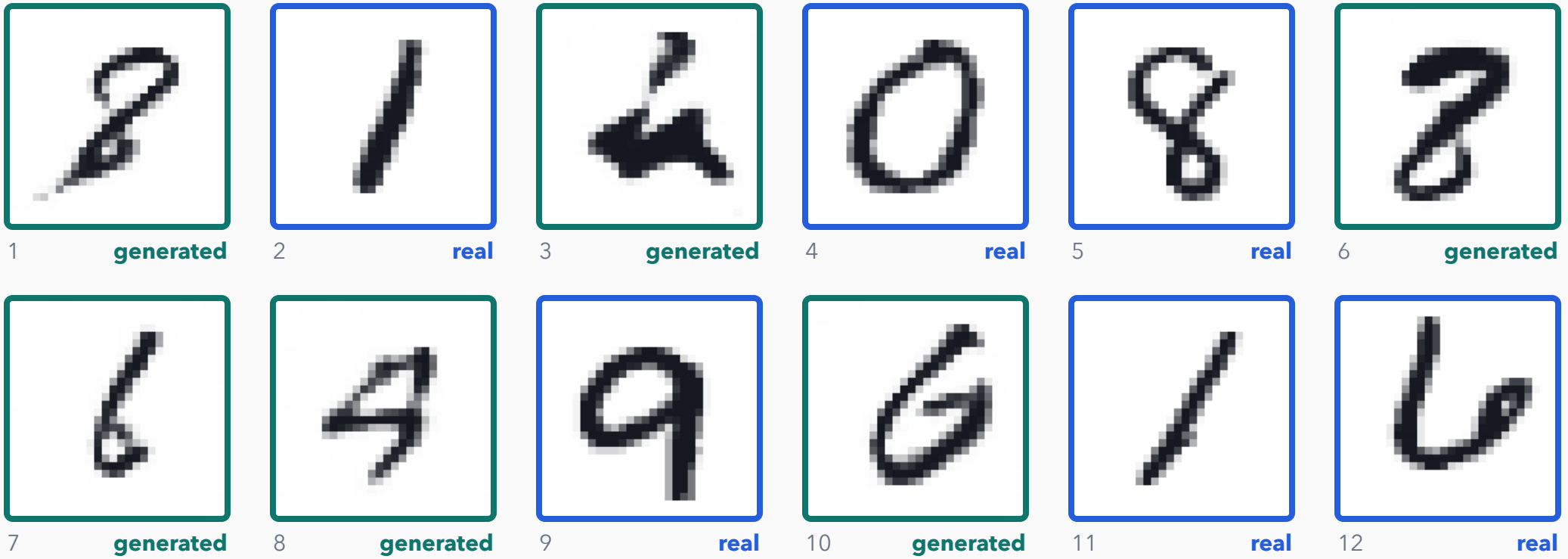
# Real or generated?




Six are MNIST test digits, six come from a network. Write down which six are fake.

YOUR TURN TO BE THE JUDGE

# Real or generated?

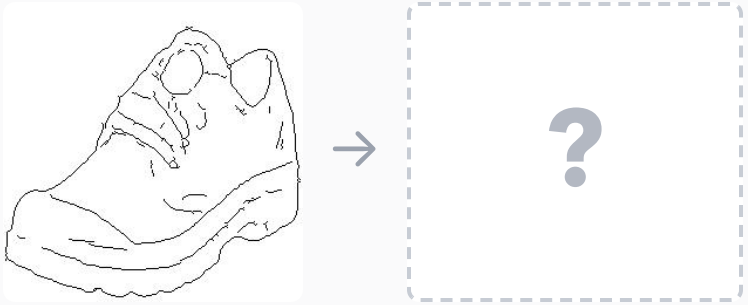



Generated: 1, 3, 6, 7, 8, 10 (teal frames). Six from a small GAN, trained in about 9 minutes: the topic of the next lecture.

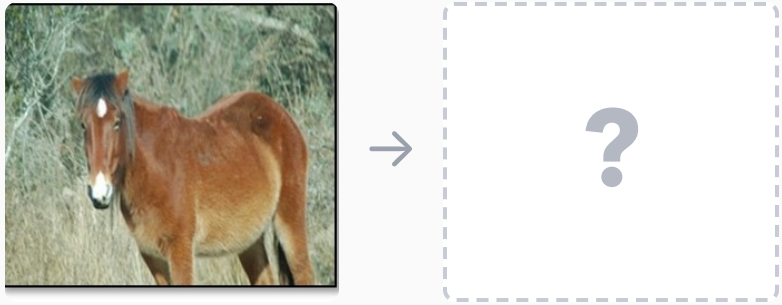
GENERATION IS NOT ONLY "MAKE A RANDOM PICTURE"

# Predict the output, then look

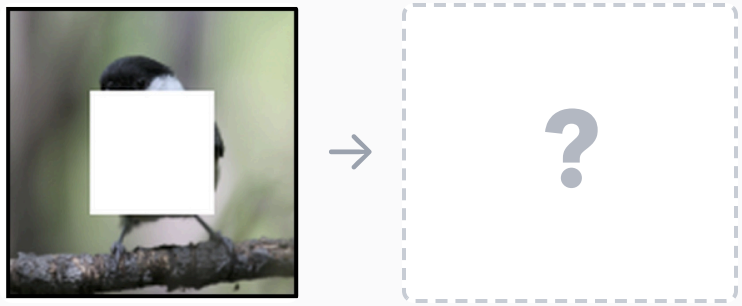
**a sketch → a photo**



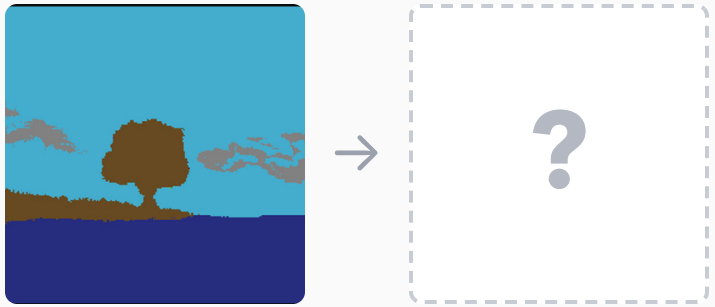

**a horse → a zebra**




**fill a hole**




**a label map → a landscape**




**What should come out on the right? Is there exactly one right answer?**

pix2pix (Isola et al. 2017) · CycleGAN (Zhu et al. 2017) · Context Encoders (Pathak et al. 2016) · SPADE (Park et al. 2019). All published model outputs.

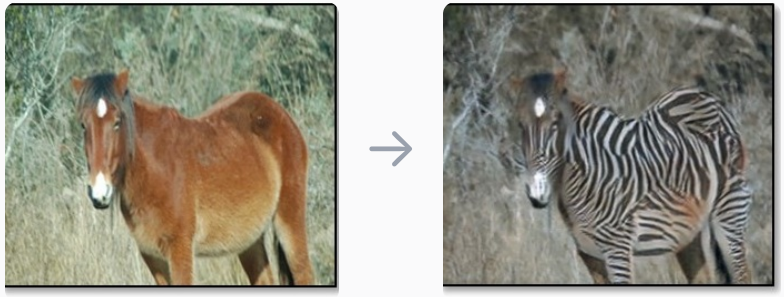
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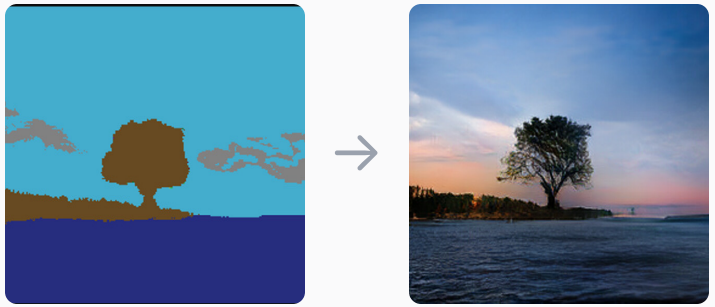
**a horse → a zebra**



**fill a hole**



**a label map → a landscape**



> Each input allows many plausible outputs. A generator must pick one that looks real.

pix2pix (Isola et al. 2017) · CycleGAN (Zhu et al. 2017) · Context Encoders (Pathak et al. 2016) · SPADE (Park et al. 2019). All published model outputs.

ONE INPUT, MANY ANSWERS

# The same label map, three different photographs



the condition

SPADE / GauGAN, Park et al. 2019, teaser figure: here the variation comes from a style image given to the generator.

ONE INPUT, MANY ANSWERS

# The same label map, three different photographs



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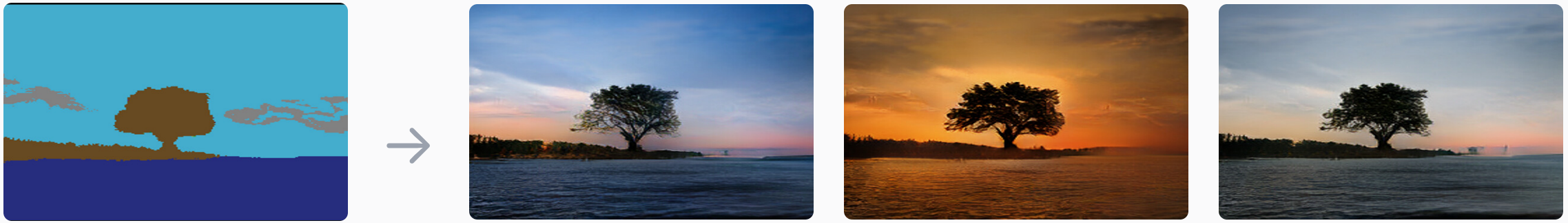


sample 1

SPADE / GauGAN, Park et al. 2019, teaser figure: here the variation comes from a style image given to the generator.

ONE INPUT, MANY ANSWERS

# The same label map, three different photographs



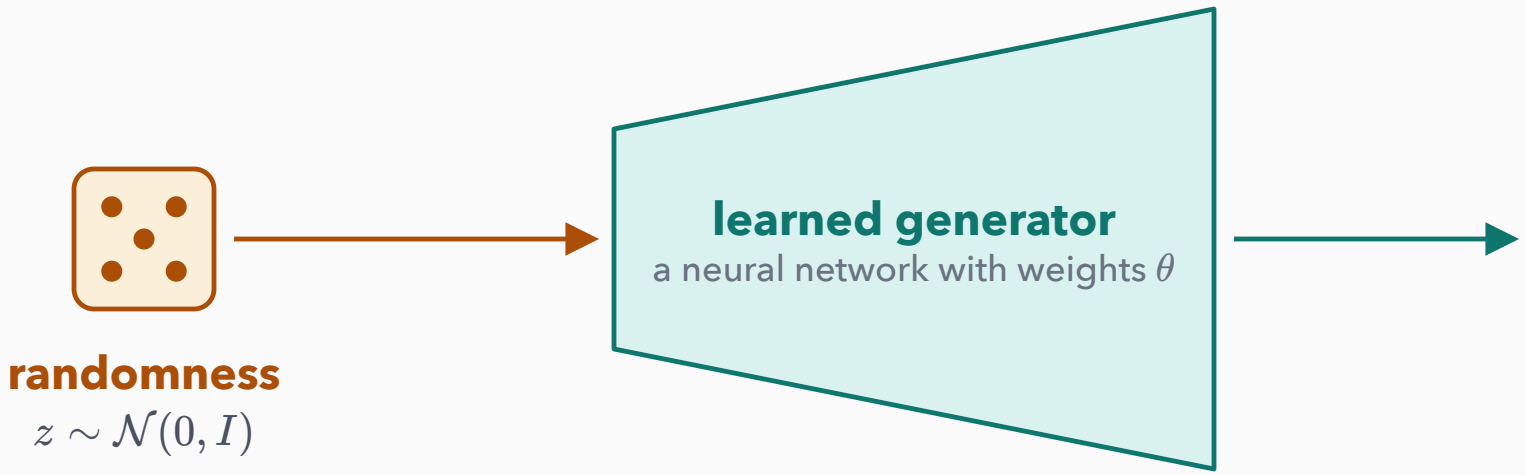
the condition | sample 1 | sample 2 | sample 3

**A generator does not learn one answer. It learns a distribution of answers and draws from it.**

SPADE / GauGAN, Park et al. 2019, teaser figure: here the variation comes from a style image given to the generator.

WHAT ALL OF THESE SHARE

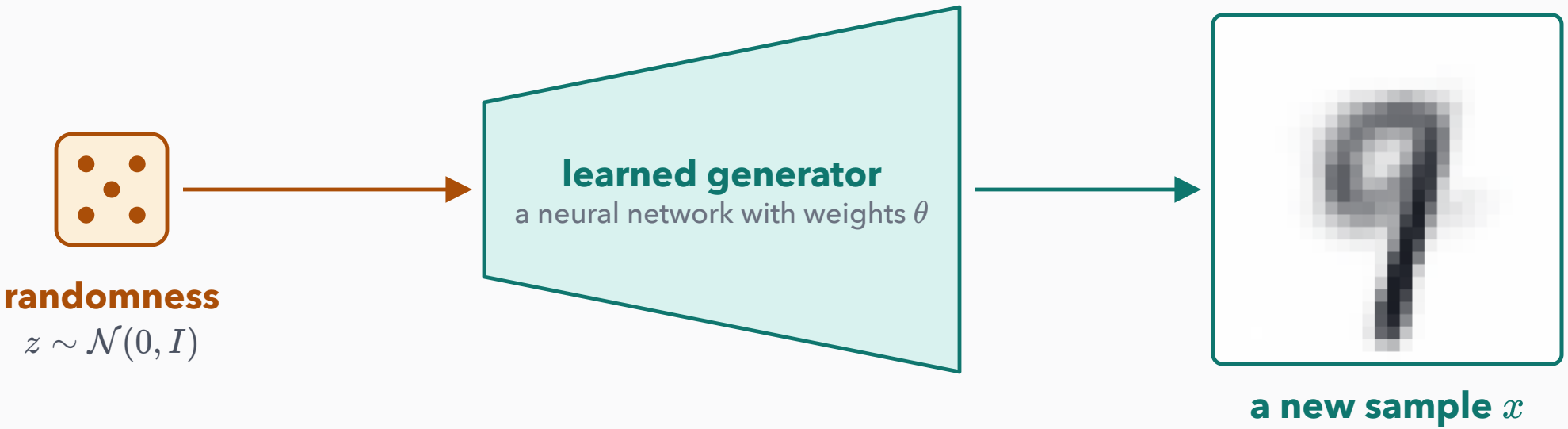
# Randomness in, a learned network, a sample out




Training means: adjust $\theta$ until the samples look like the data. But what is the loss?

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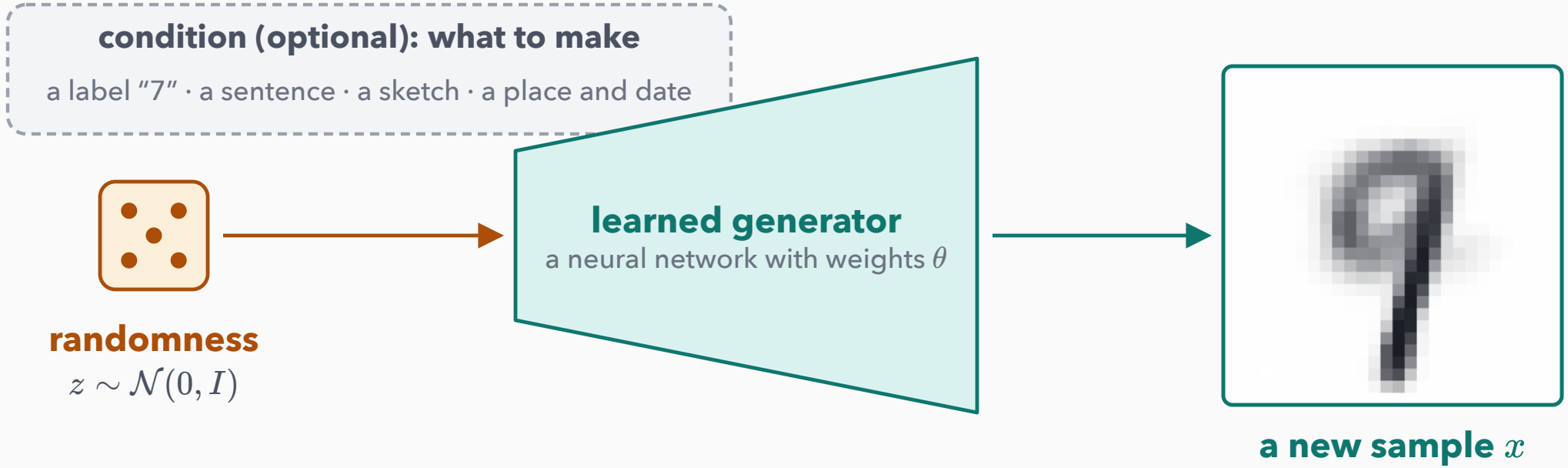
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## WHAT CAN THE CONDITION BE?

# Tell the generator what you want

**a label** $c = 7$



our own conditional VAE: eight different random $z$, all asked for a 7

## WHAT CAN THE CONDITION BE?

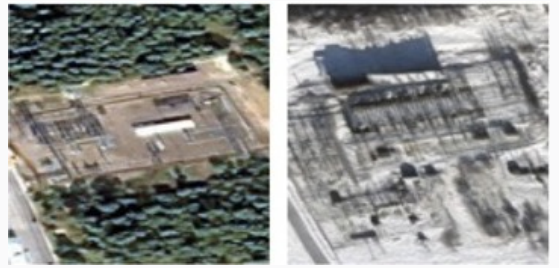
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**a sentence + a date** "`a satellite image of an electric substation in Finland`", August vs January



DiffusionSat (Khanna et al., 2024): same caption, only the month changes

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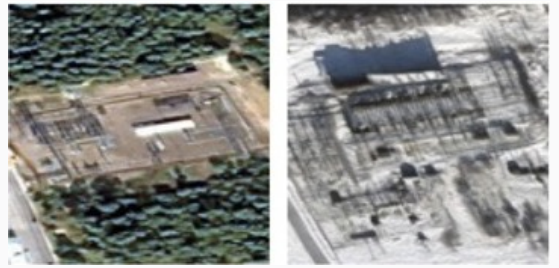
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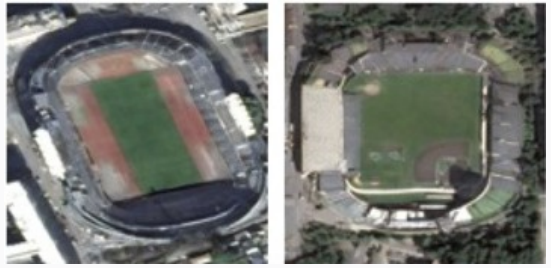
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DiffusionSat (Khanna et al., 2024): same caption, only the month changes

**a sentence + a place** "`a satellite image of a stadium`", France vs USA coordinates



DiffusionSat: same caption, only the latitude and longitude change

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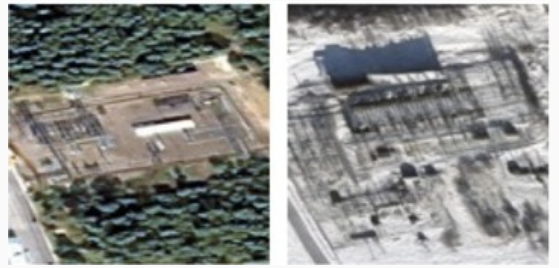
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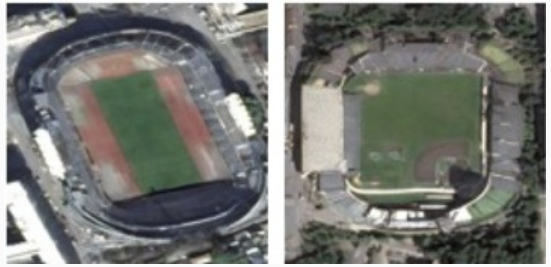
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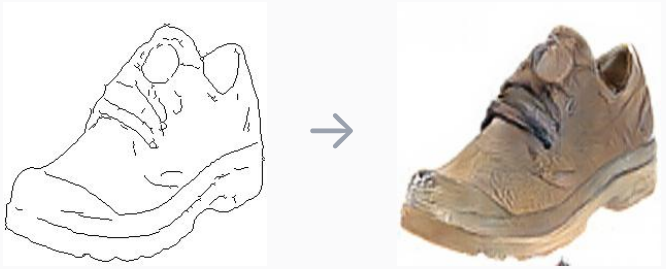
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DiffusionSat: same caption, only the latitude and longitude change

**an image** a sketch, a label map, a photo with a hole



pix2pix, from the earlier slide

> **The condition fixes *what*; the randomness still decides *which one* of the many matching outputs you get.**

OUR CONDITIONAL VAE, ALL LABELS AT ONCE

# The label picks the digit; the random $z$ picks the style




Each **row**: one label $c$, from 0 to 9.

Each **column**: one random $z$, the same all the way down.

OUR CONDITIONAL VAE, ALL LABELS AT ONCE

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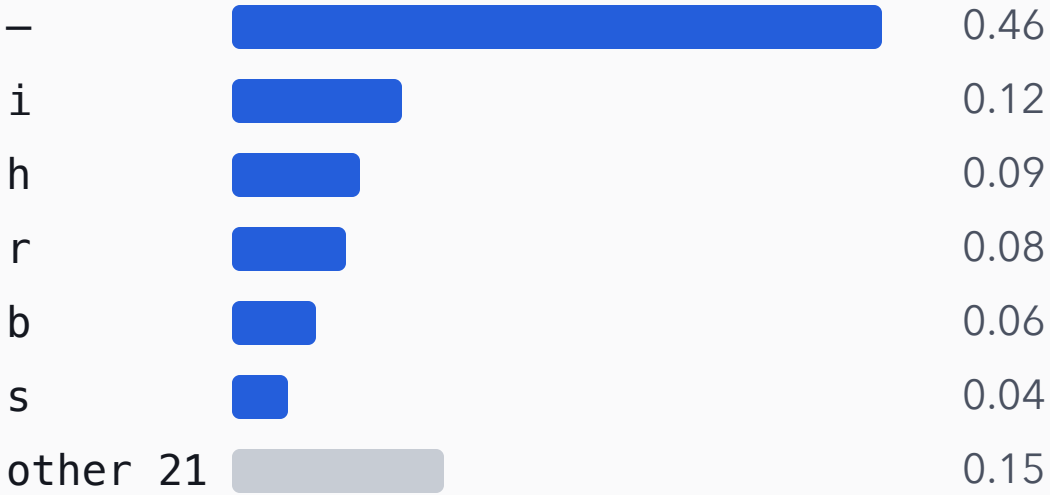
Read down a column: every digit is written with the same slant and thickness. The code $z$ carries the handwriting; the label carries the identity.

EARLIER IN THIS COURSE

# You have already built a generator

Our name model reads three characters and gives a probability for every next character.

after `a a b`, the trained model says:



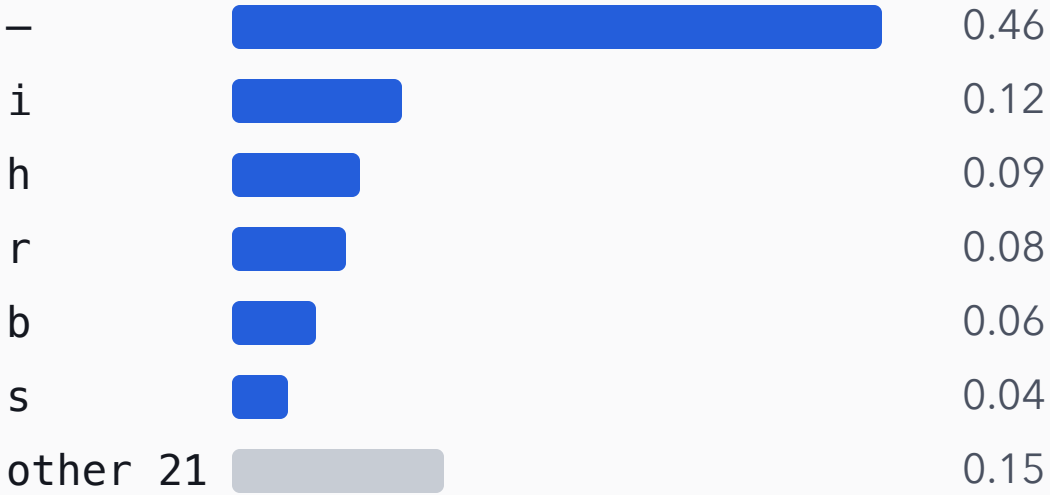

**Same idea today: learn a distribution, then sample from it. Only now the samples are images.**

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**Generate:** sample a character, append it, repeat.



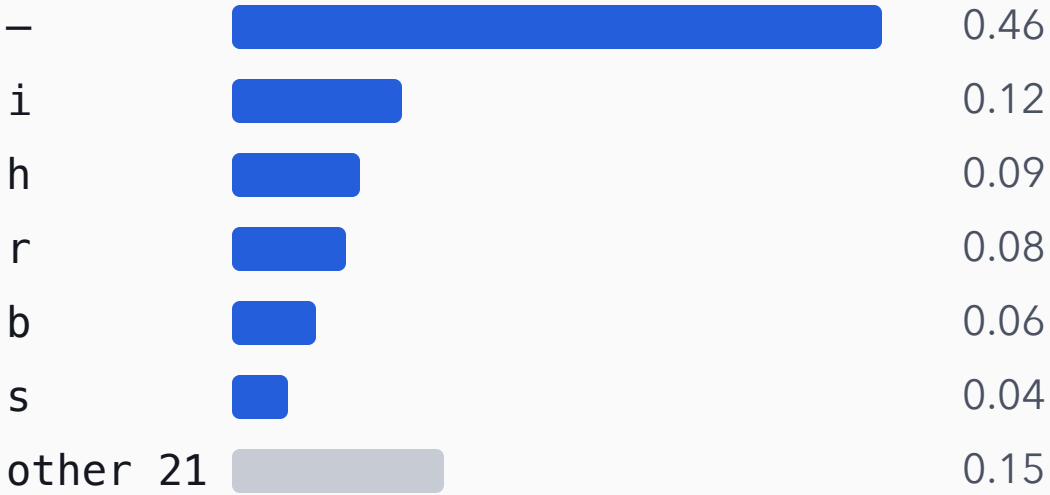

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**Generate:** sample a character, append it, repeat.




A different random draw at any step gives a different name. That is a generative model: it gives probabilities, and we sample from them.

**Same idea today: learn a distribution, then sample from it. Only now the samples are images.**

EARLIER IN THIS COURSE

# A whole name gets one probability

| context | next | probability | −log p |
|---|---|---|---|
| . . . | a | 0.08 | 2.53 |
| . . a | a | 0.08 | 2.53 |
| . a a | b | 0.02 | 3.91 |
| a a b | i | 0.12 | 2.12 |
| a b i | d | 0.09 | 2.41 |
| b i d | . | 0.23 | 1.47 |

EARLIER IN THIS COURSE

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| **aabid** | | **$3.2 \times 10^{-7}$** | **14.96** |

$$p_\theta(\text{aabid}) = \prod_t p_\theta(x_t \mid x_{<t}) \qquad -\log p_\theta(\text{aabid}) = \sum_t -\log p_\theta(x_t \mid x_{<t})$$

EARLIER IN THIS COURSE

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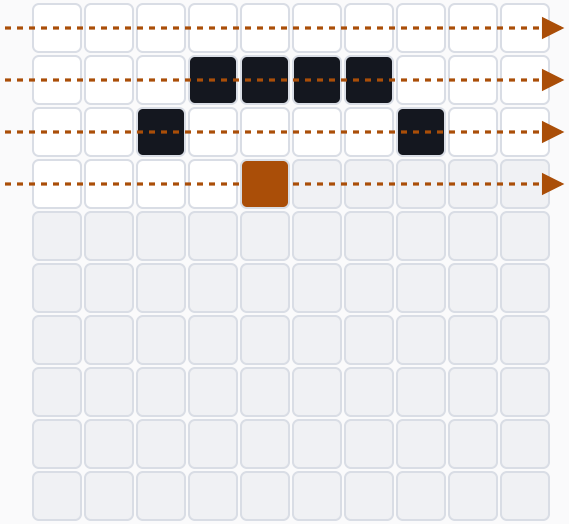
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> **Training made this number large for every name in the data. That is maximum likelihood, and it is what we want for images too.**

# Two ways to generate a 28 × 28 digit

### A · one pixel at a time

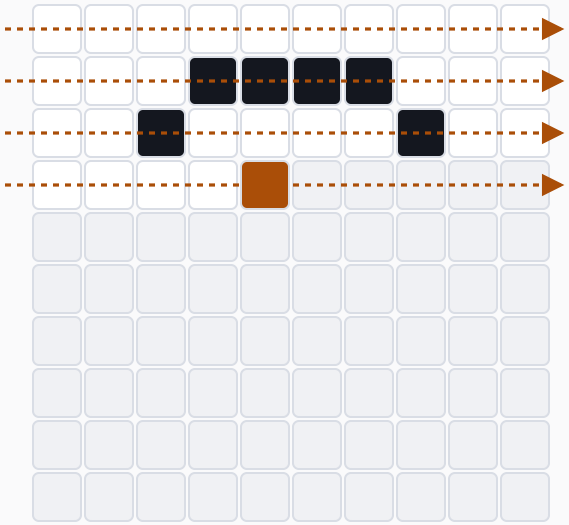


784 predictions in a row, like our names

$$p(x) = \prod_i p(x_i \mid x_{<i})$$

# Two ways to generate a 28 × 28 digit

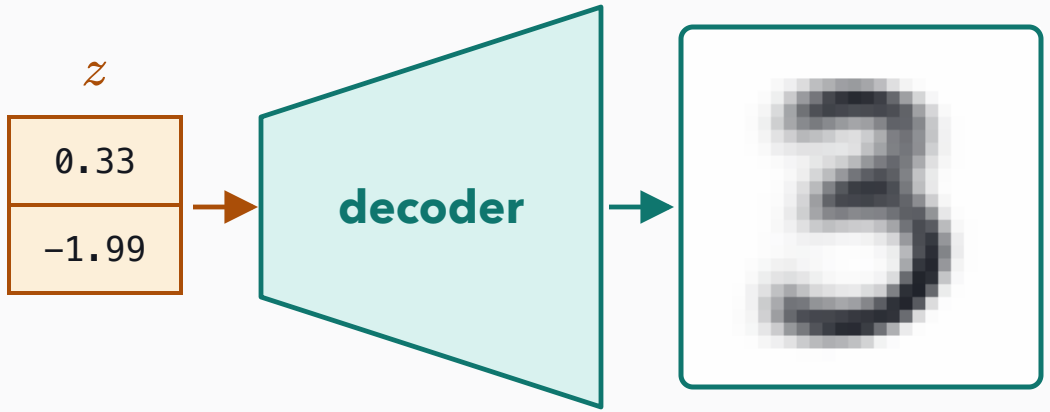
### A · one pixel at a time



784 predictions in a row, like our names

$$p(x) = \prod_i p(x_i \mid x_{<i})$$

### B · all pixels from one random vector



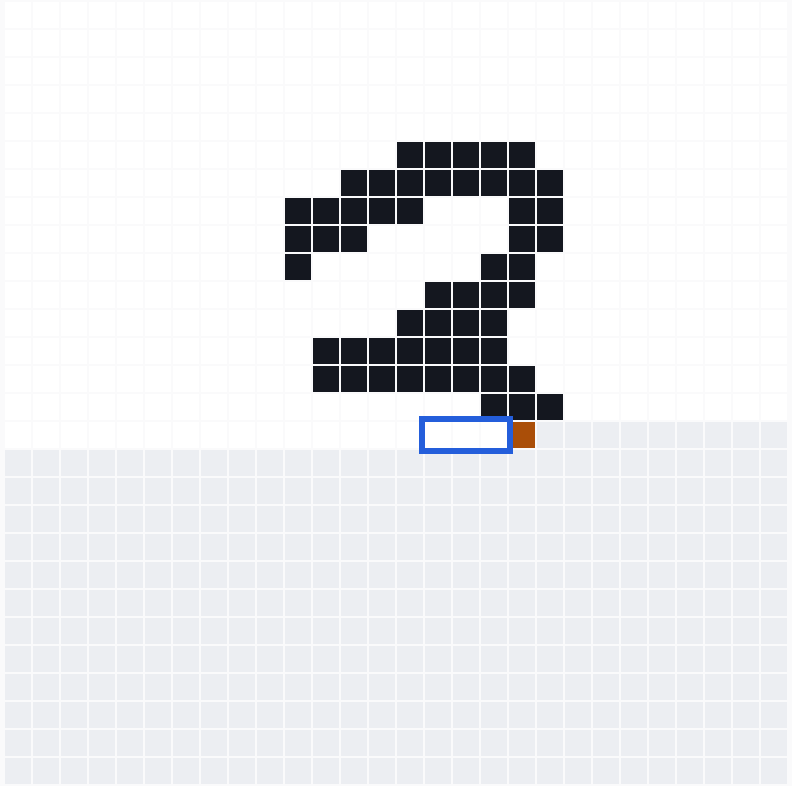

one pass: 2 numbers in, 784 pixels out

$$x = \text{decoder}(z)$$

**Today: route B. Two questions: where does $z$ come from, and how do we train a decoder when no one tells us the right $z$?**

ROUTE A, THE WAY OUR NAME MODEL WORKED

# Predict each pixel from the pixels already drawn

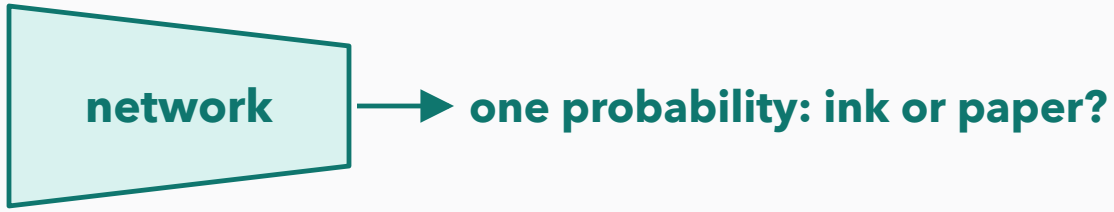


drawn so far · next pixel (orange) · not yet drawn (grey)

**context: the 3 previous pixels**

(the name model read 3 previous characters)

$$p(x_i = \text{ink} \mid 3 \text{ pixels}) = 0.24$$

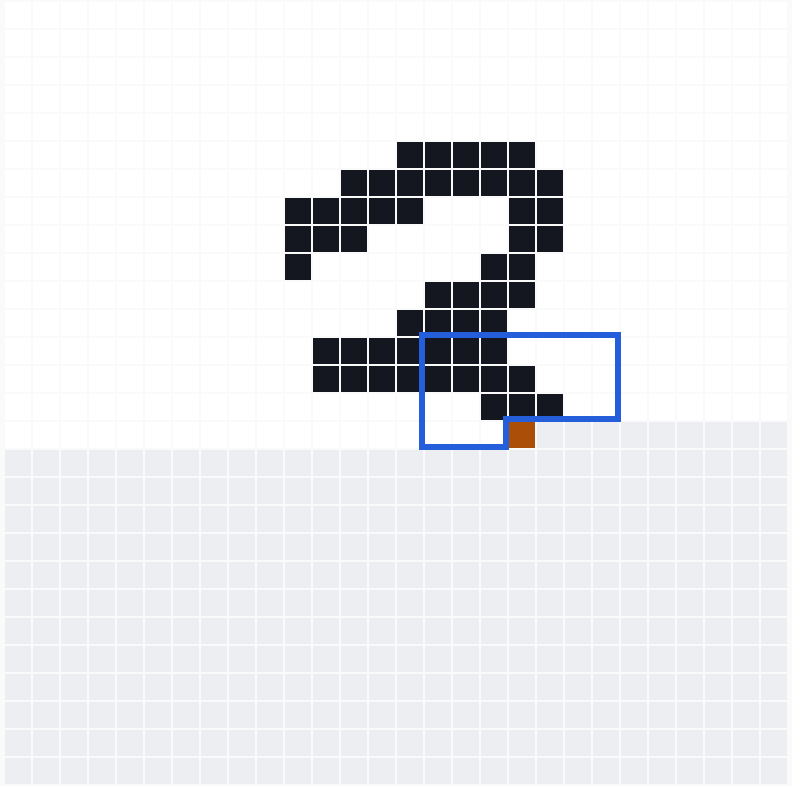



Repeat for all 784 pixels, one after another:

$$p(x) = \prod_{i=1}^{784} p(x_i \mid x_{<i})$$

ROUTE A, THE WAY OUR NAME MODEL WORKED

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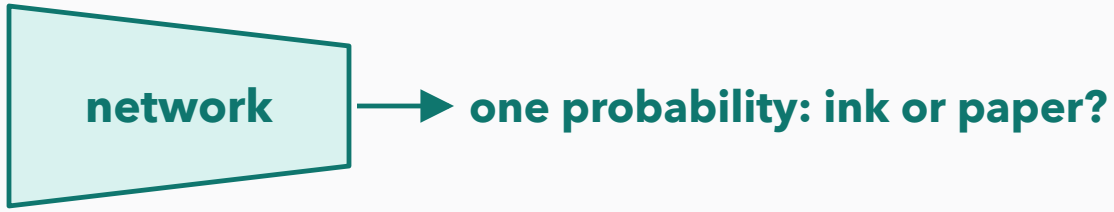


drawn so far · next pixel (orange) · not yet drawn (grey)

**context: 3 rows above + 3 to the left**

24 pixels the model may look at

$$p(x_i = \text{ink} \mid 24 \text{ pixels}) = 0.89$$




Repeat for all 784 pixels, one after another:

$$p(x) = \prod_{i=1}^{784} p(x_i \mid x_{<i})$$

ROUTE A, TRAINED ON MNIST

# What these two pixel-by-pixel models generate



context: 3 previous pixels · $\log p(\text{our } 3) = -123.7$

ROUTE A, TRAINED ON MNIST

# What these two pixel-by-pixel models generate



context: 3 previous pixels · $\log p(\text{our } 3) = -123.7$



context: 24 pixels above and left · $\log p(\text{our } 3) = -69.4$

**More context, better strokes. And route A gives an exact $\log p(x)$: just add the 784 log-probabilities.**

SO WHY ROUTE B TODAY?

# Two routes, different strengths

| | A · pixel by pixel | B · one code, one pass |
|---|---|---|
| **generate one image** | 784 network calls, in order | 1 network call |

SO WHY ROUTE B TODAY?

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SO WHY ROUTE B TODAY?

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| **a short summary** $z$ | none | **yes**: 2 to 100 numbers you can move, mix, edit |

SO WHY ROUTE B TODAY?

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| **a short summary** $z$ | none | **yes**: 2 to 100 numbers you can move, mix, edit |
| **used today in** | language models; image models over image tokens | VAEs, GANs, and the latent space of diffusion models |

> **Route A is the name model you already know. Today is route B: what the code is, and how to train a decoder when nobody tells us the right code.**

THE PLAN

# Today's route, in two sessions

## SESSION 1

1. **Distributions and fitting**
   a 1-D toy: fit, sample, how many bumps
2. **Autoencoders**
   plain, denoising, sparse, contractive; why they cannot generate
3. **The probabilistic view**
   prior, likelihood, an intractable posterior

## SESSION 2

4. **KL divergence**
   how far apart are two distributions?
5. **Variational inference and the ELBO**
   a loss that never computes $p(x)$
6. **Reparameterisation**
   backprop through a random draw
7. **The VAE on MNIST**
   one digit, end to end, with real numbers

> **One running image throughout: test digit #90, a handwritten 3. GANs get their own lecture.**

PART 1 OF 7

# Learn a distribution from examples, then sample from it

We have seen what generators can do. Before any network, the whole idea on **one number**: what does "learning" mean, and how do we then make new samples?

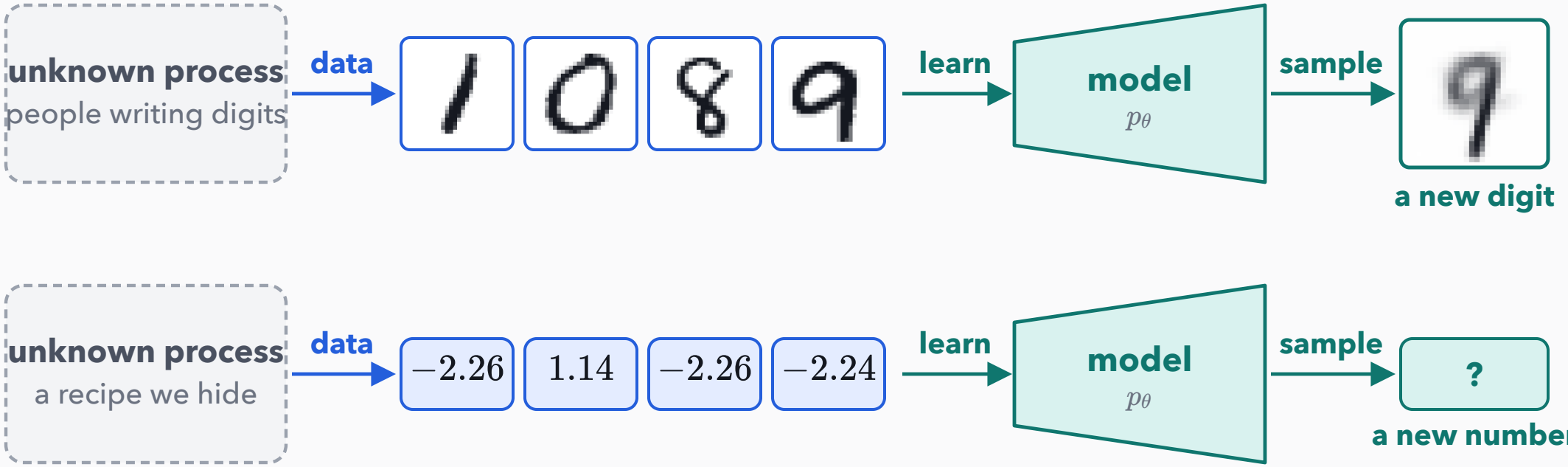

1 **Distributions and fitting**
2 Autoencoders
3 The probabilistic view
4 KL divergence
5 Variational inference and the ELBO
6 Reparameterisation
7 The VAE on MNIST

## WHAT ARE WE TRYING TO DO?

# Examples in, a model out, new examples from the model

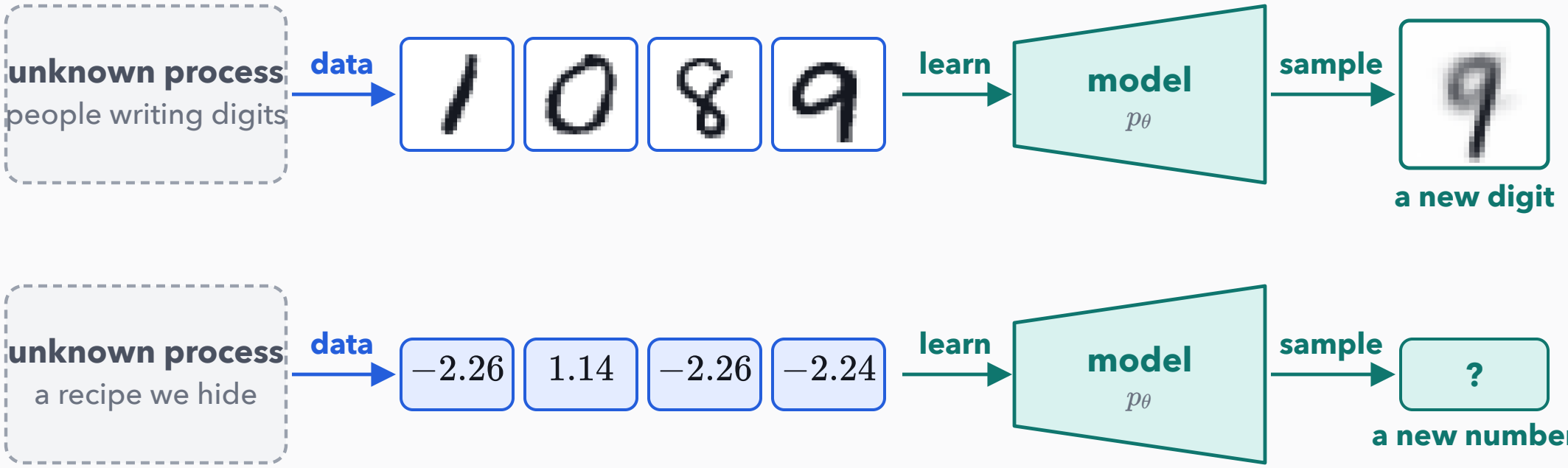


# Examples in, a model out, new examples from the model



WHAT ARE WE TRYING TO DO?

# Examples in, a model out, new examples from the model




**1 · Learn**

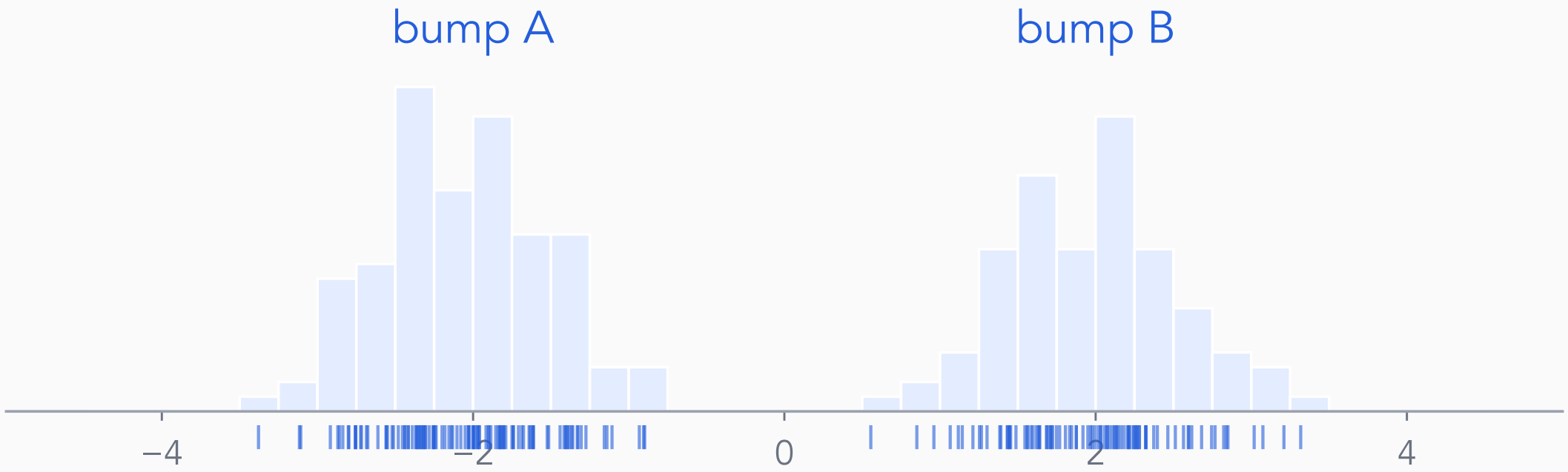
Choose the model's parameters $\theta$ so that the examples we have are **probable** under it.

**2 · Generate**

Draw **new** samples from the learned model $p_\theta$.

OUR TOY DATA

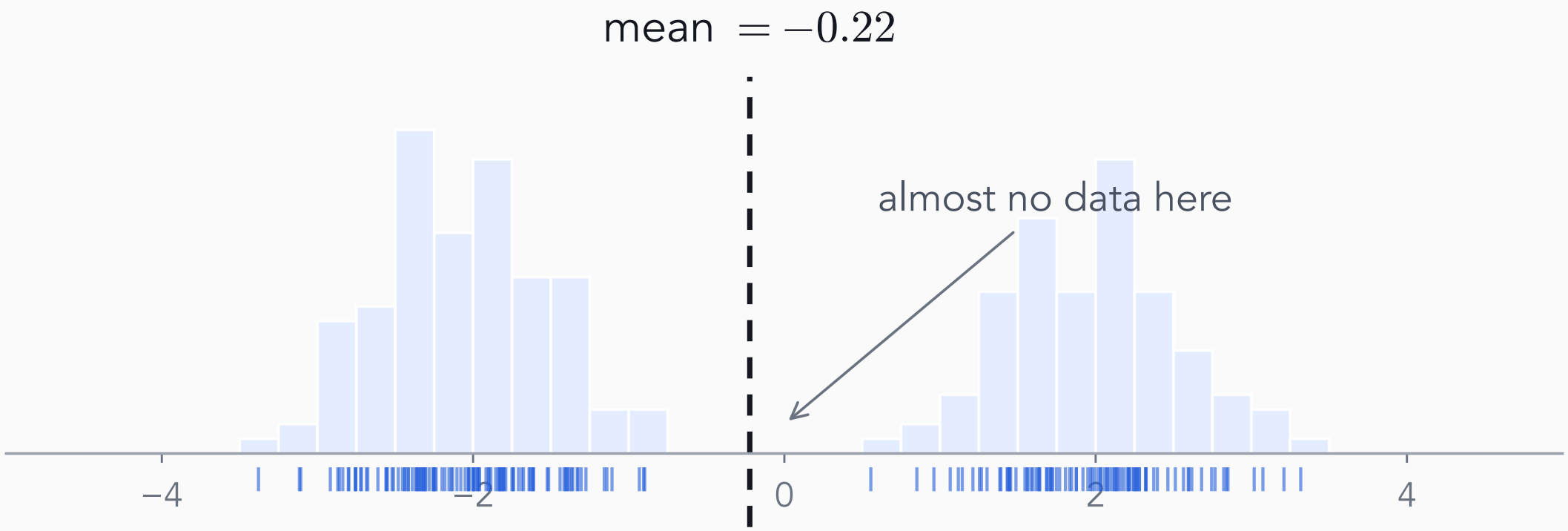
# 200 numbers from a recipe we keep hidden




> The model sees only these 200 numbers, not the recipe. Goal: learn from them, then produce a 201st that looks like it belongs.

A TEMPTING ANSWER

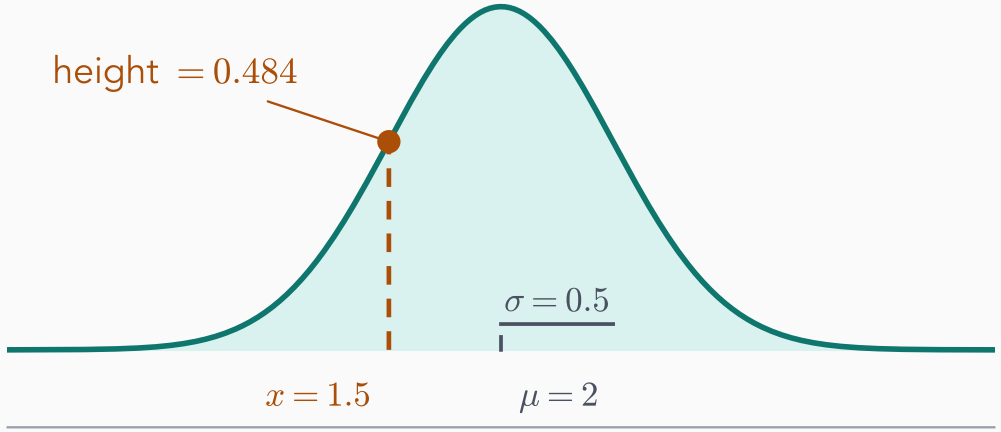
# The average is a terrible sample




> **The mean is a number that almost never occurs. Generating needs the whole spread of the data, not one summary of it.**

# How probable is $x = 1.5$ under a bump at 2?

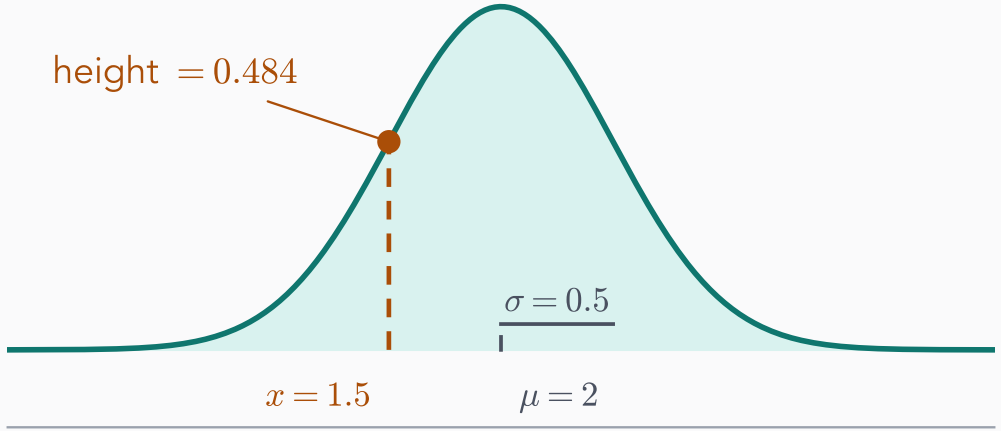
$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left( -\frac{(x-\mu)^2}{2\sigma^2} \right)$$



# How probable is $x = 1.5$ under a bump at 2?

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left( -\frac{(x-\mu)^2}{2\sigma^2} \right)$$

$$\frac{x-\mu}{\sigma} = \frac{1.5-2}{0.5} = -1$$

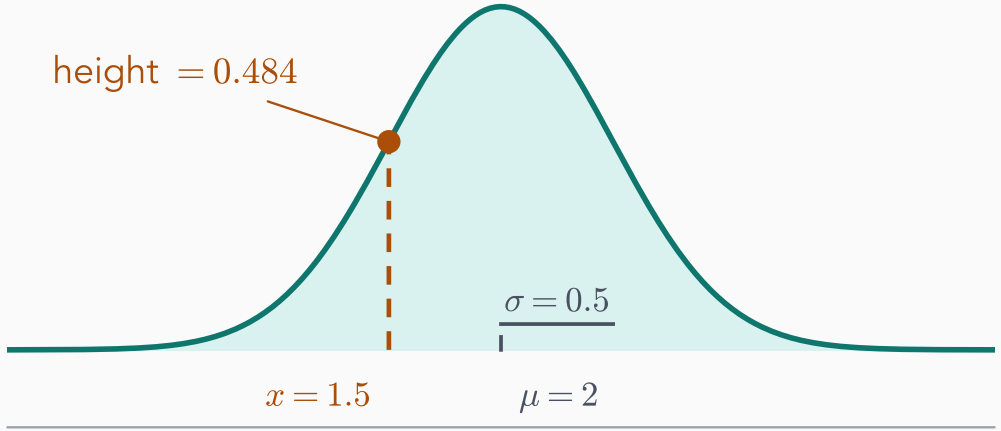


# How probable is $x = 1.5$ under a bump at 2?

$$\mathcal{N}(x;\mu,\sigma^2) = \frac{1}{\sigma\sqrt{2\pi}}\exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big)$$

$$\frac{x-\mu}{\sigma} = \frac{1.5-2}{0.5} = -1$$

$$\exp\Big(-\tfrac{1}{2}\cdot(-1)^2\Big) = e^{-0.5} = 0.607$$



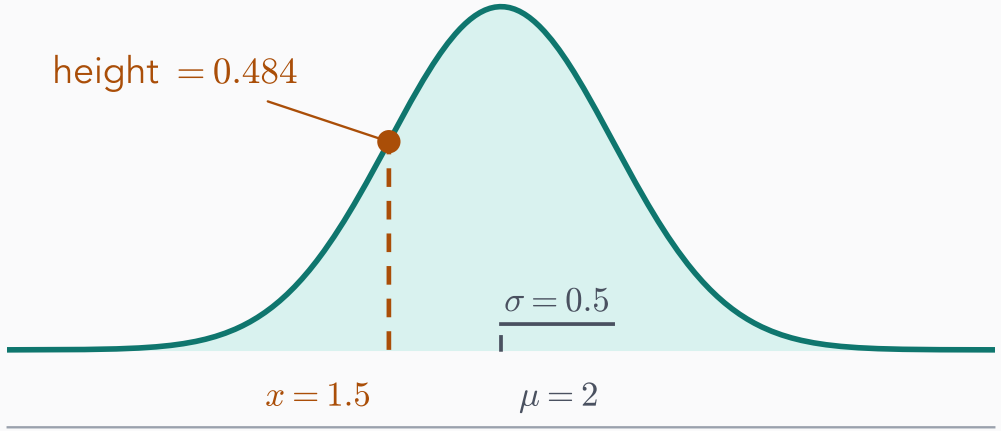
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$$\frac{x-\mu}{\sigma} = \frac{1.5-2}{0.5} = -1$$

$$\exp\Big(-\tfrac{1}{2}\cdot(-1)^2\Big) = e^{-0.5} = 0.607$$

$$\frac{1}{0.5\sqrt{2\pi}} = 0.798 \qquad \Rightarrow \qquad 0.798 \times 0.607 = \mathbf{0.484}$$




> The model's two parameters, $\mu$ and $\sigma$, decide how high the curve is at every $x$. Higher means more probable.

STEP 1 · LEARN

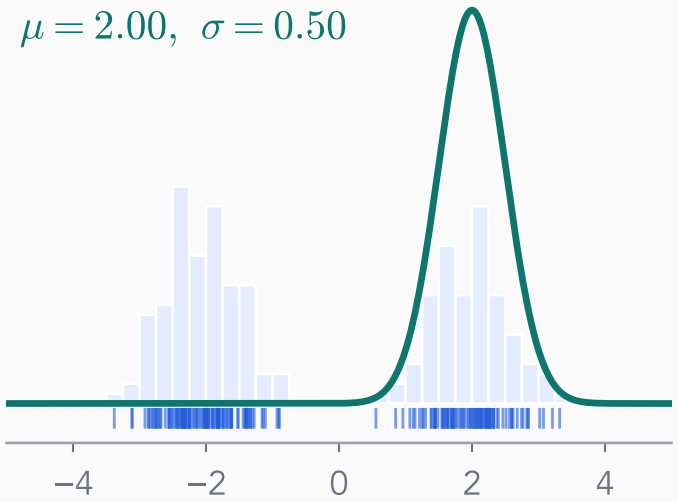
# Score a choice of $\mu, \sigma$ on all 200 numbers

$$\log p_\theta(\text{data}) = \sum_{i=1}^{200} \log \mathcal{N}(x_i; \mu, \sigma^2), \qquad \theta = (\mu, \sigma)$$

STEP 1 · LEARN

# Score a choice of $\mu, \sigma$ on all 200 numbers

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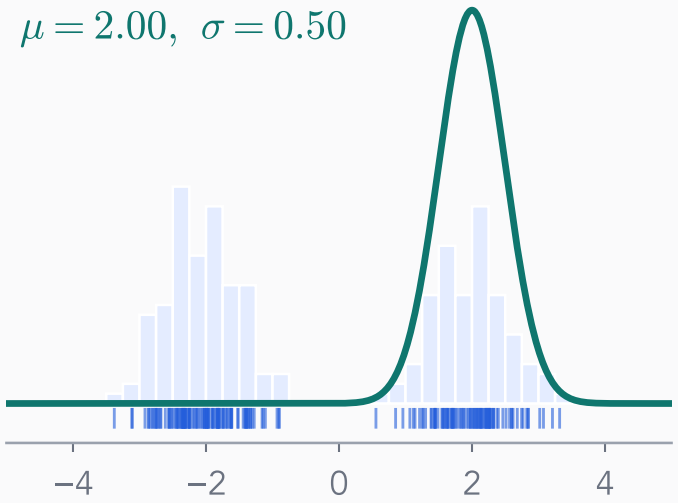



$$\log p = -3746.0$$

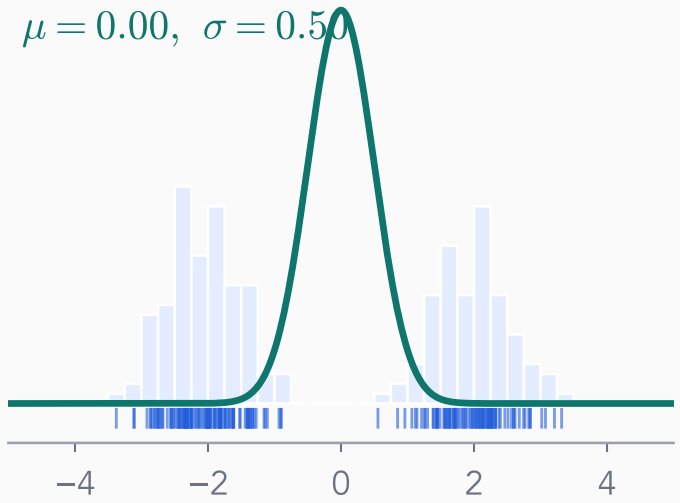
STEP 1 · LEARN

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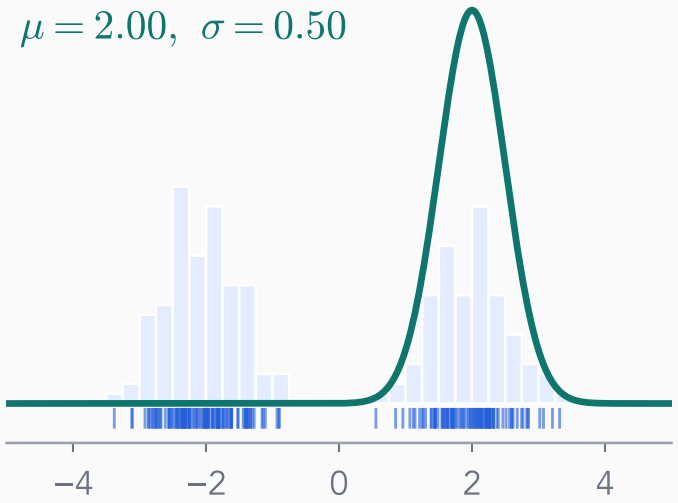



$\log p = -1790.0$

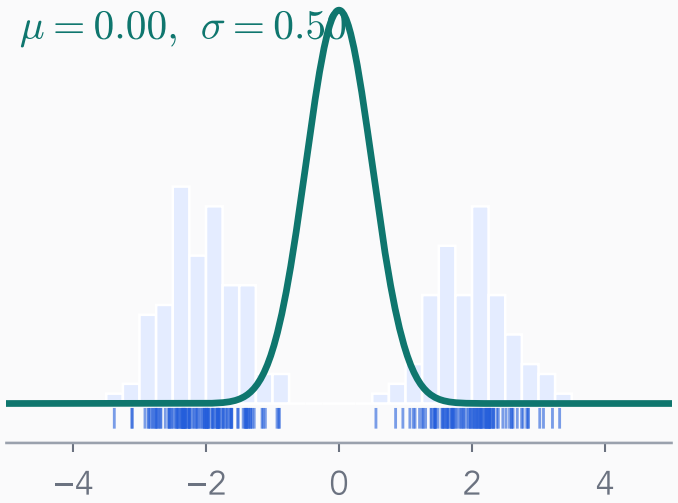
STEP 1 · LEARN

# Score a choice of $\mu, \sigma$ on all 200 numbers

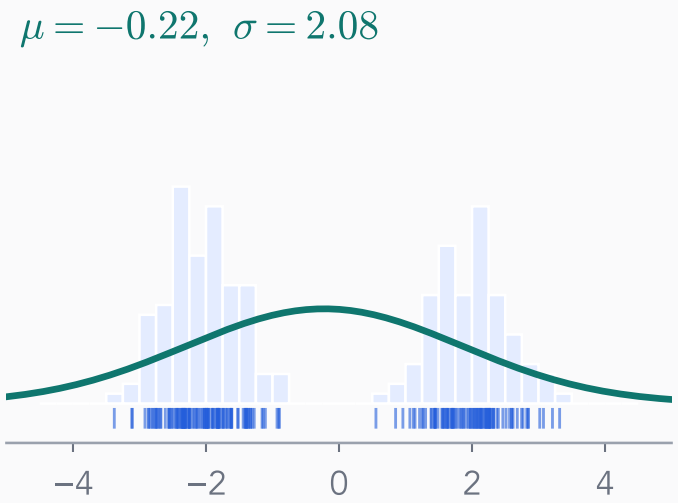
$$\log p_\theta(\text{data}) = \sum_{i=1}^{200} \log \mathcal{N}(x_i; \mu, \sigma^2), \qquad \theta = (\mu, \sigma)$$




$\log p = -3746.0$



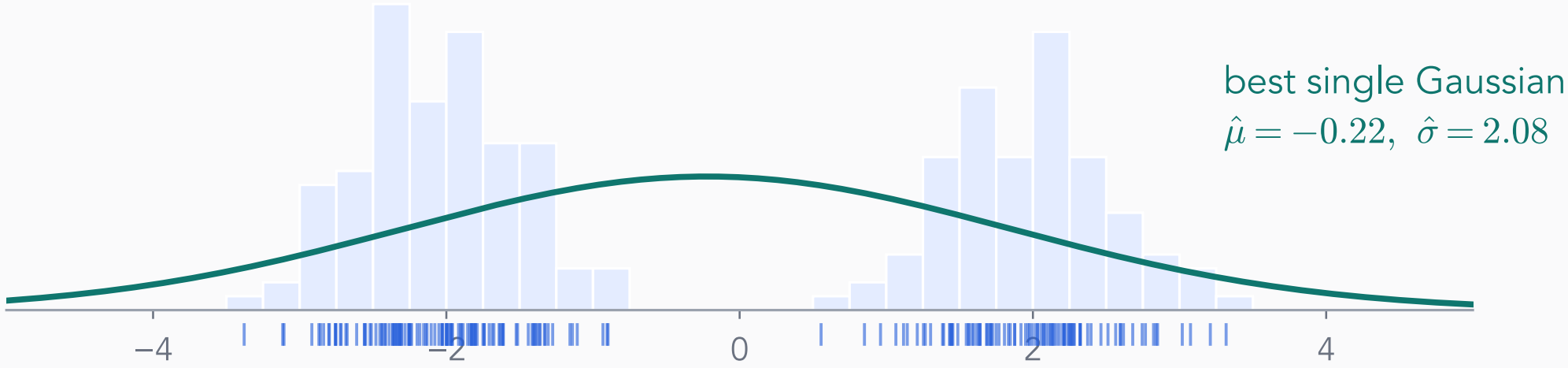

$\log p = -1790.0$




$\log p = -430.0$

> Learning = keep the $\theta$ with the highest score. That is **maximum likelihood**, the same rule that trained the name model.

# The best Gaussian has a formula; then we sample from it

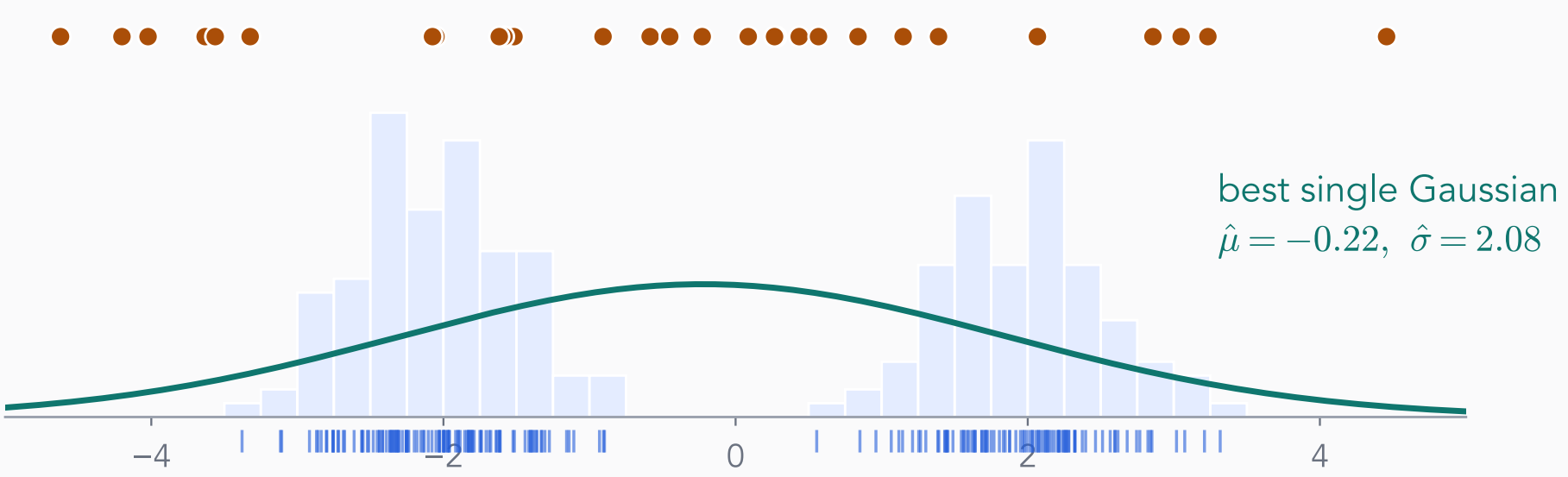



$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i = -0.22 \qquad \hat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{\mu})^2 = 4.31$$

**Setting the derivative of the score to zero gives these two numbers: the same answer as the third candidate.**

STEP 1 · LEARN, THEN STEP 2 · GENERATE

# The best Gaussian has a formula; then we sample from it




$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} x_i = -0.22 \qquad \hat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{\mu})^2 = 4.31 \qquad x_{\text{new}} = \hat{\mu} + \hat{\sigma}\,\varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1)$$

**Step 2: 11 of 30 samples land in the empty gap. Learning worked; the model was too simple.**

THE SAME FIT, IN PYTORCH

# Learning = gradient descent on the negative log-likelihood

```
x = torch.tensor(data)                    # the 200 numbers
mu        = torch.tensor(1.0, requires_grad=True)
log_sigma = torch.tensor(0.0, requires_grad=True)
opt = torch.optim.Adam([mu, log_sigma], lr=0.05)

for step in range(300):
    dist = Normal(mu, log_sigma.exp())
    loss = -dist.log_prob(x).mean()
    opt.zero_grad(); loss.backward(); opt.step()

x_new = dist.sample((30,))                # step 2: generate
```

**The parameters** $\theta = (\mu, \log \sigma)$, started anywhere. We learn $\log \sigma$ so that $\sigma = e^{\log \sigma}$ is always positive.

THE SAME FIT, IN PYTORCH

# Learning = gradient descent on the negative log-likelihood

```
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    opt.zero_grad(); loss.backward(); opt.step()

x_new = dist.sample((30,))                # step 2: generate
```

**The loss** is the score from the last slide with a minus sign, averaged: $-\frac{1}{N}\sum_i \log \mathcal{N}(x_i; \mu, \sigma^2)$.

THE SAME FIT, IN PYTORCH

# Learning = gradient descent on the negative log-likelihood

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x = torch.tensor(data)                  # the 200 numbers
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x_new = dist.sample((30,))              # step 2: generate
```

**The update:** nudge $\mu$ and $\log \sigma$ downhill, 300 times. The same loop trains every network in this course.

THE SAME FIT, IN PYTORCH

# Learning = gradient descent on the negative log-likelihood

```
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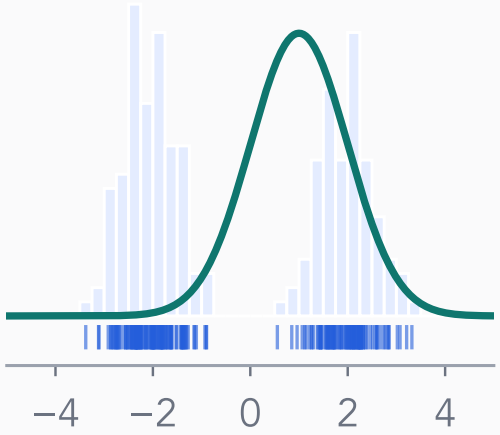
**Result:** $\mu = -0.22$, $\sigma = 2.08$, exactly the formula's answer. Then `sample` is step 2.

WATCH THE FIT HAPPEN

# Four moments of that loop

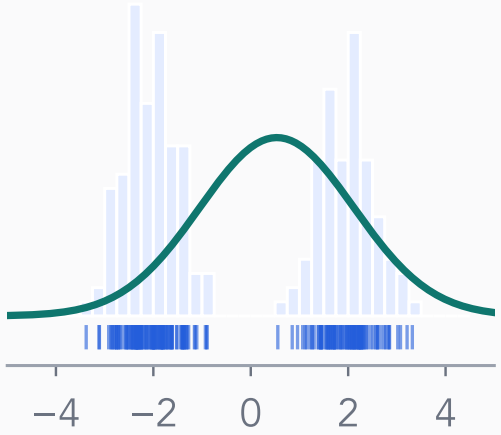
step 0: $\mu = 1.00,\ \sigma = 1.00$

loss $= 3.82$



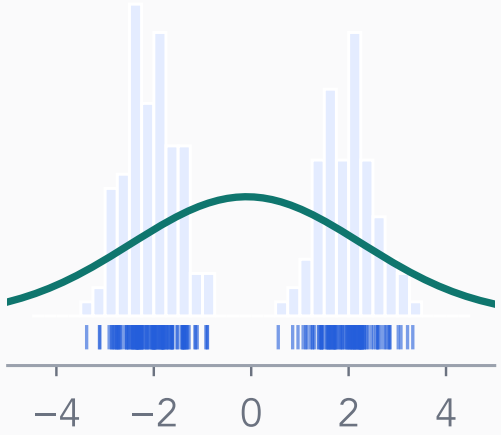

step 10: $\mu = 0.54,\ \sigma = 1.59$

loss $= 2.35$



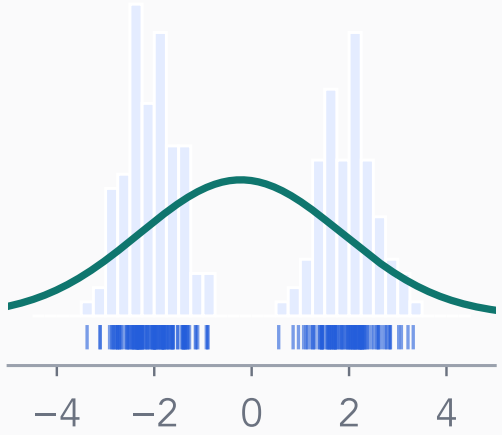

step 40: $\mu = -0.09,\ \sigma = 2.37$

loss $= 2.17$




step 300: $\mu = -0.22,\ \sigma = 2.08$

loss $= 2.15$




**The curve starts in the wrong place, widens to cover both clumps, and stops at the best single bump. The loss falls from 3.82 to 2.15.**

CHECKPOINT

# So far: learn and generate work; one bump is not enough

| | what we did | result |
|---|---|---|
| **Learn** | chose $\mu, \sigma$ to maximise $\log p_\theta(\text{data})$ | $\hat{\mu} = -0.22,\ \hat{\sigma} = 2.08$ |
| **Generate** | sampled $x = \hat{\mu} + \hat{\sigma}\varepsilon$ | many samples in the gap |

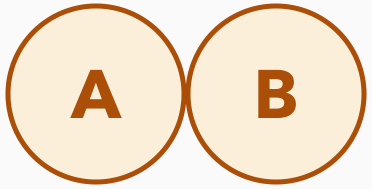
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| **Generate** | sampled $x = \hat{\mu} + \hat{\sigma}\varepsilon$ | many samples in the gap |
| **Fix** | a richer model that can make two clumps | next slide |

# Hidden choice first, then the number

**step 1 · hidden choice**



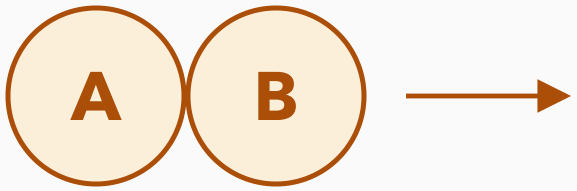

flip a fair coin:

$$p(z{=}\mathrm{A}) = p(z{=}\mathrm{B}) = \tfrac{1}{2}$$

> This is how we will generate: run the two steps. The coin is never written down in the data, so it is a hidden (latent) variable $z$.

# Hidden choice first, then the number

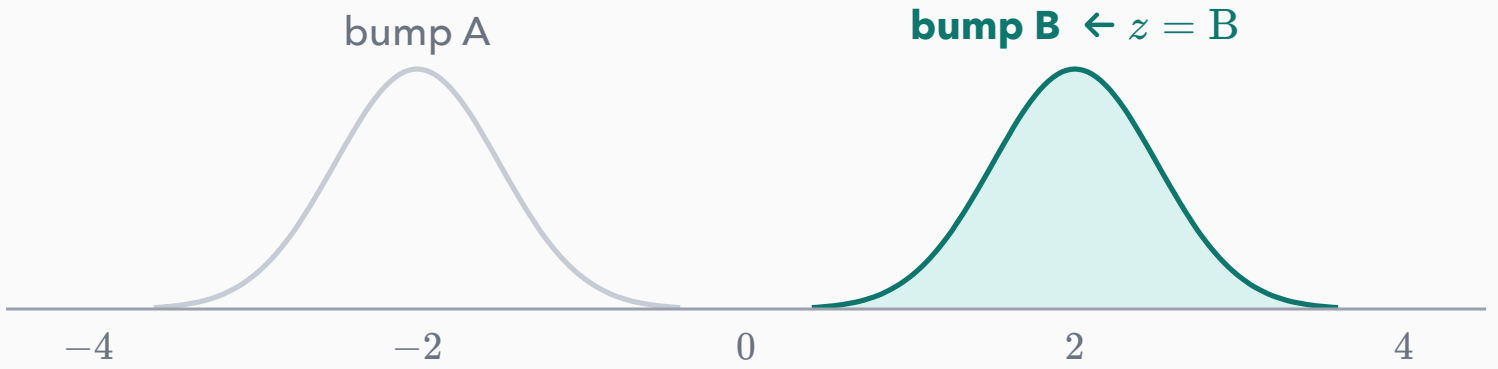
**step 1 · hidden choice**




flip a fair coin:

$p(z{=}\mathrm{A}) = p(z{=}\mathrm{B}) = \frac{1}{2}$

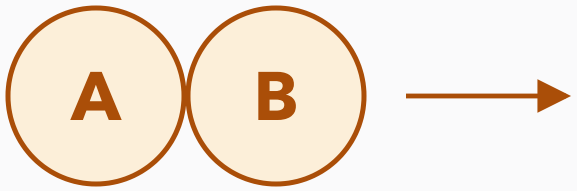
**step 2 · draw x from the chosen bump**




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# Hidden choice first, then the number

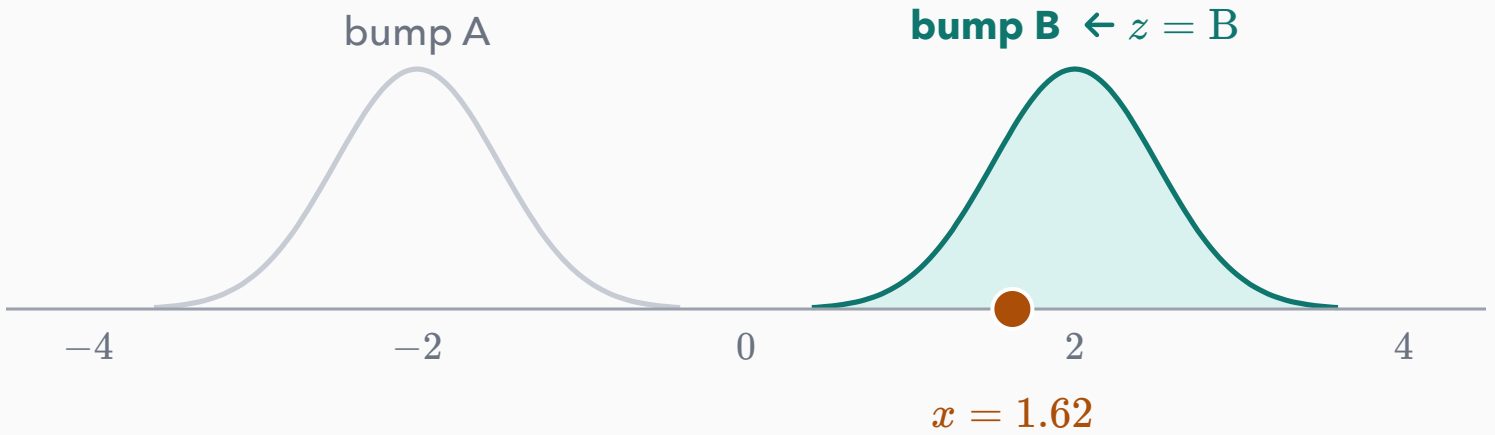
**step 1 · hidden choice**




flip a fair coin:

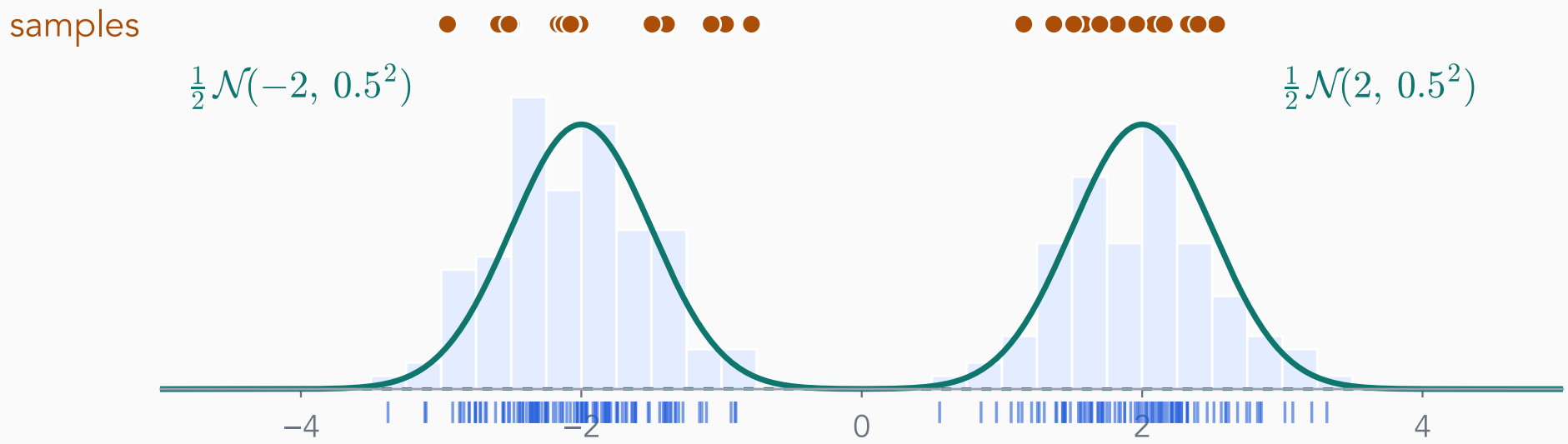
$p(z{=}\mathrm{A}) = p(z{=}\mathrm{B}) = \frac{1}{2}$

**step 2 · draw x from the chosen bump**




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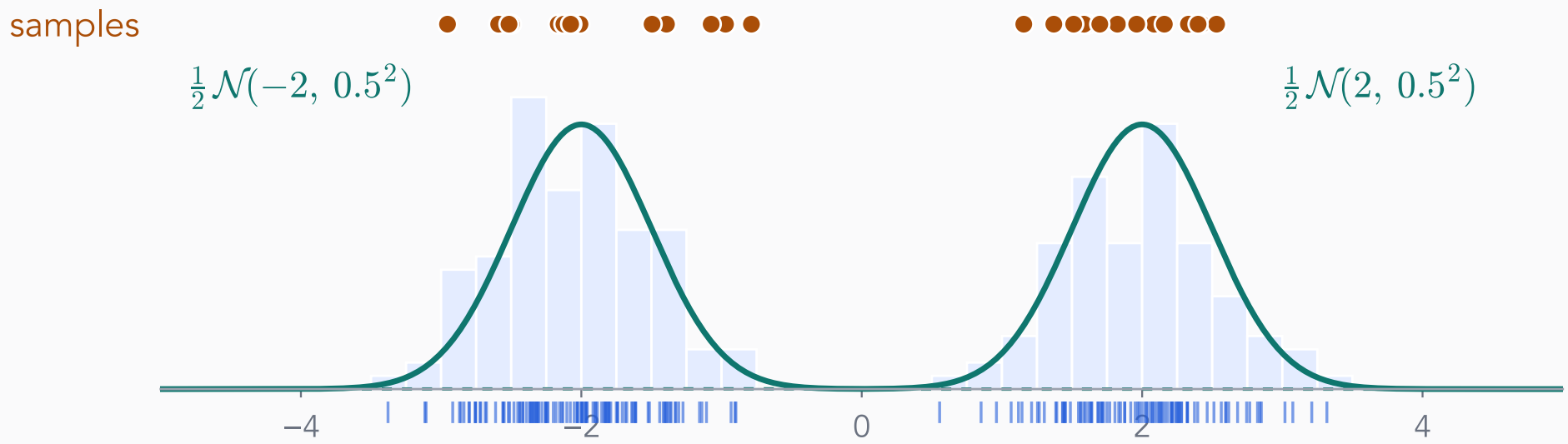
# To score $x$, add up every way it could have been made




$$p(x) = \underbrace{\tfrac{1}{2}\,\mathcal{N}(x;\,-2, 0.5^2)}_{\text{made by bump A}} + \underbrace{\tfrac{1}{2}\,\mathcal{N}(x;\,2, 0.5^2)}_{\text{made by bump B}} = \sum_z p(z)\,p(x \mid z)$$

> One hidden coin flip made the model rich enough: the samples now fall in the bumps, not in the gap.

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$$\text{at } x = 1.5: \quad \tfrac{1}{2}\cdot 0.000 + \tfrac{1}{2}\cdot 0.484 = 0.242$$

> One hidden coin flip made the model rich enough: the samples now fall in the bumps, not in the gap.

# Same loop, a richer `log_prob`

```
logits = torch.zeros(K, requires_grad=True)   # bump weights
means  = torch.randn(K, requires_grad=True)
log_sd = torch.zeros(K, requires_grad=True)

def log_p(x):                                 # log p(x), shape [N]
    comp = Normal(means, log_sd.exp()).log_prob(x[:, None])   # [N, K]
    return torch.logsumexp(log_softmax(logits) + comp, dim=1)

loss = -log_p(x).mean()        # then the same 3-line update
```

**Three parameters per bump:** a weight, a centre, a width. `softmax` turns the weights into probabilities that sum to 1.

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loss = -log_p(x).mean()        # then the same 3-line update
```

**Every number under every bump:** $\log p(x_i \mid z = k)$ for all $i$ and $k$, a table of $N \times K$ log-densities.

# Same loop, a richer `log_prob`

```
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    return torch.logsumexp(log_softmax(logits) + comp, dim=1)

loss = -log_p(x).mean()        # then the same 3-line update
```

**The sum over the hidden choice:** $\log \sum_k p(k)\, p(x \mid k)$. `logsumexp` does it without underflow.

# Same loop, a richer `log_prob`

```
logits = torch.zeros(K, requires_grad=True)   # bump weights
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    return torch.logsumexp(log_softmax(logits) + comp, dim=1)

loss = -log_p(x).mean()         # then the same 3-line update
```

**Training** is unchanged. With $K = 2$ it finds centres −2.06 and 1.98, widths 0.52 and 0.54.

STEP 2 · GENERATE, FROM THE TWO-BUMP FIT

# Sampling runs the story: choose a bump, then a number

```
weights = softmax(logits)
k = torch.multinomial(weights, 1000,
                      replacement=True)    # the coin
x_new = means[k] + log_sd.exp()[k] * torch.randn(1000)
```

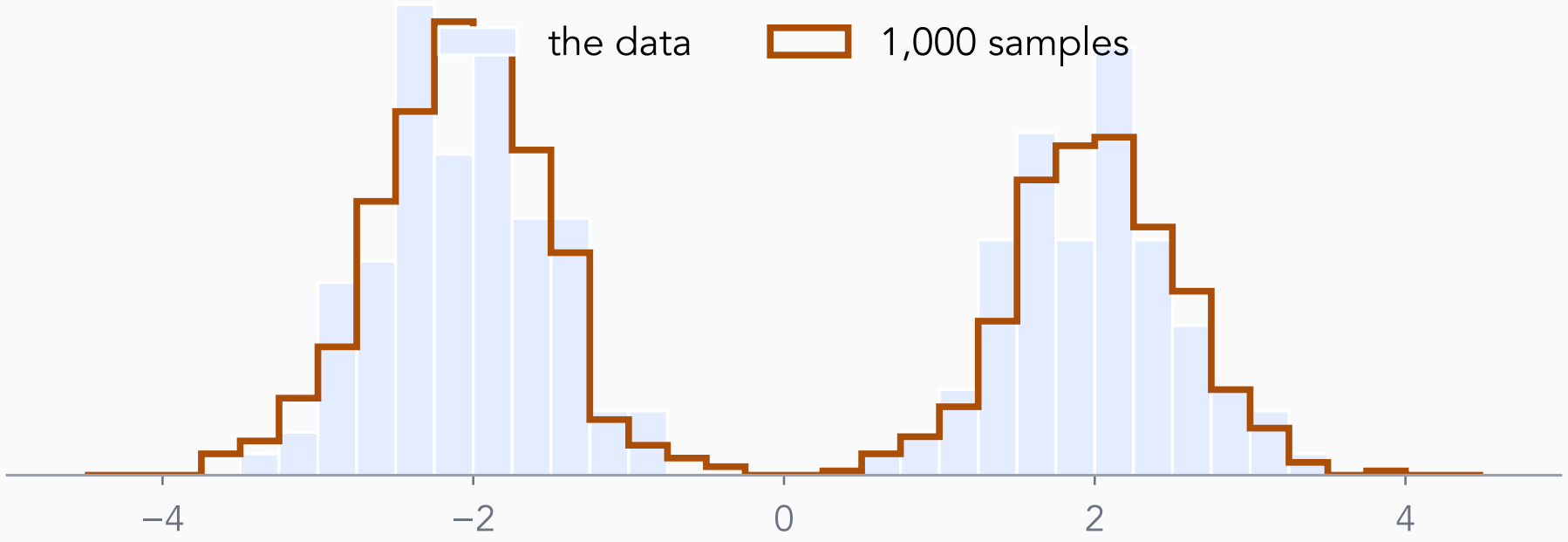
**Step 1:** pick a bump for each sample, with the learned weights.

STEP 2 · GENERATE, FROM THE TWO-BUMP FIT

# Sampling runs the story: choose a bump, then a number

```
weights = softmax(logits)
k = torch.multinomial(weights, 1000,
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```

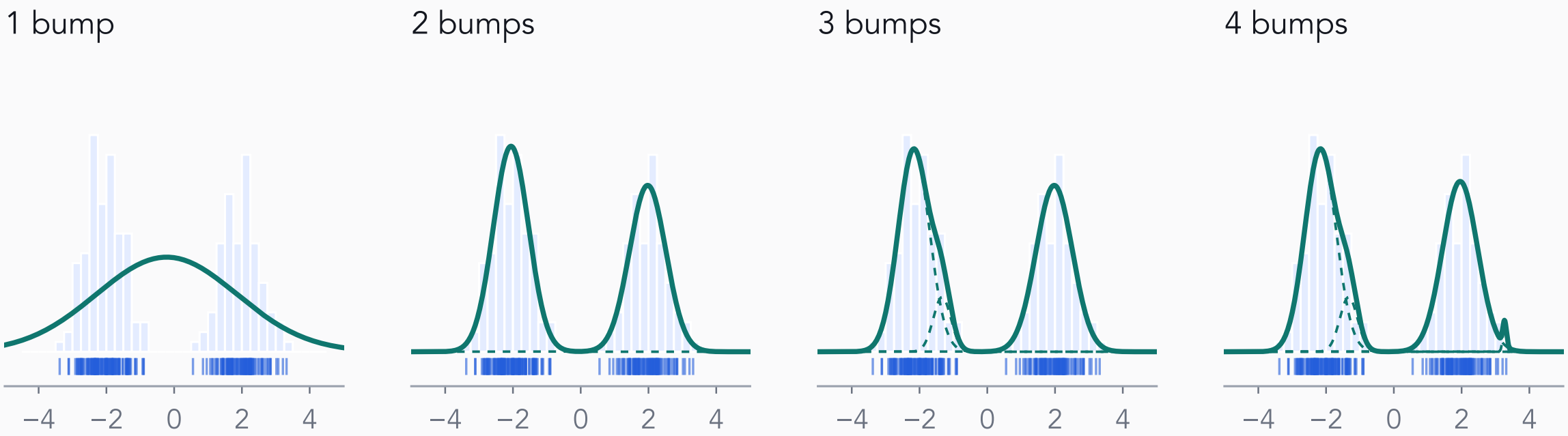
**Step 2:** draw a number from that bump: its centre plus its width times standard noise.




**1,000 new numbers that follow the data, and none of them is a copy of a training number.**

BUT HOW MANY BUMPS SHOULD WE FIT?

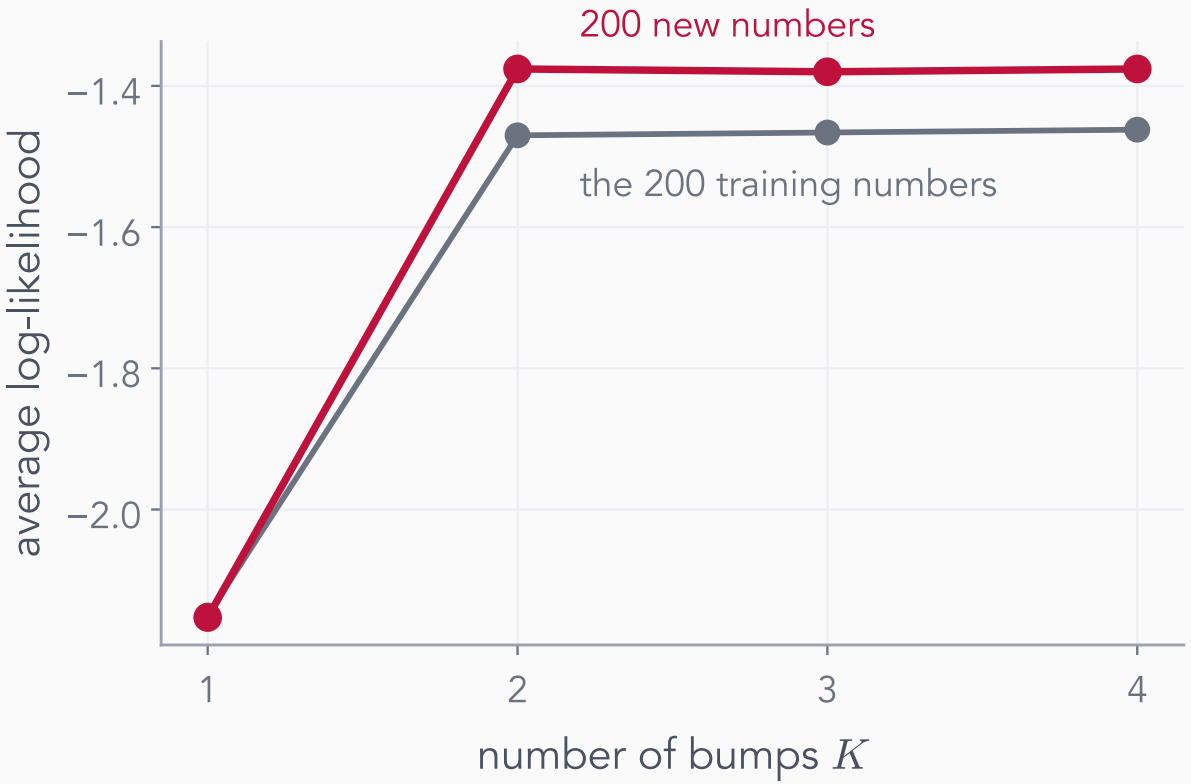
# More bumps always fit the training data better




**1, 2, 3, 4 bumps, each fitted with the same loop. The training score rises every time. So which one do we keep?**

SCORE ON DATA THE FIT HAS NEVER SEEN

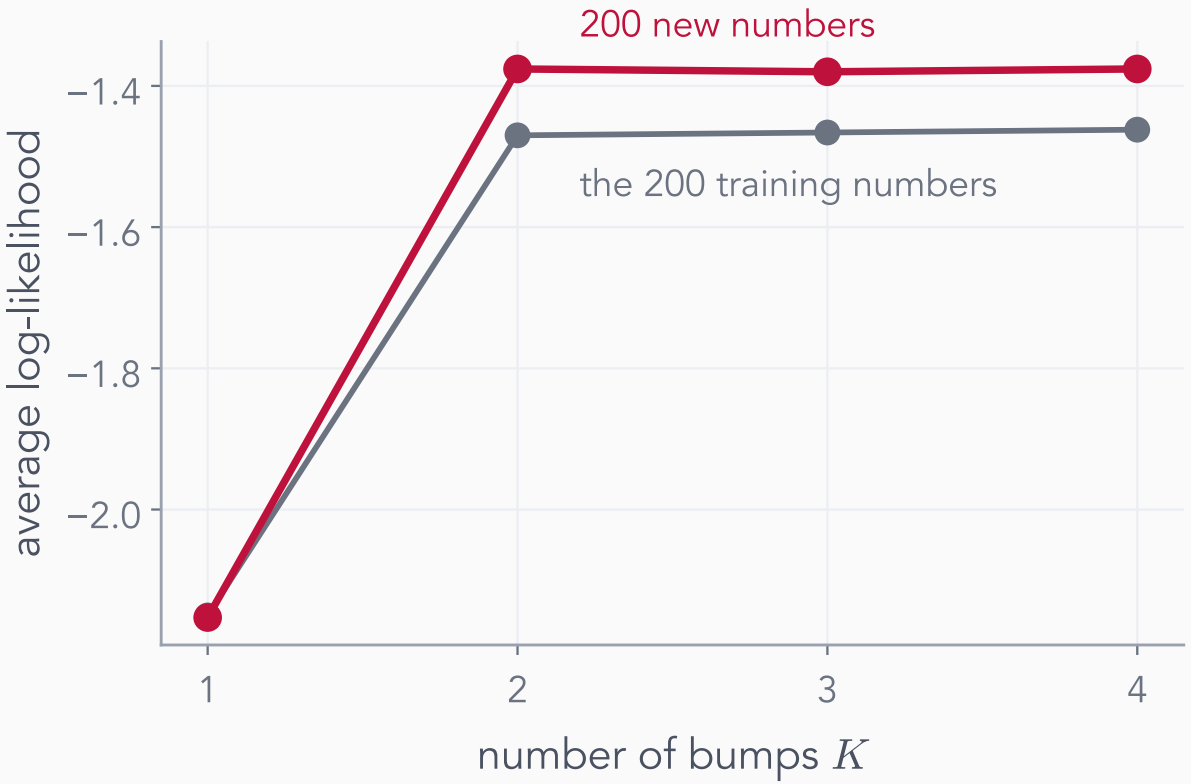
# New data decides: two bumps




Draw **200 new numbers** from the same hidden recipe and score each fit on them.

SCORE ON DATA THE FIT HAS NEVER SEEN

# New data decides: two bumps

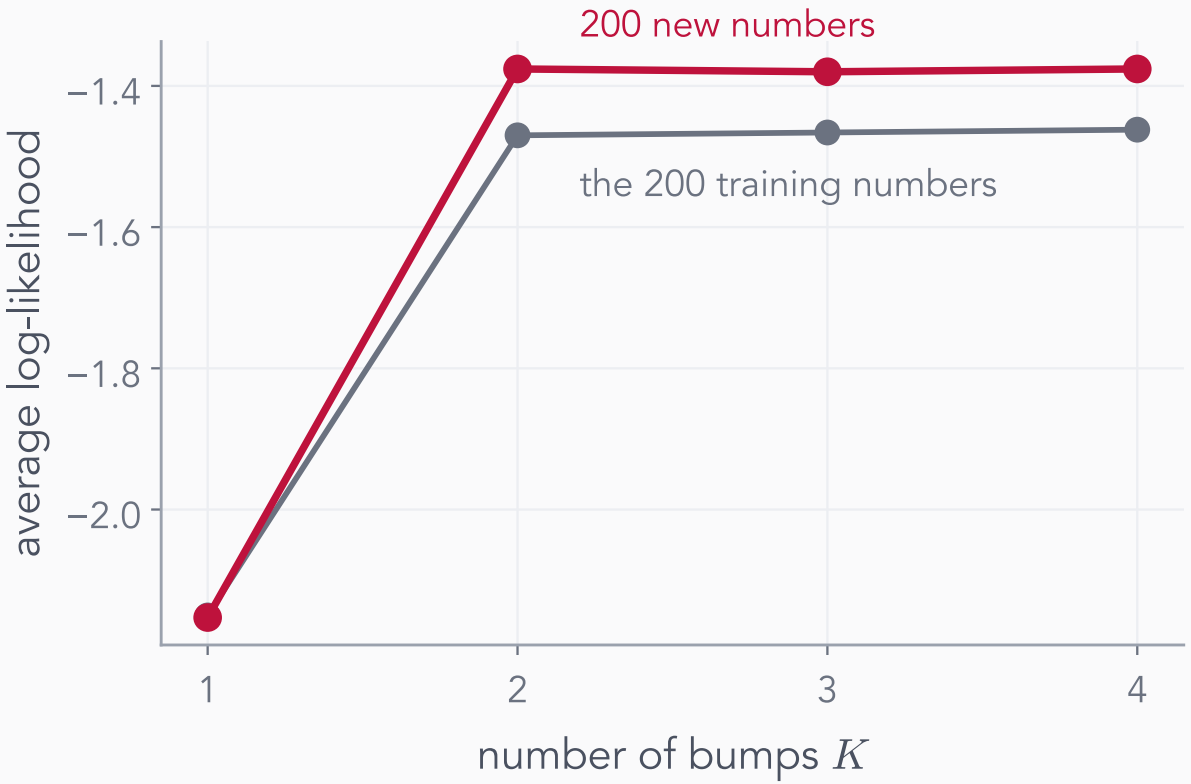



Draw **200 new numbers** from the same hidden recipe and score each fit on them.

1 → 2 bumps: $-2.153 \rightarrow -1.376$ per number. A real improvement.

SCORE ON DATA THE FIT HAS NEVER SEEN

# New data decides: two bumps




Draw **200 new numbers** from the same hidden recipe and score each fit on them.

1 → 2 bumps: $-2.153 \rightarrow -1.376$ per number. A real improvement.

3 or 4 bumps: the training score creeps up, the new-data score does not. The extra bumps fit noise in the 200 training numbers.

> **Choose the size of a model by how well it scores data it has not seen. That holds for bumps, and for networks.**

STEP 1 · LEARN, LOOKED AT CLOSELY

# Hidden in the fit: which bump made each point?

**If we knew the coin for every point**

fitting is easy: average the A-points for bump A's centre, the B-points for bump B's.

STEP 1 · LEARN, LOOKED AT CLOSELY

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**We don't, so ask for each point**

given this $x$, how probable is it that bump B made it? That is $p(\mathrm{B} \mid x)$.

STEP 1 · LEARN, LOOKED AT CLOSELY

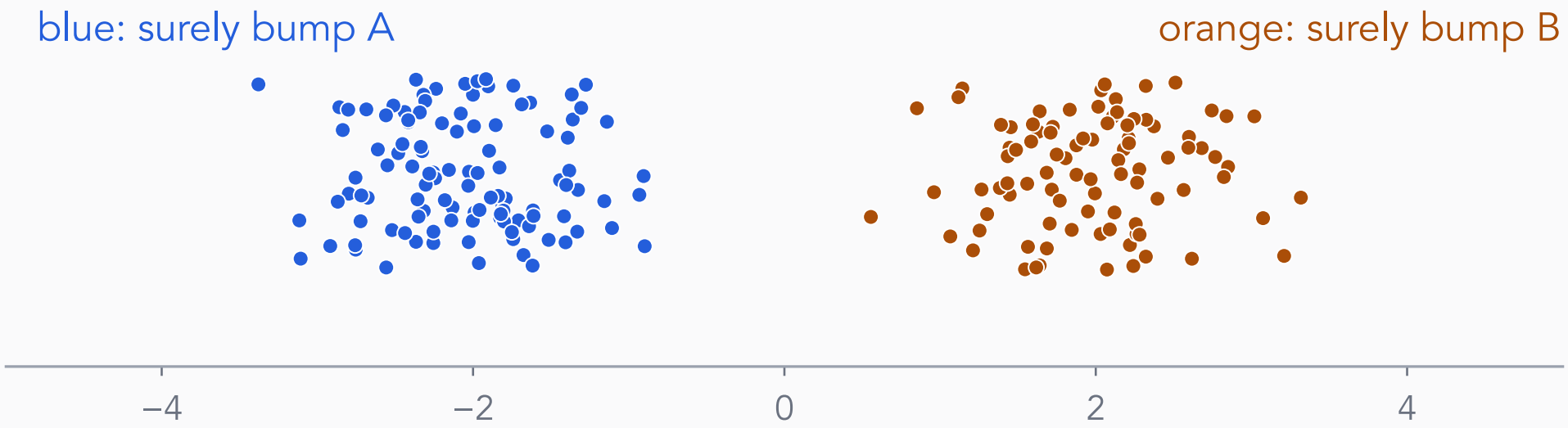
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**We don't, so ask for each point**

given this $x$, how probable is it that bump B made it? That is $p(\mathrm{B} \mid x)$.




**Running the story backwards, from a number to its likely hidden cause, is called inference. The gradient descent we just ran was doing it all along: four slides on, we will see where.**

THE BACKWARDS STEP, WORKED BY HAND

# You see $x = 0.1$. Which bump made it?

| bump $z$ | prior $p(z)$ | likelihood $p(x \mid z)$ | prior × likelihood | ÷ total = posterior |
|---|---|---|---|---|
| A, centre $-2$ | 0.5 | | | |
| B, centre $2$ | 0.5 | | | |

THE BACKWARDS STEP, WORKED BY HAND

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| bump $z$ | prior $p(z)$ | likelihood $p(x \mid z)$ | prior × likelihood | ÷ total = posterior |
|---|---|---|---|---|
| A, centre $-2$ | 0.5 | $1.18 \times 10^{-4}$ | | |
| B, centre $2$ | 0.5 | $5.84 \times 10^{-4}$ | | |

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| **total** $= p(x)$ | | | $3.51 \times 10^{-4}$ | |

THE BACKWARDS STEP, WORKED BY HAND

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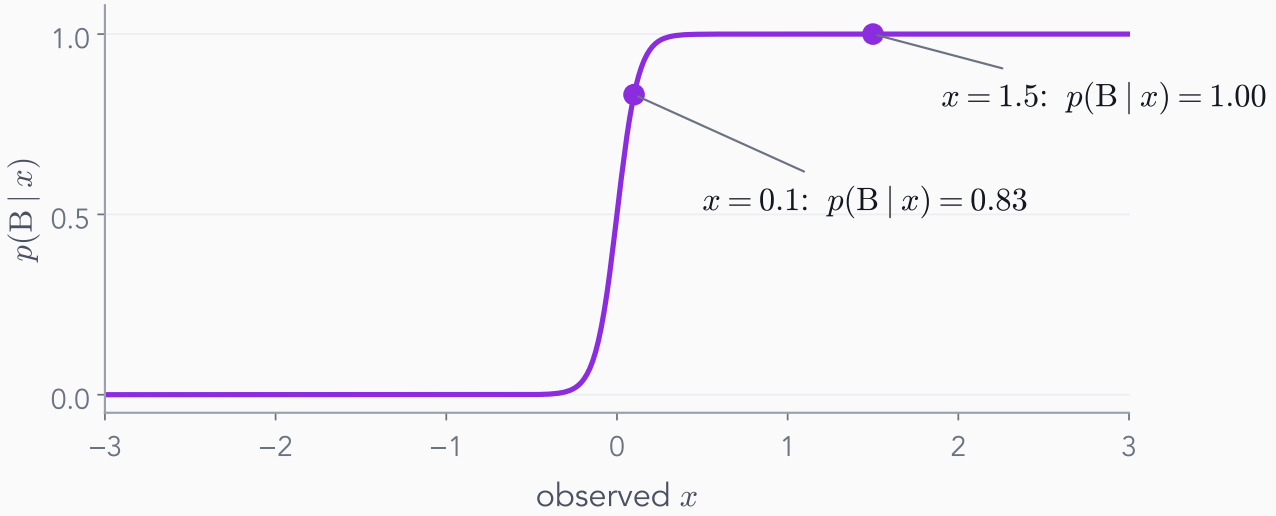
| bump $z$ | prior $p(z)$ | likelihood $p(x \mid z)$ | prior × likelihood | ÷ total = posterior |
|---|---|---|---|---|
| A, centre $-2$ | 0.5 | $1.18 \times 10^{-4}$ | $5.89 \times 10^{-5}$ | 0.168 |
| B, centre $2$ | 0.5 | $5.84 \times 10^{-4}$ | $2.92 \times 10^{-4}$ | **0.832** |
| **total** $= p(x)$ | | | $3.51 \times 10^{-4}$ | **1.000** |

$$p(\mathrm{B} \mid x) = \frac{p(\mathrm{B})\, p(x \mid \mathrm{B})}{p(x)} = 0.832$$

> **Bayes' rule is just this table: multiply, then divide by the total. Note the total is exactly the $p(x)$ of the two-bump formula.**

DO IT FOR EVERY $x$

# The backwards step, as a curve

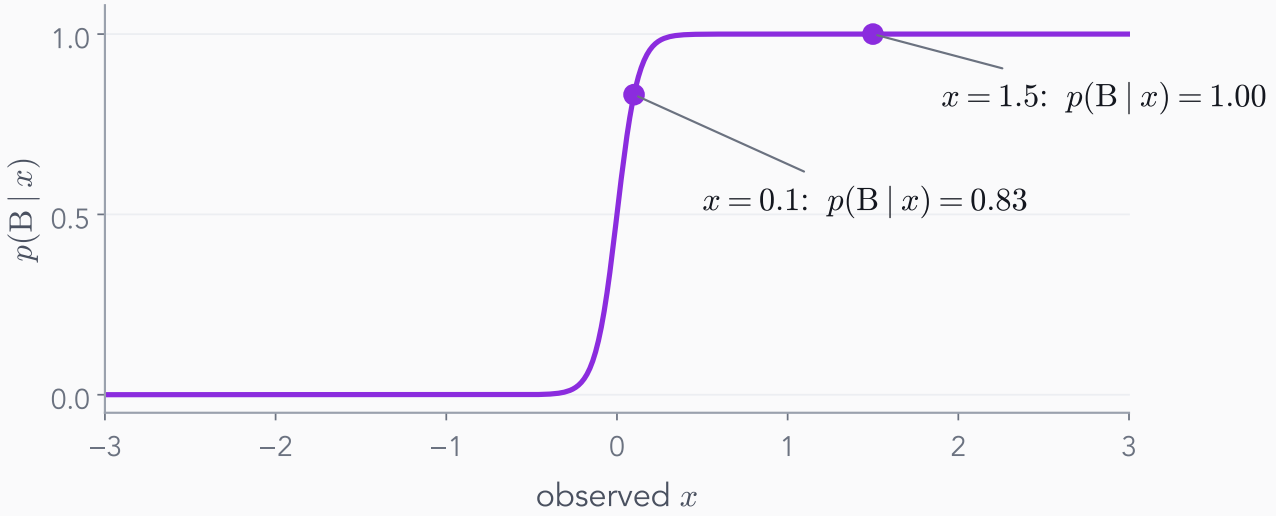



Repeat the table for every $x$ and plot $p(\mathrm{B} \mid x)$.

**These are the colours on the 200 points two slides ago.**

DO IT FOR EVERY $x$

# The backwards step, as a curve



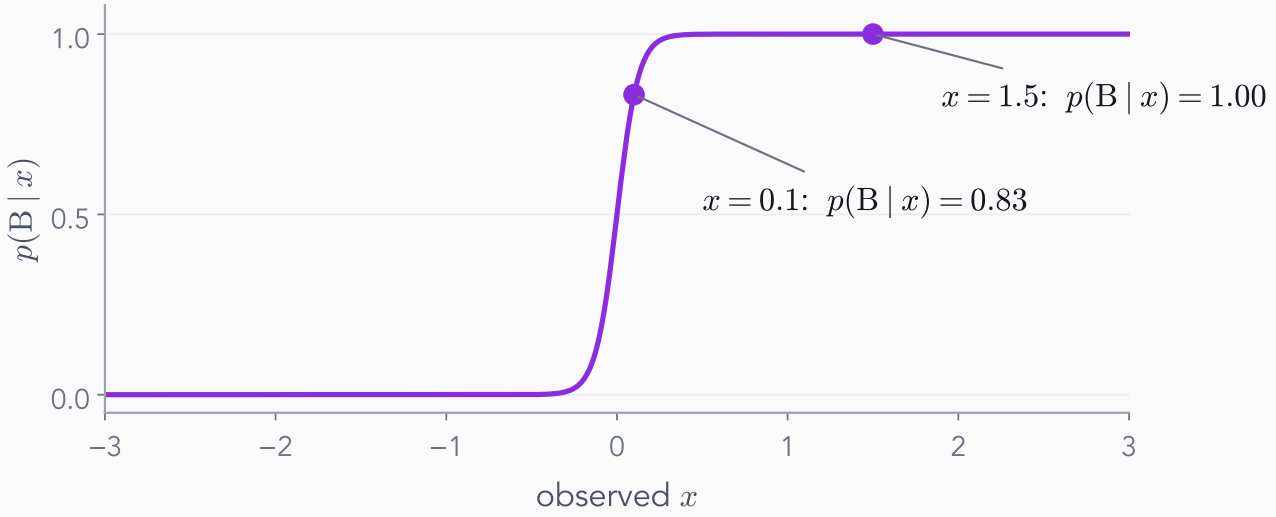

Repeat the table for every $x$ and plot $p(\mathrm{B} \mid x)$.

Far right: certainly B. Far left: certainly A. Near 0: unsure.

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DO IT FOR EVERY $x$

# The backwards step, as a curve




Repeat the table for every $x$ and plot $p(\mathrm{B} \mid x)$.

Far right: certainly B. Far left: certainly A. Near 0: unsure.

For these two bumps the curve is exactly $\mathrm{sigmoid}(16x)$: a logistic regression in disguise.

**These are the colours on the 200 points two slides ago.**

THE BACKWARDS STEP, IN ONE LINE

# Posterior = softmax of (log prior + log likelihood)

```
log_joint = log_softmax(logits) + Normal(means, log_sd.exp()).log_prob(x)   # [K]
posterior = softmax(log_joint)                  # divide by the total: Bayes
```

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log_joint = log_softmax(logits) + Normal(means, log_sd.exp()).log_prob(x)   # [K]
posterior = softmax(log_joint)                    # divide by the total: Bayes
```

With the fitted bumps, at $x = 0.1$: $p(\mathrm{A} \mid x) = 0.094$, $p(\mathrm{B} \mid x) = 0.906$. (With the true recipe it was 0.832: the fitted bumps are slightly different.)

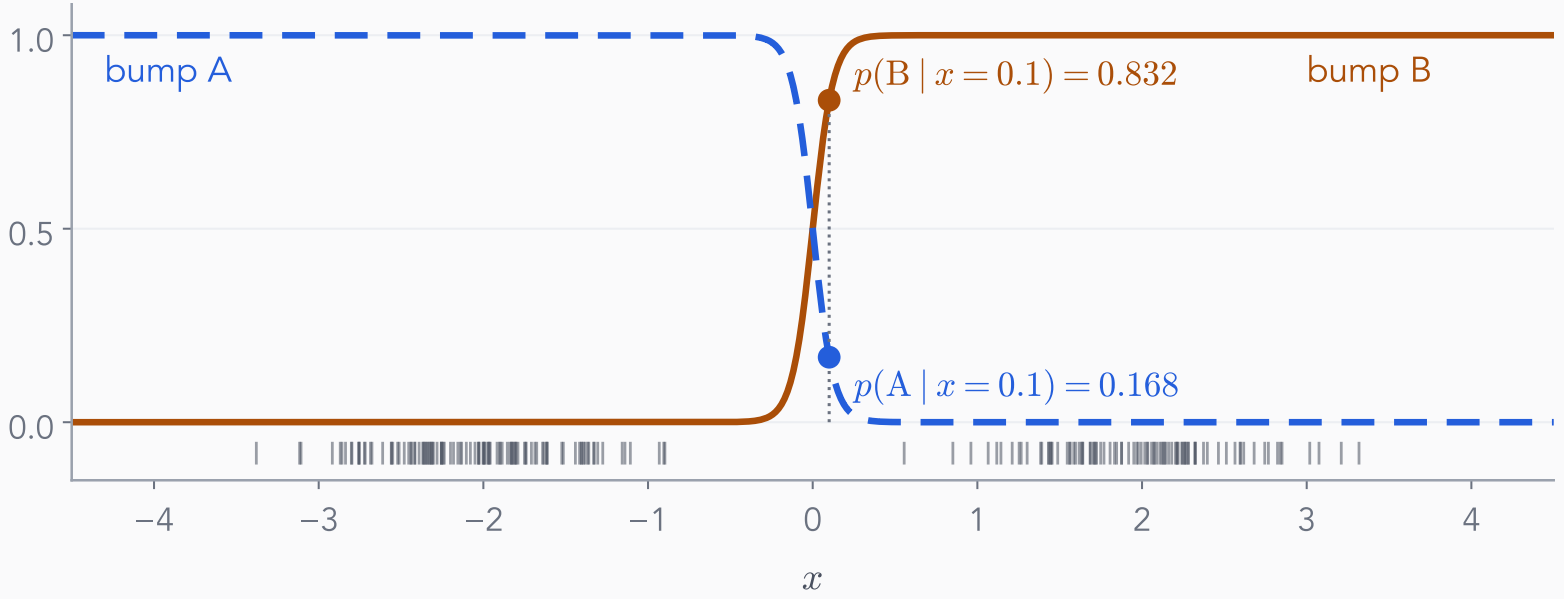
**Softmax is "exponentiate, then divide by the total": exactly the last column of the Bayes table.**

# The gradient of $\log p(x)$ is weighted by the posterior

$$\frac{\partial \log p(x)}{\partial \mu_k} = p(k \mid x) \cdot \frac{x - \mu_k}{\sigma_k^2}$$

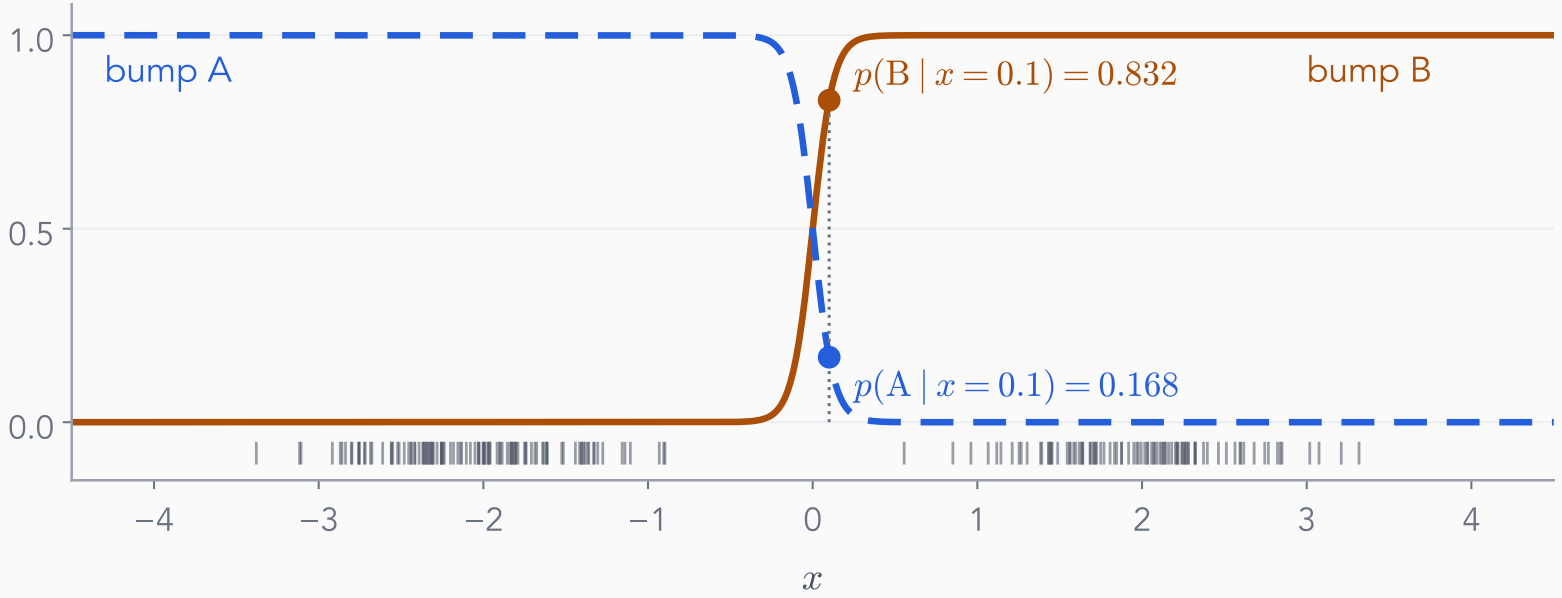
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At $x = 0.1$: bump B's centre is pulled by $0.832 \times (0.1 - 2)/0.25 = -6.32$, bump A's by $0.168 \times (0.1 + 2)/0.25 = 1.41$.

> **Each point pulls each bump in proportion to how likely that bump made it. Learning needs the posterior for every point: a 2-row table here, an intractable patch for images.**

FOUR WORDS FOR THE REST OF THE LECTURE

# Prior, likelihood, evidence, posterior

| name | symbol | question it answers | at $x = 0.1$ |
|---|---|---|---|
| **prior** | $p(z)$ | what do we believe about $z$ before seeing $x$? | 0.5, 0.5 |
| | | | |
| | | | |
| | | | |

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| **likelihood** | $p(x \mid z)$ | if $z$ were known, how probable is this $x$? | $5.84 \times 10^{-4}$ for B |

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| **evidence** | $p(x)$ | how probable is $x$ overall, summing over $z$? | $3.51 \times 10^{-4}$ |

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| **evidence** | $p(x)$ | how probable is $x$ overall, summing over $z$? | $3.51 \times 10^{-4}$ |
| **posterior** | $p(z \mid x)$ | having seen $x$, which $z$ explains it? | B: 0.832 |

**Generating uses the prior and the likelihood. Learning needs the evidence and the posterior.**

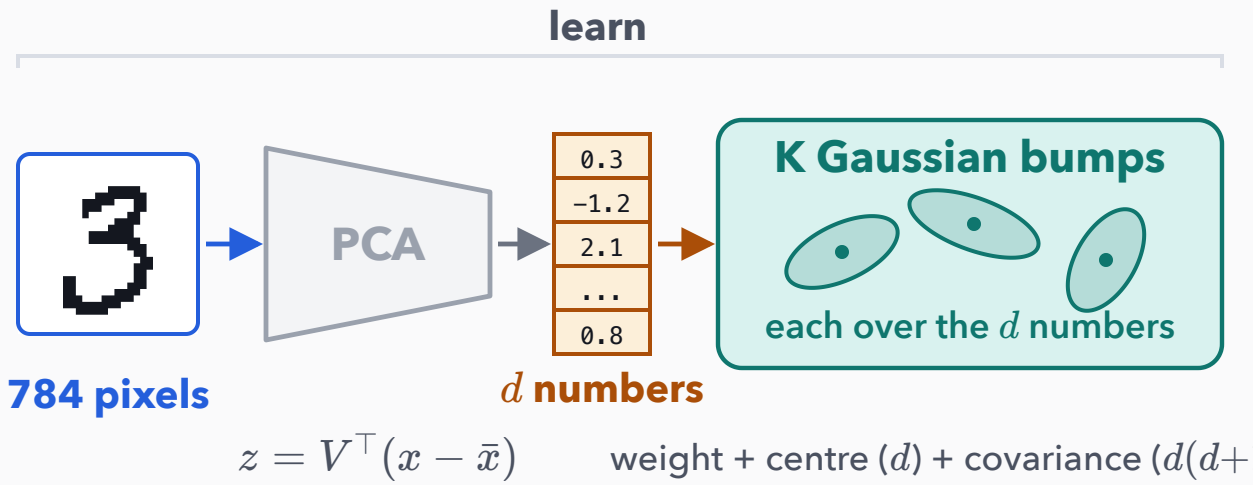
PART 1, SECOND HALF

# The same recipe, on real images

Learn a distribution, then sample from it: can Gaussian bumps generate handwritten digits? How far does this simple toolkit get us, and where does it run out?

THE PLAN, AS A PICTURE

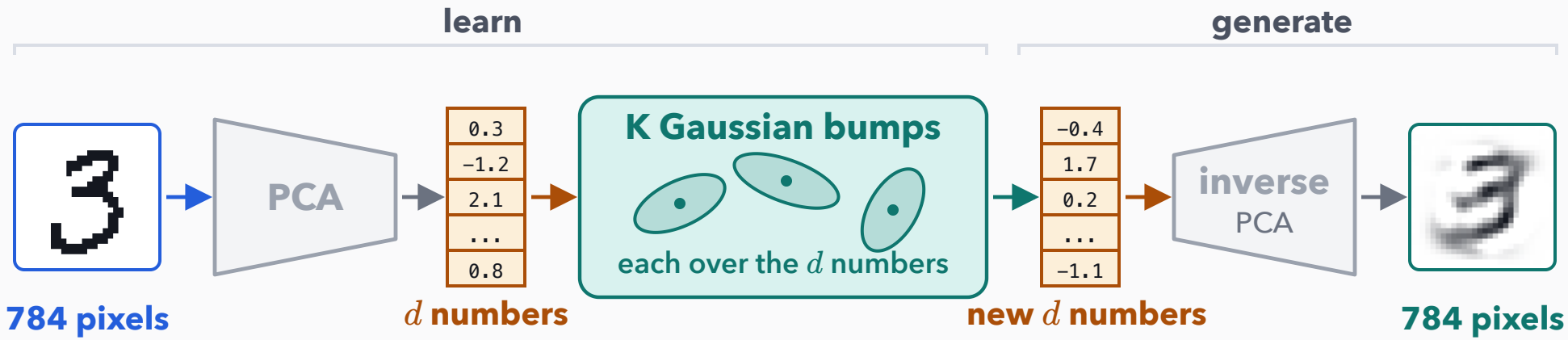
# Squeeze, fit bumps, sample, un-squeeze




$z = V^\top (x - \bar{x})$ weight + centre ($d$) + covariance ($d(d+1)/2$)

THE PLAN, AS A PICTURE

# Squeeze, fit bumps, sample, un-squeeze




Two questions before any code: each bump is a Gaussian over **what**? And how many bumps?

WHICH DIMENSION IS EACH GAUSSIAN?

# One bump, three possible sizes

| each bump is a Gaussian over | centre | covariance | numbers per bump, with its weight |
|---|---|---|---|
| 1 number (the toy) | 1 | 1 (a width) | 3 |
| | | | |
| | | | |

WHICH DIMENSION IS EACH GAUSSIAN?

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| each bump is a Gaussian over | centre | covariance | numbers per bump, with its weight |
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| 784 pixels, every pixel on its own | 784 | 784 widths | 1,569 |



Samples from one 784-pixel Gaussian with independent pixels: each pixel jitters on its own, so you get static around the average digit. Strokes need pixels that move **together**.

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| 784 pixels, with all pixel pairs | 784 | 307,720 | 308,505 |
| **50 PCA numbers (what we fit)** | **50** | **1,275** | **1,326** |



Samples from one 784-pixel Gaussian with independent pixels: each pixel jitters on its own, so you get static around the average digit. Strokes need pixels that move **together**.

> **Keep the covariance, shrink the dimension: a full Gaussian over 50 numbers costs about 230 times less than one over 784 pixels.**

# PCA and inverse PCA: generation needs both

**PCA**, image to $d$ numbers: coordinates along the $d$ directions in which digits vary most.

$$\underbrace{z}_{d} = \underbrace{V^\top}_{d \times 784} (\underbrace{x - \bar{x}}_{784})$$

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**Inverse PCA**, $d$ numbers back to an image: add the directions back onto the mean digit.

$$\underbrace{\hat{x}}_{784} = \bar{x} + \underbrace{V}_{784 \times d} \underbrace{z}_{d}$$

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top: test digits, our 3 among them · bottom: PCA to 50 numbers, then inverse PCA

> **The bumps only ever see $z$. Every generated sample is $d$ numbers until the inverse PCA turns it into pixels. With $d = 50$, 82.5% of the variation survives the round trip.**

SQUEEZE ALL THE WAY TO 2 NUMBERS

# In 2-D you can see every bump

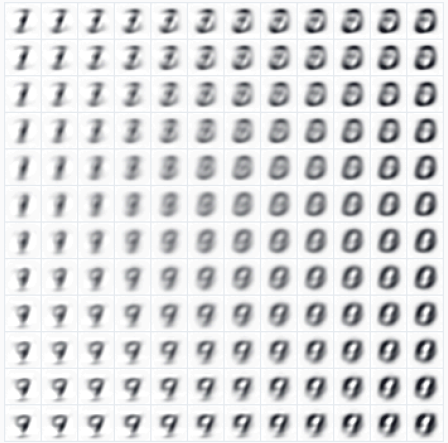



4,000 test digits by their 2 PCA numbers, coloured by label (the model never saw labels). Black: 10 fitted bumps, 1 and 2 widths. Red ×: 24 samples.

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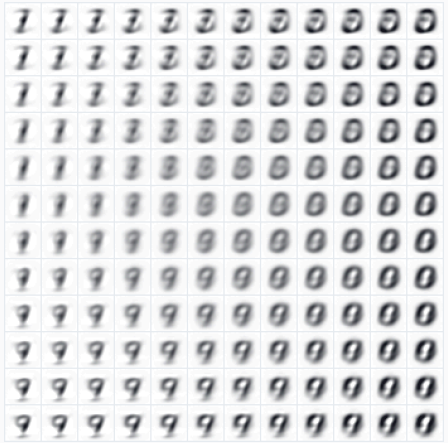
Inverse PCA of a grid over the plane: every point is a blend of a few blurry shapes.

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Inverse PCA of a grid over the plane: every point is a blend of a few blurry shapes.



the 24 red × after inverse PCA

**2 numbers keep only 16.8% of the variation. Every bump is visible, every sample is a ghost: interpretable is not the same as good.**

THE TWO-BUMP CODE, SCALED UP

# PCA, fit $K$ bumps, sample, inverse PCA

```
x = mnist_train.flatten(1)                  # [60000, 784], grey levels
mean = x.mean(0)
_, _, Vt = torch.linalg.svd(x - mean, full_matrices=False)
V = Vt[:d].T                               # [784, d]
z = (x - mean) @ V                          # PCA: [60000, d]

gmm = MixtureSameFamily(Categorical(logits=logits),
                        MultivariateNormal(means, scale_tril=L))
loss = -gmm.log_prob(z_batch).mean()        # the same 3-line update

z_new = gmm.sample((24,))                   # step 2: generate
x_new = (mean + z_new @ V.T).clamp(0, 1)    # inverse PCA: [24, 784]
```

**PCA:** one SVD gives the directions; each image becomes $d$ numbers.

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```

**The mixture:** `logits` [K], `means` [K, d], `L` [K, d, d] lower-triangular, so each covariance $LL^\top$ is always valid.

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```

**Learning:** the loop that fitted the two bumps, unchanged: minus the average log-probability, Adam.

THE TWO-BUMP CODE, SCALED UP

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```

**Generating:** pick a bump by its weight, draw $d$ numbers from it, inverse PCA to 784 pixels.

TRY IT: EVERY COMBINATION BELOW WAS FITTED

# Slide the dimension and the number of bumps

PCA numbers $d$

**50**

bumps $K$

**100**

variation kept by $d$ numbers: **82.5%**
parameters: **132,599**
score on new images: **−16.5**
new images prefer $K$ = **100** at this $d$



test digits

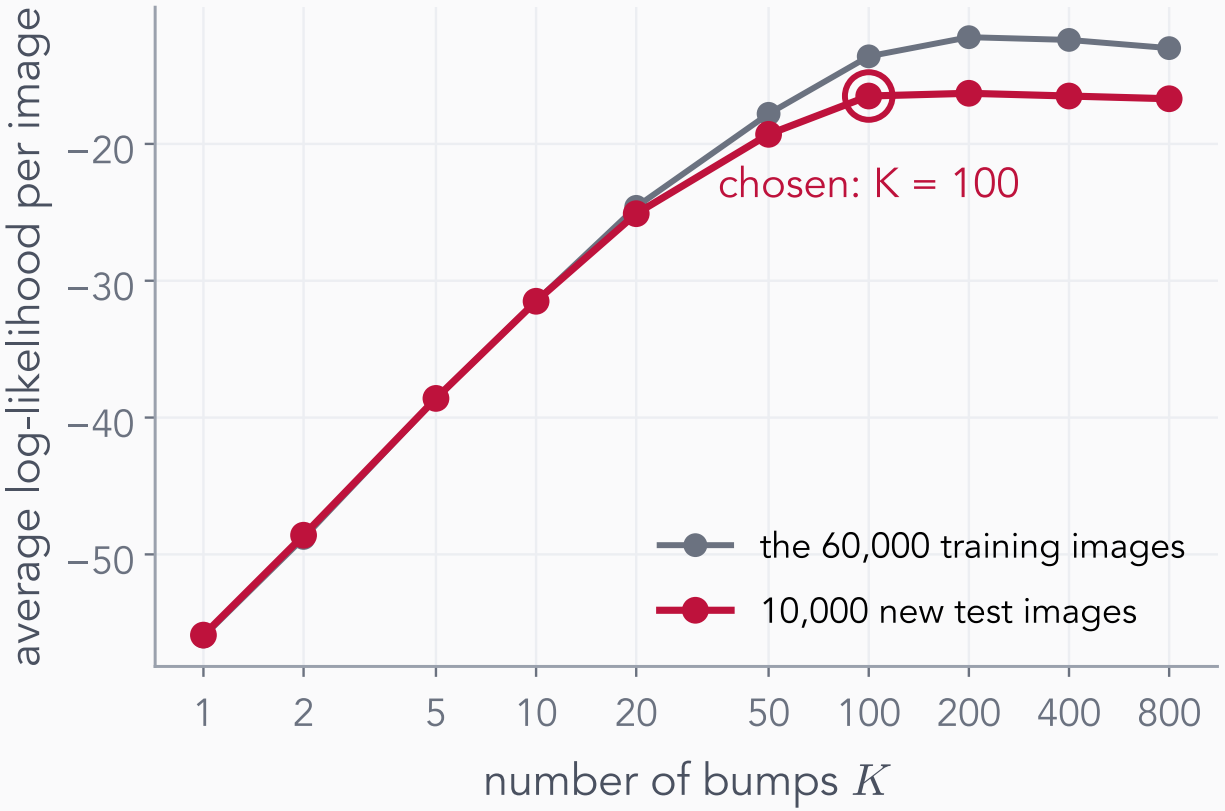


after PCA to $d$ numbers and back



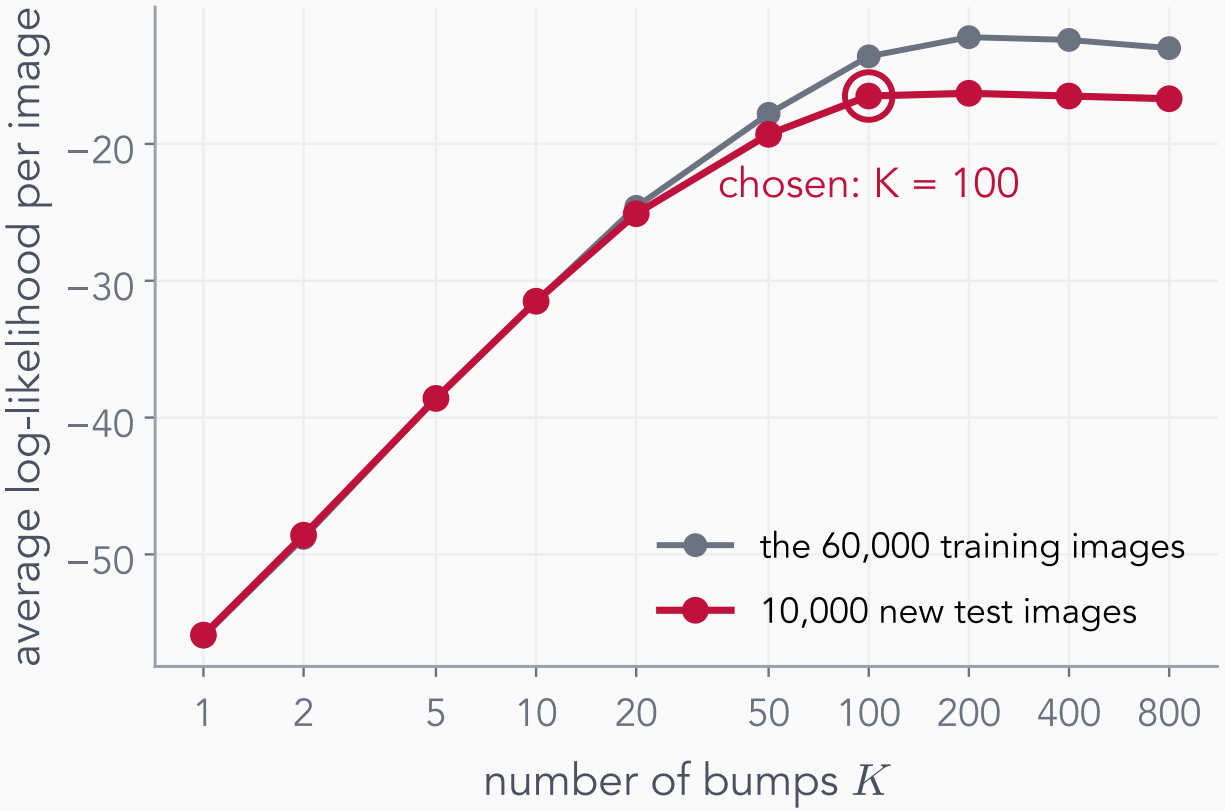
24 samples from the $K$-bump fit, after inverse PCA

# The toy's rule: score on images the model has not seen




$d = 50$. Fit $K$ = 1, 2, 5, 10, 20, 50, 100, 200, 400, 800 bumps to the 60,000 training images.

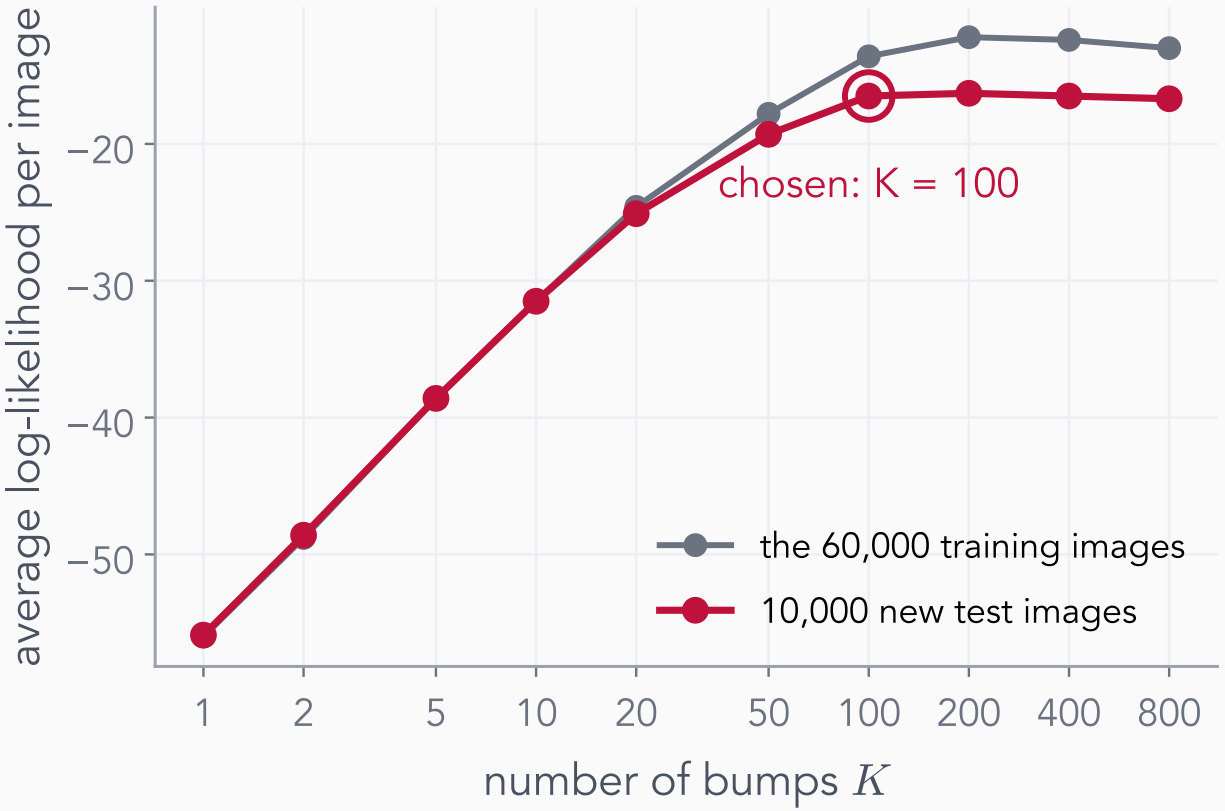
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Training images: more bumps fit them better, as with the toy.

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Training images: more bumps fit them better, as with the toy.

10,000 new test images: the score stops improving at $K = 100$. Doubling to 200 bumps changes the score by +0.2 nats. At 800 even the training score drops: that many bumps are more than 2,000 steps can fit.

**Pick the smallest $K$ that does as well as any on new images: $K = 100$, 132,599 parameters, each bump a Gaussian over 50 numbers.**

# AIC and BIC: charge for every parameter

$$\mathrm{AIC} = 2p - 2\log L \qquad\qquad \mathrm{BIC} = p\,\log N - 2\log L$$

$\log L$: the training log-likelihood, summed over all $N$ = 60,000 images · $p$: the number of parameters · smaller is better

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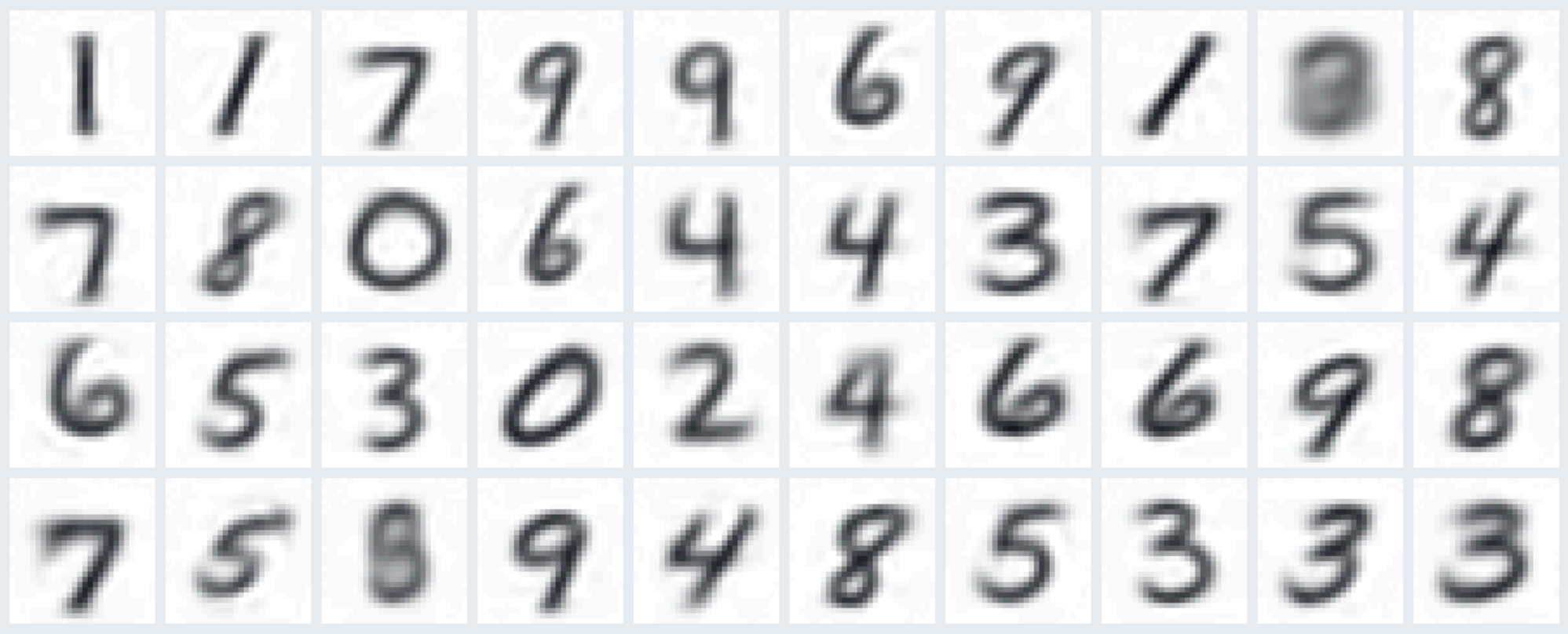
$\log L$: the training log-likelihood, summed over all $N$ = 60,000 images · $p$: the number of parameters · smaller is better

| rule | needs | price per parameter | picks |
|---|---|---|---|
| score on new images | 10,000 extra images | none, it is measured | $K = 100$ |
| AIC | training data only | 2 | $K = 100$ |
| BIC | training data only | $\log 60{,}000 = 11.0$ | $K = 50$ |

**New images and AIC agree on $K = 100$; BIC, which charges more per parameter, keeps only 50. With plenty of data, measure on new data; with little, AIC or BIC.**

WHAT THE BUMPS LEARNED

# Each bump's centre is a style of digit



the centres of the 100-bump fit, after inverse PCA, heaviest bump first (the heaviest has weight 3.9%)

> **The hidden coin now has 100 sides, one prototype each, found without the labels. The backwards step $p(z \mid x)$ is still just a 100-row table.**

STEP 2 · GENERATE

# Samples, as the number of bumps grows

1 bump: a single Gaussian over the 50 numbers of every digit



STEP 2 · GENERATE

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10 bumps



STEP 2 · GENERATE

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1 bump: a single Gaussian over the 50 numbers of every digit



10 bumps



100 bumps, the size chosen on new images



OR SKIP PCA: MODEL THE 784 PIXELS DIRECTLY

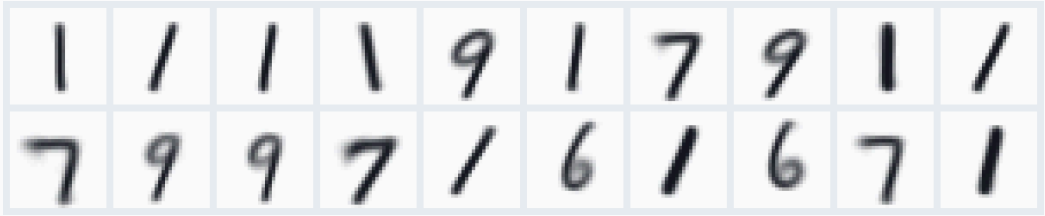
# A mixture of Bernoullis: each bump is 784 ink probabilities

$$\log p(x \mid k) = \sum_{i=1}^{784} \Big[ x_i \log \pi_{ki} + (1 - x_i) \log(1 - \pi_{ki}) \Big]$$

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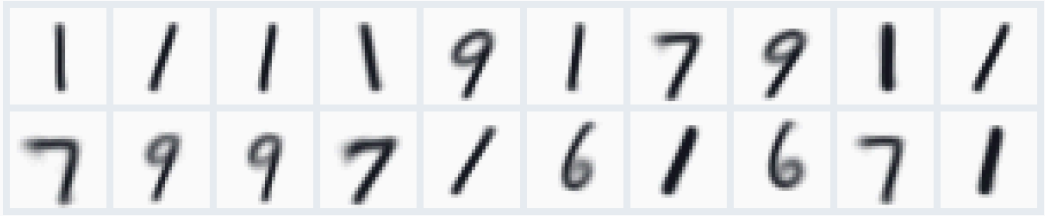


the 20 heaviest of 200 bumps: $\pi_k$, ink probability per pixel

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the 20 heaviest of 200 bumps: $\pi_k$, ink probability per pixel



samples: pick a bump, flip each pixel's coin

**Inside a bump every pixel is on its own, so samples are speckled; only the choice of bump ties pixels together. Same lesson as the 784-pixel Gaussian.**

AN HONEST LOOK

# New digits, not copies, and not yet sharp



top: 8 samples · bottom: the closest of the 60,000 training images

AN HONEST LOOK

# New digits, not copies, and not yet sharp



top: 8 samples · bottom: the closest of the 60,000 training images

**What worked**

The Part 1 loop, unchanged, learns a distribution over images and samples digit-like shapes, close to real digits but not copies of them.

AN HONEST LOOK

# New digits, not copies, and not yet sharp



top: 8 samples · bottom: the closest of the 60,000 training images

### What worked

The Part 1 loop, unchanged, learns a distribution over images and samples digit-like shapes, close to real digits but not copies of them.

### What did not

Blur from the straight-line squeeze, and each bump can only stretch its template along fixed directions, not bend a stroke.

WHY WE CANNOT STOP HERE

# This recipe does not scale

| image | numbers | covariance per bump |
|---|---|---|
| MNIST, 28 × 28 grey | 784 | 307,720 |
| | | |
| | | |
| | | |

WHY WE CANNOT STOP HERE

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| image | numbers | covariance per bump |
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| 32 × 32 colour | 3,072 | 4.7 million |
| 64 × 64 colour face | 12,288 | 75.5 million |
| 256 × 256 colour photo | 196,608 | 19.3 billion |

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**Squeeze first?** A straight-line squeeze needs more and more numbers for richer images, and still blurs.

**More bumps?** Each bump is one template plus a Gaussian wiggle. Every pose, style and lighting needs its own templates.

**Nothing is shared:** the bump for a slanted 7 learns nothing from the bump for an upright 7.

> **We need a squeeze that can bend, learned by a network, and parts that are shared. Part 2: the autoencoder.**

END OF PART 1

# What we did, and what comes next

## WHAT WE DID

1. **Learn** = choose parameters that make the data probable.
2. **Generate** = sample from the learned model.
3. One bump was too simple; a hidden coin $z$ fixed it.
4. Scoring $x$ sums over $z$; learning needs the backwards step $p(z \mid x)$.
5. In code: one loop, `loss = -log_p(x).mean()`; choose the size on new data.
6. On MNIST: PCA to 50 numbers, 100 Gaussian bumps over them, inverse PCA: digit-like samples, but blurry, and the recipe does not scale.

END OF PART 1

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### NEXT: IMAGES

PCA squeezed each image along straight lines. Next, learn the squeeze with a network: encode each image into a few numbers and rebuild it. That is the autoencoder.

Part 3 returns to this probabilistic story, with a code $z$ in place of the coin.

PART 2 OF 7

# First attempt: an autoencoder

Part 1 modelled single numbers; an image has 784. Before any probability, a simpler idea: squeeze each image into a few numbers with an **encoder**, rebuild it with a **decoder**, and see how far that gets us towards generating.


✓ Distributions and fitting

2 **Autoencoders**

3 The probabilistic view

4 KL divergence

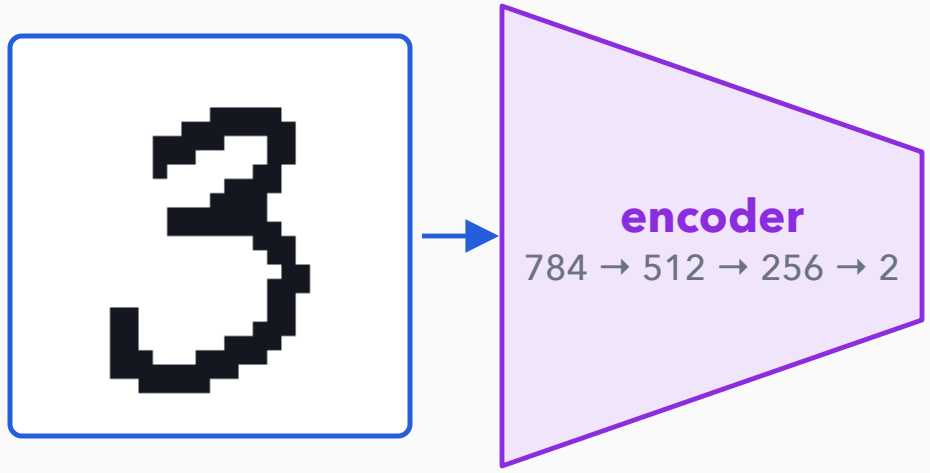
5 Variational inference and the ELBO

6 Reparameterisation

7 The VAE on MNIST

SQUEEZE, THEN REBUILD

# An autoencoder on our 3

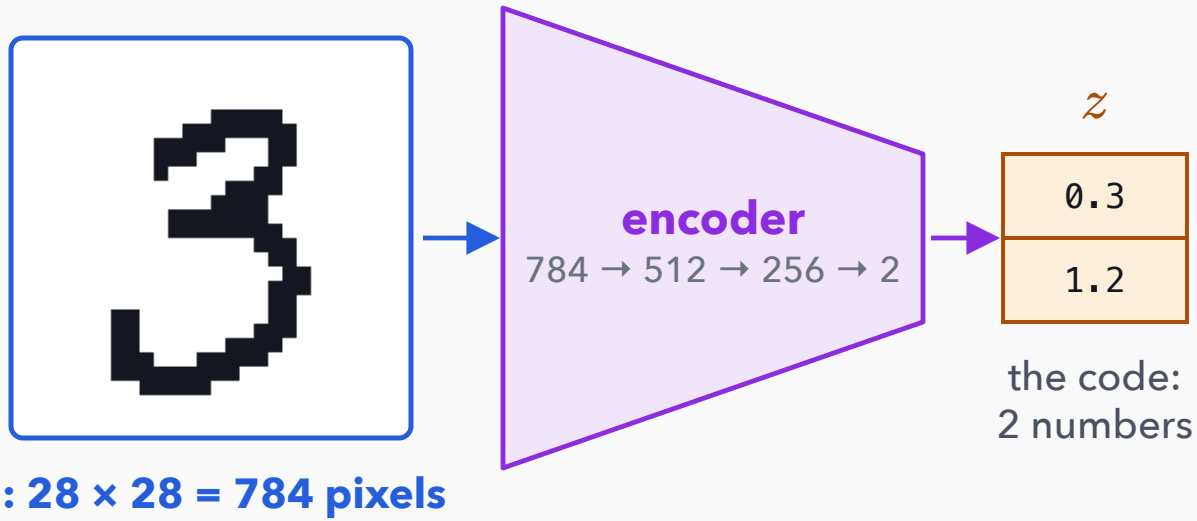



$x$: 28 × 28 = 784 pixels

> No labels needed: the image is its own target. The 2 numbers in the middle are all the decoder gets.

SQUEEZE, THEN REBUILD

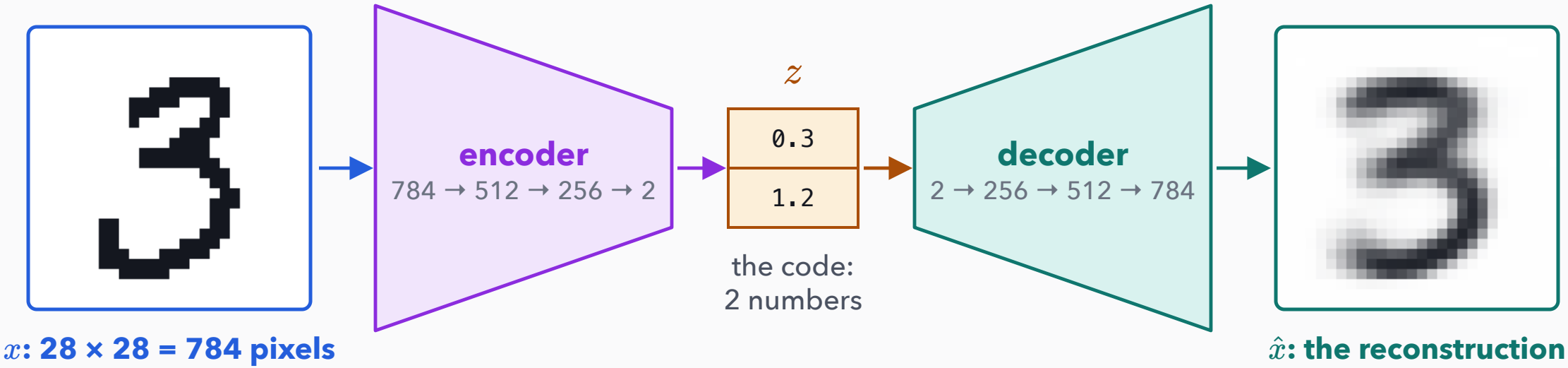
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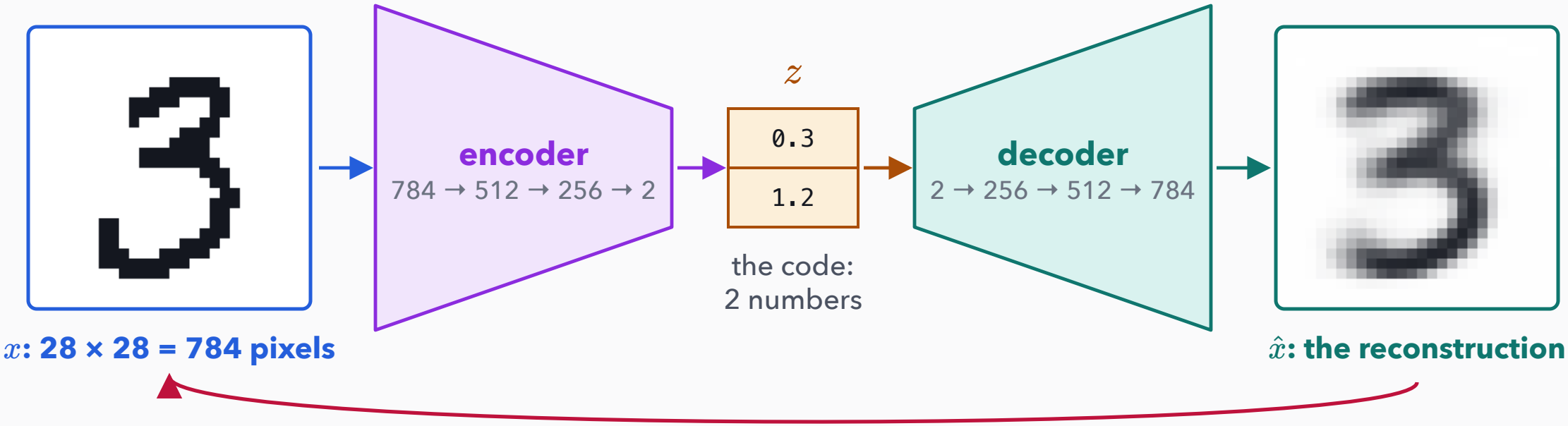
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SQUEEZE, THEN REBUILD

# An autoencoder on our 3




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COMPARING THE REBUILD WITH THE ORIGINAL

# Binary cross-entropy, one pixel at a time

$$\text{loss}(x, \hat{x}) = -\sum_{i=1}^{784} \Big[ x_i \log \hat{x}_i + (1 - x_i) \log(1 - \hat{x}_i) \Big]$$

| pixel | true value | predicted | loss |
|---|---|---|---|
| ink, predicted well | 1 | 0.90 | $-\log 0.90 = 0.11$ |
| | | | |
| | | | |

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| paper, predicted well | 0 | 0.10 | $-\log 0.90 = 0.11$ |

COMPARING THE REBUILD WITH THE ORIGINAL

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| pixel | true value | predicted | loss |
|---|---|---|---|
| ink, predicted well | 1 | 0.90 | $-\log 0.90 = 0.11$ |
| paper, predicted well | 0 | 0.10 | $-\log 0.90 = 0.11$ |
| ink, missed | 1 | 0.20 | $-\log 0.20 = 1.61$ |

**Pixels are 0 or 1 here (we binarise MNIST). Later we will see this is exactly a log-likelihood.**

# Two small networks and one loss

```
encoder = nn.Sequential(nn.Flatten(),
    nn.Linear(784, 512), nn.ReLU(),
    nn.Linear(512, 256), nn.ReLU(), nn.Linear(256, d))
decoder = nn.Sequential(nn.Linear(d, 256), nn.ReLU(),
    nn.Linear(256, 512), nn.ReLU(), nn.Linear(512, 784))

z      = encoder(x)                    # [B, d]   the code
logits = decoder(z)                    # [B, 784] one per pixel
loss = F.binary_cross_entropy_with_logits(
          logits, x.flatten(1), reduction="none").sum(1).mean()
```

**Encoder:** 784 pixels in, $d$ numbers out. Each layer narrower than the last.

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**Decoder:** the mirror image, $d$ numbers back up to 784.

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```

**Forward pass:** image to code to one logit per pixel; sigmoid of a logit is the probability of ink.

# Two small networks and one loss

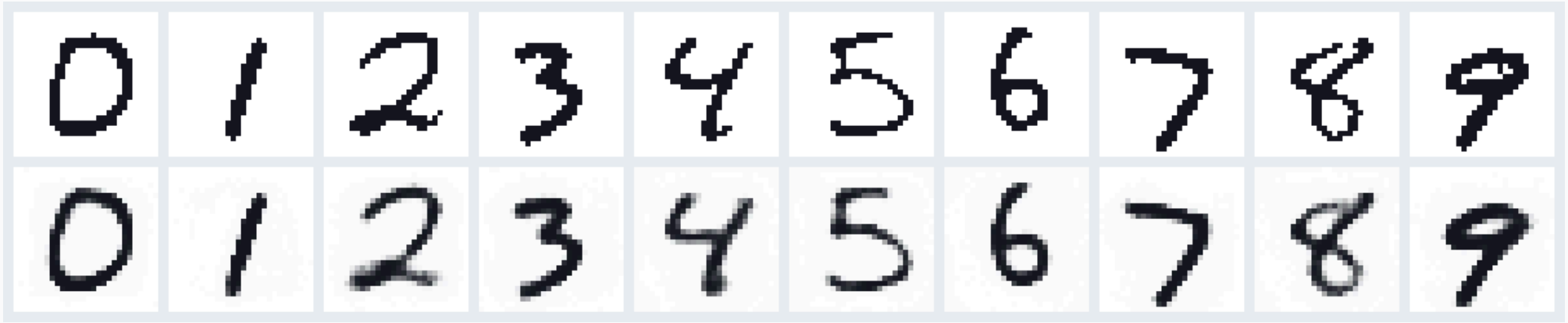
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```

**Loss:** binary cross-entropy summed over the 784 pixels of each image, then averaged over the batch. Train with the same three-line update as the bumps.

TRAINED ON 60,000 DIGITS

# It reconstructs very well



top: test digits it never saw

bottom: decoded from a 16-number code

**784 pixels squeezed into 16 numbers and back. Average loss 44.9 nats per image. Can we now make new digits?**

HOW NARROW CAN THE MIDDLE BE?

# 2 numbers or 16 numbers per image

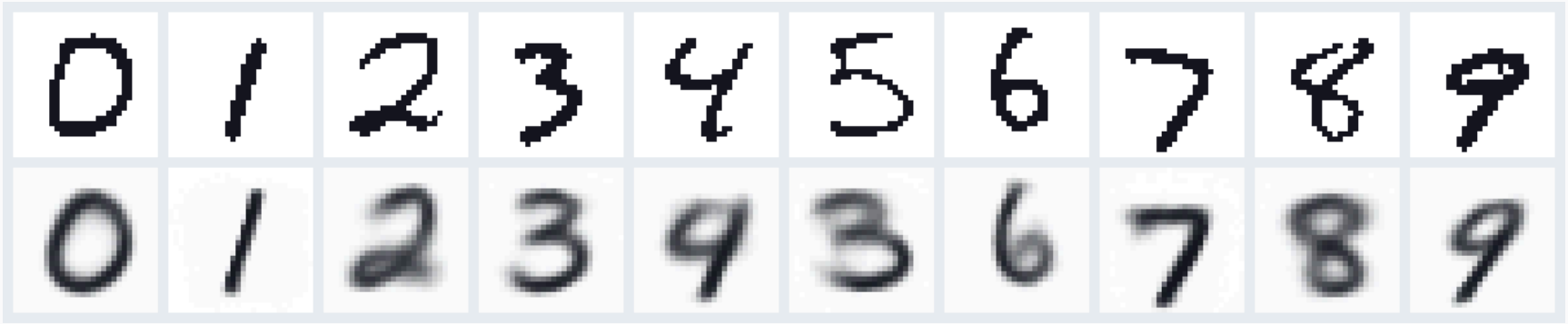
a 2-number code · loss 128.1 nats per image



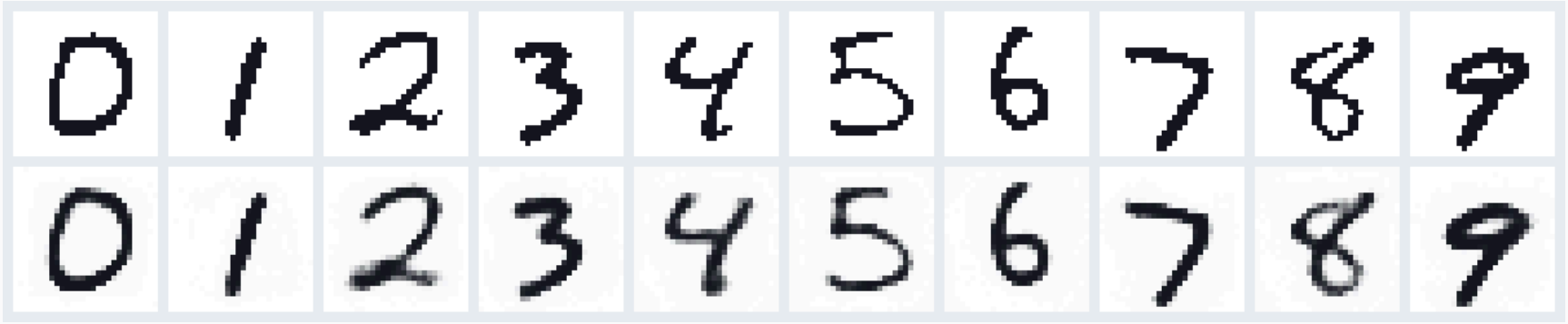
HOW NARROW CAN THE MIDDLE BE?

# 2 numbers or 16 numbers per image

a 2-number code · loss 128.1 nats per image



a 16-number code · loss 44.9 nats per image



**Each pair of rows: test digits on top, reconstructions below. Two numbers keep the digit's identity; sixteen also keep its slant and stroke.**

ONE IDEA, SEVERAL VARIATIONS

# The autoencoder family

| variant | what changes | what it encourages |
|---|---|---|
| **plain** | nothing: compress, rebuild | a short code that keeps what matters |
| | | |
| | | |
| | | |
| | | |

Following Lilian Weng, "From Autoencoder to Beta-VAE" (2018)

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| **sparse** | penalty: only a few code units may be on | each unit means one part of a digit |
| **contractive** | penalty: the code barely moves when the input moves a little | smooth, robust codes |
| **variational** | each image becomes a **cloud** of codes near $\mathcal{N}(0, I)$ | a code space you can **sample** from |

> **The first four change the loss or the input. The last one changes what a code is, and it is where today is going.**

Following Lilian Weng, "From Autoencoder to Beta-VAE" (2018)

VARIANT 1 · DENOISING AUTOENCODER

# Feed it a damaged image, ask for the clean one

input: 20% of pixels flipped



```
noisy = flip_pixels(x, rate=0.2)                    # the only new line
loss  = bce_with_logits(decoder(encoder(noisy)), x)  # target: the CLEAN x
```

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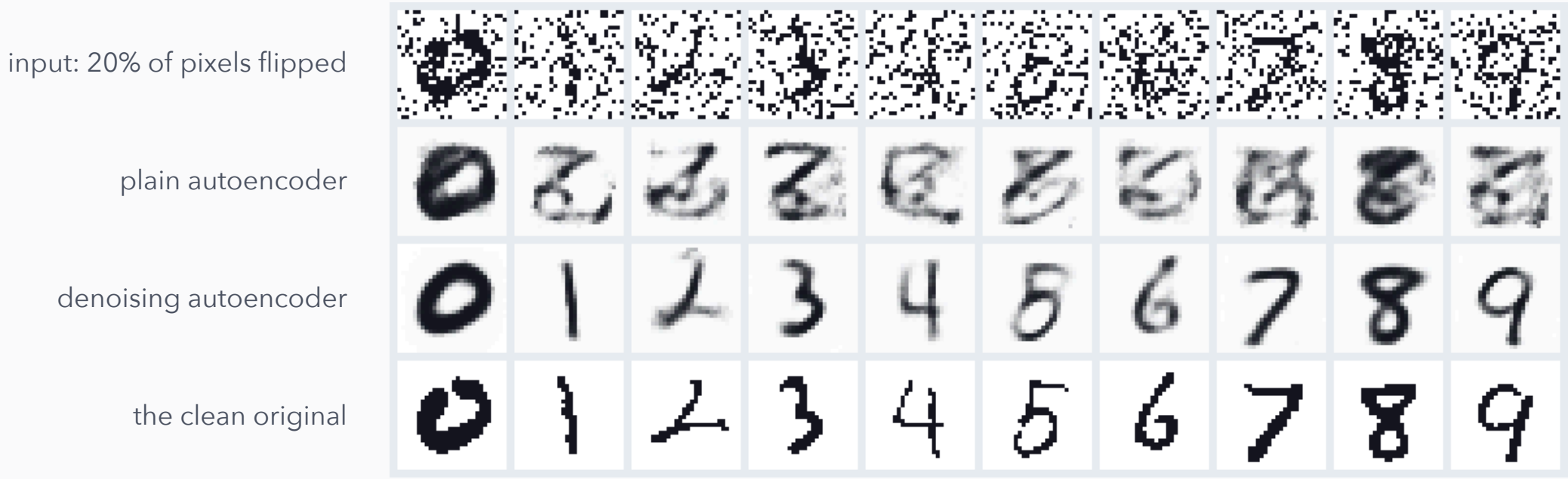
plain autoencoder



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VARIANT 1 · DENOISING AUTOENCODER

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```

**The plain autoencoder copies the noise. Trained to undo damage, the code must hold the digit, not the specks.**

VARIANT 2 · SPARSE AUTOENCODER

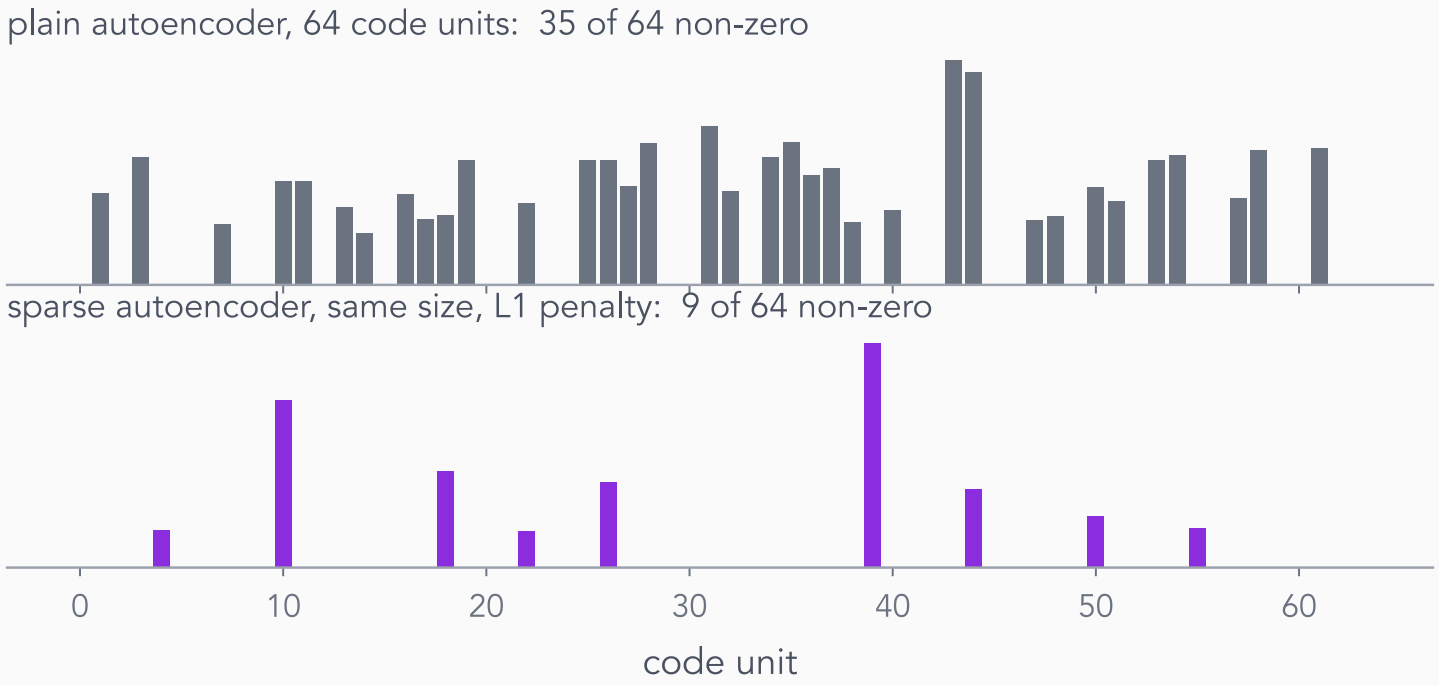
# A wide code, but only a few units allowed on

$$\text{loss} = \text{reconstruction} + \lambda \sum_{j=1}^{64} |h_j|, \qquad h = \text{ReLU}(\text{encoder}(x))$$

VARIANT 2 · SPARSE AUTOENCODER

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$$\text{loss} = \text{reconstruction} + \lambda \sum_{j=1}^{64} |h_j|, \qquad h = \text{ReLU}(\text{encoder}(x))$$




**The 64-number code of our 3. Averaged over test digits: 34.9 units on without the penalty, 8.8 with it.**

VARIANT 3 · CONTRACTIVE AUTOENCODER (NOT TRAINED HERE)

# Nudge the input; the code should barely move

$$\text{loss} = \text{reconstruction} + \lambda \left\| \frac{\partial h}{\partial x} \right\|_F^2 = \text{reconstruction} + \lambda \sum_{j,i} \left( \frac{\partial h_j}{\partial x_i} \right)^2$$

VARIANT 3 · CONTRACTIVE AUTOENCODER (NOT TRAINED HERE)

# Nudge the input; the code should barely move

$$\text{loss} = \text{reconstruction} + \lambda \left\| \frac{\partial h}{\partial x} \right\|_F^2 \;=\; \text{reconstruction} + \lambda \sum_{j,i} \left( \frac{\partial h_j}{\partial x_i} \right)^2$$

**Read the penalty**

for every code unit $h_j$ and every pixel $x_i$: how much does $h_j$ change if $x_i$ changes a little? Add up the squares.

VARIANT 3 · CONTRACTIVE AUTOENCODER (NOT TRAINED HERE)

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**Read the penalty**

for every code unit $h_j$ and every pixel $x_i$: how much does $h_j$ change if $x_i$ changes a little? Add up the squares.

**What it buys**

two slightly different 3s get almost the same code. Nearby inputs, nearby codes: a first step towards a smooth code space.

WHAT AUTOENCODERS ARE USED FOR

# Compress, clean, search, and spot the odd one out

**Compress:** 784 pixels → 16 numbers, 49× fewer.

**Clean:** the denoising slide.

**Search:** similar images have nearby codes, so search the codes.

WHAT AUTOENCODERS ARE USED FOR

# Compress, clean, search, and spot the odd one out

**Compress:** 784 pixels → 16 numbers, 49× fewer.

**Clean:** the denoising slide.

**Search:** similar images have nearby codes, so search the codes.

**Spot anomalies:** an image the model has never seen anything like gets a large reconstruction error.



top: typical test digits, loss ≈ 40.5

bottom: the 8 worst of 10,000, loss up to 218.0

> **Useful, all of it. But none of it generates a new image. Can the decoder do that on its own?**

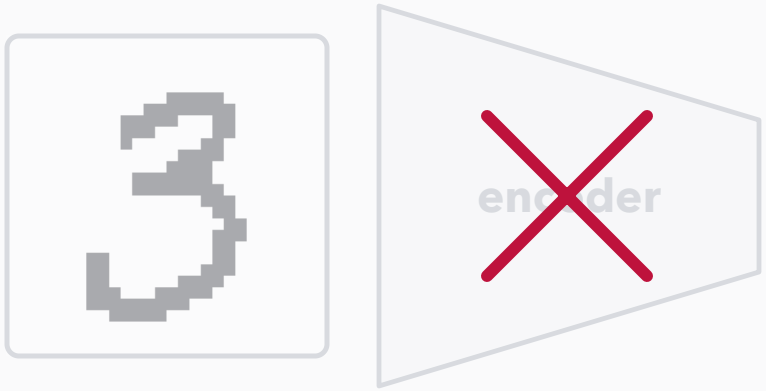
PART 2, SECOND HALF

# Can an autoencoder generate?

The decoder turns codes into digits. If we pick a code ourselves, do we get a new digit?

THE OBVIOUS IDEA

# Keep the decoder, feed it a random code

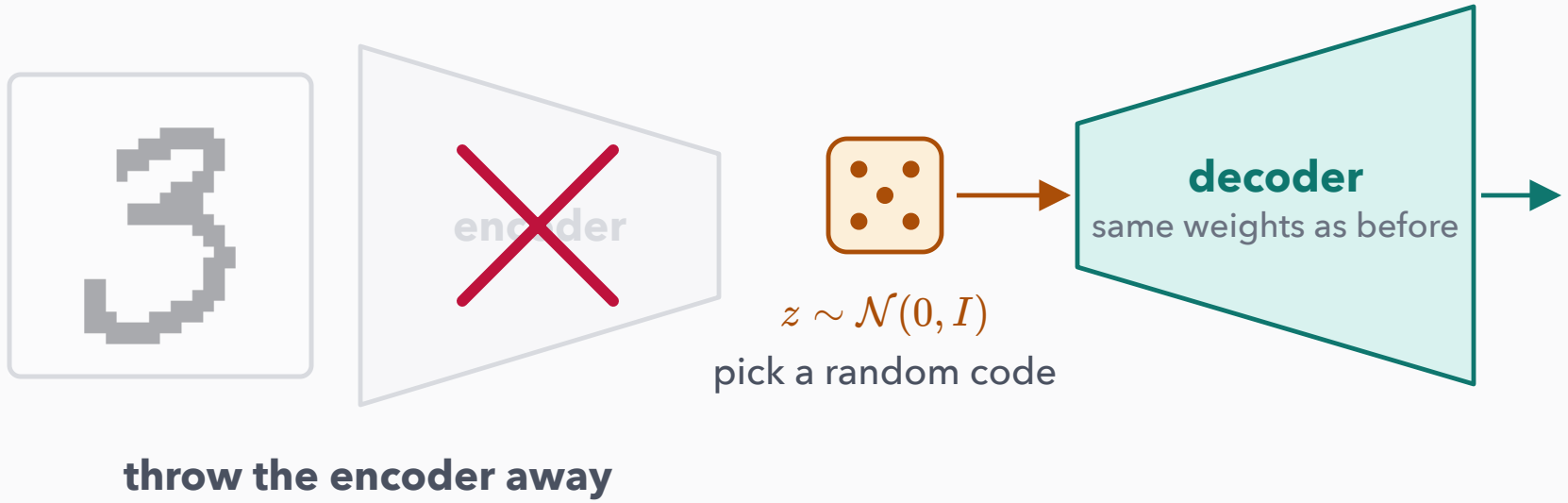



throw the encoder away

> The decoder turns codes into digits. So a random code should give a random digit. Right?

THE OBVIOUS IDEA

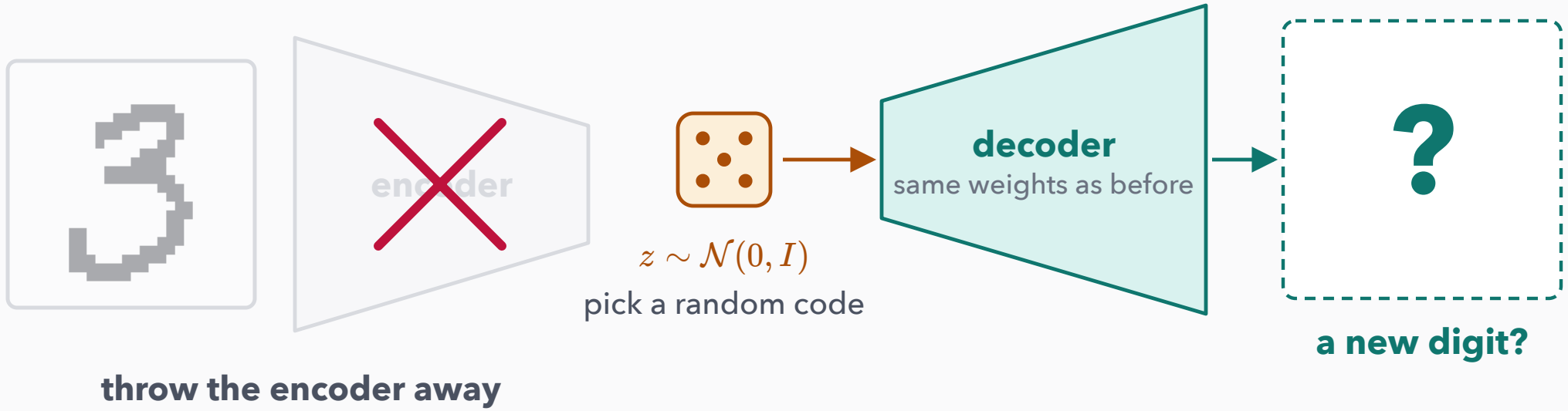
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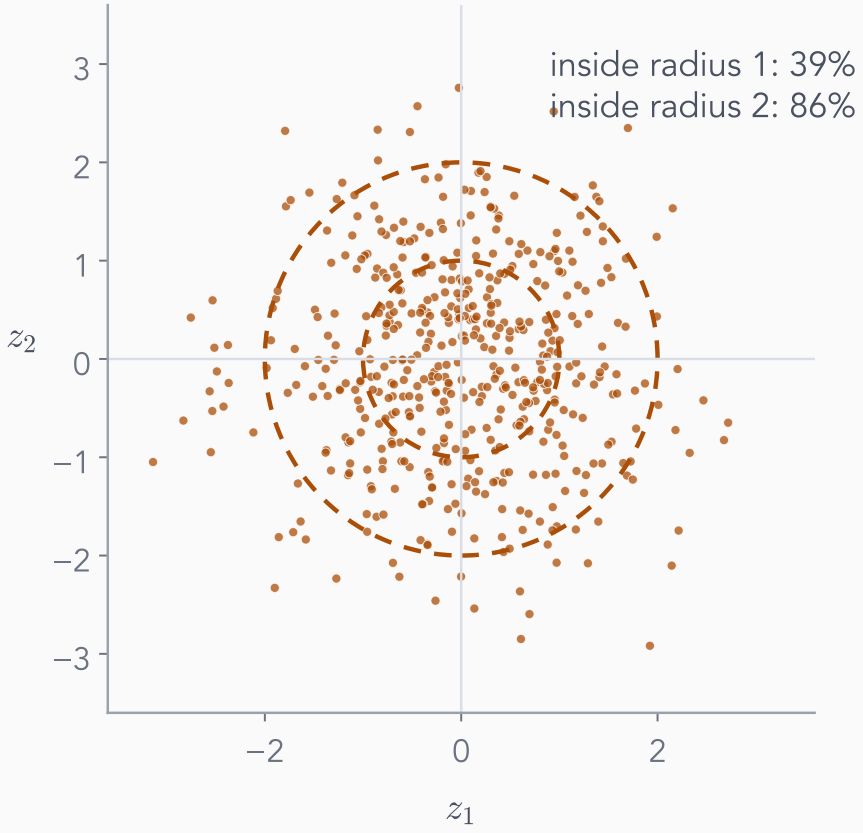
THE OBVIOUS IDEA

# Keep the decoder, feed it a random code



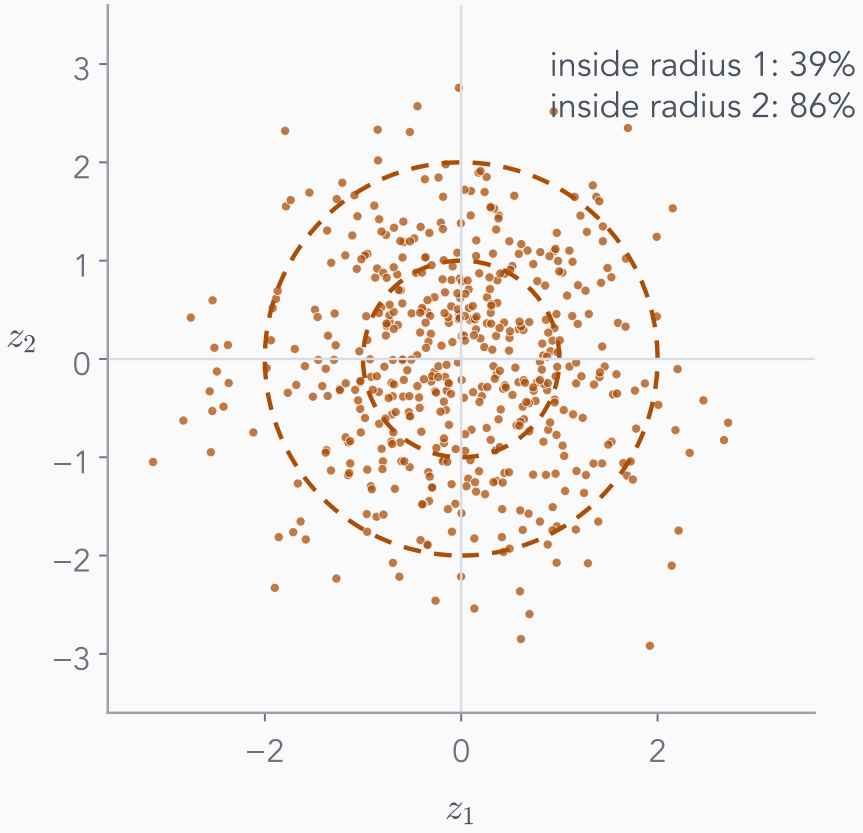

The decoder turns codes into digits. So a random code should give a random digit. Right?

# $z \sim \mathcal{N}(0, I)$: two independent bell curves




Draw $z_1$ from a standard bell curve $\mathcal{N}(0, 1)$: mean 0, spread 1. Draw $z_2$ the same way, independently.

# $z \sim \mathcal{N}(0, I)$: two independent bell curves

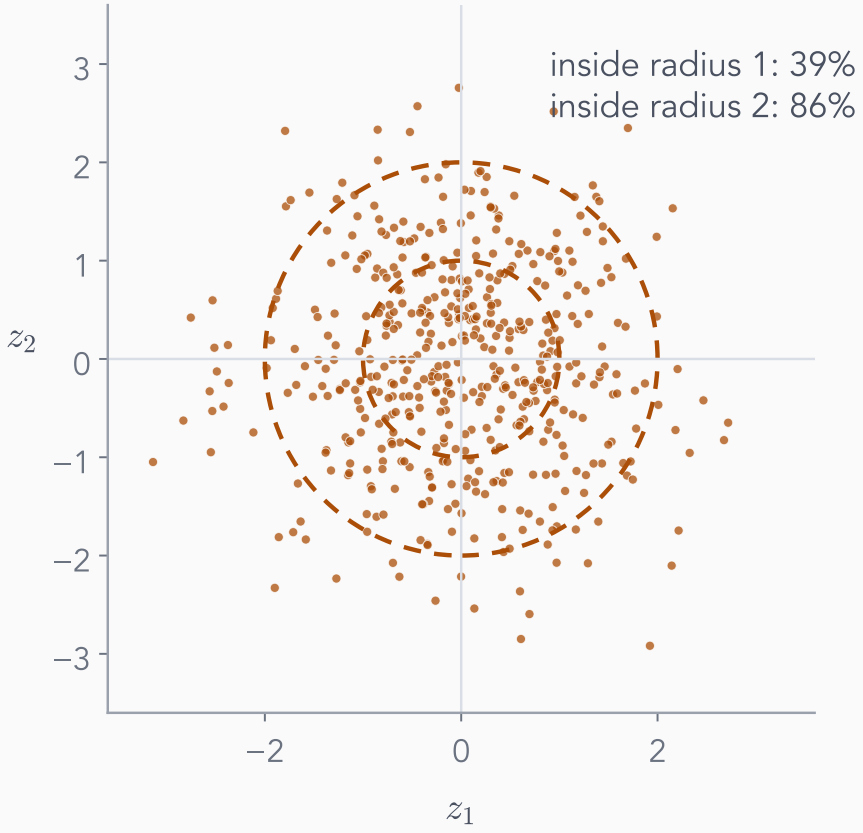



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Together that is written $\mathcal{N}(0, I)$: the **0** says the centre is $(0, 0)$; the **I** (identity matrix) says each number has spread 1 and the two do not depend on each other.

## WHAT IS A "RANDOM CODE"?

# $z \sim \mathcal{N}(0, I)$: two independent bell curves




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Together that is written $\mathcal{N}(0, I)$: the **0** says the centre is $(0, 0)$; the **I** (identity matrix) says each number has spread 1 and the two do not depend on each other.

500 draws, plotted: most land within 2 of the origin, almost none beyond 3.

> **Why this choice? It is the simplest distribution we can sample from. The decoder, though, was never told about it.**

WHAT ACTUALLY COMES OUT

# Random codes give scribbles

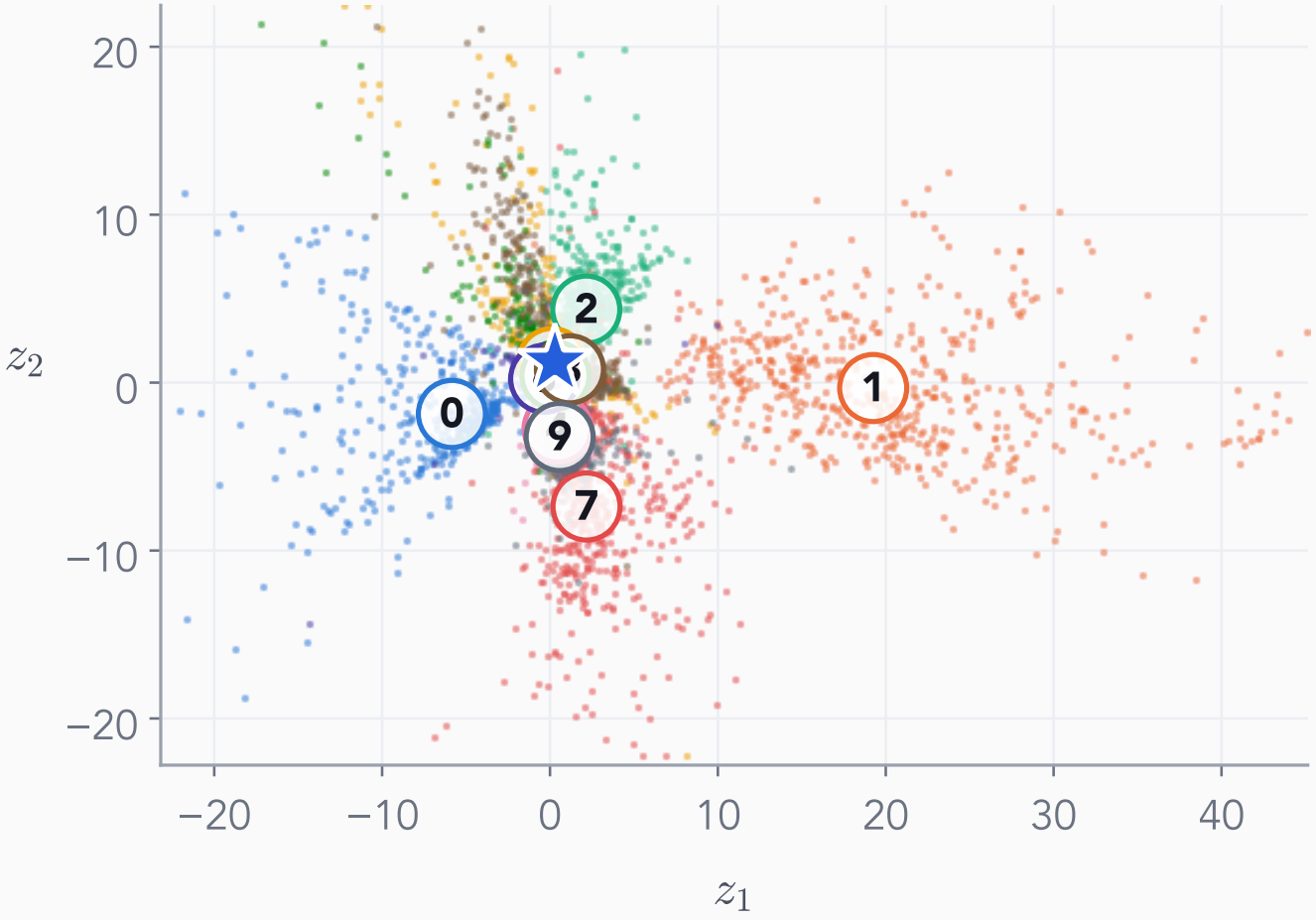


ten codes drawn from $\mathcal{N}(0, I)$ in 16 dimensions, through the trained decoder

**The decoder only ever saw codes made by the encoder. Where do those codes live?**

LOOK INSIDE (2-NUMBER VERSION)

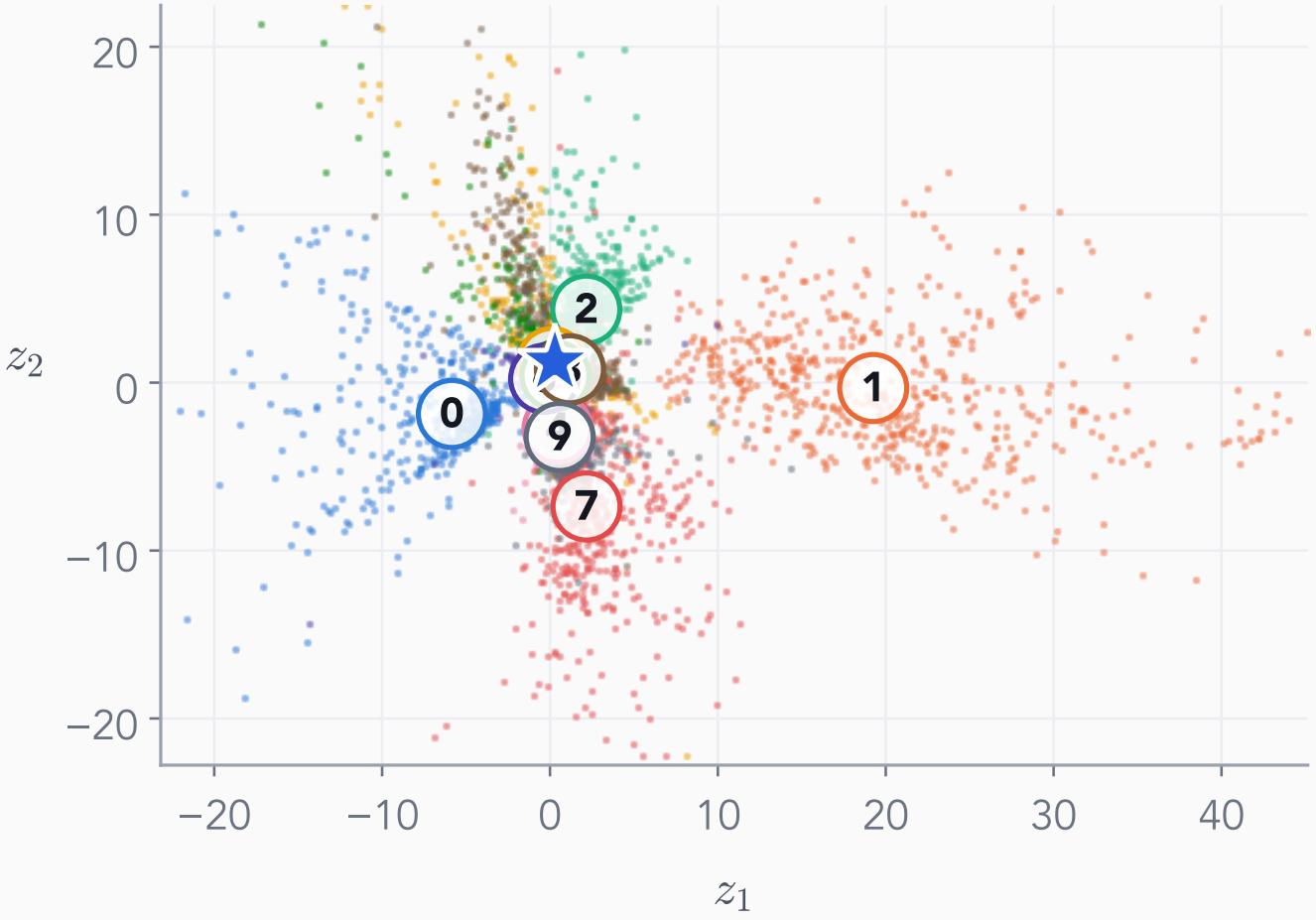
# Where 5,000 test digits land




Each dot is one test digit's code, coloured by its true label. The star is our 3.

LOOK INSIDE (2-NUMBER VERSION)

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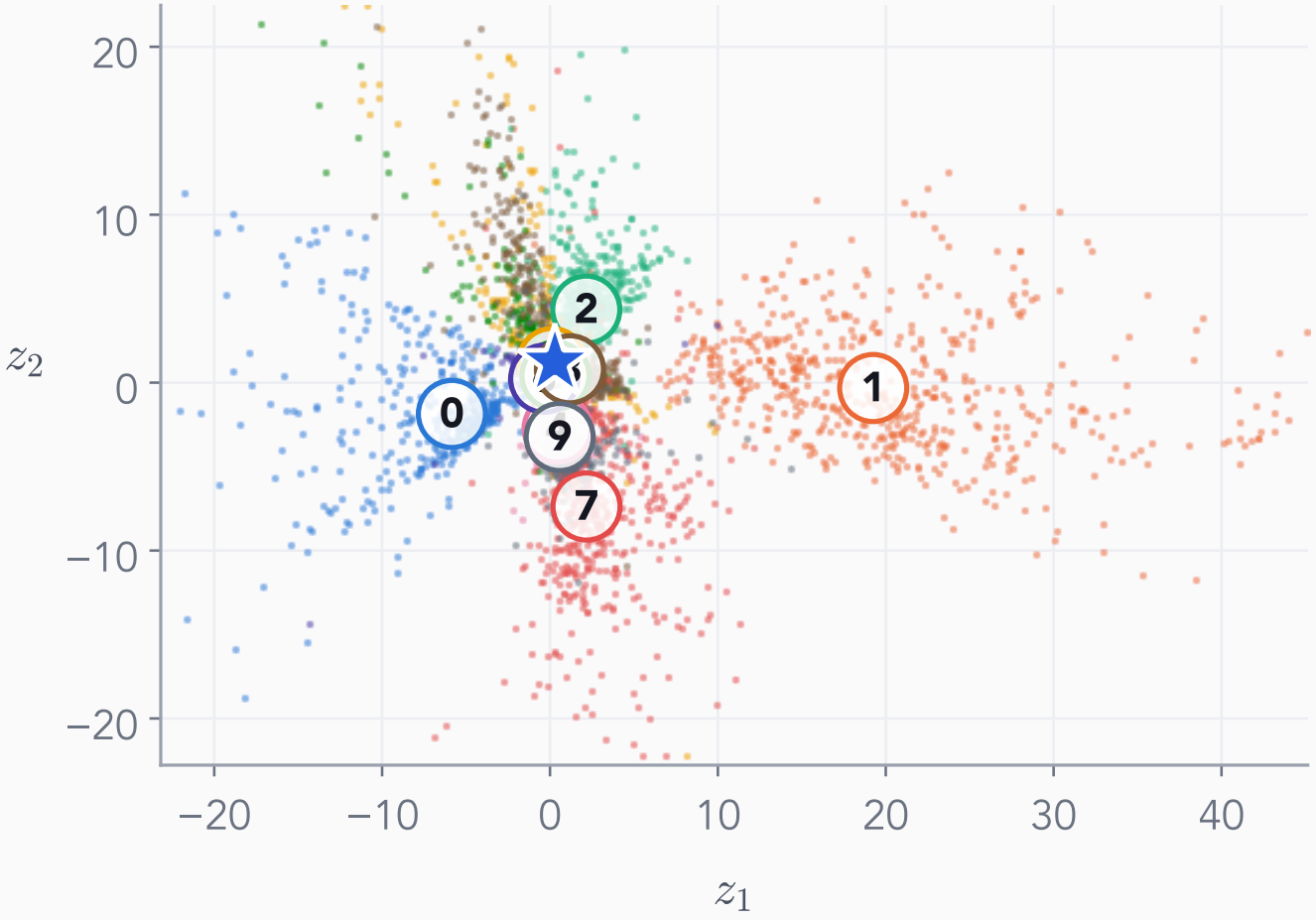



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Same digits cluster together. Nobody asked for that; it just helps reconstruction.

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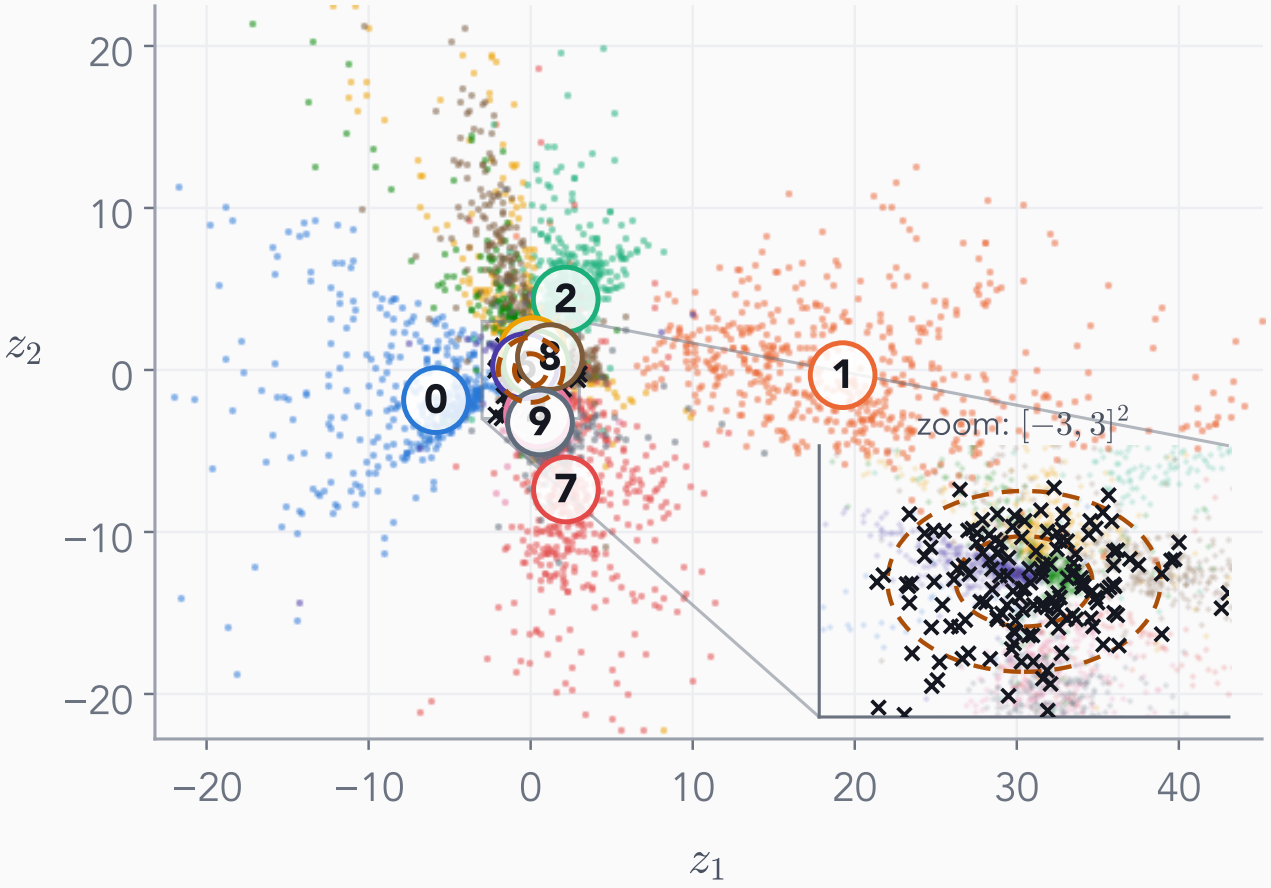
Same digits cluster together. Nobody asked for that; it just helps reconstruction.

Look at the axes: $z_1$ runs from −18.5 to 40.4, $z_2$ from −19.7 to 19.4. Nothing made the codes stay near the origin.

**The code space has an arbitrary shape, scale and position.**

OVERLAY RANDOM CODES

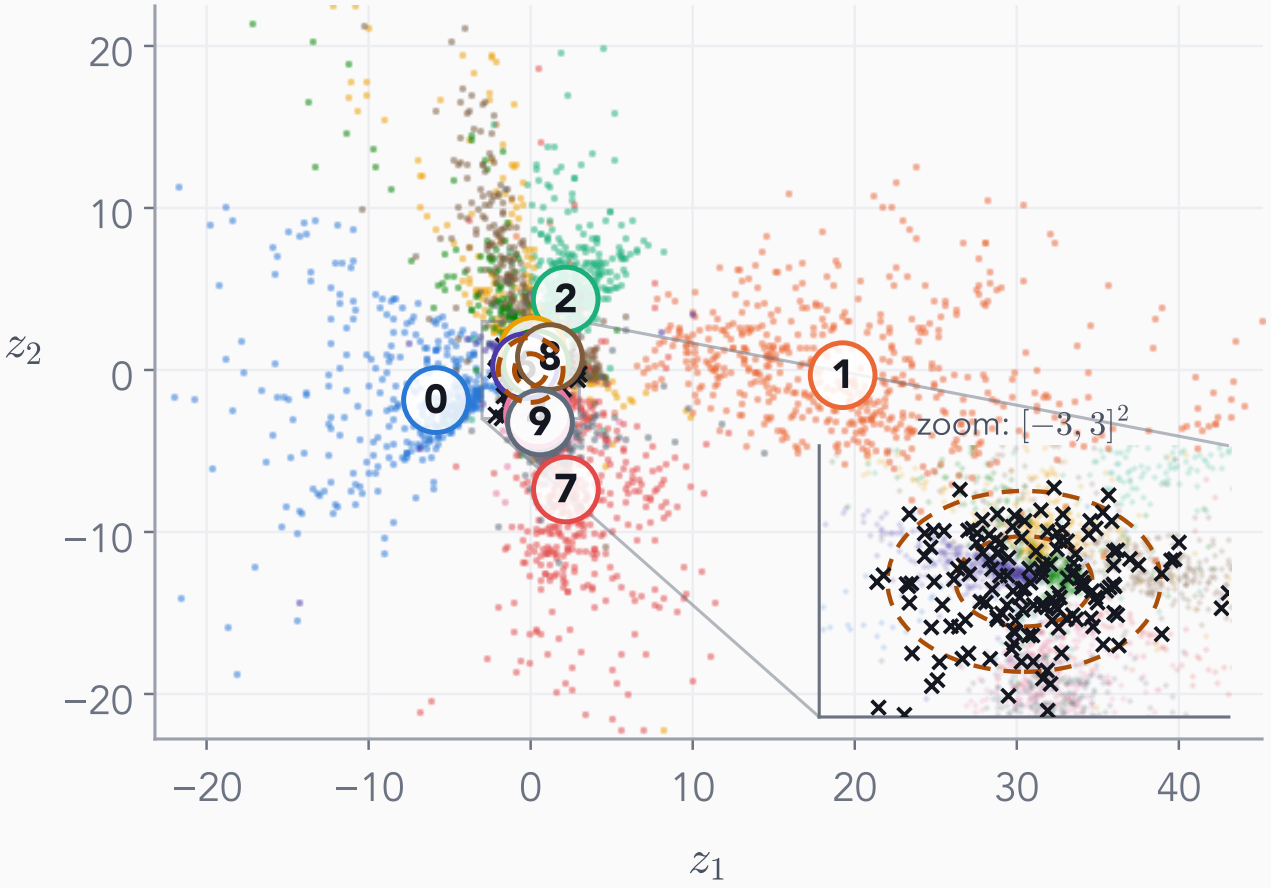
# Random draws all land in one small spot




Now add 150 random codes from $\mathcal{N}(0, I)$ (black ×), the kind we fed the decoder two slides ago. Dashed circles: radius 1 and 2. On this map they all fall into one small crowded patch (zoomed, bottom right).

OVERLAY RANDOM CODES

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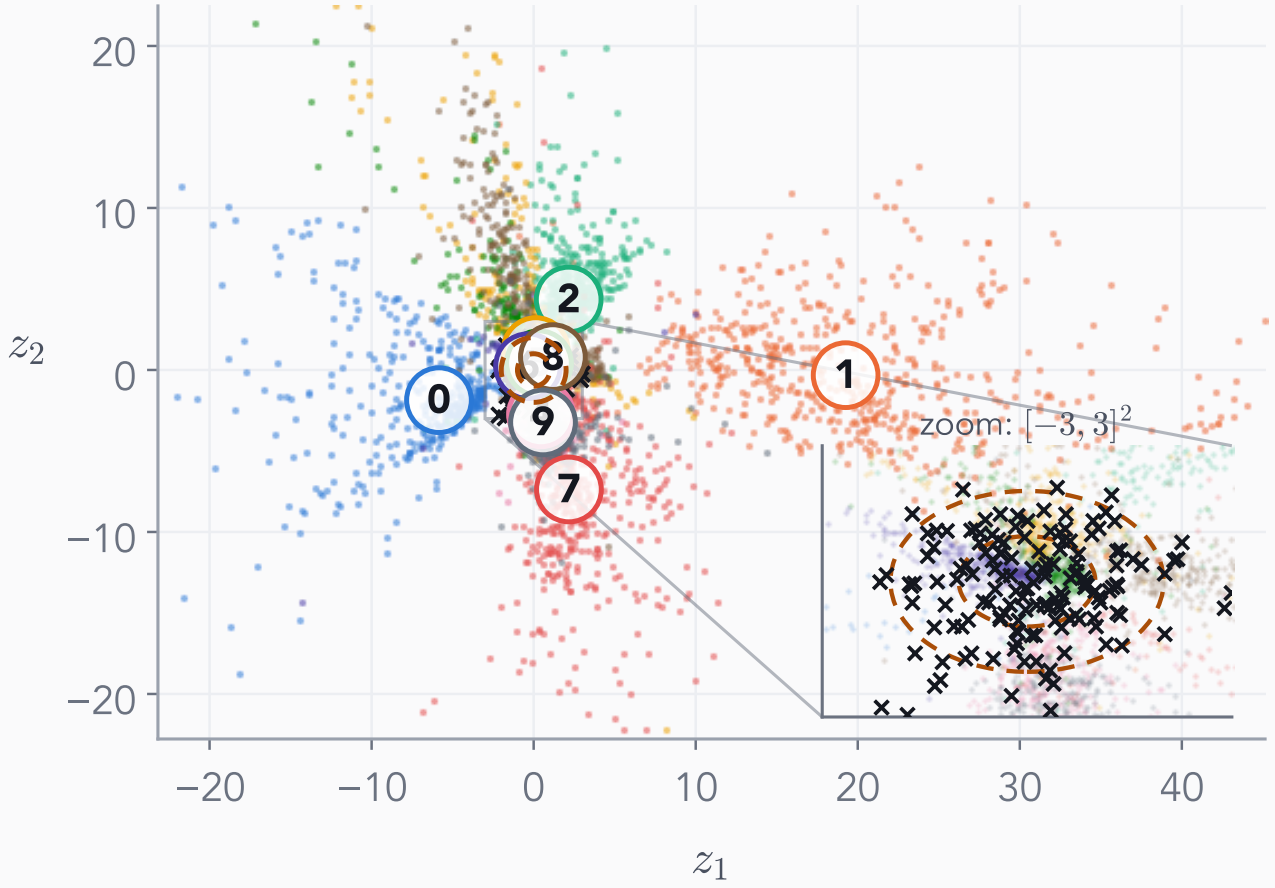

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ten of them decoded: some digits, some smudges

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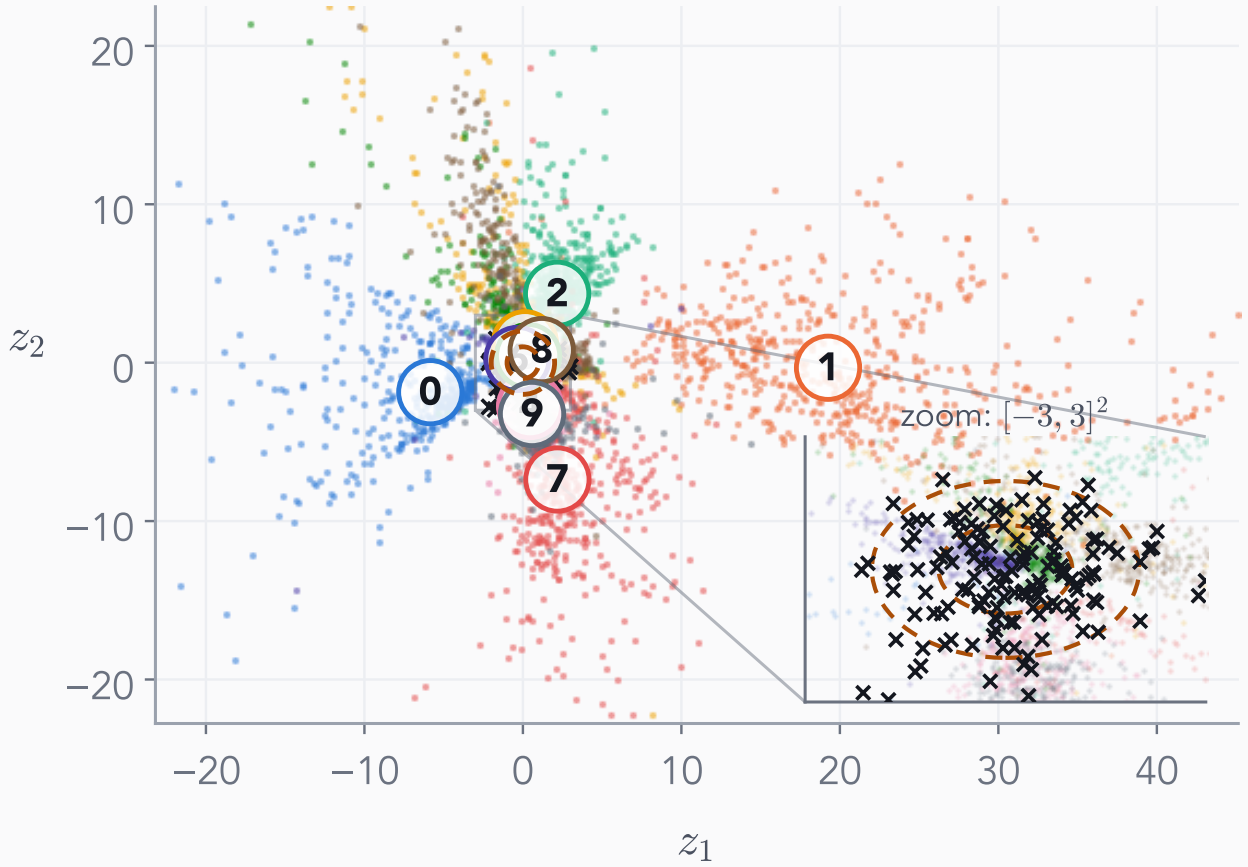
ten of them decoded: some digits, some smudges

And we can never generate a 1: the 1s live near $z_1 \approx 20$.

**To sample, we need to know where the codes are, and every point there must decode to something sensible.**

TRY IT: THE TRAINED AUTOENCODER, IN YOUR BROWSER

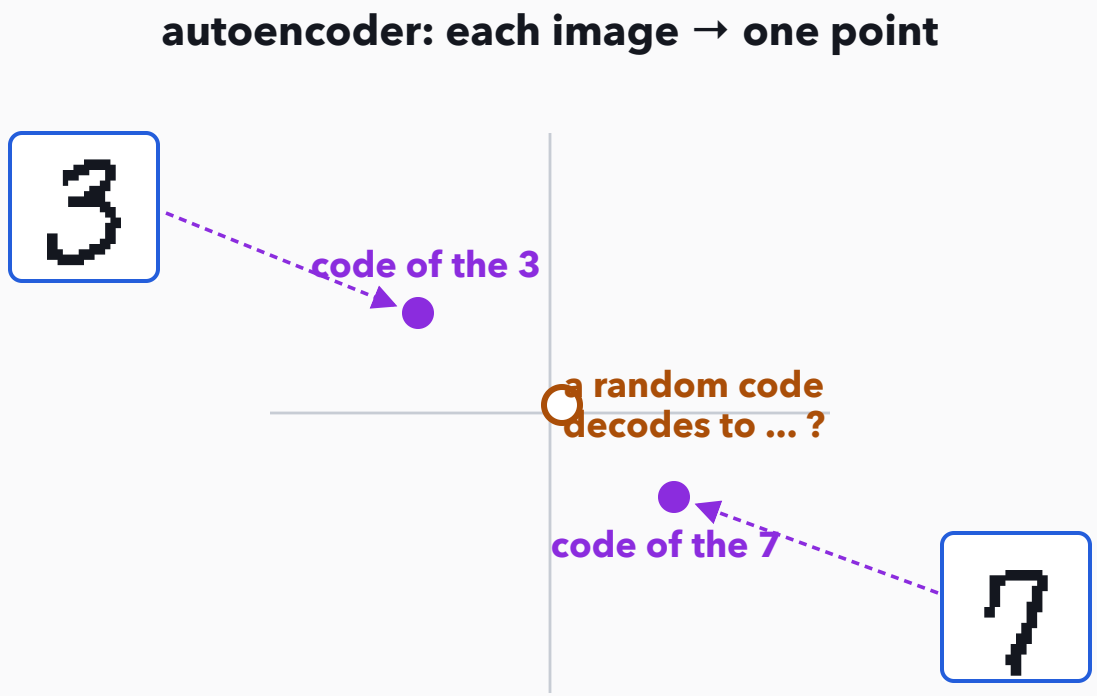
# Pick a code, any code, and decode it




A real digit's code decodes well. A random code usually gives a smudge: the decoder only learned the dotted regions.

WHAT WE WANT INSTEAD

# Encode to a cloud, not a point

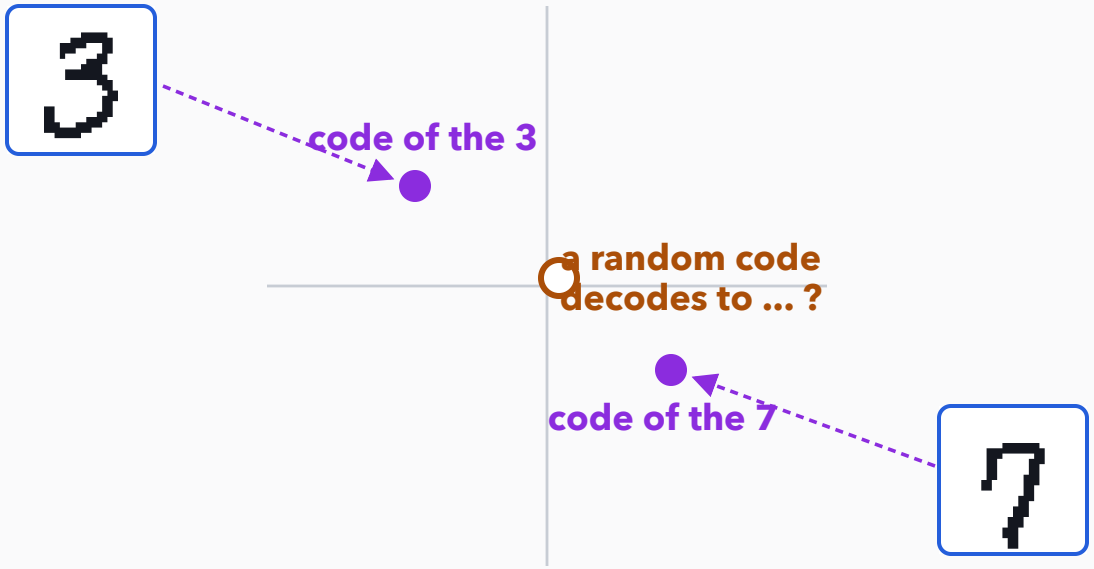



training only ever used the two purple points

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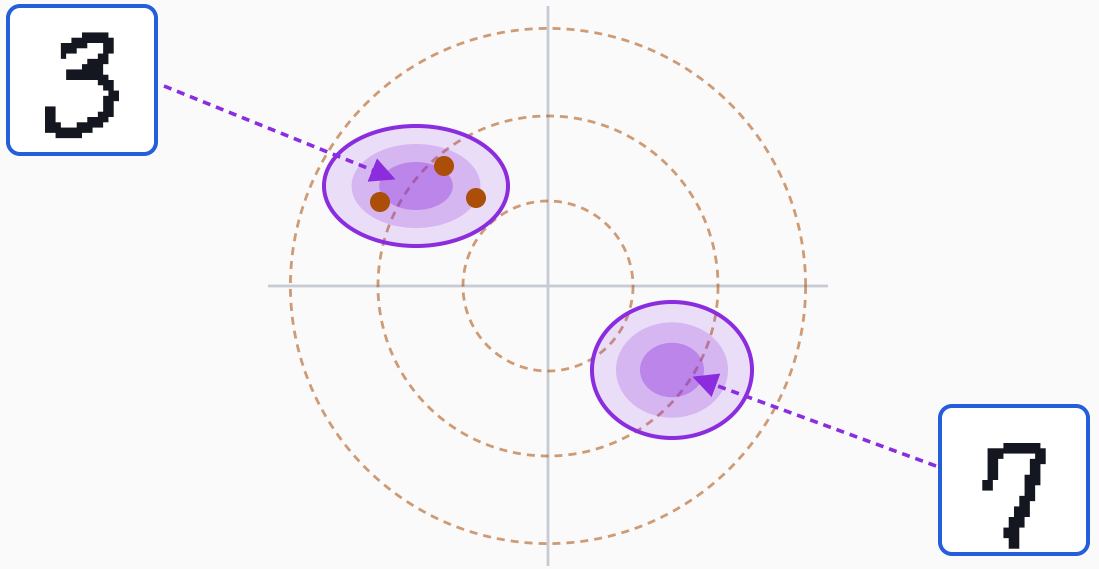
# Encode to a cloud, not a point

**autoencoder: each image → one point**




training only ever used the two purple points

**VAE: each image → a small cloud**



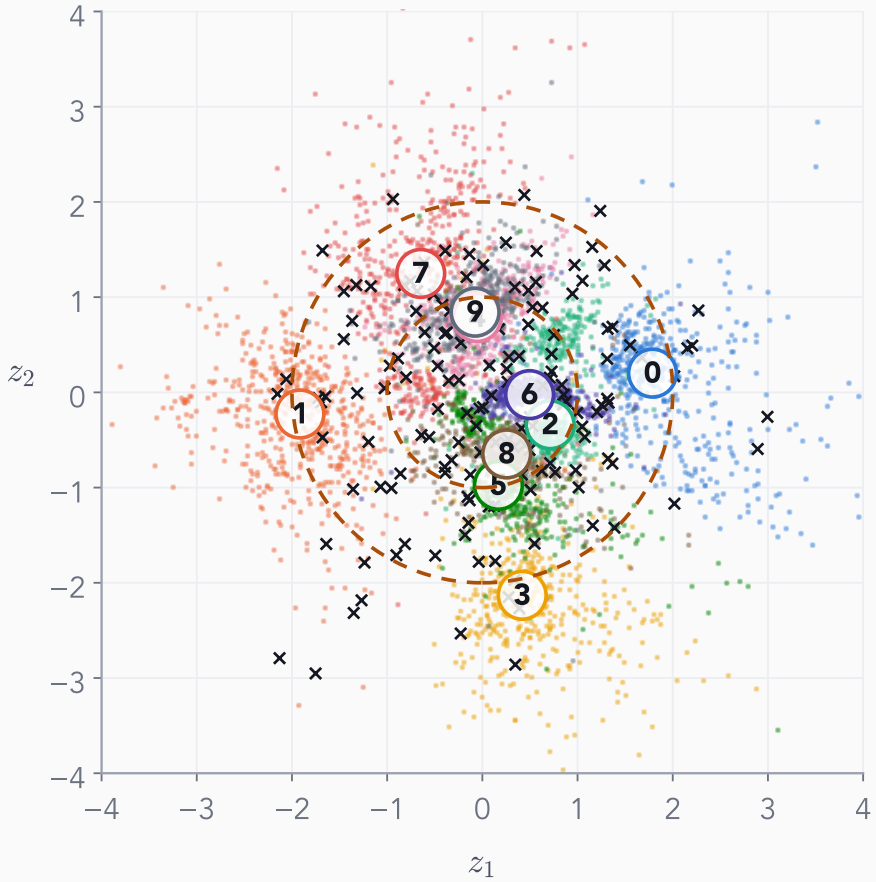

training: every code in the 3's cloud must decode back to the 3

**Known:** pull every cloud towards $\mathcal{N}(0, I)$, so random draws land on real codes.

**Smooth:** every point in a cloud must decode to that image, so neighbourhoods are meaningful.

A PREVIEW OF WHERE WE ARE GOING

# The same plot for a VAE

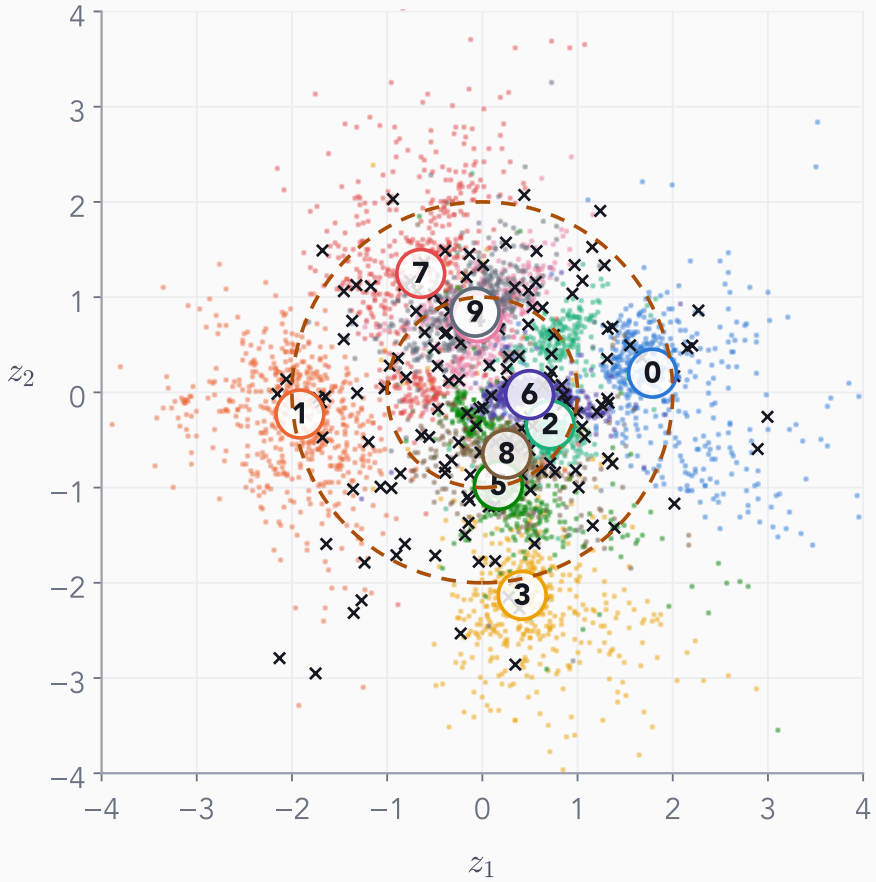



Same data, same 2-number code, same random draws. Now the codes sit inside $\mathcal{N}(0, I)$.

**To get here: make the model probabilistic (Part 3), measure how close two distributions are (Part 4), and derive the loss (Part 5).**

A PREVIEW OF WHERE WE ARE GOING

# The same plot for a VAE




Same data, same 2-number code, same random draws. Now the codes sit inside $\mathcal{N}(0, I)$.



the same kind of random codes, decoded

**To get here: make the model probabilistic (Part 3), measure how close two distributions are (Part 4), and derive the loss (Part 5).**

END OF PART 2

# What we did, and what comes next

### WHAT WE DID

1. An encoder squeezes an image into a short code; a decoder rebuilds it. The image is its own target.
2. Random codes from $\mathcal{N}(0, I)$ decode to scribbles.
3. The codes spread over an arbitrary range, and the decoder never saw the space between them.
4. Variants change the input (denoising) or add a penalty (sparse, contractive).
5. The wish: encode each image to a **cloud**, and keep every cloud near $\mathcal{N}(0, I)$.

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### NEXT: THE PROBABILISTIC VIEW

Turn the decoder into a probability model with a prior over codes, and ask what it would take to train it by maximum likelihood, as in Part 1.

PART 3 OF 7

# The probabilistic view: a decoder is a likelihood

The autoencoder could not generate because nothing said where codes live. Part 1 had exactly what is missing: a **prior** over the hidden choice. Put Part 1's story and Part 2's decoder together.


✓ Distributions and fitting

✓ Autoencoders

3 **The probabilistic view**

4 KL divergence

5 Variational inference and the ELBO

6 Reparameterisation

7 The VAE on MNIST

# Same story, bigger pieces

| | two bumps (so far) | images (from now on) |
|---|---|---|
| **hidden** $z$ | a coin: A or B | |
| $p(x \mid z)$ | a bump at $-2$ or $+2$ | |
| **observed** $x$ | one number | |
| **sum over** $z$ | 2 terms | |

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| **observed** $x$ | one number | 784 pixels, each ink (1) or paper (0) |
| **sum over** $z$ | 2 terms | |

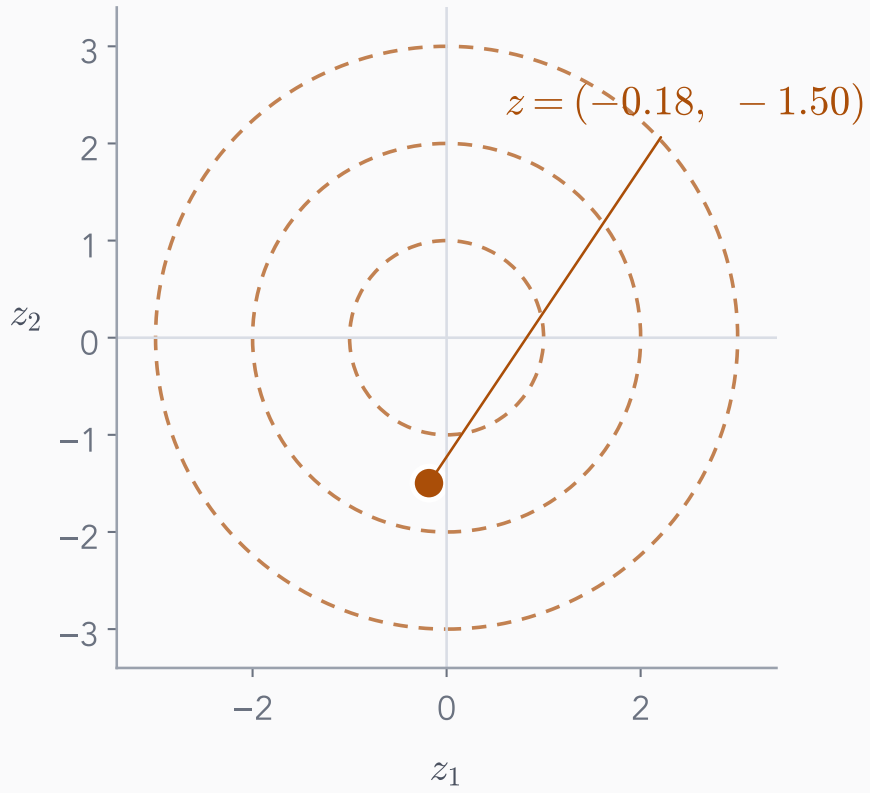
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| **observed** $x$ | one number | 784 pixels, each ink (1) or paper (0) |
| **sum over** $z$ | 2 terms | every point in a plane: infinitely many |

**Next three slides: generate one image this way, one step at a time.**

GENERATING AN IMAGE · STEP 1 OF 3

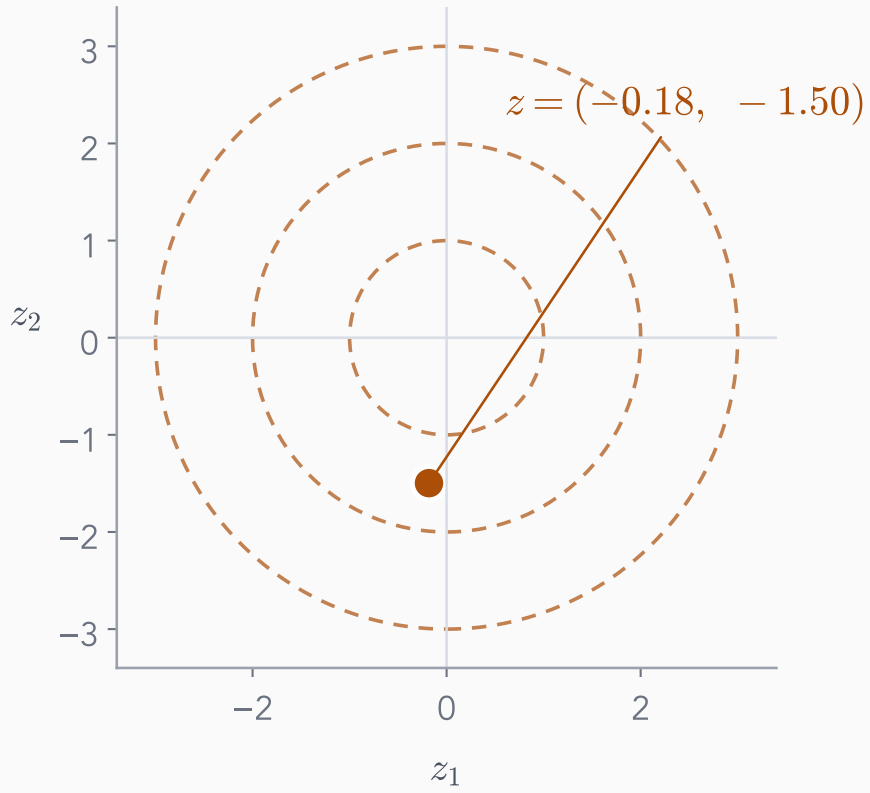
# Draw a code




$$z \sim \mathcal{N}(0, I)$$

Two numbers, each from a standard bell curve: like flipping the coin, but the "coin" can land anywhere in the plane.

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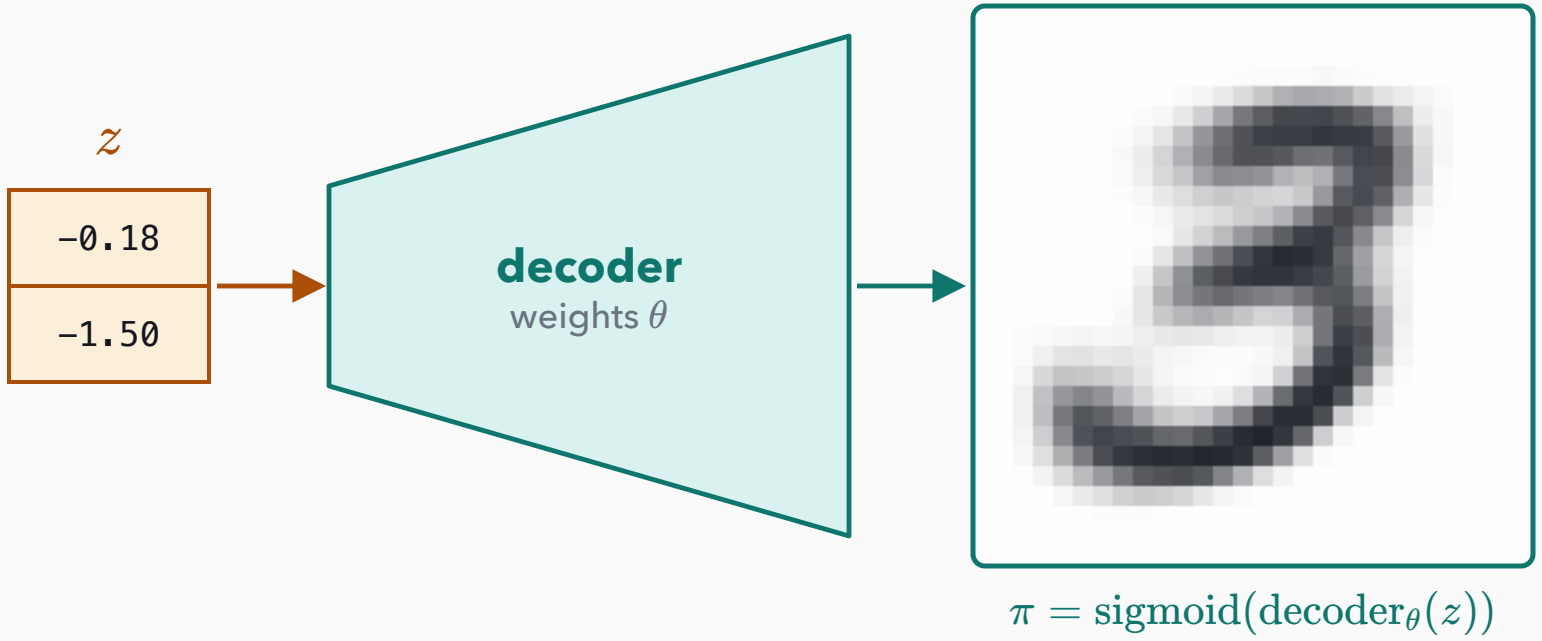
Two numbers, each from a standard bell curve: like flipping the coin, but the "coin" can land anywhere in the plane.

This draw: $z = (-0.18, \ -1.5)$.

**On its own, $z$ means nothing. The decoder will give it meaning.**

GENERATING AN IMAGE · STEP 2 OF 3

# Decode it into 784 pixel probabilities



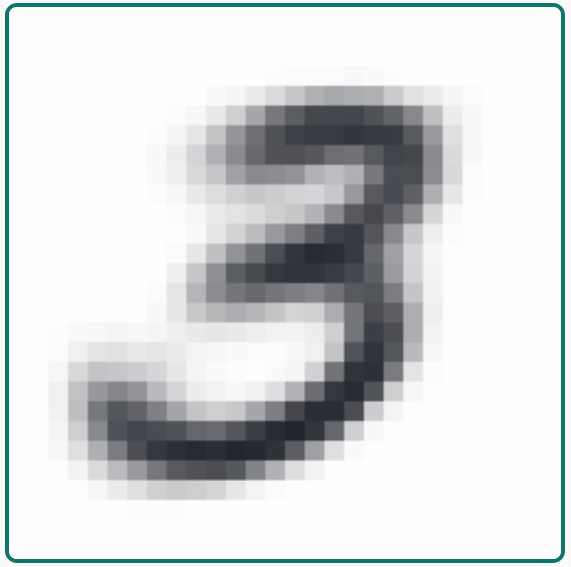

each $\pi_i$ is between 0 and 1

$\pi_i$ = probability that pixel $i$ is ink

$\pi_i$ is the probability that pixel $i$ is ink. Dark grey = probably ink; white = probably paper.

GENERATING AN IMAGE · STEP 3 OF 3

# Flip one coin per pixel



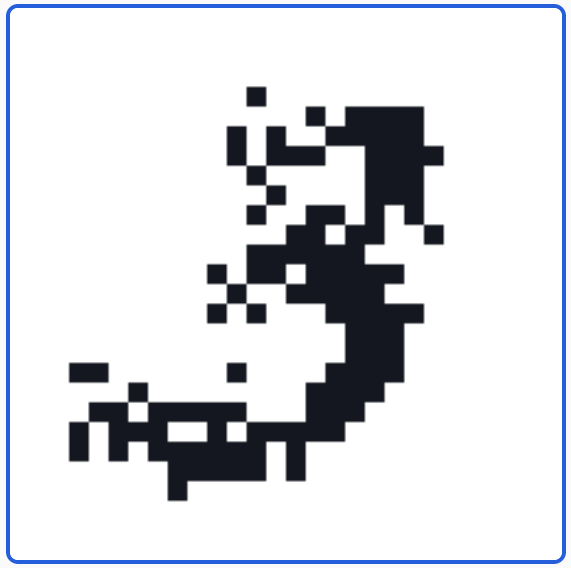
probabilities $\pi$

GENERATING AN IMAGE · STEP 3 OF 3

# Flip one coin per pixel



probabilities $\pi$



one sample $x$

$$x_i \sim \text{Bernoulli}(\pi_i)$$

Each pixel independently: ink with probability $\pi_i$.

**That is a generated image. In practice we usually display $\pi$ itself, because it looks cleaner.**

THE SAME TWO STEPS AS BEFORE

# Generating is easy. Learning is the catch.

**Generate ✓**

draw $z \sim \mathcal{N}(0, I)$, decode, flip pixels. Three slides, done.

THE SAME TWO STEPS AS BEFORE

# Generating is easy. Learning is the catch.

**Generate ✓**

draw $z \sim \mathcal{N}(0, I)$, decode, flip pixels. Three slides, done.

**Learn ?**

choose the decoder's weights $\theta$ to make the training images probable:

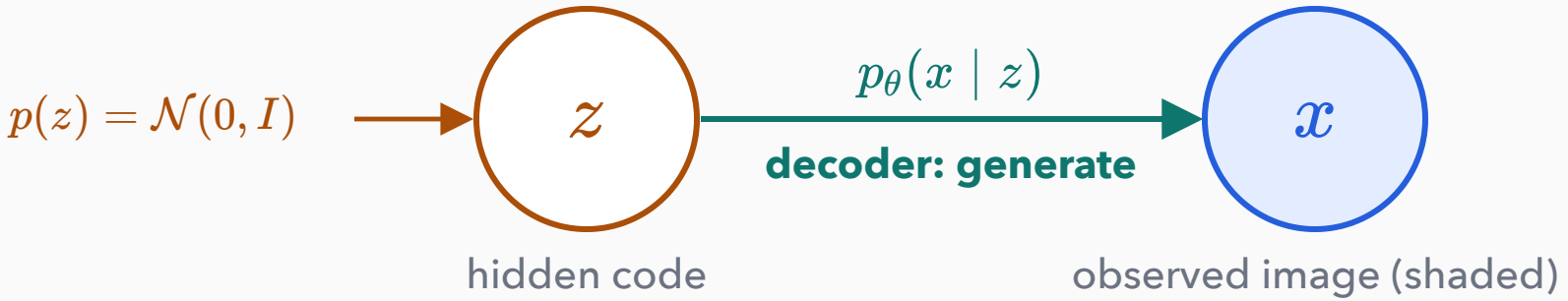
$$p_\theta(x) = \int p(z)\, p_\theta(x \mid z)\, dz$$

the two-bump sum, now over every code in the plane.

**The rest of this part: why this integral is so hard, and the three ways people get around it.**

THE MODEL, AS A PICTURE

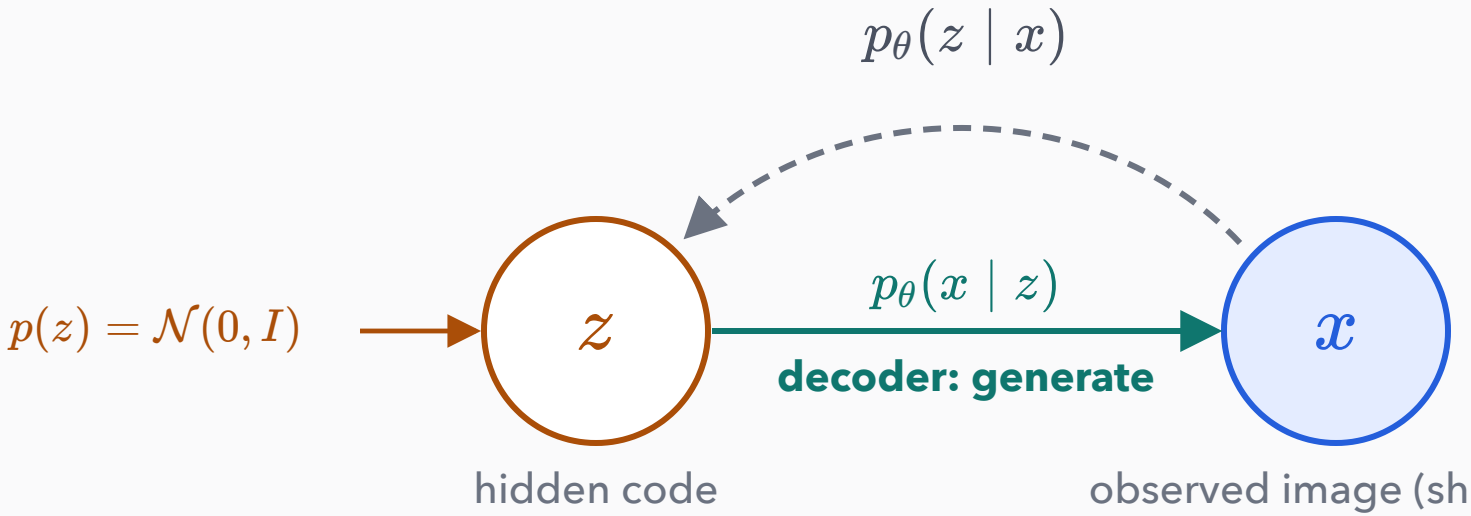
# Two arrows: generate down, infer up




$$p_\theta(x, z) = p(z)\, p_\theta(x \mid z)$$

THE MODEL, AS A PICTURE

# Two arrows: generate down, infer up

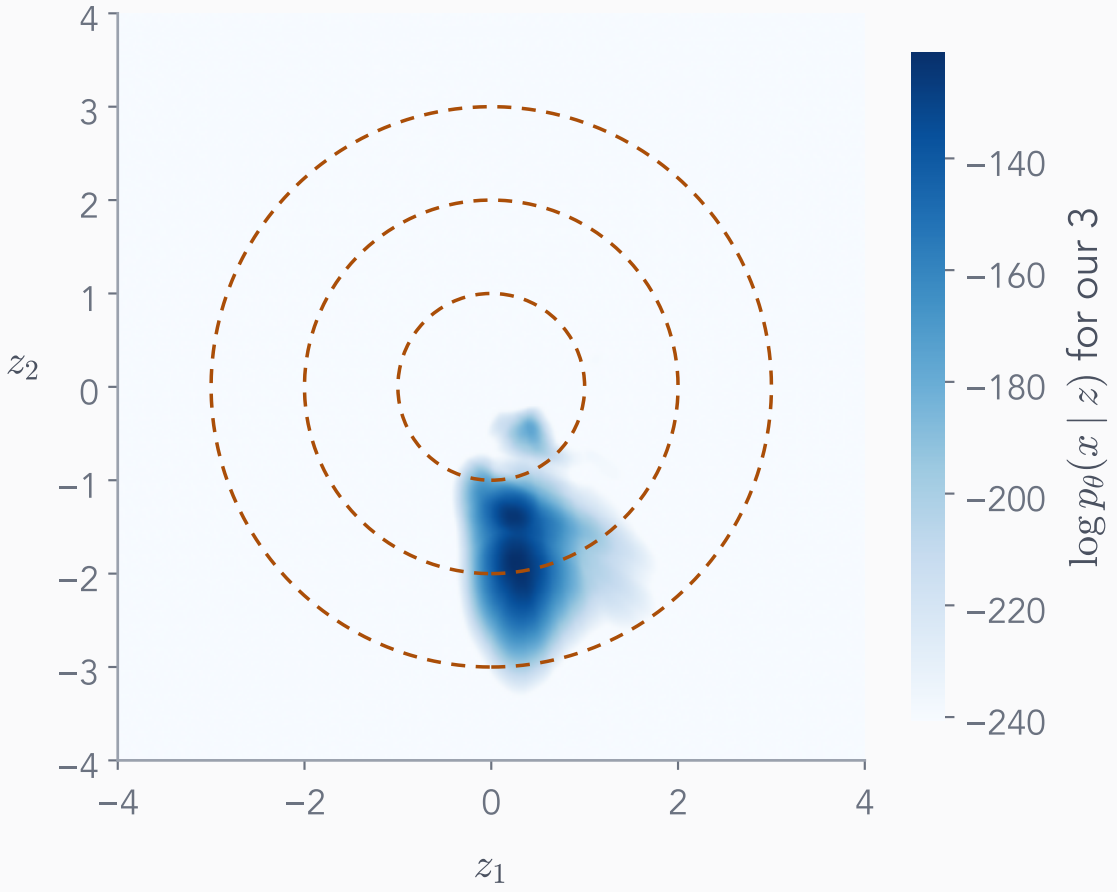



**infer: which $z$ made this $x$? Bayes, but it divides by $p_\theta(x)$ (dashed: out of reach)**

$$p_\theta(x, z) = p(z)\, p_\theta(x \mid z)$$

MAXIMUM LIKELIHOOD FOR OUR 3

# How probable is our 3 under the model?

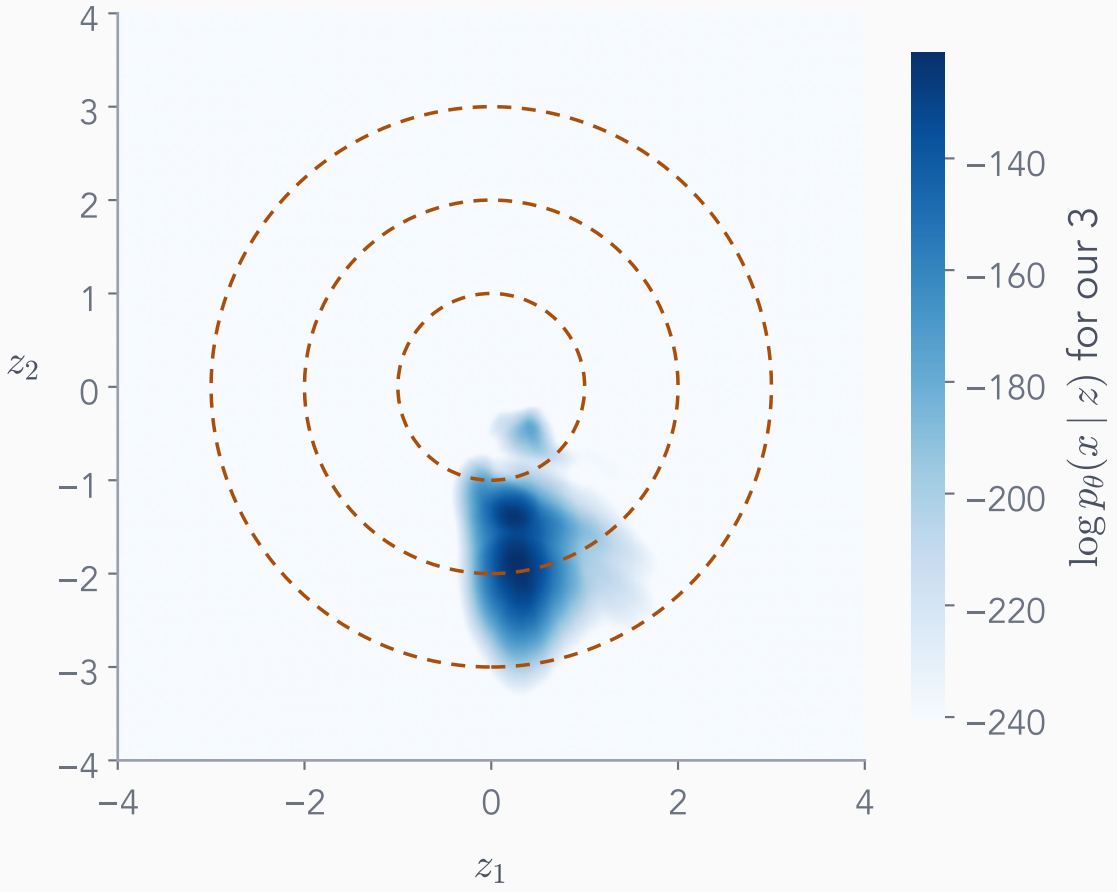



With the coin we summed over two values:

$$p(x) = \sum_{z \in \{\mathrm{A},\mathrm{B}\}} p(z)\, p(x \mid z)$$

MAXIMUM LIKELIHOOD FOR OUR 3

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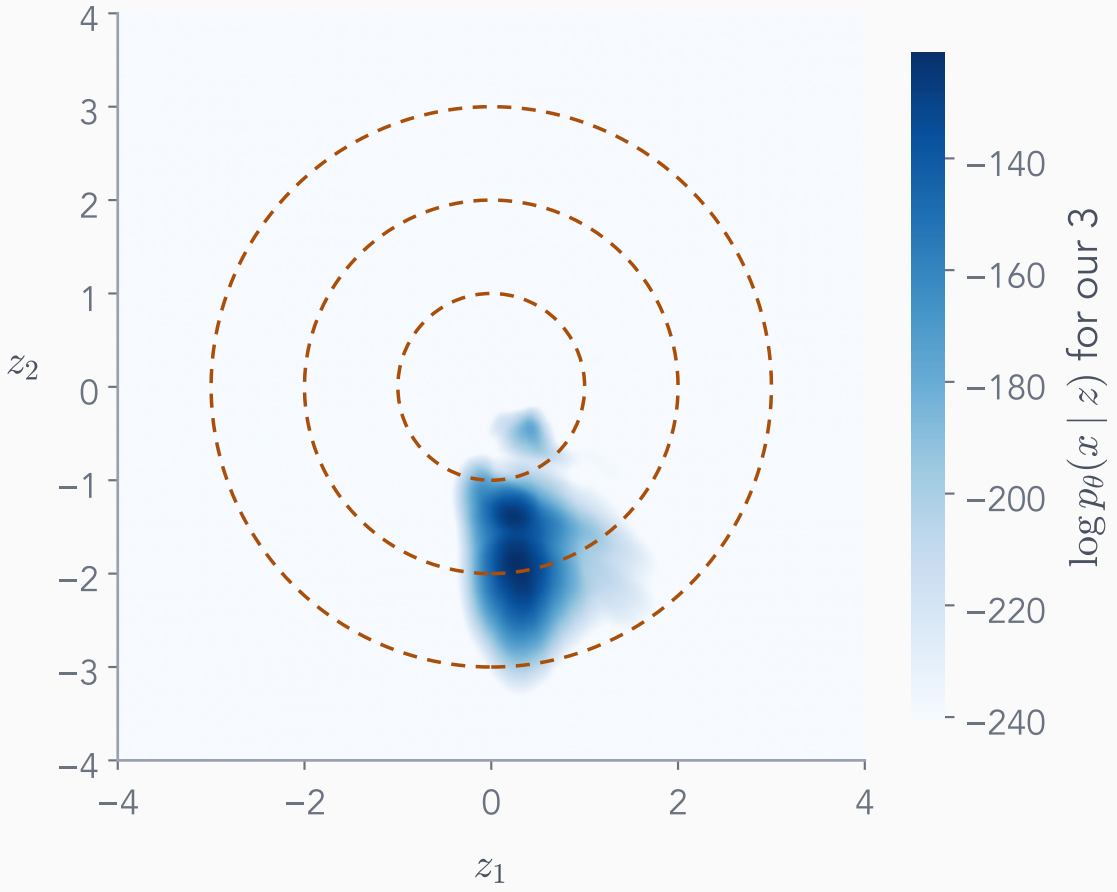
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A code can be any point in the plane, so the sum becomes an integral:

$$p_\theta(x) = \int p(z)\; p_\theta(x \mid z)\, dz$$

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2 numbers per code, so we can afford a 901 × 901 grid of decoder calls: $\log p_\theta(x) = -128.7$ nats.

## WHY NOT JUST COMPUTE $p_\theta(x)$?

# The integral is intractable

**No formula.** With two bumps, $p(x)$ was a sum of 2 terms. Here it is $\int p(z)\, p_\theta(x \mid z)\, dz$ with a neural network inside: no algebra gives the answer.

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**Too many terms to add up.** Put a grid of 901 points on every axis of the code and add up the decoder's answers:

| numbers in $z$ | decoder calls |
|---|---|
| 1 | 901 |
| 2 (our 3) | $8.1 \times 10^5$ · a second |
| 4 | $6.6 \times 10^{11}$ · a week |
| 8 | $4.3 \times 10^{23}$ |
| 16 | $1.9 \times 10^{47}$ |

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**"Intractable"** means exactly this: no formula, and far too many terms to sum.

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**And we need $p_\theta(x)$ for every image, at every training step.**

IDEA: ESTIMATE THE INTEGRAL BY SAMPLING

# Random codes almost never hit the right patch

$$p_\theta(x) = \underbrace{\mathbb{E}_{z \sim p(z)}\left[\, p_\theta(x \mid z) \,\right]}_{\text{the average over codes from the prior}} \approx \frac{1}{S} \sum_{s=1}^{S} p_\theta(x \mid z_s)$$

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**100,000**

random codes $z_s \sim \mathcal{N}(0, I)$

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**0.07%**

land within 2 widths of our 3's patch

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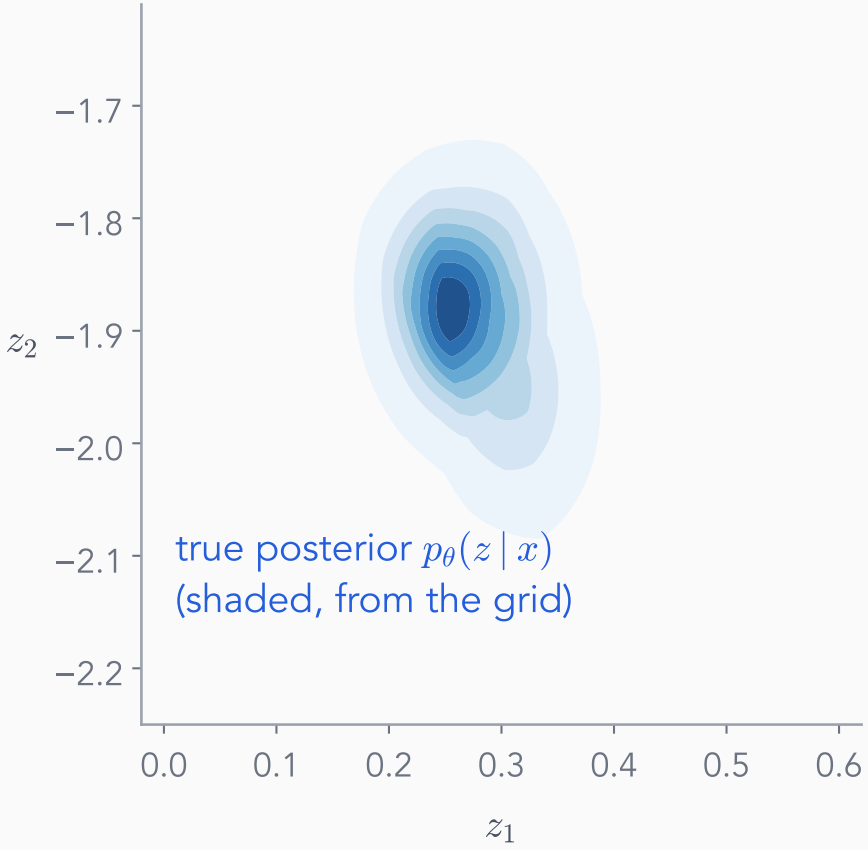
**0.07%**
land within 2 widths of our 3's patch

**−363.9**
median $\log p(x \mid z)$ of a random code

**−121.0**
the best of the 100,000: $\log p(x \mid z)$

> **A typical random code makes our 3 about $e^{243}$ times less probable than the best code. Blind sampling wastes almost every draw.**

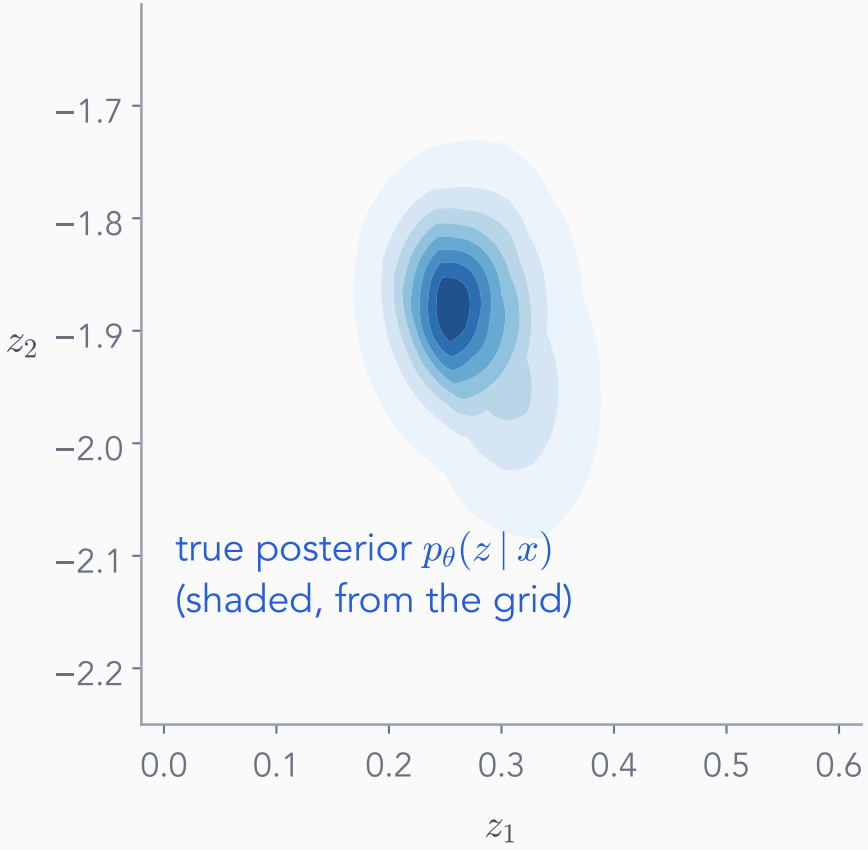
# The backwards step: which codes could have made our 3?




Part 1's Bayes table, with a plane of codes instead of two bumps:

$$p_\theta(z \mid x) = \frac{p(z)\, p_\theta(x \mid z)}{p_\theta(x)}$$

# The backwards step: which codes could have made our 3?

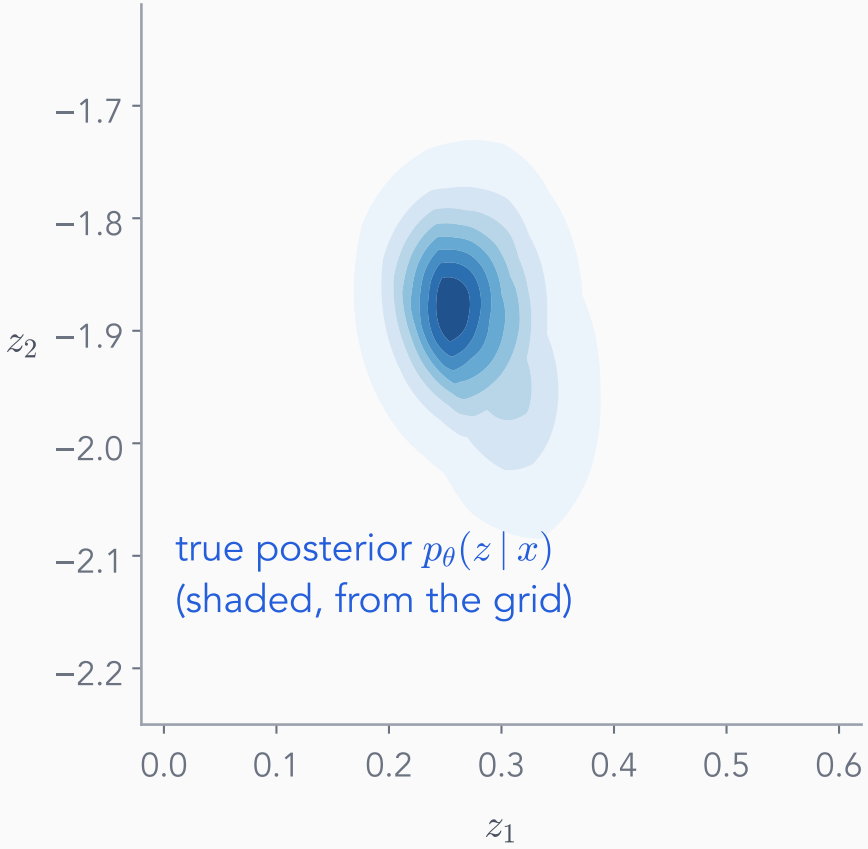



Part 1's Bayes table, with a plane of codes instead of two bumps:

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The bottom is the evidence: the same impossible integral. So the posterior is out of reach too.

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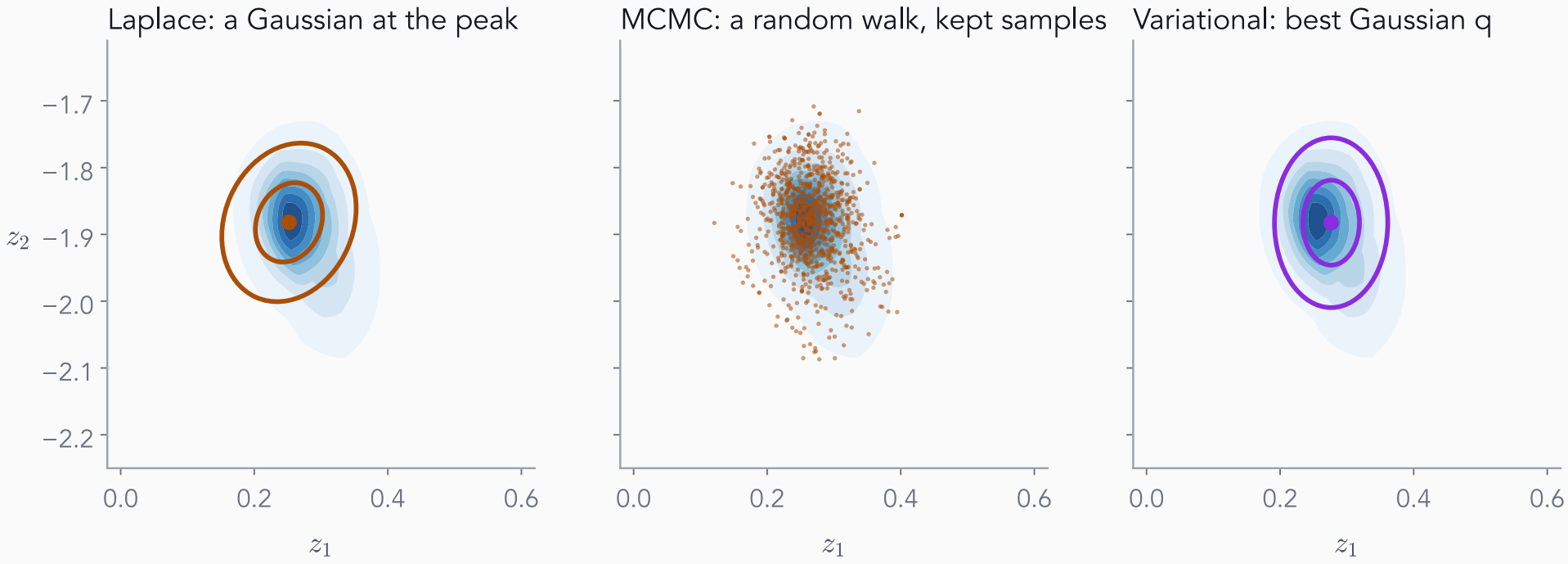
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The bottom is the evidence: the same impossible integral. So the posterior is out of reach too.

From the 2-number grid we can still look at it: a small, slightly tilted patch near $(0.25,\ -1.88)$, about 0.05 wide. The prior is 1 wide: one image pins its code down tightly.

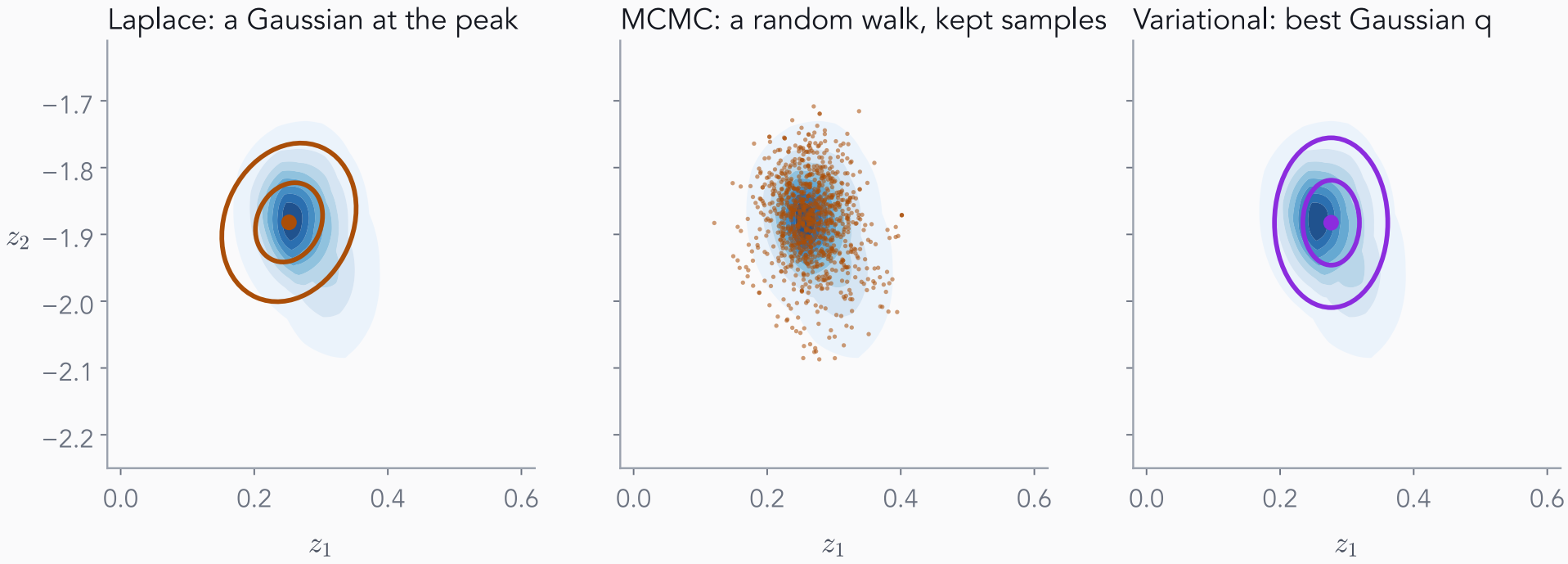
> **Training needs this patch for every image. Next: three ways to approximate a posterior we cannot compute.**

# Three ways, all run on our 3's real posterior




**Laplace:** find the peak, fit a Gaussian to the curvature there. Fast, local.

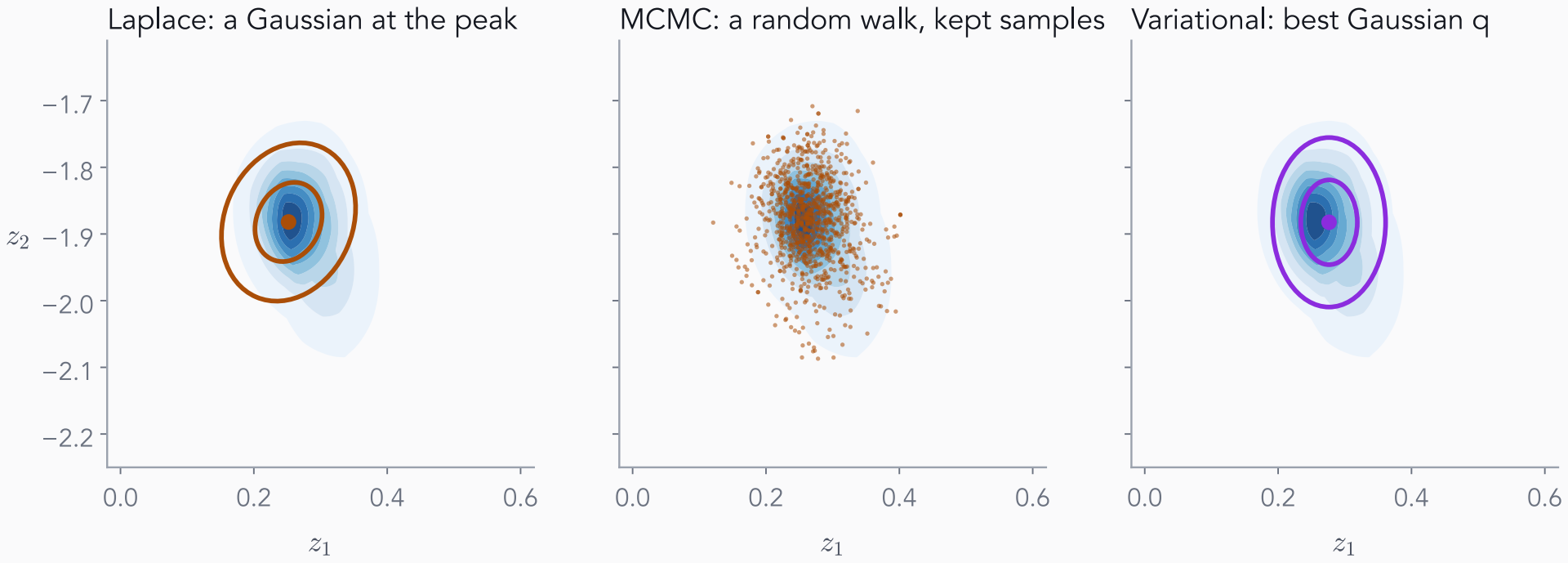
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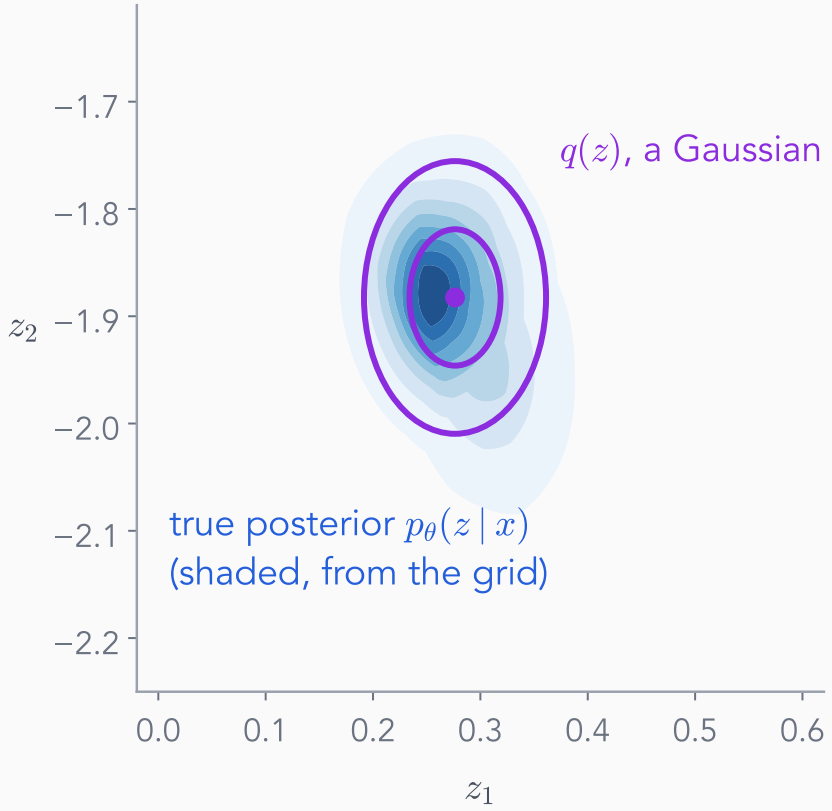
**MCMC:** a random walk that visits regions in proportion to the posterior. Exact in the long run, slow.

**Variational:** choose a simple family $q$ and optimise it to be as close as possible. Fast, trainable.

> **Today: only the variational one. It turns inference into optimisation, which is what neural networks are good at.**

THE VARIATIONAL IDEA, FOR OUR 3

# Propose a simple $q$, then pull it towards the true posterior

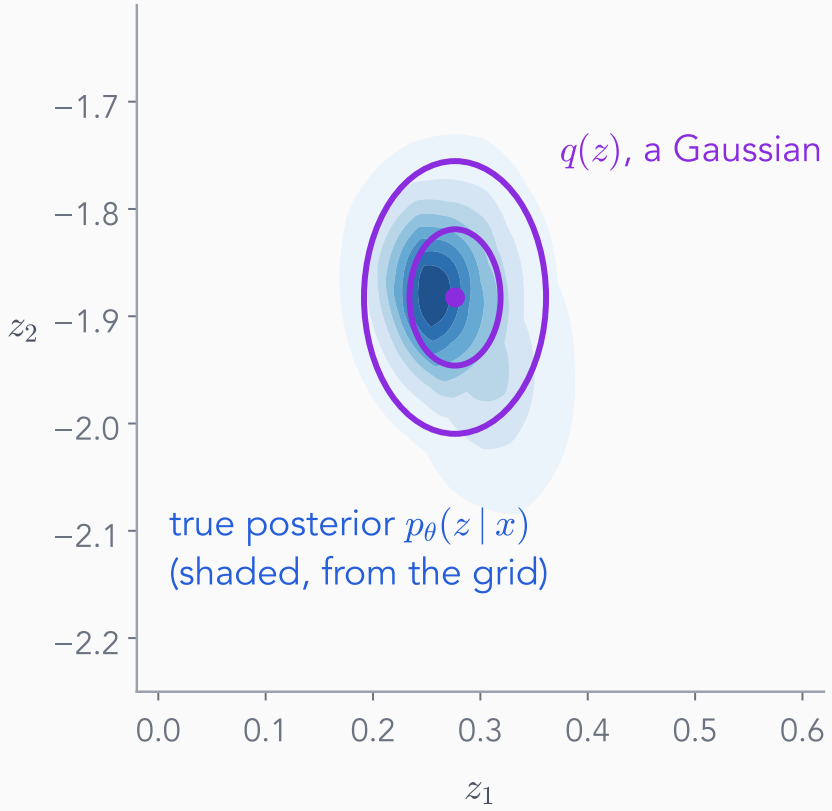



$$q(z) = \mathcal{N}\big(m,\ \mathrm{diag}(s_1^2, s_2^2)\big)$$

Four numbers to choose: a centre $m = (m_1, m_2)$ and two widths $s_1, s_2$.

THE VARIATIONAL IDEA, FOR OUR 3

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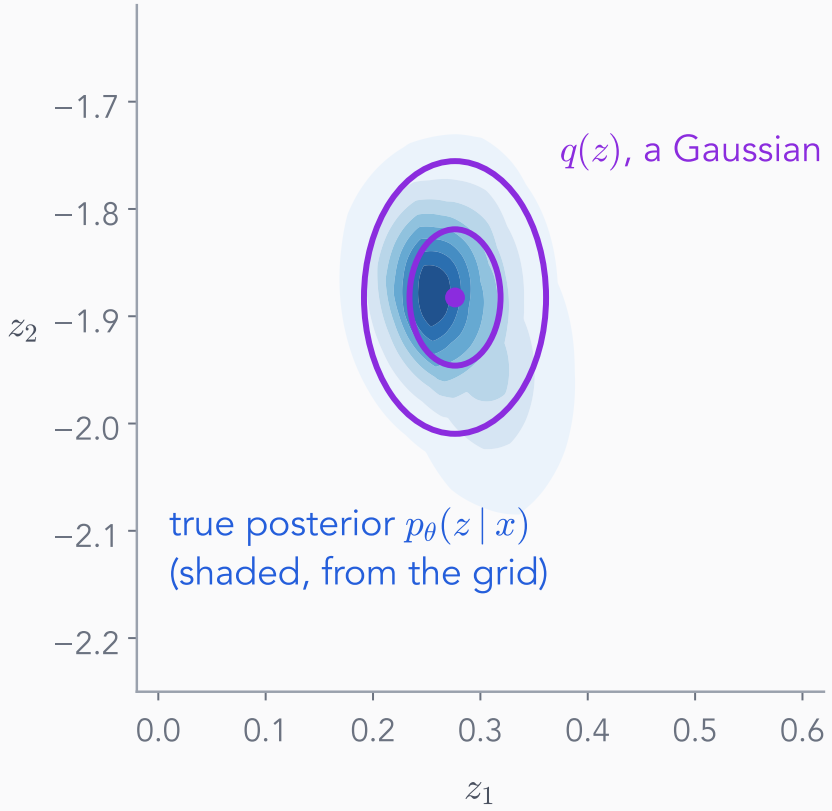

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Goal: make $q(z)$ as close as possible to $p_\theta(z \mid x)$, the shaded patch.

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"As close as possible" needs a **number** for how far apart two distributions are.

**That number is the KL divergence: the first thing in Session 2.**

END OF PART 3 · END OF SESSION 1

# What we did, and what comes next

### WHAT WE DID

1. Learn = make the data probable; generate = sample. Code: one loop.
2. Autoencoders compress and rebuild (plain, denoising, sparse, contractive), but cannot generate.
3. As a probability model: prior $p(z) = \mathcal{N}(0, I)$, likelihood $p_\theta(x \mid z)$ from the decoder.
4. Training needs $p_\theta(x)$ and the posterior $p_\theta(z \mid x)$: both intractable.
5. Variational idea: a simple $q$, pulled close to the posterior.

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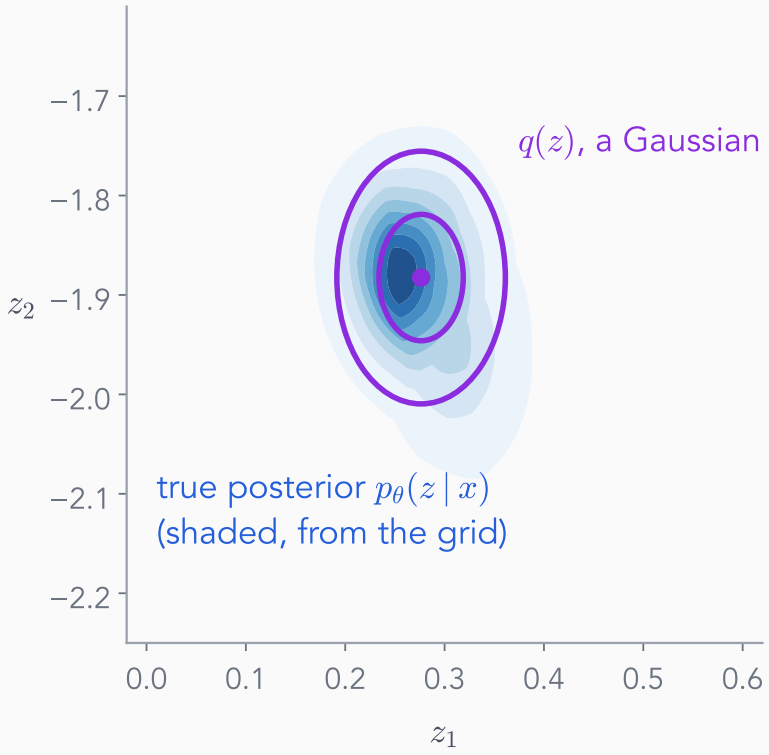
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### SESSION 2

A distance between distributions (KL), the loss that pulls $q$ close without ever computing $p_\theta(x)$ (the ELBO), gradients through random draws, and the VAE on MNIST.

SESSION 2 · WHERE WE STOPPED

# Last time: we need $q$ close to a posterior we cannot compute




A decoder plus a prior is a probability model: $p(z)\,p_\theta(x \mid z)$.

Training it needs $p_\theta(x) = \int p(z)\,p_\theta(x \mid z)\,dz$: intractable.

So does the posterior $p_\theta(z \mid x)$, the shaded patch for our 3.

**Variational idea:** a simple Gaussian $q$ (purple), pulled close to it.

**Today: how to measure "close" (KL), the loss that does the pulling (the ELBO), and the VAE.**

PART 4 OF 7

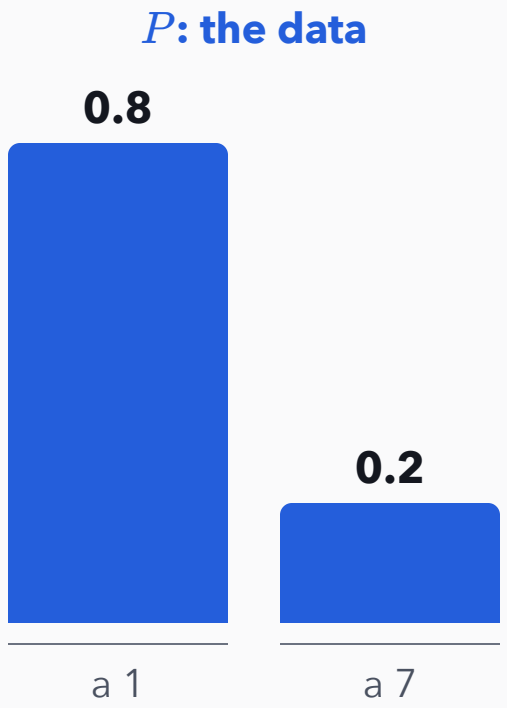
# KL divergence: how far apart are two distributions?

We want $q$ **close** to the posterior. Before two clouds in a plane, the simplest possible pair: two bar charts over the digits 1 and 7. Slowly, from a betting game.


✓ Distributions and fitting
✓ Autoencoders
✓ The probabilistic view
4 **KL divergence**
5 Variational inference and the ELBO
6 Reparameterisation
7 The VAE on MNIST

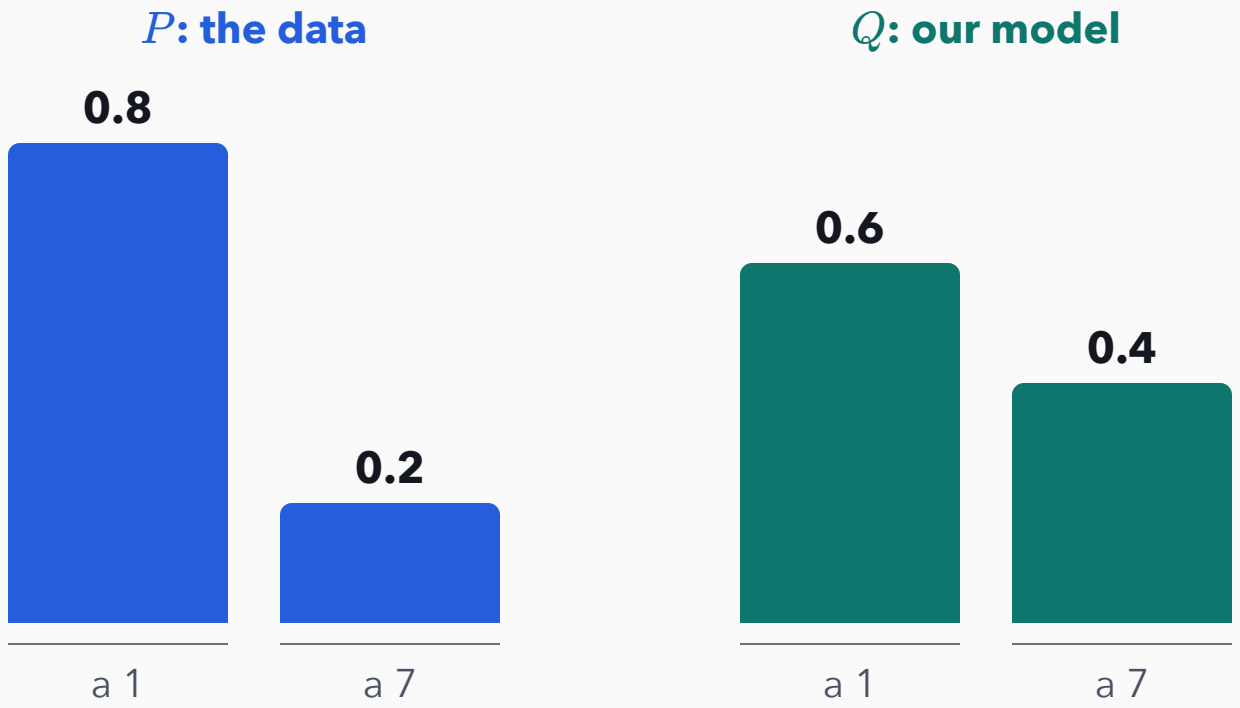
TWO DISTRIBUTIONS OVER THE SAME TWO OUTCOMES

# The data says 80/20. The model says 60/40.



TWO DISTRIBUTIONS OVER THE SAME TWO OUTCOMES

# The data says 80/20. The model says 60/40.




A dataset of 1s and 7s, 80% 1s. The model believes 60/40. How wrong is it, as one number?

A BETTING GAME

# Which model makes the real data more likely?

Ten digits from the data: eight 1s, two 7s. Each model gives each digit a probability; multiply them.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $P$ **(the truth)** | **1** 0.8 | **1** 0.8 | **1** 0.8 | **7** 0.2 | **1** 0.8 | **1** 0.8 | **1** 0.8 | **1** 0.8 | **7** 0.2 | **1** 0.8 |

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$$0.8^8 \times 0.2^2 = 0.0067$$

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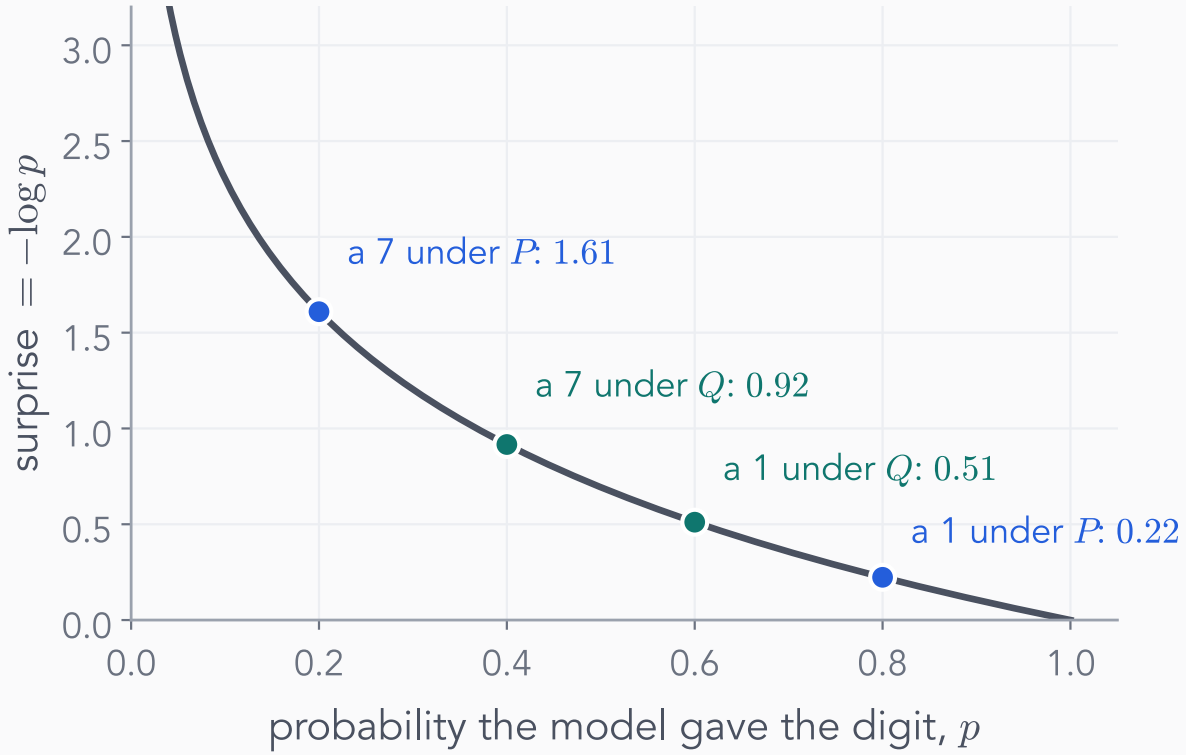
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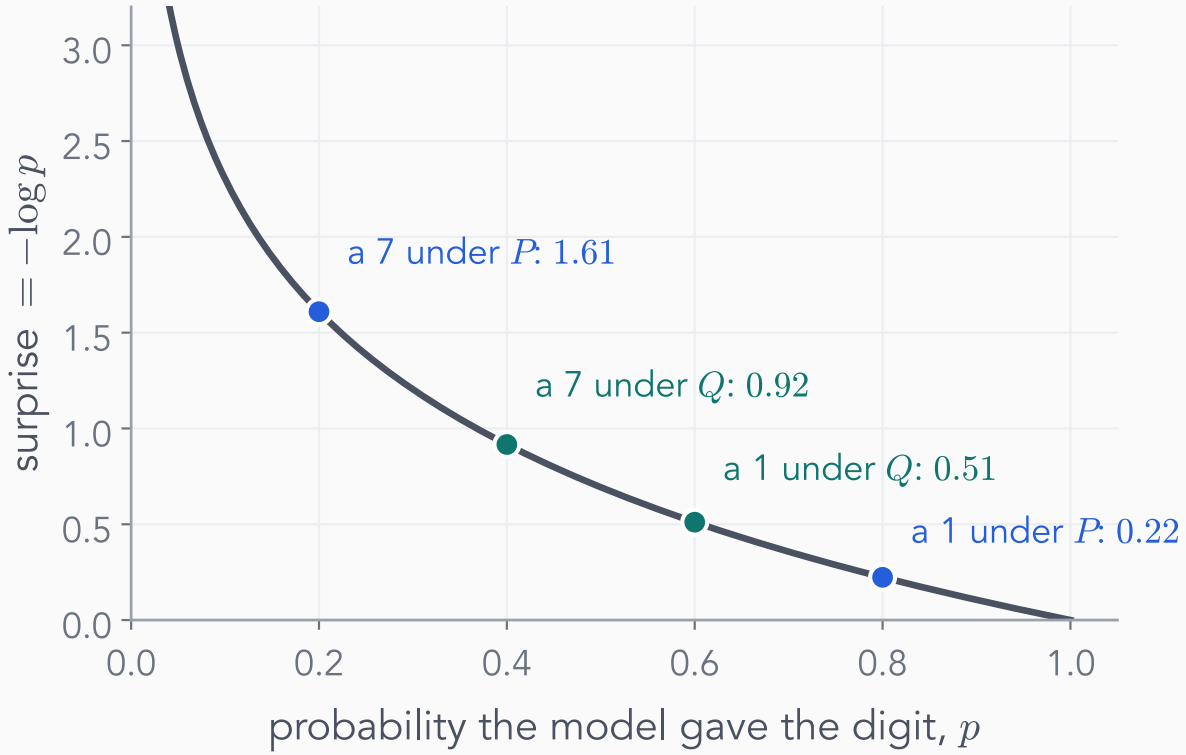
**The truth wins, by a factor of $0.0067/0.0027 = 2.5$. KL will measure this gap, per digit.**

# Surprise $= -\log p$, and why the log




What should "surprise" do?

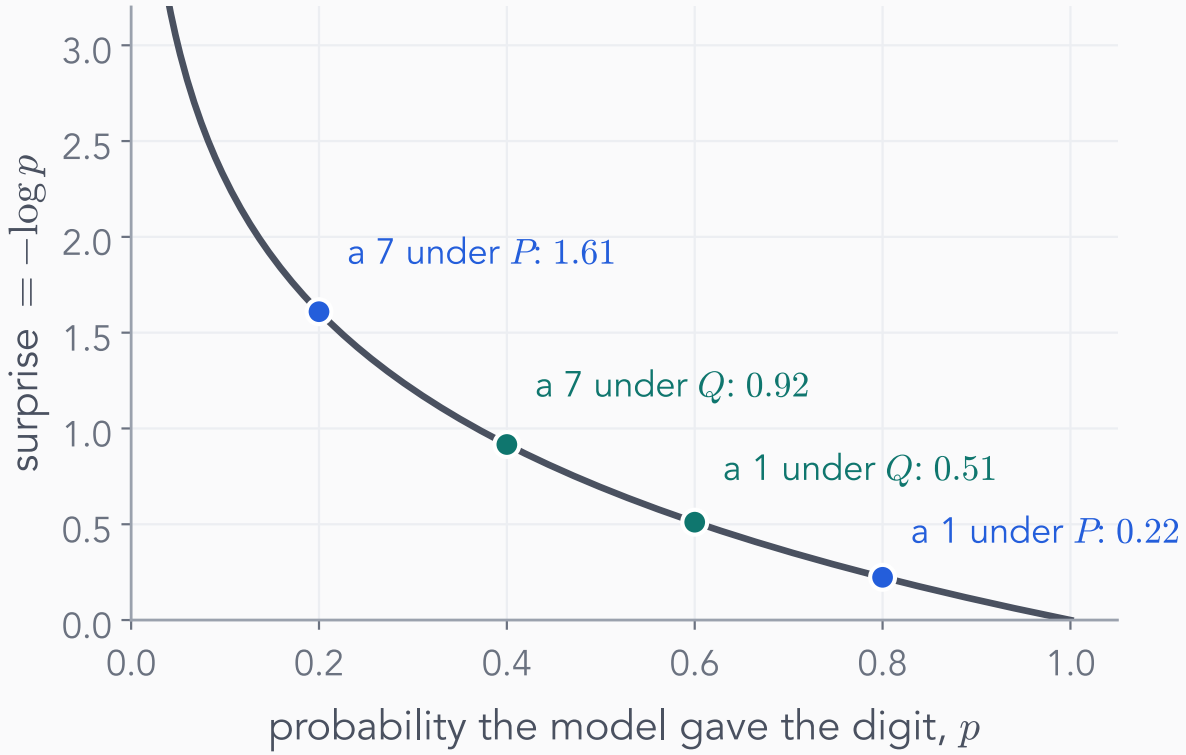
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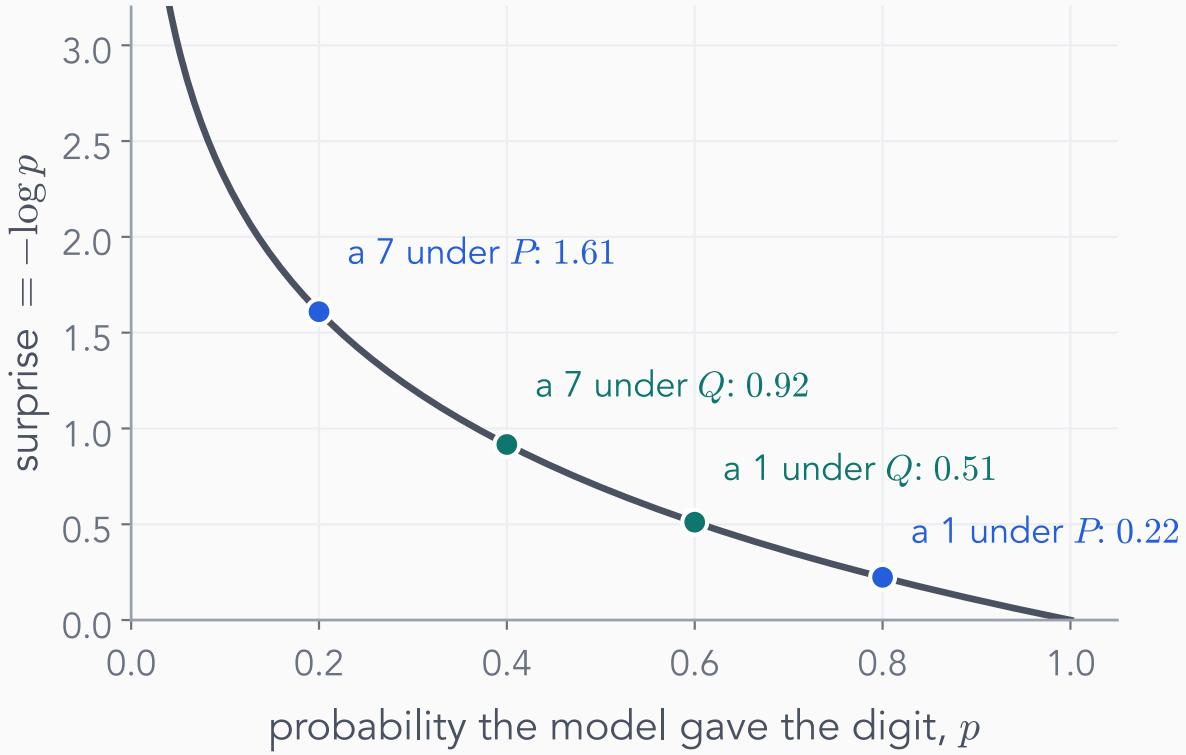



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2 · rarer means more surprising. $-\log p$ grows as $p$ shrinks.

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2 · rarer means more surprising. $-\log p$ grows as $p$ shrinks.

3 · two independent digits: probabilities **multiply**, surprises should **add**. Only the log turns products into sums.

> **So the betting-game product becomes a sum of surprises, one per digit. Natural log, so the unit is the nat.**

# The surprise table

| outcome | $P$ | $Q$ | surprise if you believe $P$ | surprise if you believe $Q$ |
|---|---|---|---|---|
| a 1 | 0.8 | 0.6 | $-\log 0.8 = 0.223$ | |
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OUR TWO DIGITS, SCORED BOTH WAYS

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Each row is one kind of digit. The next two slides average each column, weighted by how often the digit really appears.

Believing $Q$, a 1 surprises us more than it should ($0.511$ vs $0.223$), a 7 less ($0.916$ vs $1.609$).

AVERAGE SURPRISE, WHEN YOU BELIEVE THE TRUTH

# Entropy: $H(P) = 0.500$ nats per digit

$$H(P) = \sum_i P_i \left(-\log P_i\right) = \underbrace{0.8}_{\text{how often a 1}} \times 0.223 + \underbrace{0.2}_{\text{how often a 7}} \times 1.609 = 0.500$$

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The weights are how often each digit really appears: eight in ten, two in ten. The batch of ten digits from the game, averaged.

**Even the perfect model is surprised by 0.500 nats per digit on average: the data are random.**

SAME DATA, A DIFFERENT SCORER

## Cross-entropy: $H(P, Q) = 0.592$ nats per digit

$$H(P, Q) = \sum_i P_i \left(-\log Q_i\right) = 0.8 \times 0.511 + 0.2 \times 0.916 = 0.592$$

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**Still weighted by $P$**

the digits still come from the real data: eight 1s, two 7s.

**Scored by $Q$**

each digit's surprise now uses the model's probability.

**Believing the wrong model costs more surprise on the same data: 0.592 instead of 0.500.**

THE EXTRA SURPRISE

# KL divergence = cross-entropy − entropy

$$H(P) = 0.500$$

unavoidable: the data are random

together: $H(P, Q) = 0.592$ (more precisely $0.5004 + 0.0915 = 0.5919$)

THE EXTRA SURPRISE

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|---|---|
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$$\mathrm{KL}(P \,\|\, Q) = H(P, Q) - H(P) = \sum_i P_i \log \frac{P_i}{Q_i} = 0.092 \text{ nats}$$

**KL is the part of the surprise that comes only from using the wrong model.**

# Where the 0.092 comes from

| digit | how often $P_i$ | extra surprise $\log(P_i/Q_i)$ | contribution |
|---|---|---|---|
| a 1 | 0.8 | $\log(0.8/0.6) = 0.2877$ | $0.2301$ |

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DIGIT BY DIGIT, AND BACK TO THE GAME

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| **KL** | | | $0.0915$ |

$$\text{the game:} \quad \tfrac{1}{10} \log \frac{0.0067}{0.0027} = \tfrac{1}{10} \log 2.497 = 0.0915$$

> One row is negative: $Q$ over-predicts 7s, so a 7 surprises it less. The total is still positive, and it is exactly the betting-game gap, per digit.

NOTATION WE WILL USE ALL LECTURE

# $\mathbb{E}$ means "average over"

$$\mathbb{E}_{x \sim P}\big[f(x)\big] = \sum_x P(x)\, f(x)$$

exact, for a sum over outcomes

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$$\mathbb{E}_{x \sim P}\big[f(x)\big] \approx \frac{1}{S} \sum_{s=1}^{S} f(x_s), \quad x_s \sim P$$

estimated from samples

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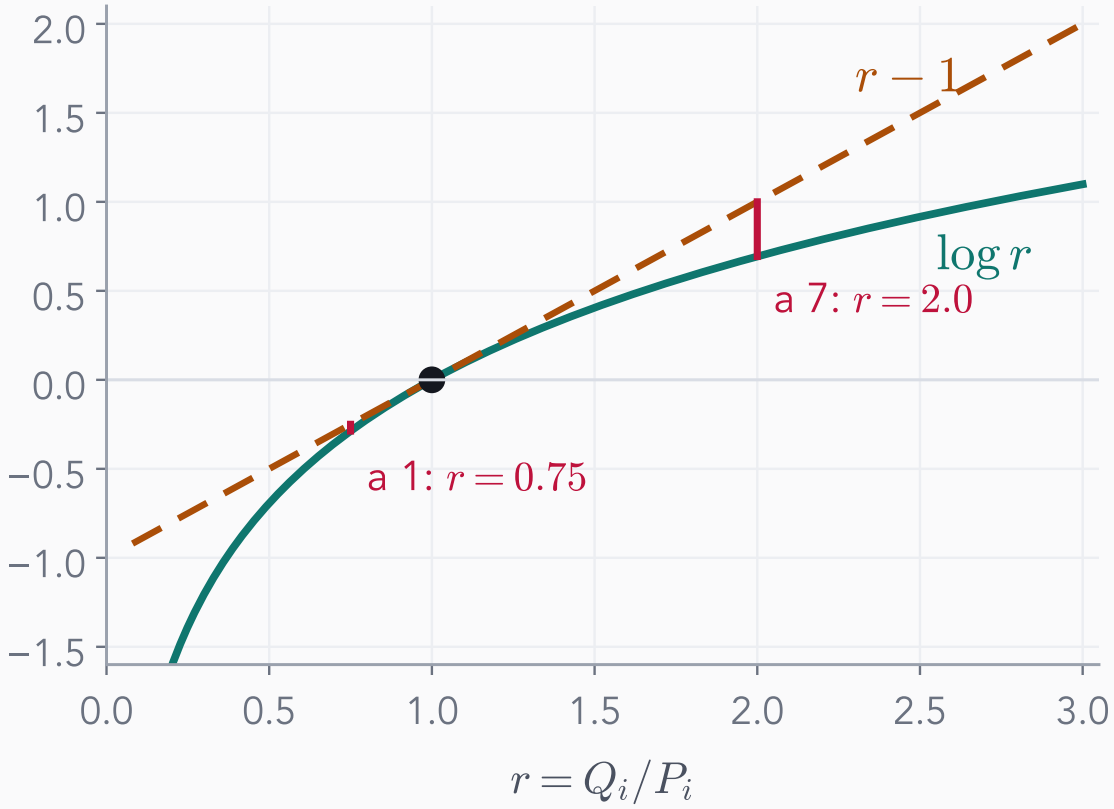
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KL is an average: $\mathbf{KL}(P\|Q) = \mathbb{E}_{x\sim P}\big[\log \frac{P(x)}{Q(x)}\big]$. Five draws 1, 1, 7, 1, 1 give $\frac{1}{5}(4 \times 0.288 - 0.693) = 0.092$.

WHY KL CAN NEVER BE NEGATIVE

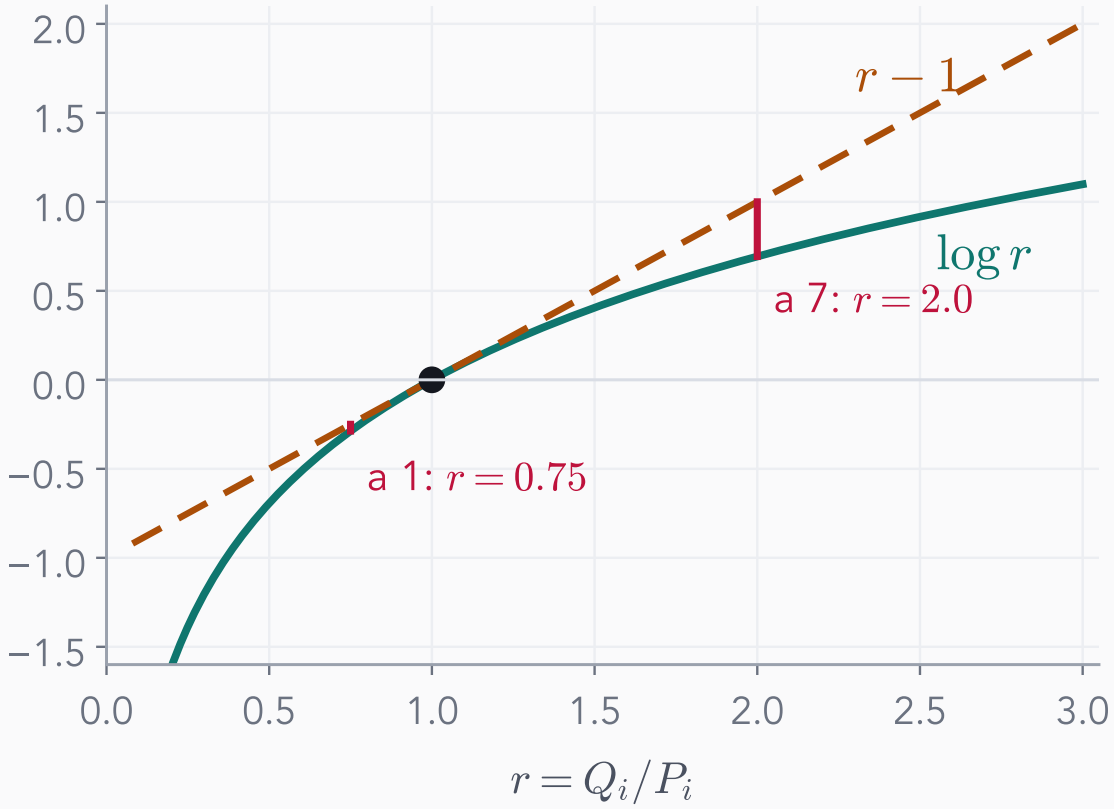
# One picture: $\log r$ lies under its tangent




Let $r = Q_i/P_i$. The curve $\log r$ never rises above the line $r - 1$.

WHY KL CAN NEVER BE NEGATIVE

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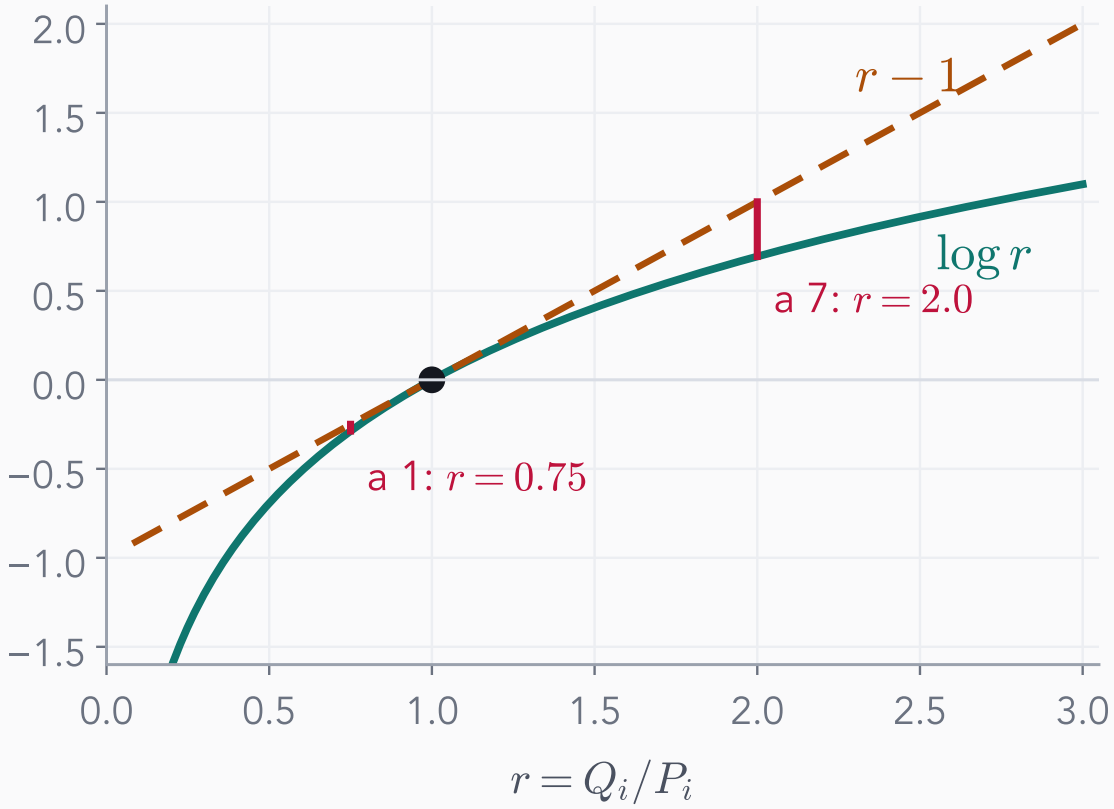



Let $r = Q_i/P_i$. The curve $\log r$ never rises above the line $r - 1$.

$$\mathrm{KL} = \sum_i P_i \left(-\log r_i\right) \;\geq\; \sum_i P_i \left(1 - r_i\right)$$

WHY KL CAN NEVER BE NEGATIVE

# One picture: $\log r$ lies under its tangent




Let $r = Q_i/P_i$. The curve $\log r$ never rises above the line $r - 1$.

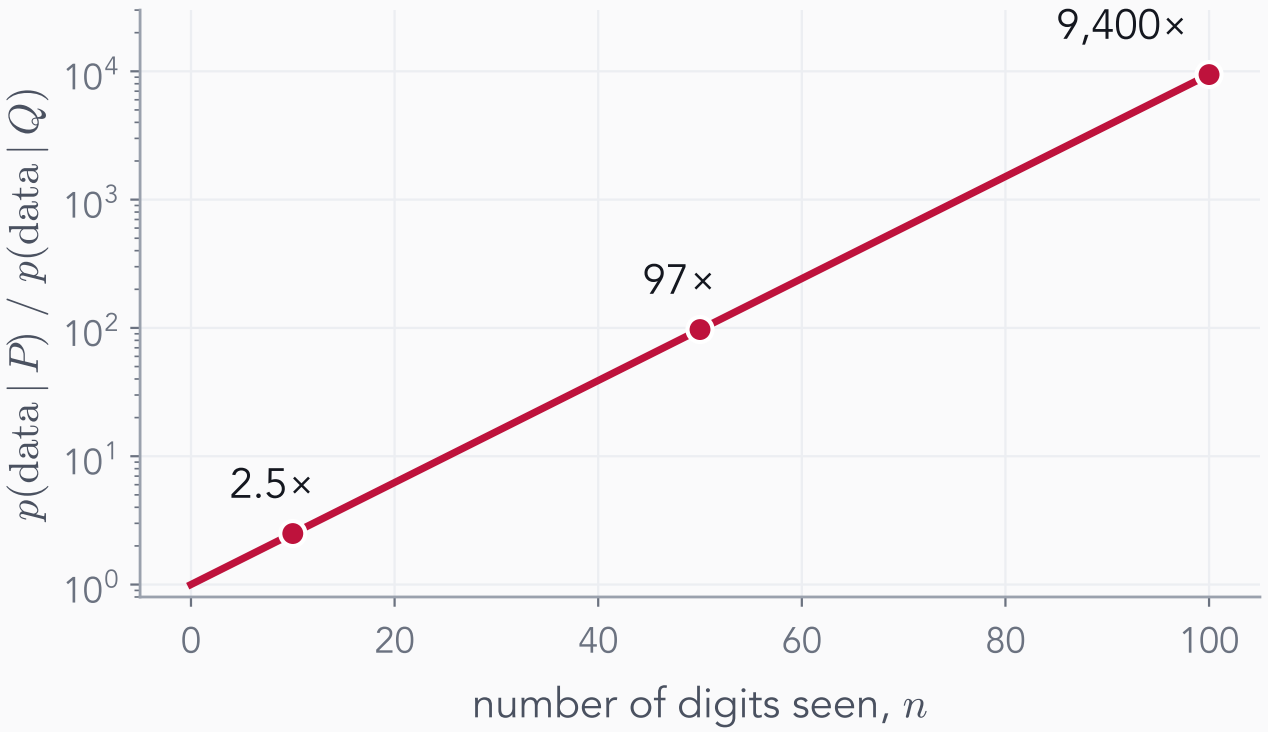
$$\mathrm{KL} = \sum_i P_i \left(-\log r_i\right) \;\geq\; \sum_i P_i \left(1 - r_i\right)$$

$$= \sum_i P_i - \sum_i Q_i = 1 - 1 = 0$$

**Equality only when every $r_i = 1$: $Q = P$. No model beats the truth on the truth's own data.**

WHAT THE NUMBER MEANS

# KL is how fast evidence piles up




$$\frac{p(\text{data} \mid P)}{p(\text{data} \mid Q)} \approx e^{n\,\mathrm{KL}(P\|Q)}$$

After 10 digits the truth is 2.5× more likely; after 50, 97×; after 100, 9,400×.

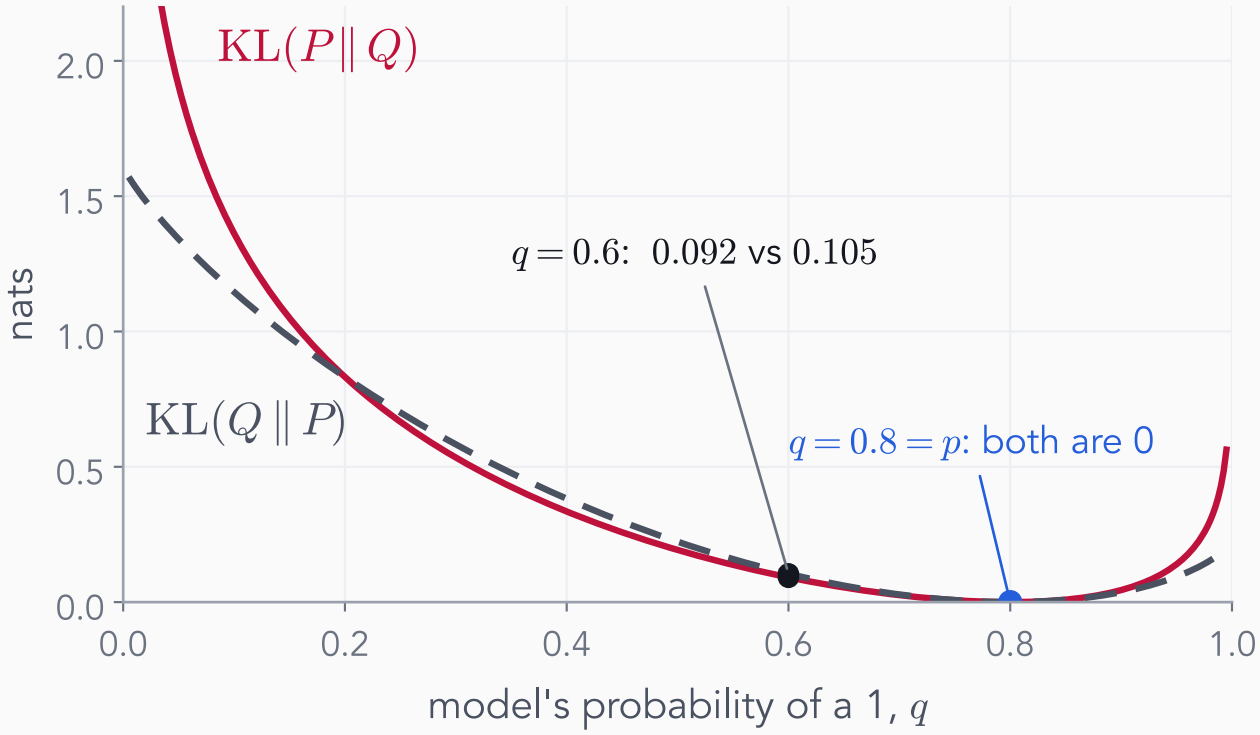
**A small KL means the two models are hard to tell apart from data; a large one means a few samples settle it.**

NOT A DISTANCE

# Swap the arguments, get a different number

$$\mathrm{KL}(Q\|P) = 0.6\log\frac{0.6}{0.8} + 0.4\log\frac{0.4}{0.2}$$

$$= -0.1726 + 0.2773 = 0.1046$$



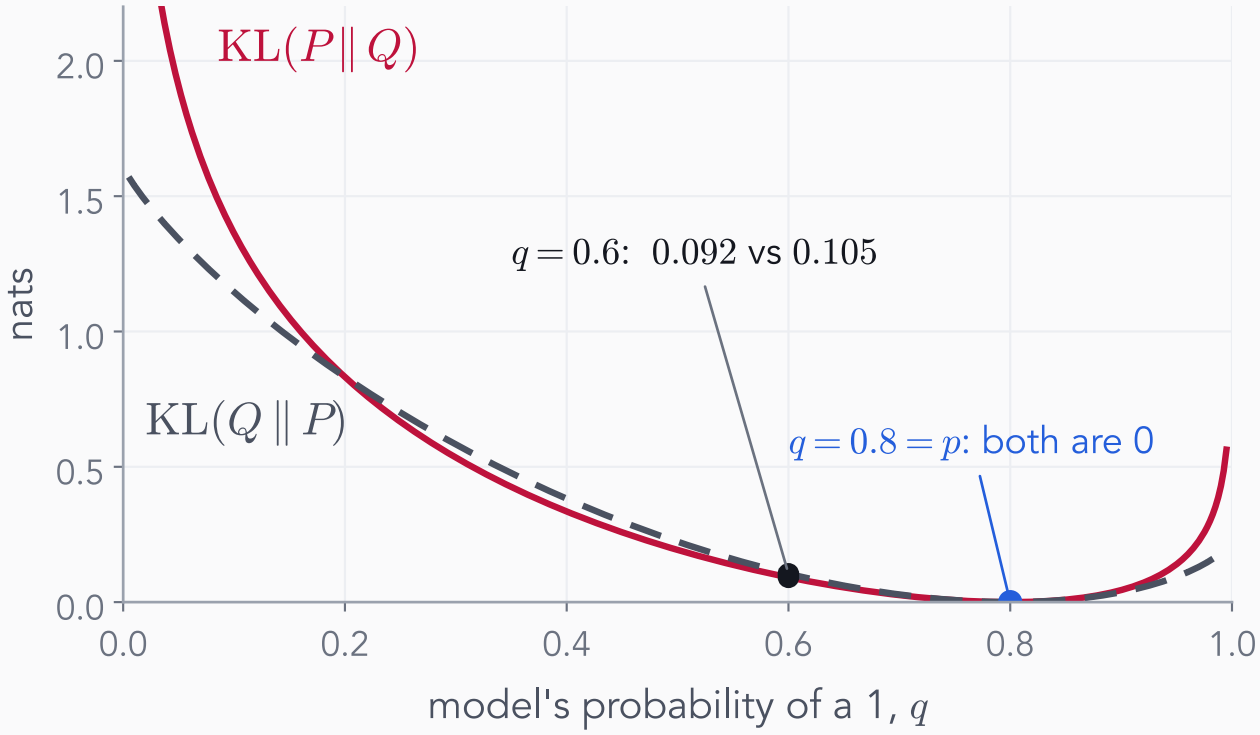
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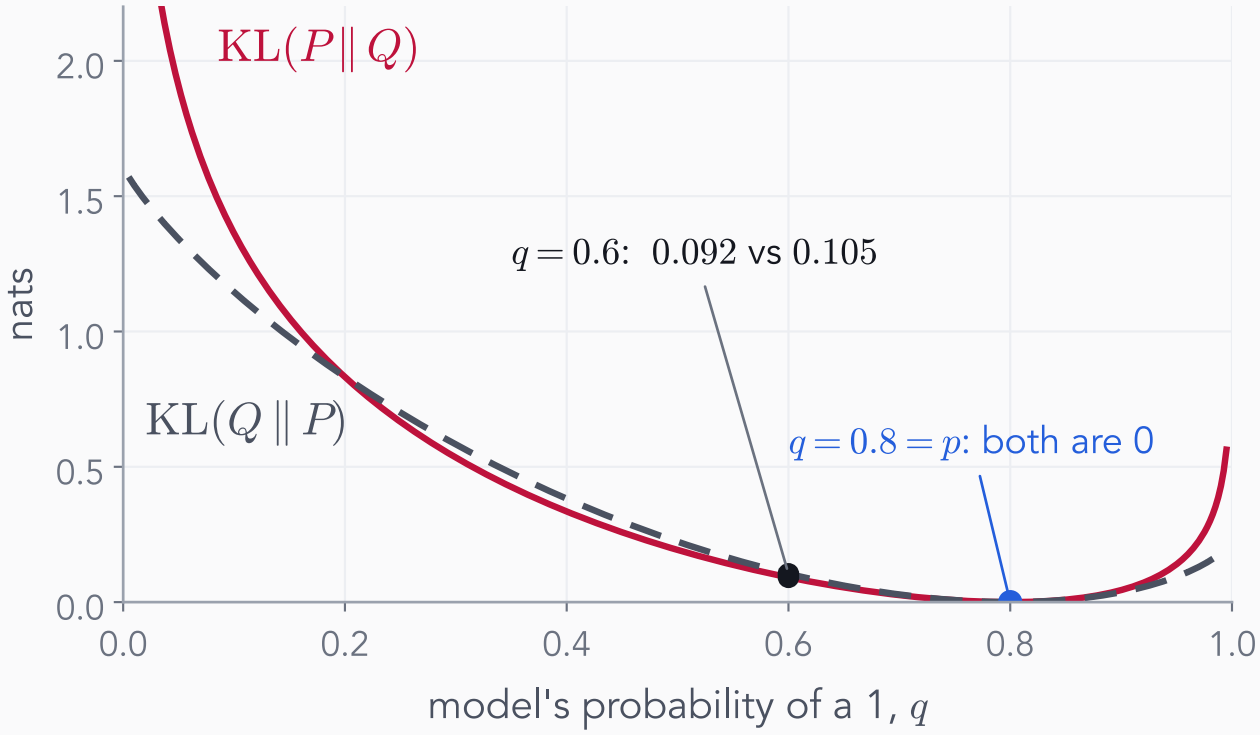
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$0.105 \neq 0.092$: KL is not symmetric. "How wrong is $Q$ about $P$" differs from "how wrong is $P$ about $Q$".

Push $q \to 0$ and $\mathrm{KL}(P\|Q) \to \infty$. Never give zero probability to something that happens.



THE ORDER MATTERS, SHARPLY

# A model that "forgets" 7s

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$$\mathrm{KL}(P\|Q') = 0.43$$

real 7s keep showing up, and each costs $\log(0.2/0.01) = 3.0$ nats

$$\mathrm{KL}(Q'\|P) = 0.18$$

averaged under $Q'$, which almost never draws a 7, so the costly digit rarely counts

**$\mathrm{KL}(P\|Q)$ punishes a model for giving tiny probability to things that really happen.**

# Cross-entropy = entropy + KL

$$H(P, Q) = -\sum_i P_i \log Q_i$$

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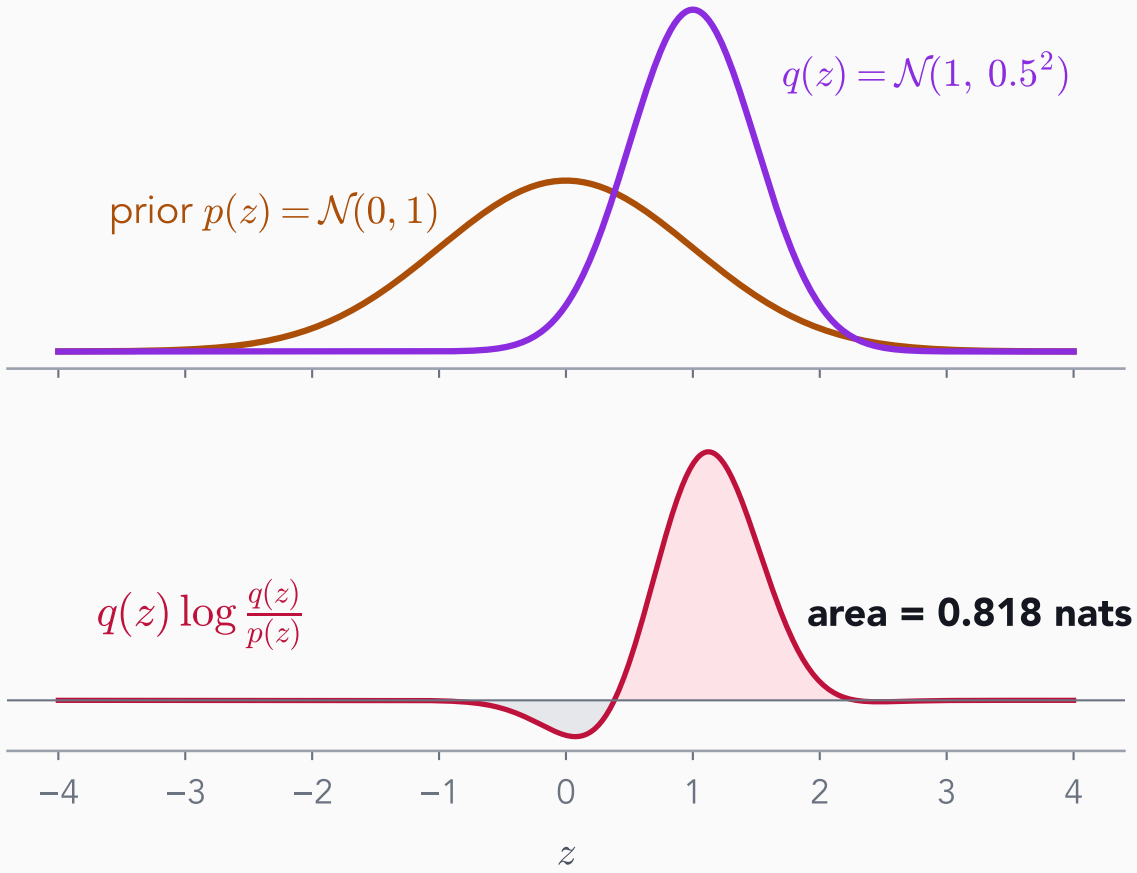
add and subtract $\sum_i P_i \log P_i$

$$= \underbrace{H(P)}_{\text{entropy: fixed by the data}} + \mathrm{KL}(P \,\|\, Q)$$

**Lowering cross-entropy can only lower the KL part. So does maximum likelihood: it minimises $\mathrm{KL}(\text{data} \,\|\, \text{model})$.**

THE VERSION THE VAE NEEDS

# KL between two bumps: an area

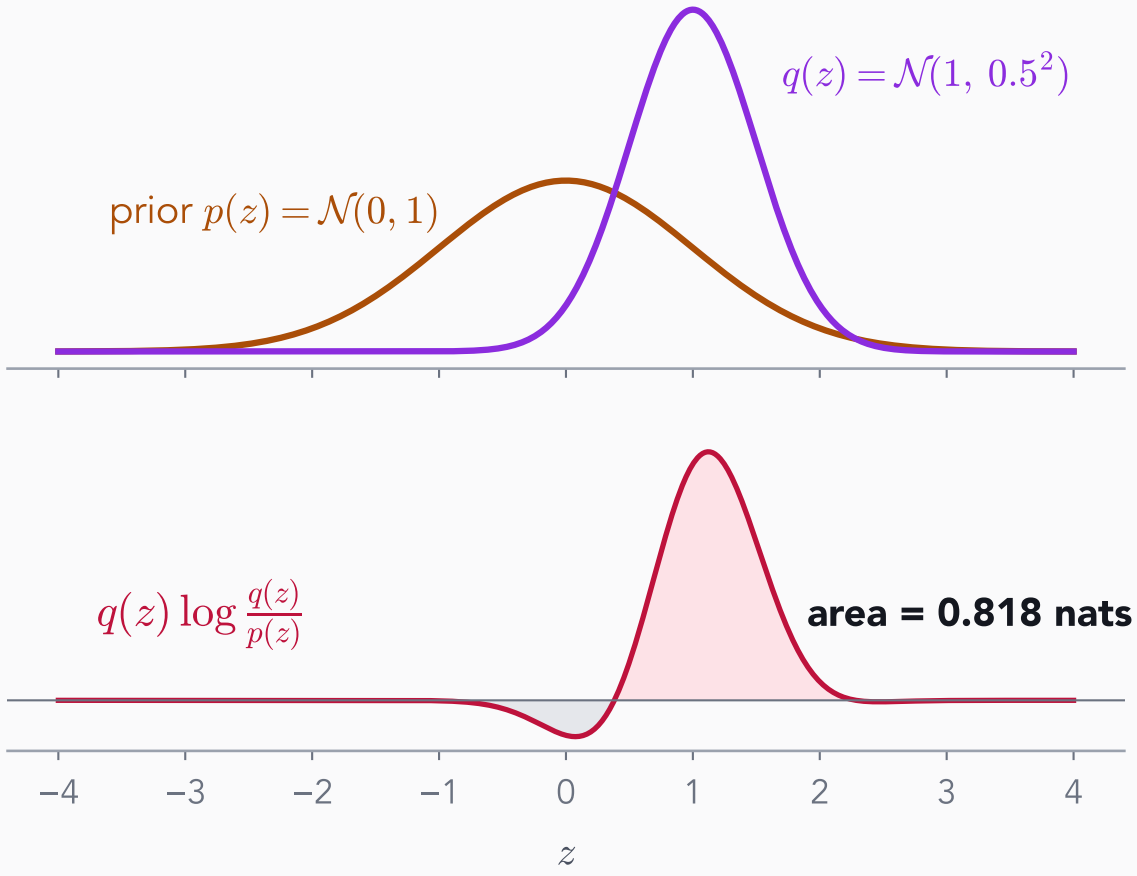



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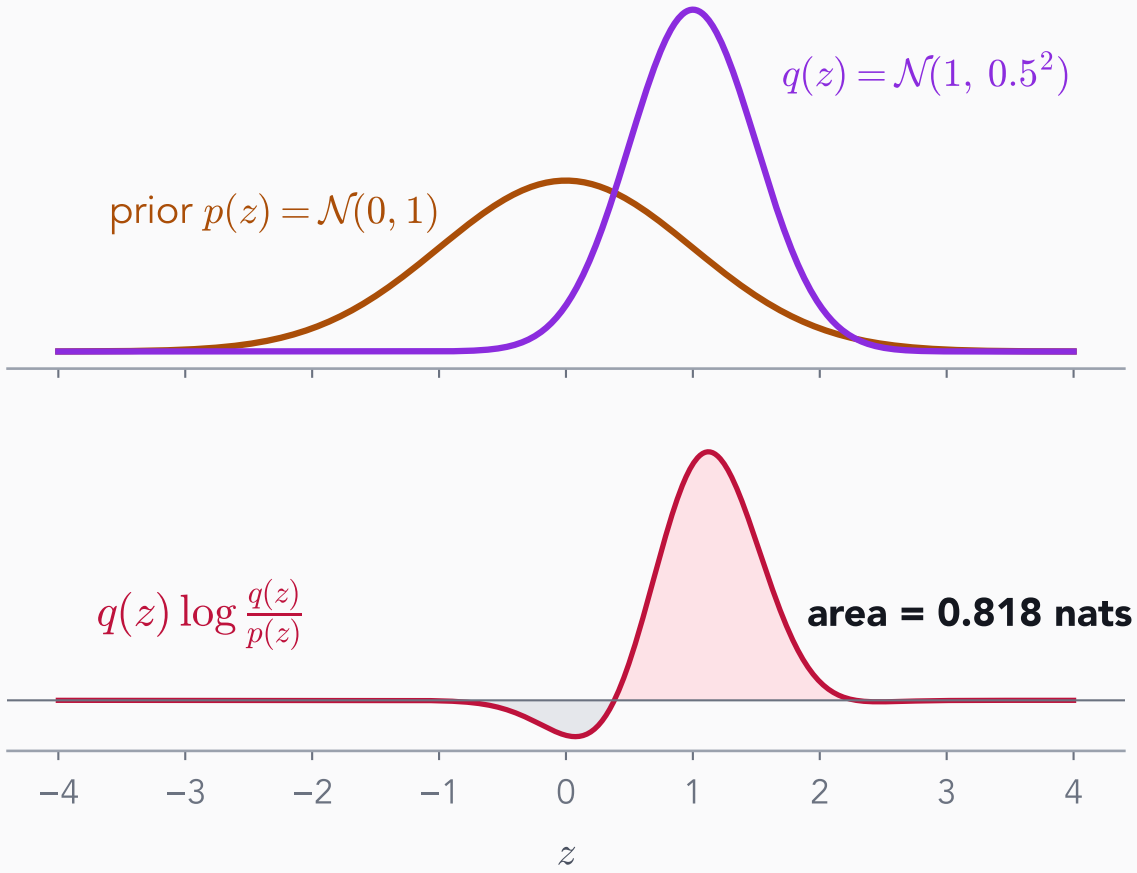

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The shaded area, computed numerically, is **0.818** nats.

**Can we get 0.818 without a computer? Yes: for two Gaussians the integral has a formula. Two steps.**

GAUSSIAN KL, STEP 1 OF 2

# Write out the log of the ratio

$$\log q(z) = -\log \sigma - \tfrac{1}{2}\log 2\pi - \frac{(z-\mu)^2}{2\sigma^2}$$

log of $\mathcal{N}(z; \mu, \sigma^2)$

GAUSSIAN KL, STEP 1 OF 2

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log of $\mathcal{N}(z; 0, 1)$

$$\log \frac{q(z)}{p(z)} = -\log \sigma - \frac{(z-\mu)^2}{2\sigma^2} + \frac{z^2}{2}$$

subtract: the $\frac{1}{2}\log 2\pi$ terms cancel

**Three terms. Next: average each one over $z \sim q$.**

GAUSSIAN KL, STEP 2 OF 2

# Average each term over the cloud $q$

$\mathbb{E}_q\left[(z-\mu)^2\right] = \sigma^2$ (that is the variance)

$\mathbb{E}_q\left[z^2\right] = \mu^2 + \sigma^2$ (mean squared + variance)

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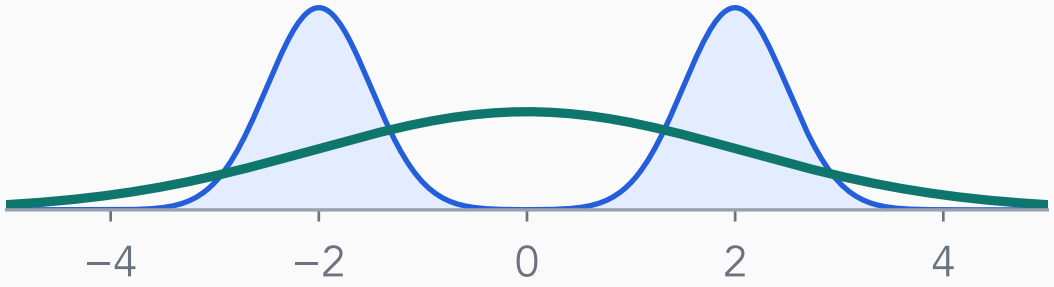
$$\mu = 1,\ \sigma = 0.5: \quad \tfrac{1}{2}\left(1 + 0.25 - 1 + 1.386\right) = 0.818\ \checkmark$$

**Per coordinate: moving the mean costs $\frac{1}{2}\mu^2$; a width other than 1 costs $\frac{1}{2}(\sigma^2 - 1 - \log\sigma^2)$, zero only at $\sigma = 1$. Several coordinates: add one term each.**

BACK TO OUR TWO BUMPS

# Fit one Gaussian: the KL direction decides what you get

best $q = \mathcal{N}(0.0,\ 2.06^2)$

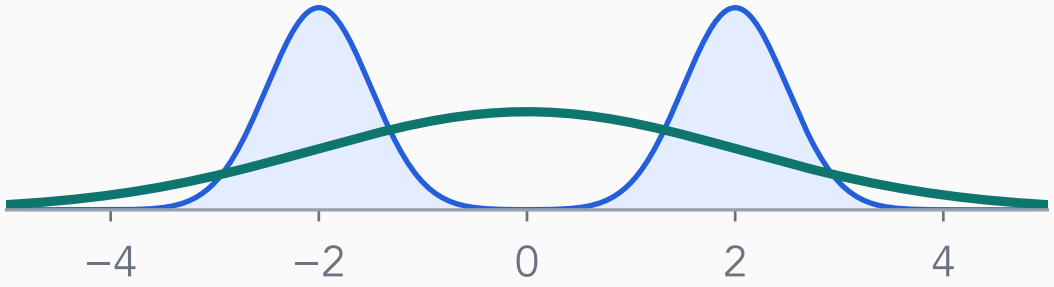



minimise $\mathrm{KL}(\mathrm{data} \,\|\, q)$: cover everything.
Part 1's maximum-likelihood fit, with infinite data.

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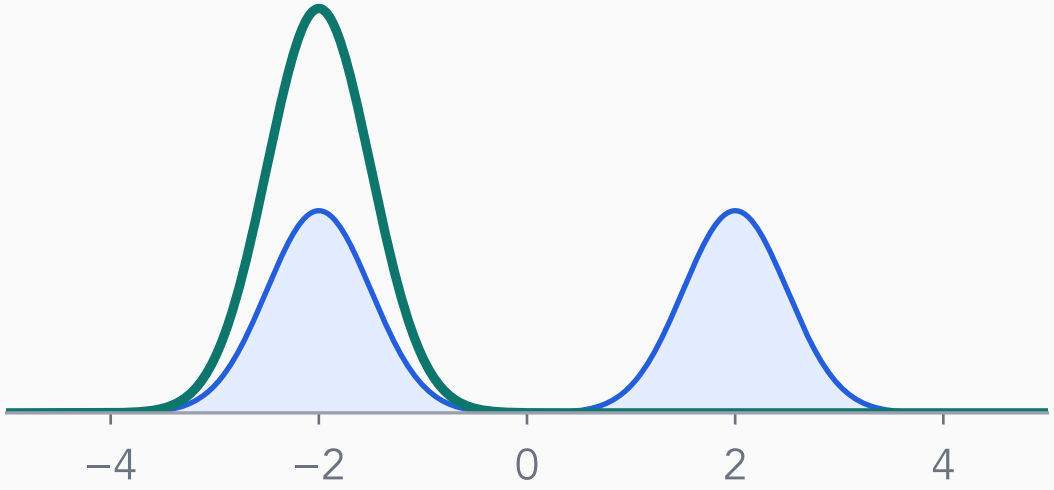
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Part 1's maximum-likelihood fit, with infinite data.

best $q = \mathcal{N}(-2.0,\ 0.50^2)$




minimise $\mathrm{KL}(q \,\|\, \mathrm{data})$: never put mass where data is absent, so pick one bump.

**Mode-covering vs mode-seeking. Variational inference will minimise $\mathrm{KL}(q \,\|\, \mathrm{posterior})$: the second kind.**

END OF PART 4

# What we did, and what comes next

### WHAT WE DID

1. Surprise $= -\log p$; entropy = average surprise under the truth.
2. Cross-entropy = the same data scored by a model; KL = the extra.
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### NEXT: MAKE $q$ CLOSE TO THE POSTERIOR

Minimise $\mathrm{KL}(q \,\|\, p_\theta(z \mid x))$. It contains the intractable $p_\theta(x)$; Part 5 rearranges it into a loss that does not.

PART 5 OF 7

# Variational inference, and the ELBO

Part 3 wanted a simple $q$ close to the posterior of our 3. Part 4 gave a number for "close". Write the goal down: one rearrangement turns it into a loss we can compute, the ELBO. Then run it, and see why it leads to an encoder.


- ✓ Distributions and fitting
- ✓ Autoencoders
- ✓ The probabilistic view
- ✓ KL divergence
- 5 **Variational inference and the ELBO**
- 6 Reparameterisation
- 7 The VAE on MNIST

THE GOAL, WRITTEN DOWN

# Find the $q$ with the smallest KL to the posterior

$$q^\star = \arg\min_{q \in \mathcal{Q}} \mathrm{KL}\big( q(z) \,\|\, p_\theta(z \mid x) \big)$$

**$\mathcal{Q}$: the shapes we allow**

Diagonal Gaussians. For our 2-number code: 4 numbers, a centre $m$ and two widths $s$.

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### The catch

$p_\theta(z \mid x) = p_\theta(x, z) \,/\, p_\theta(x)$, and $p_\theta(x)$ is the integral we cannot do. We cannot even evaluate this KL.

**Four short moves get $p_\theta(x)$ out of the way.**

# Write out the KL

$$\mathrm{KL}\big(q(z) \,\|\, p_\theta(z \mid x)\big)$$

$$= \mathbb{E}_{z\sim q}\Big[\, \log q(z) - \log p_\theta(z \mid x) \,\Big]$$

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The definition from Part 4: the average, under the first argument, of the log of the ratio. The highlighted term is the one we cannot compute.

# Bayes: the posterior is the joint over the evidence

$$\mathbb{E}_q\big[\log q(z) - \log p_\theta(z \mid x)\big]$$

$$= \mathbb{E}_q\big[\log q(z) - \log p_\theta(x, z)\big] + \log p_\theta(x)$$

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$\log p_\theta(z \mid x) = \log p_\theta(x, z) - \log p_\theta(x)$. The evidence has no $z$ in it: the same number for every draw, so it comes out of the average.

**The joint $p_\theta(x, z) = p(z)\, p_\theta(x \mid z)$ is easy: prior times decoder. All the difficulty is now in one number.**

# Move the evidence to the other side

$$\mathrm{KL}\big(q \,\|\, p_\theta(z \mid x)\big) = \mathbb{E}_q\big[\log q(z) - \log p_\theta(x, z)\big] + \log p_\theta(x)$$

$$\log p_\theta(x) = \underbrace{\mathbb{E}_q\big[\log p_\theta(x, z) - \log q(z)\big]}_{\mathrm{ELBO}(q)} + \mathrm{KL}\big(q \,\|\, p_\theta(z \mid x)\big)$$

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Rearrange, and give the highlighted average a name: the **ELBO**. Every piece of it is computable: the joint is prior times decoder, and $q$ is ours.

**Why "Evidence Lower BOund"? The last move.**

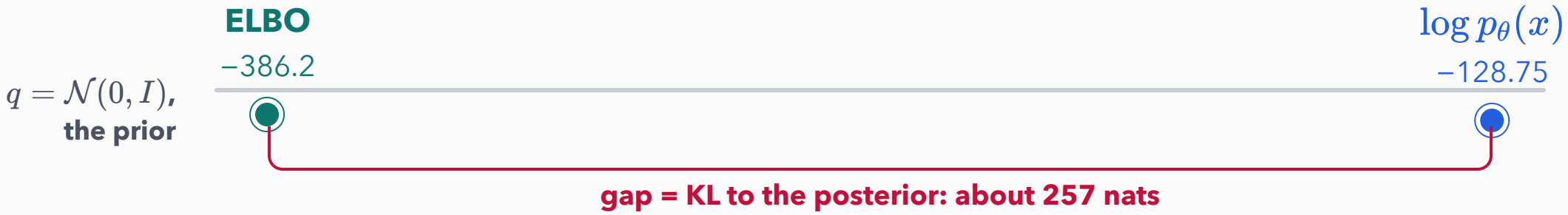
# The KL is never negative, so the ELBO is a lower bound

$$\log p_\theta(x) = \mathrm{ELBO}(q) + \underbrace{\mathrm{KL}\big(q \,\|\, p_\theta(z \mid x)\big)}_{\geq 0} \;\geq\; \mathrm{ELBO}(q)$$

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Our 3, two choices of $q$. The right end, $\log p_\theta(x)$, is fixed by the model; it does not move when $q$ changes.



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Our 3, two choices of $q$. The right end, $\log p_\theta(x)$, is fixed by the model; it does not move when $q$ changes.




> The total cannot move, so raising the ELBO must shrink the KL: maximise the ELBO and $q$ approaches the posterior, without ever computing $p_\theta(x)$.

# The joint is prior times decoder

$$\mathrm{ELBO}(q) = \mathbb{E}_q\big[\log p_\theta(x, z) - \log q(z)\big]$$

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$p_\theta(x, z) = p(z)\, p_\theta(x \mid z)$: the generative story from Part 3. The log of a product is a sum of logs.

**Why unpack? To see which parts are the decoder and which are the prior, and to get two terms with names.**

# The average of a sum is the sum of the averages

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$\mathbb{E}[a+b] = \mathbb{E}[a] + \mathbb{E}[b]$. Group the decoder term on its own, and the two terms that only involve $z$ together.

# Recognise the second average: a KL to the prior

$$\mathbb{E}_q\big[\log p(z) - \log q(z)\big]$$

$$= -\,\mathbb{E}_q\left[\log \frac{q(z)}{p(z)}\right] = -\mathrm{KL}\big(q(z)\,\|\,p(z)\big)$$

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Flip the sign to flip the fraction. Then use the definition: $\mathrm{KL}(q\|p) = \mathbb{E}_{z\sim q}\big[\log \frac{q(z)}{p(z)}\big]$, the average extra surprise.

**And for Gaussians we already have its formula: $\frac{1}{2}\sum_j(\mu_j^2 + \sigma_j^2 - 1 - \log\sigma_j^2)$.**

PUT THE PIECES TOGETHER

# The ELBO, in symbols and in words

$$\text{ELBO} = \underbrace{\mathbb{E}_{z\sim q}\big[\log p_\theta(x \mid z)\big]}_{\text{rebuild } x \text{ from codes in its cloud}} - \underbrace{\mathrm{KL}\big(q(z) \,\|\, p(z)\big)}_{\text{keep the cloud near } \mathcal{N}(0,I)}$$

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First term: the autoencoder's reconstruction score, with the code drawn from the cloud.

Second term: the Gaussian KL from Part 4, in closed form.

**The two wishes from the end of Part 2, derived rather than invented: rebuild well, and keep codes where the prior can find them.**

CHECK IT WITH NUMBERS YOU CAN DO BY HAND

# Neat or messy writer? A two-value latent

$z \in \{\text{neat}, \text{messy}\}$, prior $\frac{1}{2}, \frac{1}{2}$. Observation $x$: a thick stroke.

$$p(x \mid \text{neat}) = 0.8, \quad p(x \mid \text{messy}) = 0.2$$

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Our guess $q = (q_0,\ 1 - q_0)$:

$q_0$ = `0.60`

| quantity | value |
|---|---|
| $\mathbb{E}_q[\log p(x \mid z)]$ | −0.778 |
| $-\,\mathrm{KL}(q \,\|\, \text{prior})$ | −0.020 |
| ELBO | **−0.798** |
| $+\,\mathrm{KL}(q \,\|\, \text{posterior})$ | 0.105 |
| $\log p(x)$ | **−0.693** |

**Move $q$: the ELBO and the gap change, their sum never does. At $q$ = posterior $(0.8)$ the gap is 0.**

VARIATIONAL INFERENCE BY HAND, FOR OUR 3

# Four numbers, gradient ascent on the ELBO

```
x = our_3                                  # 784 pixels, fixed
m     = torch.zeros(2, requires_grad=True)   # q's centre
log_s = torch.zeros(2, requires_grad=True)   # q's log-widths
opt = torch.optim.Adam([m, log_s], lr=0.03)

for step in range(800):
    z = m + log_s.exp() * torch.randn(64, 2)    # 64 draws from q
    rec = -bce(decoder(z), x).sum(1).mean()    # E_q[log p(x|z)]
    kl  = 0.5 * (m**2 + (2*log_s).exp() - 1 - 2*log_s).sum()
    loss = -(rec - kl)                           # minus the ELBO
    opt.zero_grad(); loss.backward(); opt.step()
```

**Only $q$ is learned.** The decoder is a trained one, kept fixed; $x$ is our 3. Start at $q = \mathcal{N}(0, I)$: $m = 0$, $s = 1$.

VARIATIONAL INFERENCE BY HAND, FOR OUR 3

# Four numbers, gradient ascent on the ELBO

```
x = our_3                                  # 784 pixels, fixed
m     = torch.zeros(2, requires_grad=True)   # q's centre
log_s = torch.zeros(2, requires_grad=True)   # q's log-widths
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**Draw codes from $q$:** centre plus width times standard noise, exactly how Part 1 sampled from a bump. Part 6 explains why gradients flow through this line.

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```

**The two terms:** reconstruction, averaged over the 64 draws; the KL to the prior by the Gaussian formula from Part 4.

VARIATIONAL INFERENCE BY HAND, FOR OUR 3

# Four numbers, gradient ascent on the ELBO

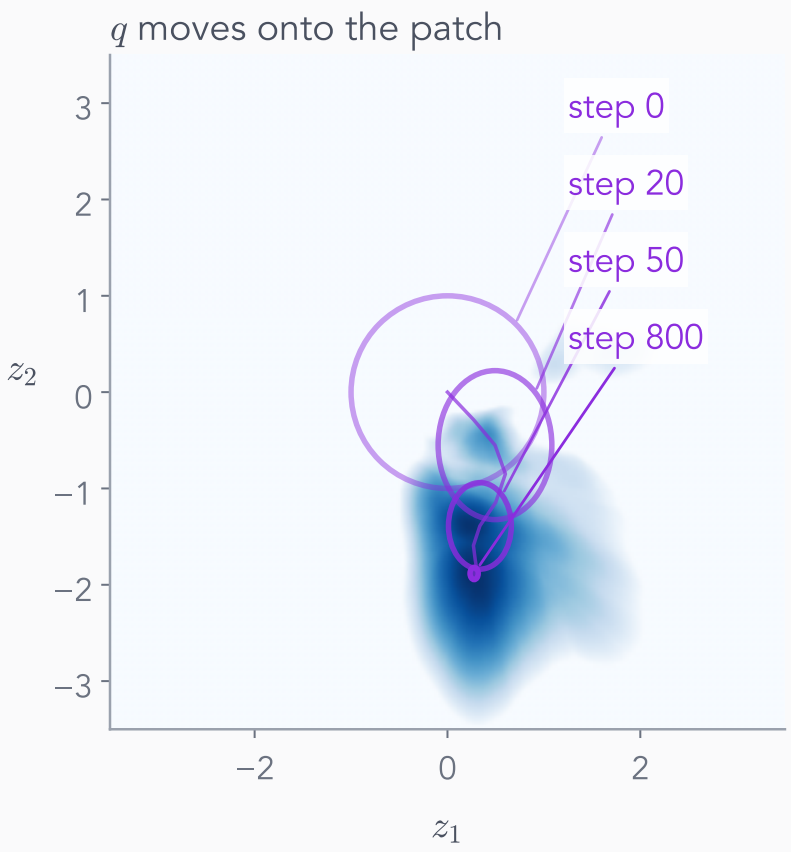
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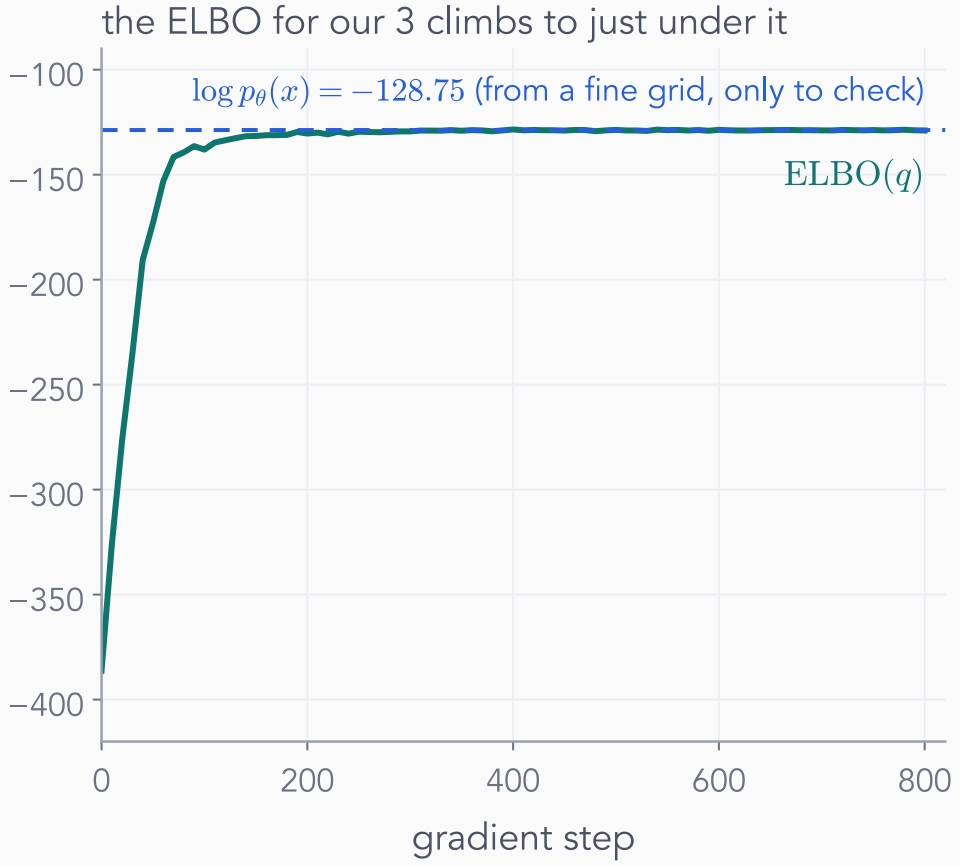
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```

**The same three-line loop** as fitting one Gaussian in Part 1. The ELBO rises from −386.0 to −128.8.

WATCH $q$ TRAVEL

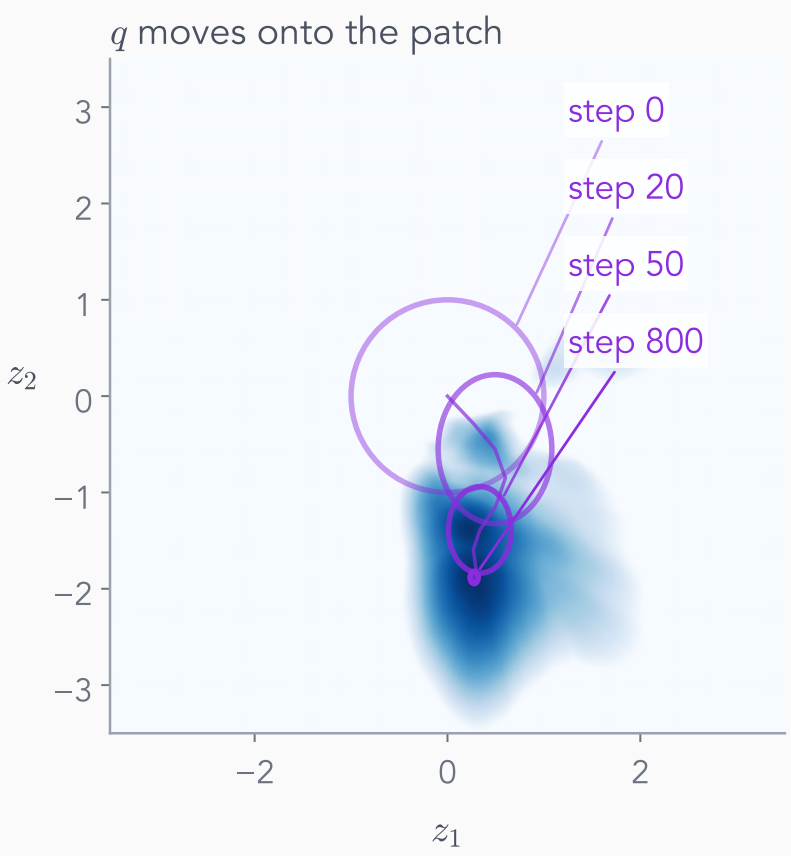
# From $\mathcal{N}(0, I)$ to our 3's patch in 800 steps

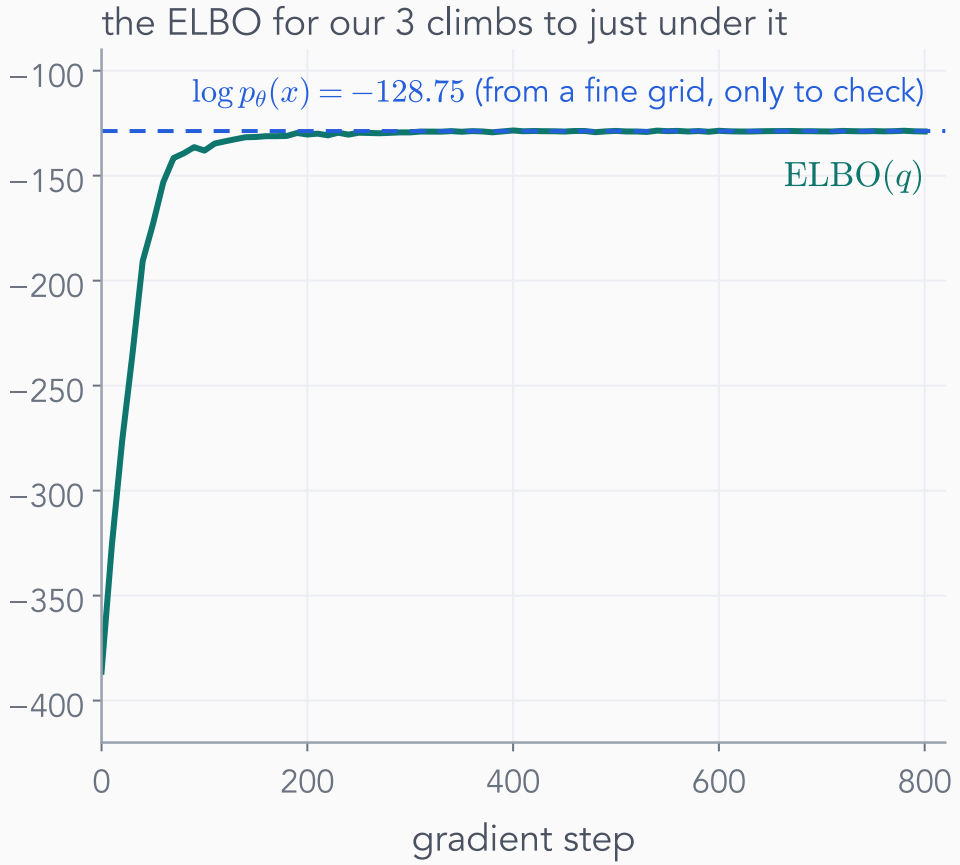


WATCH $q$ TRAVEL

# From $\mathcal{N}(0, I)$ to our 3's patch in 800 steps




**Final ELBO −128.83, just under $\log p_\theta(x) = -128.75$. The gap, $\mathrm{KL}(q \,\|\, \text{posterior})$, is 0.08 nats: $q$ is almost exactly right.**

INSIDE THOSE 800 STEPS

# The two terms pull against each other

| step | reconstruction $\mathbb{E}_q[\log p_\theta(x \mid z)]$ | $-\,\mathrm{KL}(q \,\|\, \text{prior})$ | ELBO |
|---|---|---|---|
| $0 \cdot q$ is the prior | −386.2 | — 0.0 | −386.2 |
| | | | |
| | | | |

INSIDE THOSE 800 STEPS

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| 20 | −275.9 | — 0.53 | −276.4 |
| 50 | −170.9 | — 2.1 | −173.0 |

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Reconstruction gains about 264 nats: codes from $q$ now decode to our 3.

The KL to the prior grows from 0 to 6.73: a narrow cloud far from the origin costs something.

**The ELBO is a trade: a $q$ precise enough to rebuild the image, no more precise than it needs to be.**

A BONUS: WITH $q$, EVEN $p_\theta(x)$ BECOMES EASY TO ESTIMATE

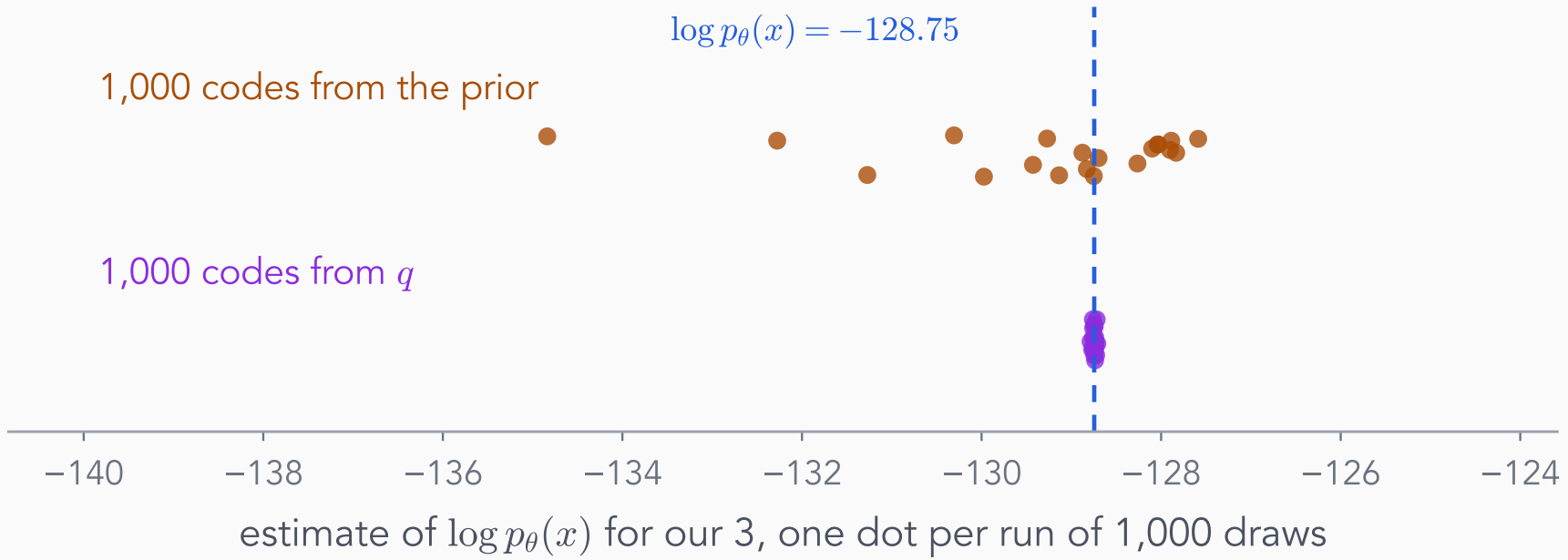
# Sample codes where the patch is, then reweight

$$p_\theta(x) = \int q(z)\,\frac{p(z)\,p_\theta(x \mid z)}{q(z)}\,dz = \mathbb{E}_{z\sim q}\left[\frac{p_\theta(x,z)}{q(z)}\right] \approx \frac{1}{S}\sum_{s=1}^{S}\frac{p_\theta(x,z_s)}{q(z_s)}$$

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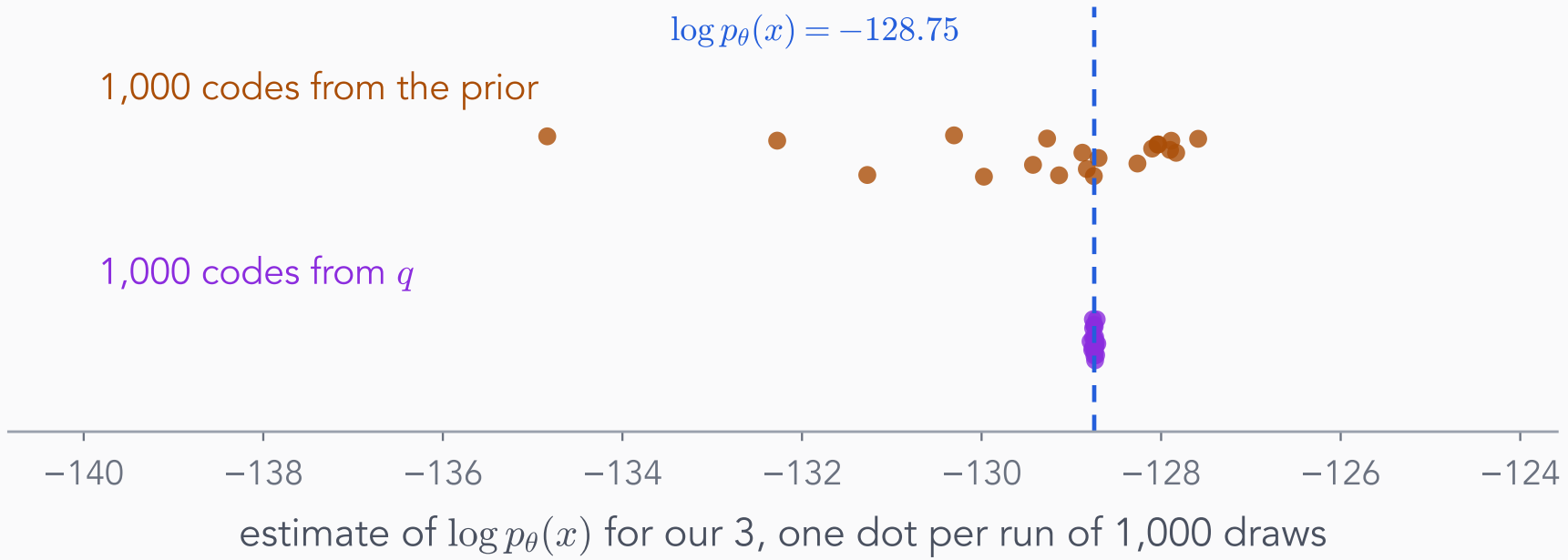
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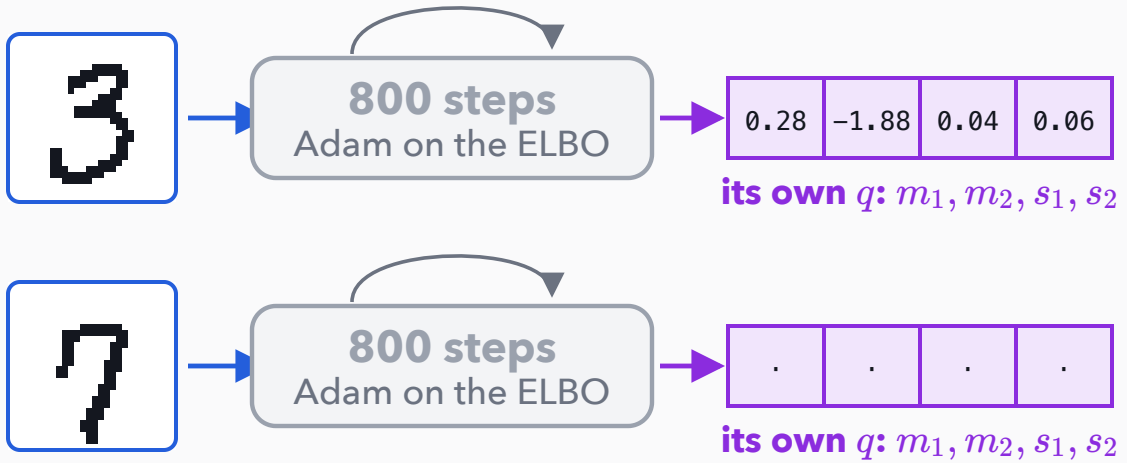



**20 runs of 1,000 draws each. From the prior, the estimates scatter from −134.8 to −127.6. From $q$: −128.74, give or take 0.017.**

800 STEPS FOR ONE IMAGE

# Amortised inference: learn to predict $q$

**VI by hand: one optimisation per image**




× 60,000 training images, redone each time the decoder changes, and again for every new image

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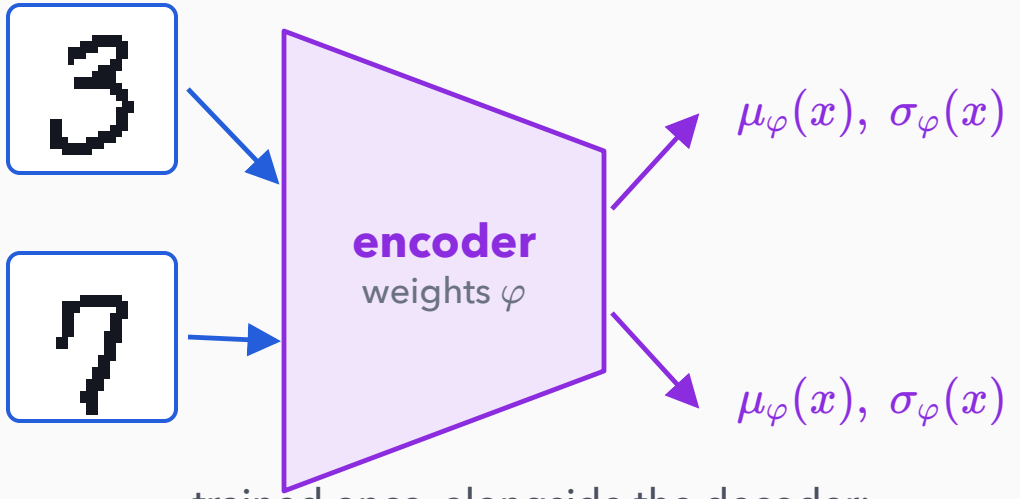
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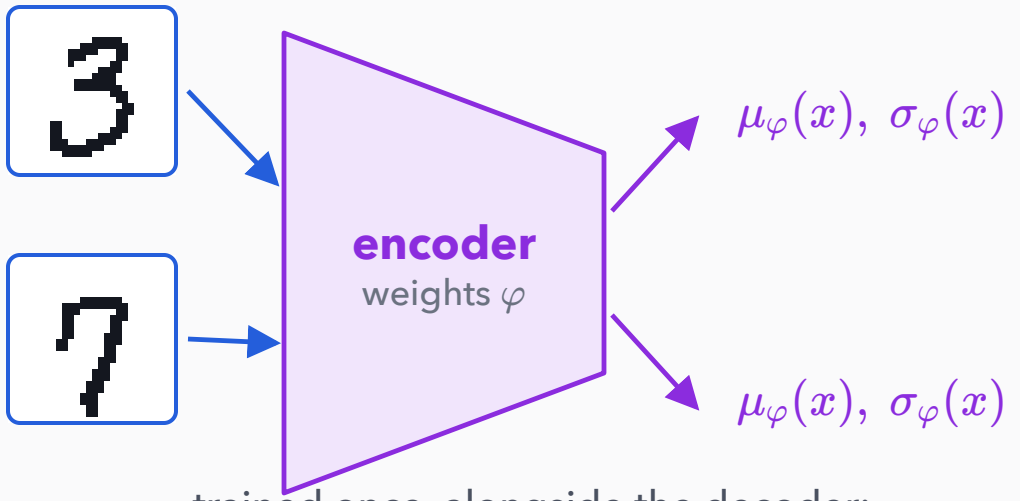
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> Our 3: ELBO −128.83 by hand in 800 steps, −129.3 from the encoder in one pass. Pay once during training; a little worse, thousands of times cheaper.

PUT THE PIECES TOGETHER

# Encoder + decoder + ELBO = the variational autoencoder

$$\max_{\theta,\varphi} \sum_{i=1}^{N} \left( \mathbb{E}_{z \sim q_\varphi(z|x_i)} \left[ \log p_\theta(x_i \mid z) \right] - \mathrm{KL}\left( q_\varphi(z \mid x_i) \,\|\, p(z) \right) \right)$$

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**"Variational"**

$q$ always depended on the image; now we write it $q_\varphi(z \mid x)$: the best Gaussian under the ELBO, predicted by a network.

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image → encoder → code → decoder → image: the shape from Part 2, now with a cloud for a code.

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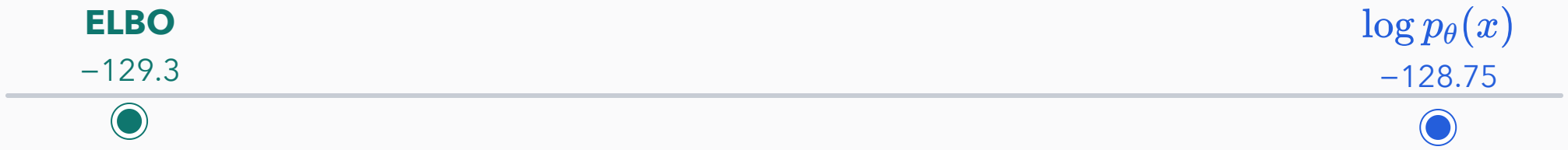
### Trained together

Pushing on $\theta$ raises the ELBO mostly by making the data more probable: a better model. Pushing on $\varphi$ shrinks the gap: a better $q$.

> **Kingma & Welling (2014) and Rezende et al. (2014): one network for the likelihood, one for the posterior, one loss for both.**

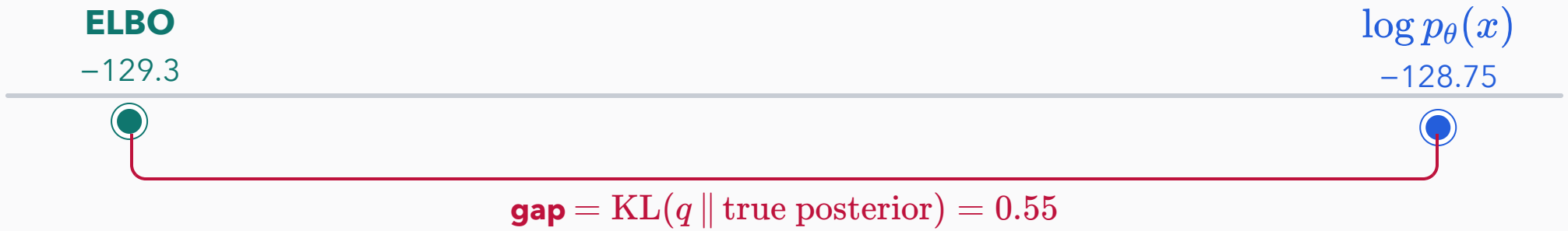
THE GAP FOR THE ENCODER'S $q$, MEASURED ON OUR 3

# Two different KLs, two different jobs



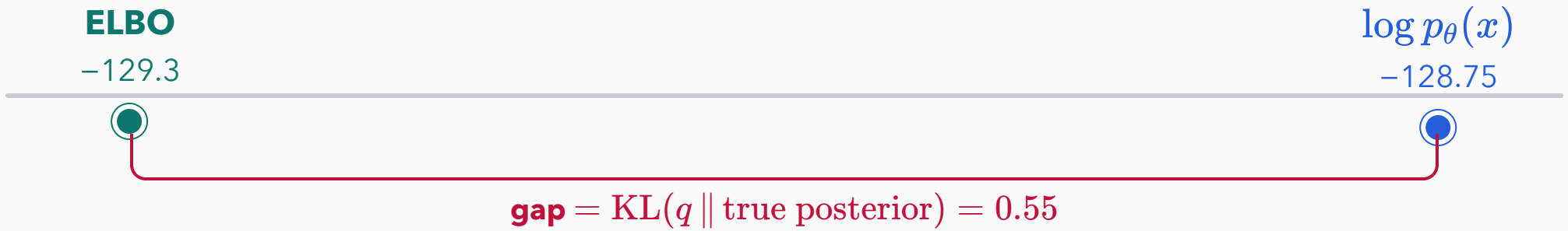
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THE GAP FOR THE ENCODER'S $q$, MEASURED ON OUR 3

# Two different KLs, two different jobs




**In the loss:** $\mathrm{KL}(q \,\|\, \text{prior}) = 6.97$ nats.
Computable by formula.

**In the gap:** $\mathrm{KL}(q \,\|\, \text{true posterior}) = 0.55$ nats.
Needs $p_\theta(x)$; we only got it from the grid.

> **VI by hand closed the gap to 0.08 nats in 800 steps; the encoder's single pass leaves 0.55. That small loss is the price of amortising.**

END OF PART 5

# What we did, and what comes next

## WHAT WE DID

1. Goal: minimise $\mathrm{KL}(q \,\|\, \text{posterior})$. It contains $p_\theta(x)$.
2. Rearranged: $\log p_\theta(x) = \mathrm{ELBO} + \mathrm{KL}$. Maximising the ELBO minimises the KL.
3. ELBO = reconstruction $-\ \mathrm{KL}(q \,\|\, \text{prior})$: both computable.
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### NEXT: GRADIENTS THROUGH A RANDOM DRAW

The loss draws $z$ from the encoder's cloud. Part 6: why writing the draw as $\mu + \sigma\varepsilon$ lets gradients reach the encoder.

PART 6 OF 7

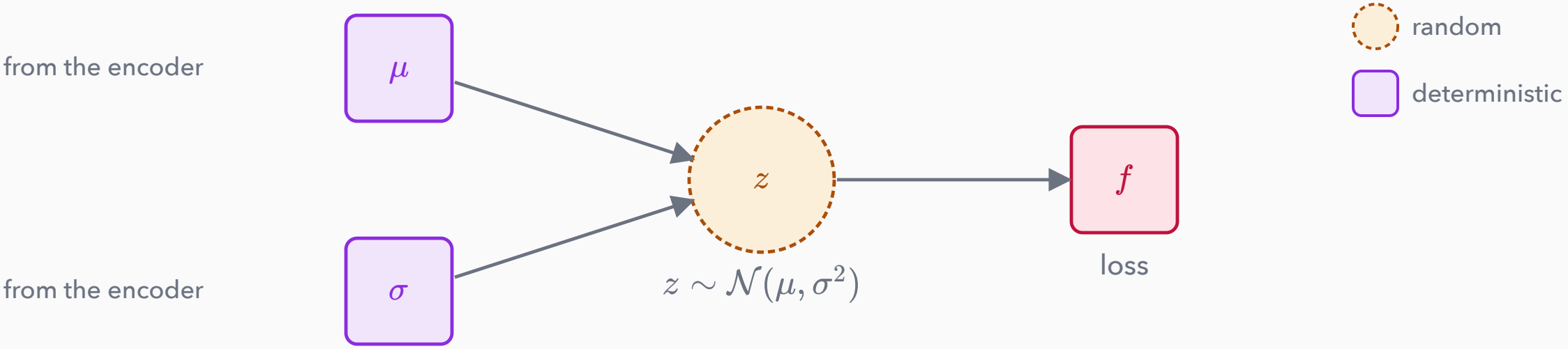
# Backpropagating through a random draw

VI by hand wrote each draw as $m + s \cdot \varepsilon$ without comment. The VAE loss also uses a $z$ drawn from the encoder's cloud, and to train the encoder, gradients must get from the loss back to $\mu$ and $\sigma$ through that draw. Here is why the way we write the draw is the whole trick.


✓ Distributions and fitting
✓ Autoencoders
✓ The probabilistic view
✓ KL divergence
✓ Variational inference and the ELBO
6 **Reparameterisation**
7 The VAE on MNIST

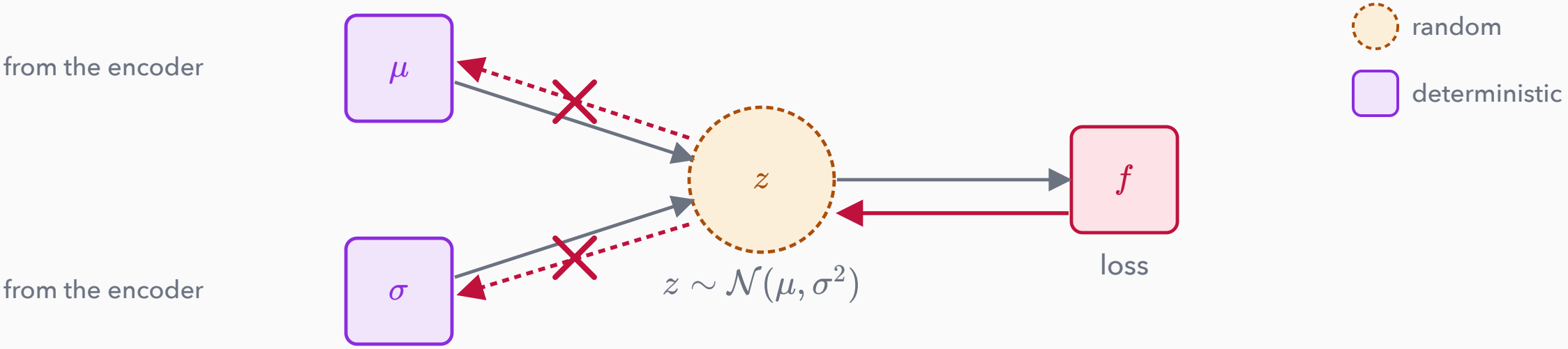
THE PROBLEM

# A sampling step sits in the middle of the graph



THE PROBLEM

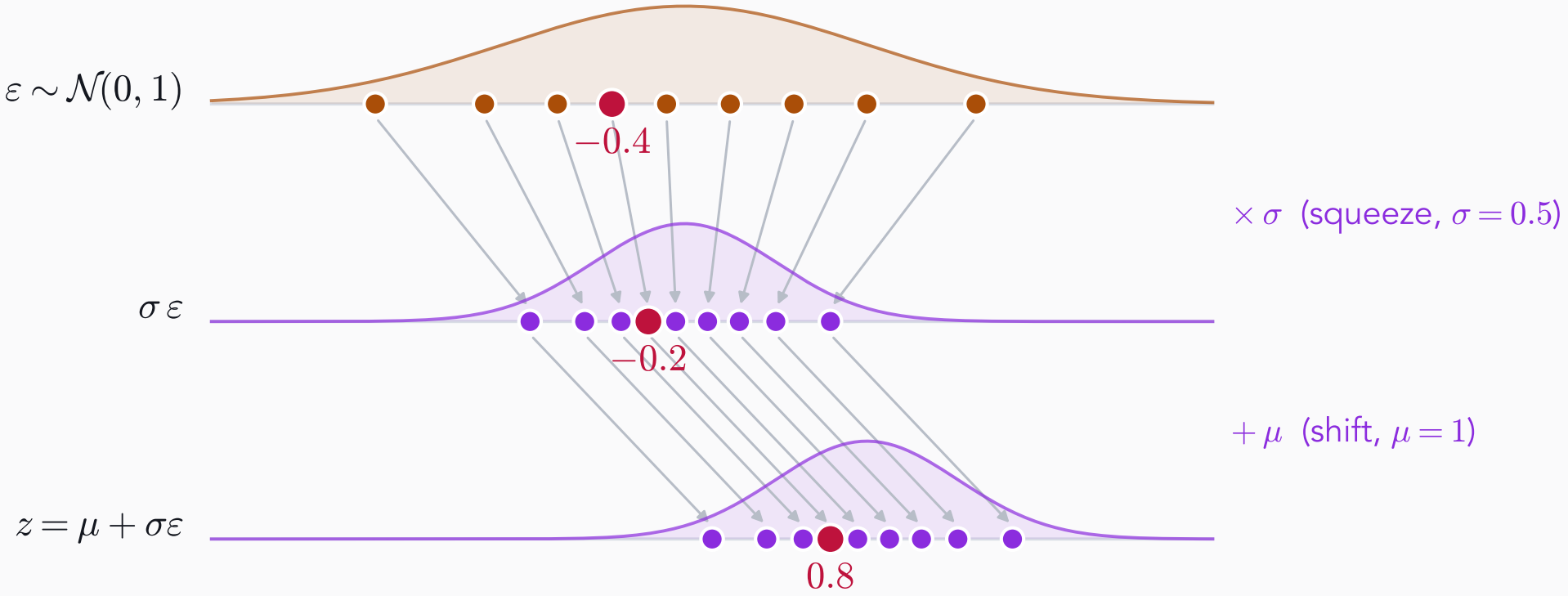
# A sampling step sits in the middle of the graph




**the loss knows how $z$ should move, but $z$ is a fresh random draw: how does it change when $\mu$ or $\sigma$ changes a little?**

THE TRICK

# Draw standard noise first, then stretch and shift it

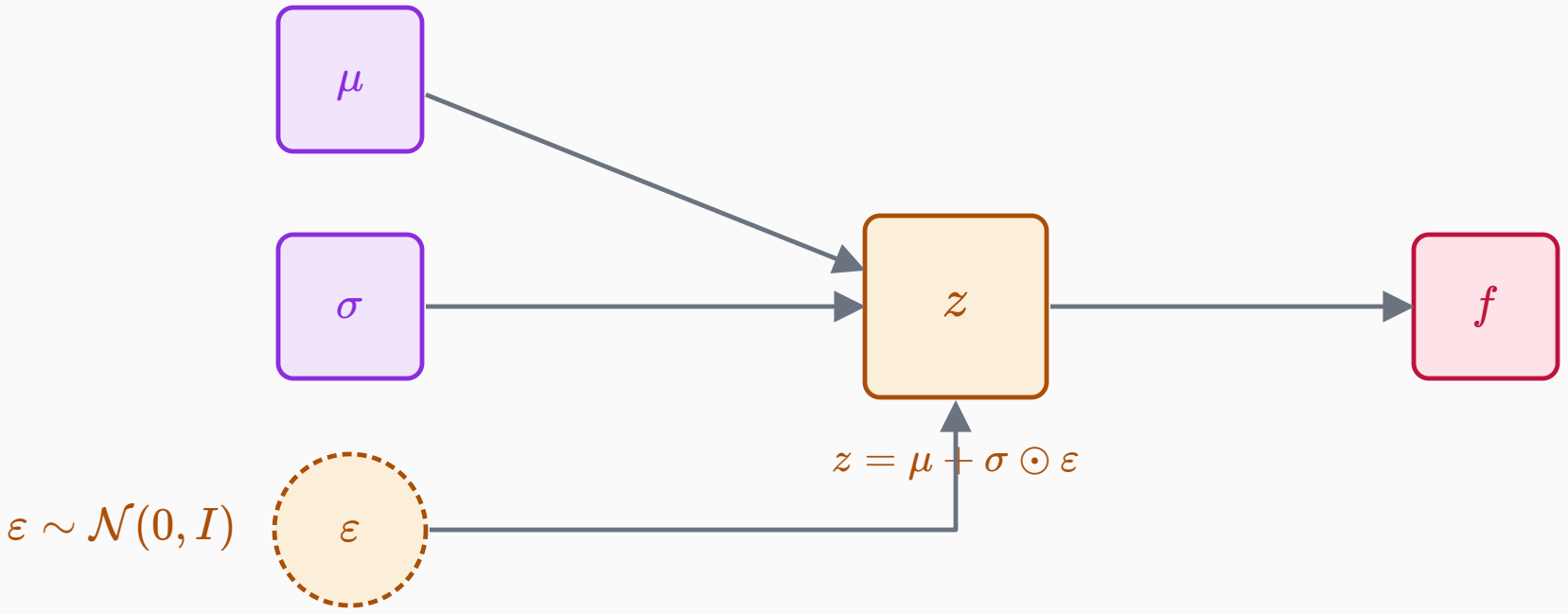



$$z = \mu + \sigma\,\varepsilon, \quad \varepsilon \sim \mathcal{N}(0,1) \quad \Longrightarrow \quad z \sim \mathcal{N}(\mu, \sigma^2)$$

> **Same distribution as before (the $\mu = 1, \sigma = 0.5$ cloud of the Gaussian KL slide). But the randomness now enters as an input, $\varepsilon$, that does not depend on any weights.**

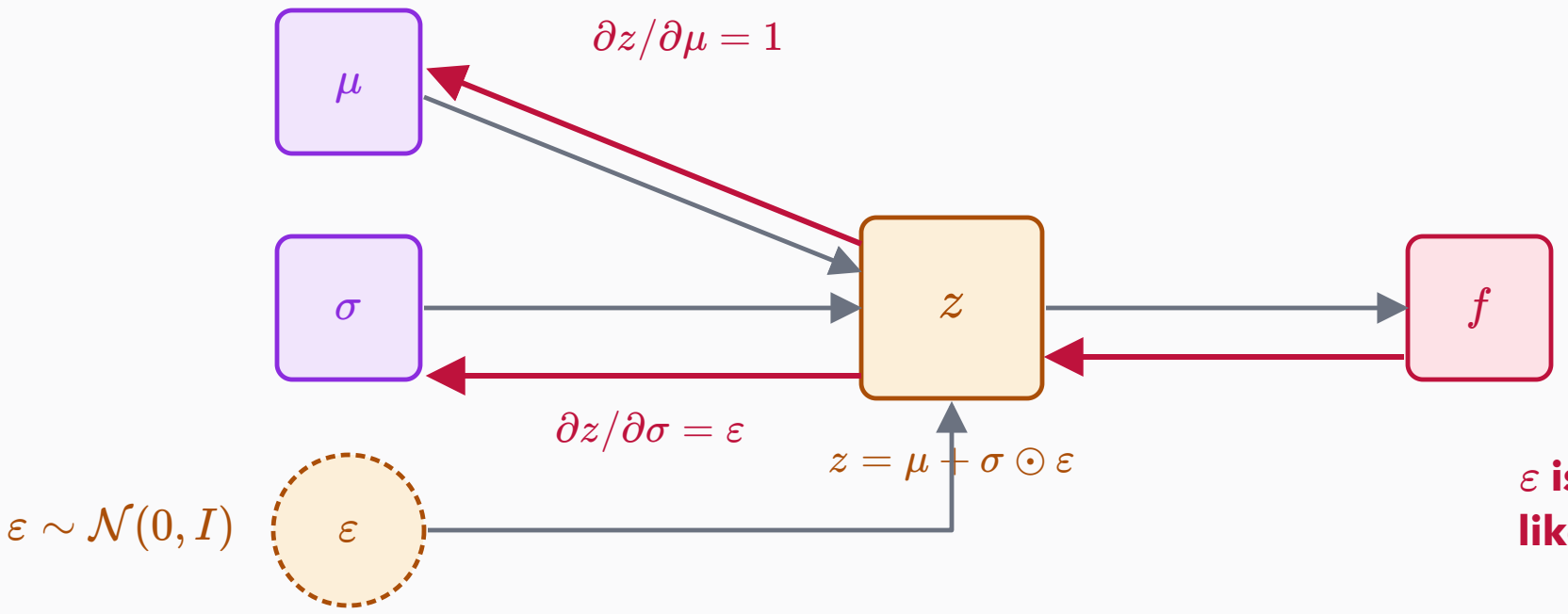
THE FIX

# Now the path from the loss to $\mu$ and $\sigma$ is ordinary arithmetic



THE FIX

# Now the path from the loss to $\mu$ and $\sigma$ is ordinary arithmetic




**$\varepsilon$ is just an input, like a pixel value**

# Take $f(z) = z^2$ as the loss

$$z = \mu + \sigma\varepsilon \qquad \Rightarrow \qquad \frac{\partial z}{\partial \mu} = 1, \qquad \frac{\partial z}{\partial \sigma} = \varepsilon, \qquad \frac{\partial f}{\partial z} = 2z$$

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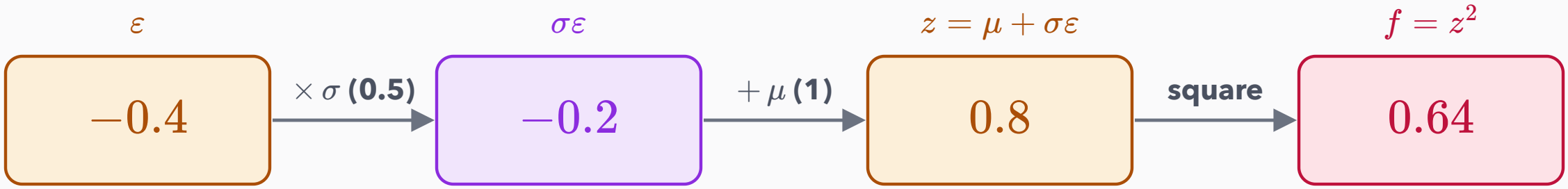
$$\frac{\partial f}{\partial \mu} = \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial \mu} = 2z \cdot 1$$

$$\frac{\partial f}{\partial \sigma} = \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial \sigma} = 2z \cdot \varepsilon$$

The noise value $\varepsilon$ appears in the $\sigma$-gradient: a draw far from the centre pushes $\sigma$ harder.

## NOW PLUG IN NUMBERS

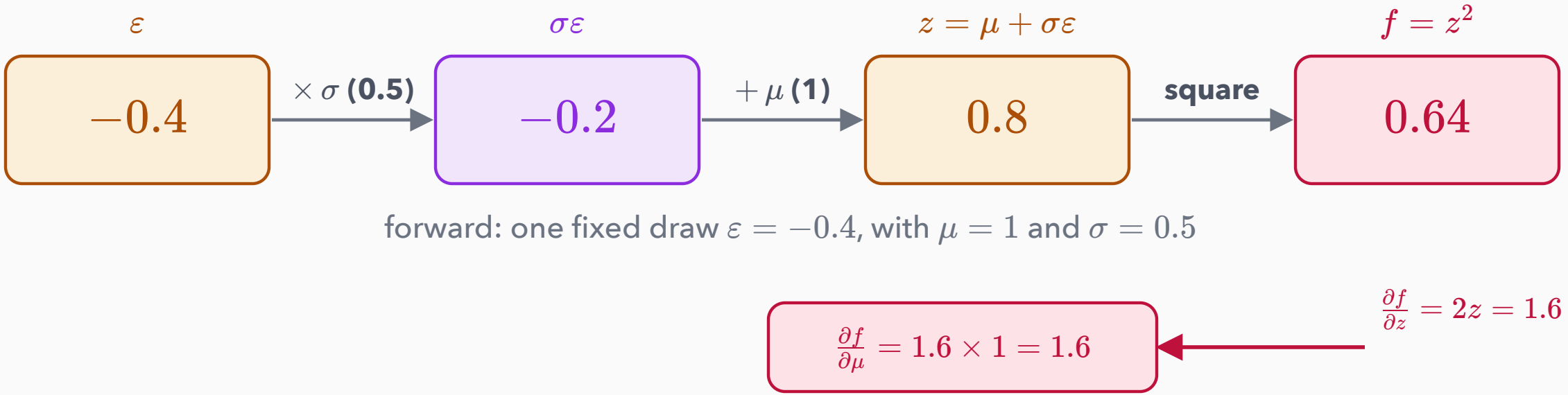
$$\mu = 1,\ \sigma = 0.5,\ \varepsilon = -0.4$$




forward: one fixed draw $\varepsilon = -0.4$, with $\mu = 1$ and $\sigma = 0.5$

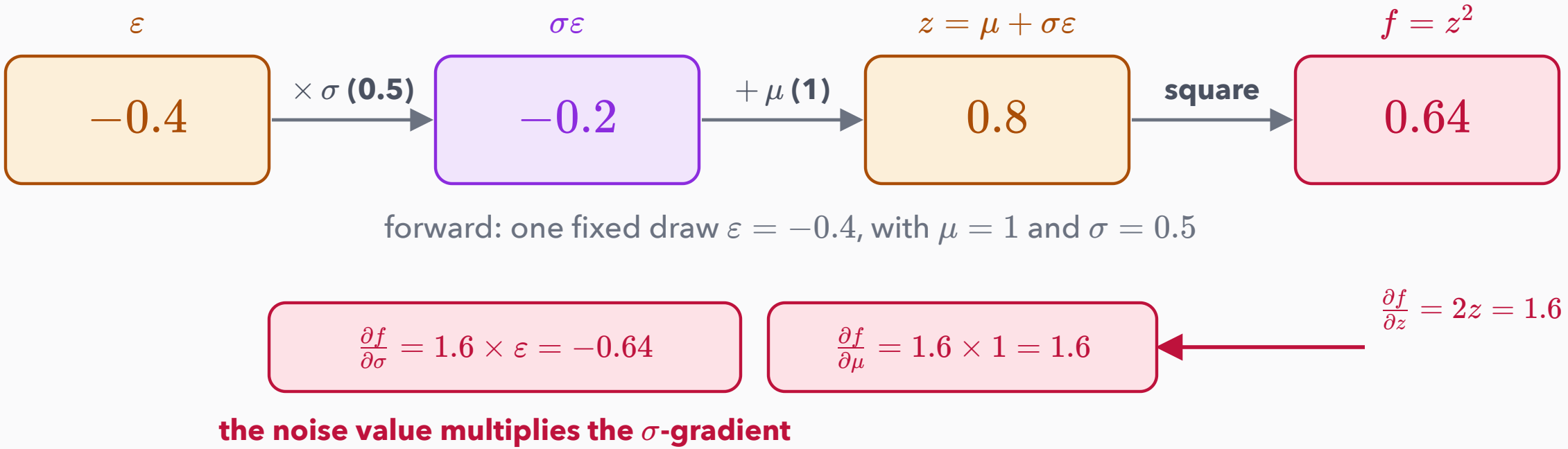
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**the noise value multiplies the $\sigma$-gradient**

ONE MORE LINK IN THE CHAIN

# The network outputs $v = \log \sigma^2$, not $\sigma$

$$\sigma = e^{v/2}$$

$v$ can be any real number; $\sigma$ is always positive

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derivative of $e^{v/2}$

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derivative of $e^{v/2}$

$$\frac{\partial f}{\partial v} = \frac{\partial f}{\partial \sigma} \cdot \frac{\partial \sigma}{\partial v} = -0.64 \times 0.25 = -0.16$$

one more chain-rule factor

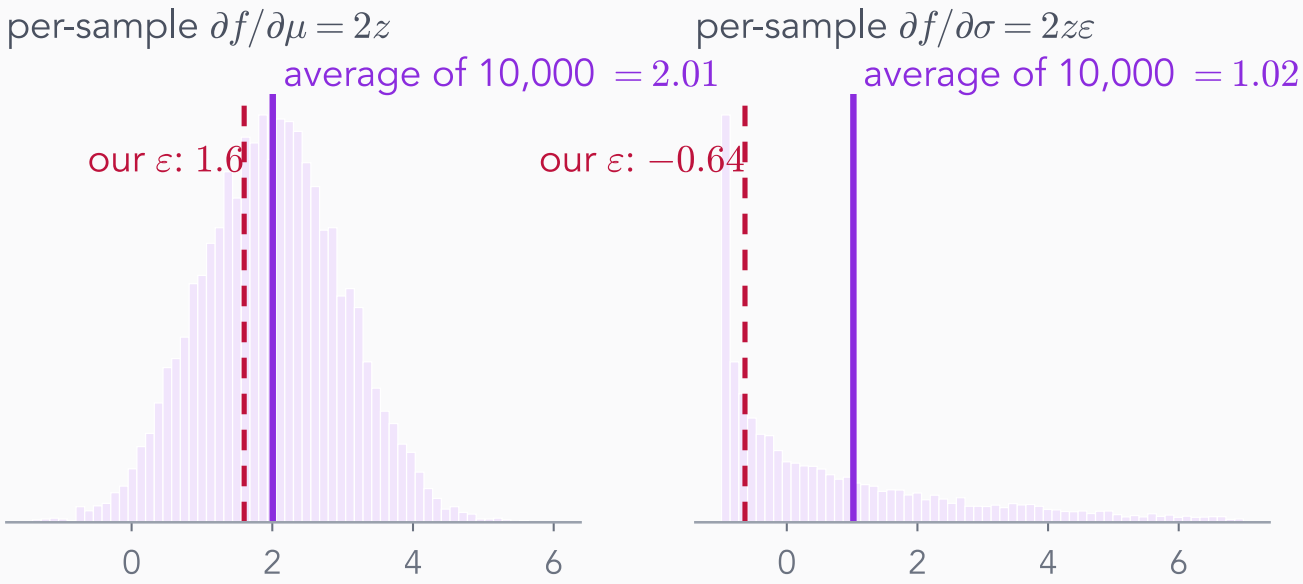
**Autograd does exactly this. The point of doing it by hand once: no step is magic.**

# One draw is noisy; the average is exactly right

$$\mathbb{E}_\varepsilon\big[f\big] = \mathbb{E}\big[(\mu+\sigma\varepsilon)^2\big] = \mu^2+\sigma^2 \quad \Rightarrow \quad \frac{\partial}{\partial\mu}\mathbb{E}[f] = 2\mu = 2, \qquad \frac{\partial}{\partial\sigma}\mathbb{E}[f] = 2\sigma = 1$$

# One draw is noisy; the average is exactly right

$$\mathbb{E}_\varepsilon\big[f\big] = \mathbb{E}\big[(\mu+\sigma\varepsilon)^2\big] = \mu^2+\sigma^2 \quad \Rightarrow \quad \frac{\partial}{\partial\mu}\mathbb{E}[f] = 2\mu = 2, \qquad \frac{\partial}{\partial\sigma}\mathbb{E}[f] = 2\sigma = 1$$




**Our single $\varepsilon$ gave 1.6 and $-0.64$. Averaged over 10,000 draws: 2.01 and 1.02. SGD only needs the right answer on average.**

IN PYTORCH

# Three lines

```
# encoder outputs, shape [B, 2] each
mu, logvar = encoder(x)
std = torch.exp(0.5 * logvar)     # sigma = exp(v / 2)
eps = torch.randn_like(std)       # standard normal, no grad
z   = mu + std * eps              # differentiable in mu, std
```

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Why output $\log \sigma^2$? It can be any real number; $\sigma$ must be positive.

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Why output $\log \sigma^2$? It can be any real number; $\sigma$ must be positive.

`Normal(mu, std).rsample()` does the same thing; `.sample()` drops the gradient.

For our 3: $\mu = (0.3,\ -1.93)$, $\varepsilon = (0.62,\ -1.15)$ gives $z = (0.33,\ -1.99)$.

END OF PART 6

# What we did, and what comes next

### WHAT WE DID

1. Sampling $z$ directly cuts the gradient path to $\mu$ and $\sigma$.
2. Write $z = \mu + \sigma\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$: same distribution, randomness as an input.
3. $\partial z/\partial \mu = 1$, $\partial z/\partial \sigma = \varepsilon$; one draw is noisy, the average is exact.
4. In PyTorch: three lines, or `rsample()`.

END OF PART 6

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4. In PyTorch: three lines, or `rsample()`.

### NEXT: EVERYTHING TOGETHER

Encoder, cloud, reparameterised sample, decoder, ELBO: train it on MNIST and follow our 3 through every step.

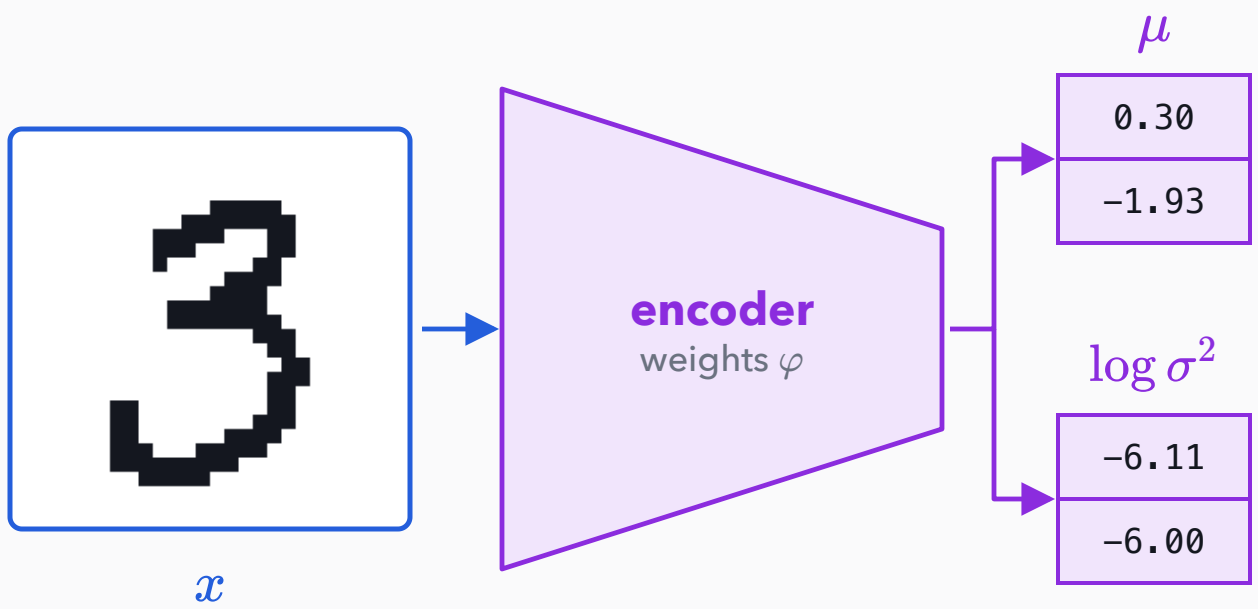
PART 7 OF 7

# The variational autoencoder, end to end

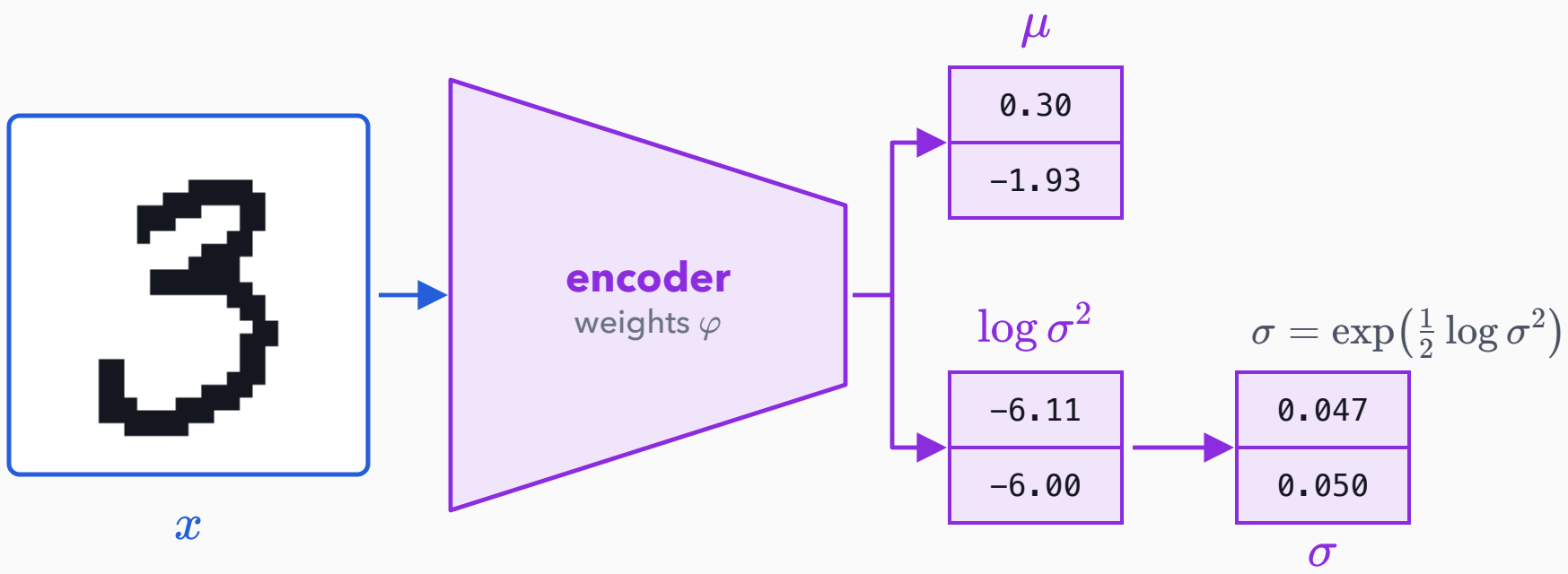
Every piece is in place: encoder, cloud, KL, ELBO, reparameterisation. Follow our 3 through the trained network, then generate.

- ✓ Distributions and fitting
- ✓ Autoencoders
- ✓ The probabilistic view
- ✓ KL divergence
- ✓ Variational inference and the ELBO
- ✓ Reparameterisation
- 7 **The VAE on MNIST**

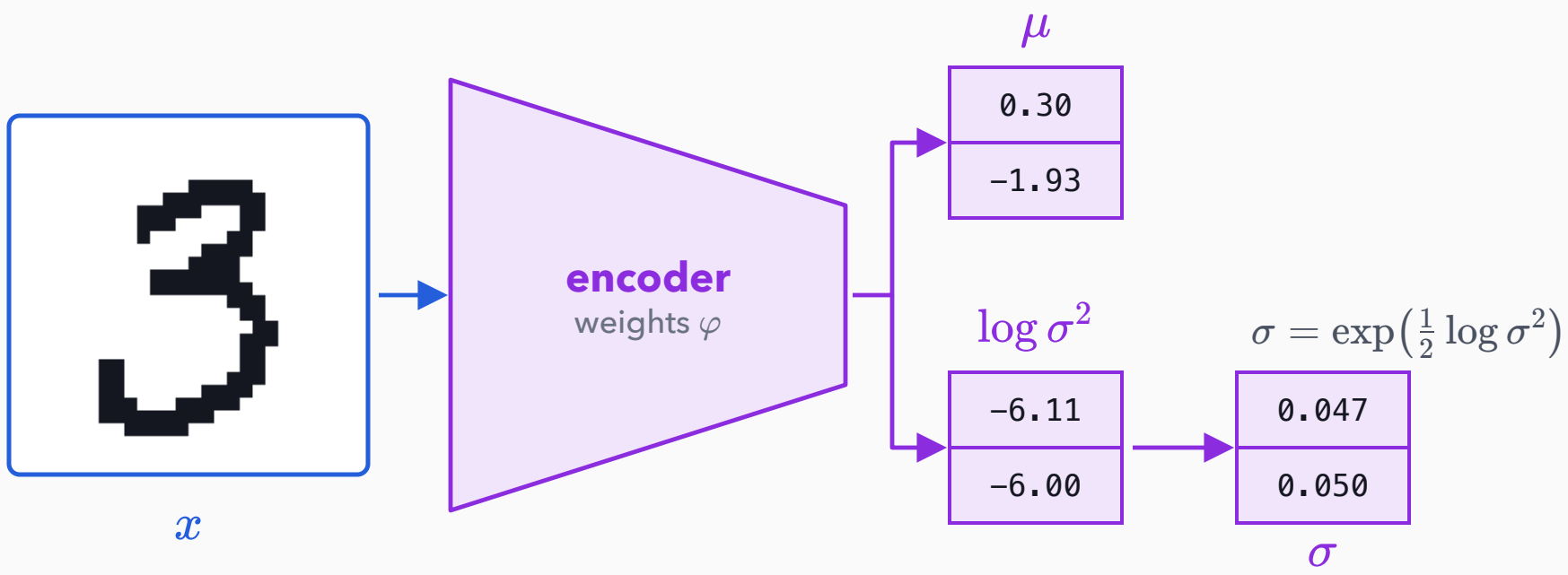
# Encode: the image becomes a cloud



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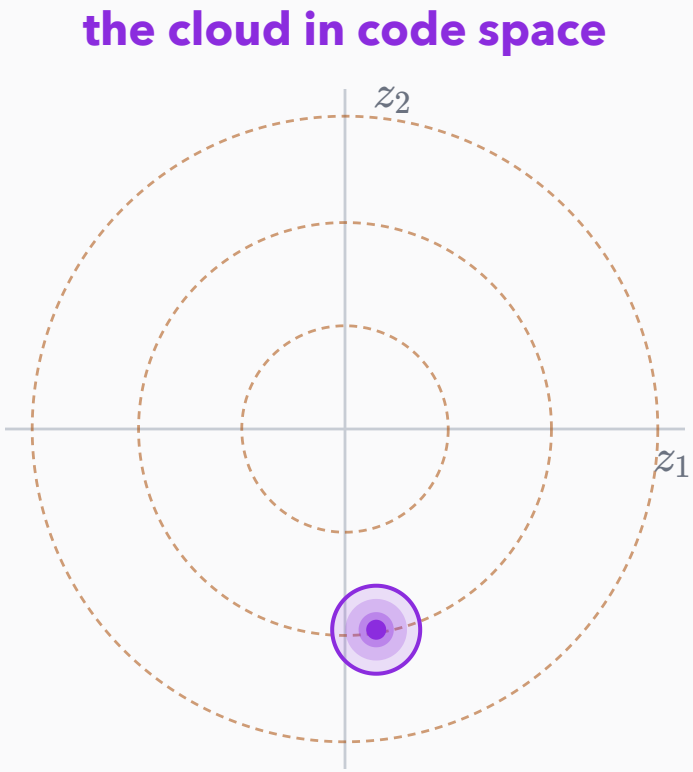



**a cloud centred at $\mu$ with width $\sigma$:**

$$q_\varphi(z \mid x) = \mathcal{N}(\mu, \sigma^2)$$

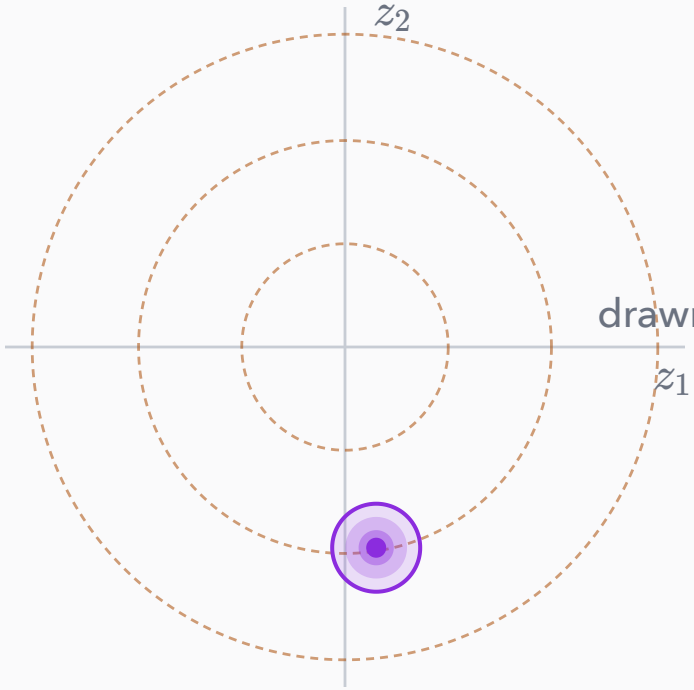
$\sigma \approx 0.05$ against the prior's $\sigma = 1$: the encoder is very sure where our 3 belongs.

# Sample: one code from inside the cloud



# Sample: one code from inside the cloud

**the cloud in code space**





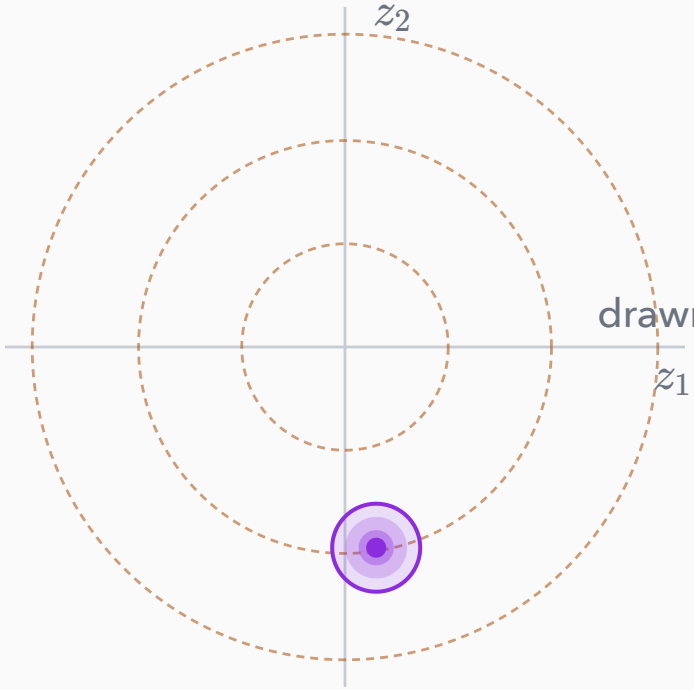


$\varepsilon = (0.62,\ -1.15)$

drawn from a standard normal once

# Sample: one code from inside the cloud

**the cloud in code space**




$\varepsilon$

$\varepsilon = (0.62,\ -1.15)$

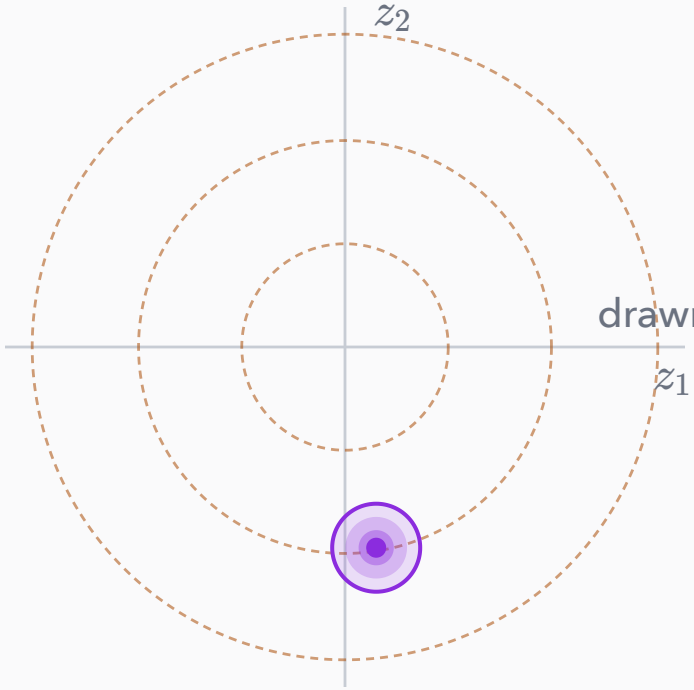
drawn from a standard normal once

$$z_1 = \mu_1 + \sigma_1 \varepsilon_1$$

$$= 0.30 + 0.047 \times 0.62 = 0.33$$

# Sample: one code from inside the cloud

**the cloud in code space**




$\varepsilon$

$\varepsilon = (0.62,\ -1.15)$

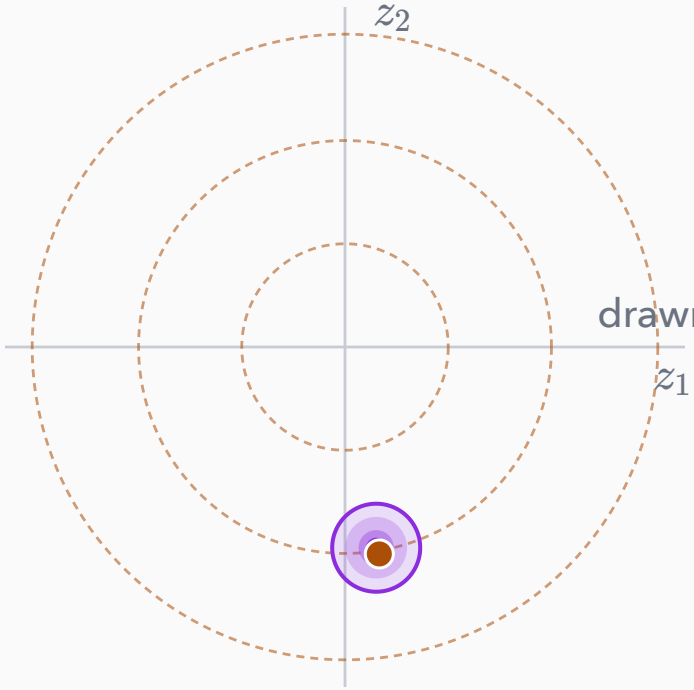
drawn from a standard normal once

$$z_1 = \mu_1 + \sigma_1 \varepsilon_1$$
$$= 0.30 + 0.047 \times 0.62 = 0.33$$

$$z_2 = \mu_2 + \sigma_2 \varepsilon_2$$
$$= -1.93 + 0.050 \times (-1.15) = -1.99$$

# Sample: one code from inside the cloud

**the cloud in code space**




$\varepsilon$

$\varepsilon = (0.62,\ -1.15)$

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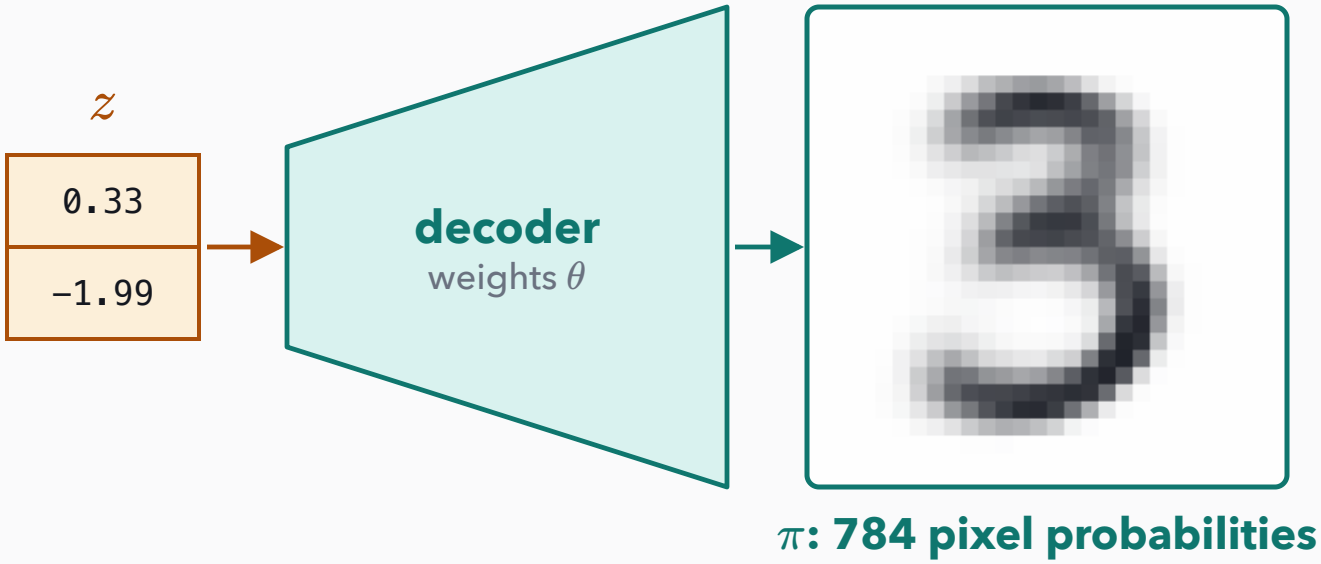
$$z_2 = \mu_2 + \sigma_2 \varepsilon_2$$
$$= -1.93 + 0.050 \times (-1.15) = -1.99$$

$z =$

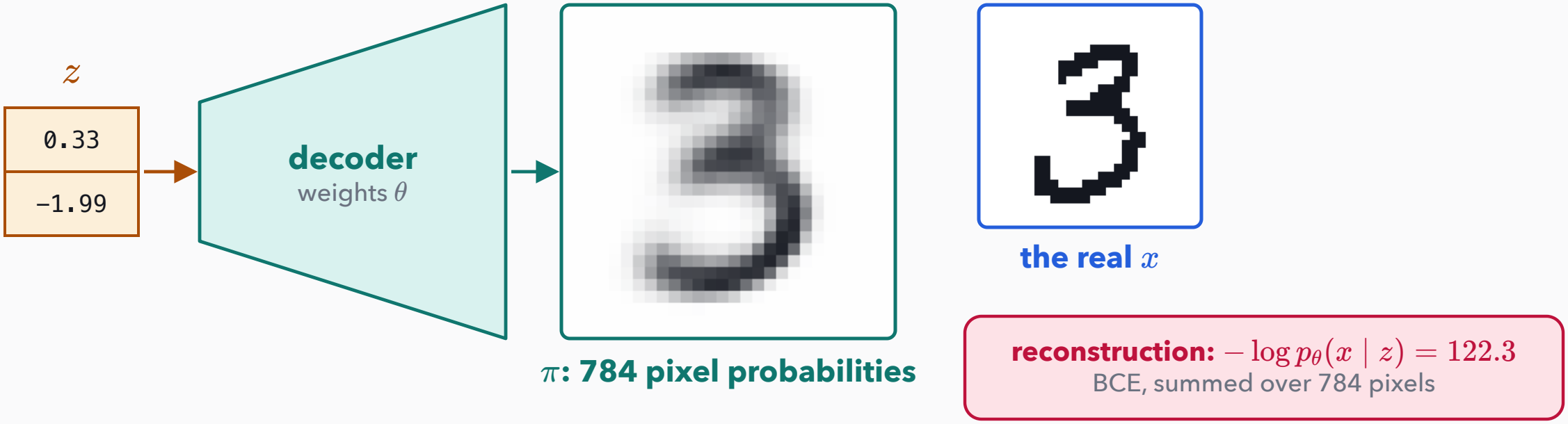
| 0.33 | -1.99 |
|---|---|

**one code from inside the cloud**

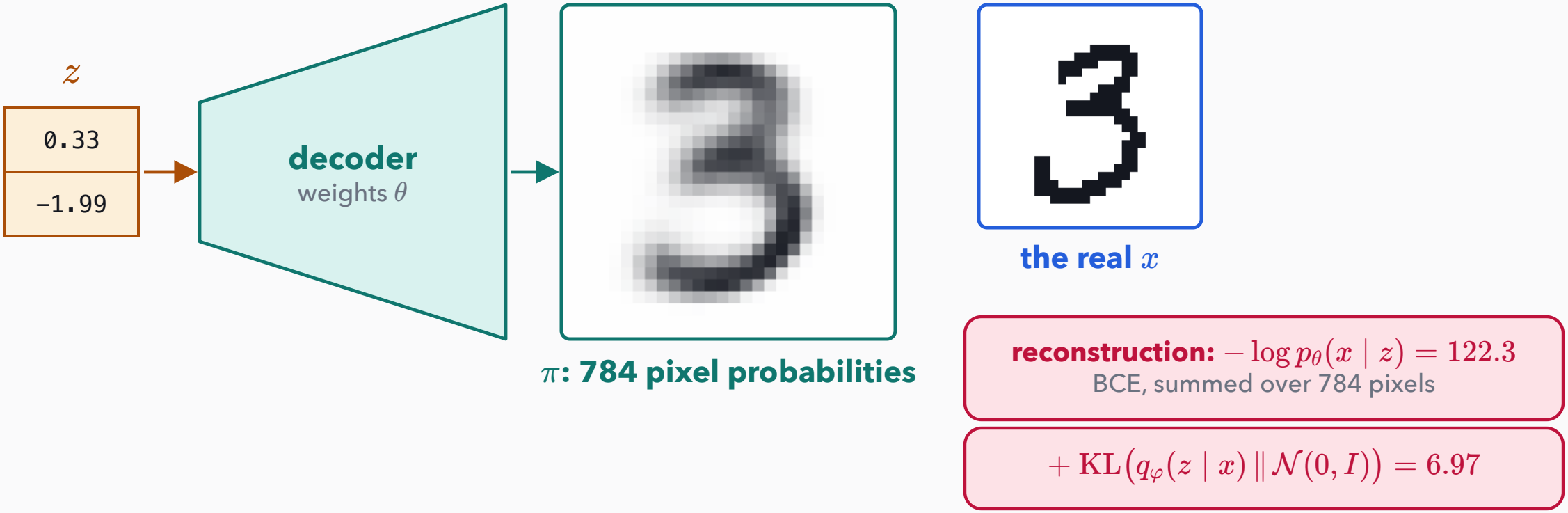
# Decode, then score against the real 3



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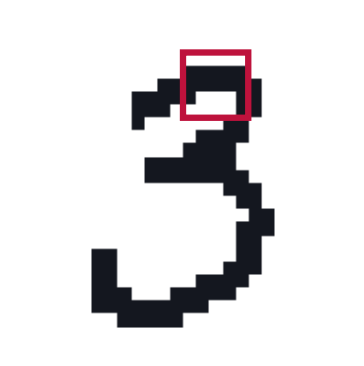
# Decode, then score against the real 3




**Loss for our 3: $122.3 + 6.97 = 129.3$ nats. Backpropagate it into both $\theta$ and $\varphi$.**

STEP 3, ZOOMED IN: THE RECONSTRUCTION TERM

# Each pixel is a coin flip with probability $\pi_i$



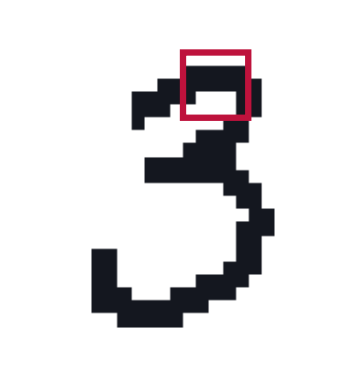
our 3, one 5 × 5 patch

observed $x_i$

| | | | | |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 |

STEP 3, ZOOMED IN: THE RECONSTRUCTION TERM

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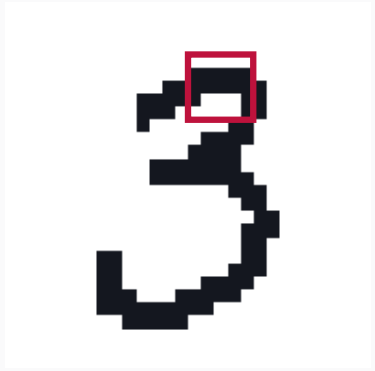
| | | | | |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 |

decoder's $\pi_i$

| | | | | |
|---|---|---|---|---|
| 0.38 | 0.28 | 0.13 | 0.05 | 0.02 |
| 0.89 | 0.82 | 0.65 | 0.41 | 0.18 |
| 0.77 | 0.79 | 0.76 | 0.67 | 0.40 |
| 0.41 | 0.52 | 0.67 | 0.66 | 0.51 |
| 0.22 | 0.35 | 0.55 | 0.60 | 0.48 |

STEP 3, ZOOMED IN: THE RECONSTRUCTION TERM

# Each pixel is a coin flip with probability $\pi_i$



our 3, one 5 × 5 patch

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$-\log p(x_i \mid z)$

| | | | | |
|---|---|---|---|---|
| 0.48 | 0.33 | 0.14 | 0.05 | 0.02 |
| 0.12 | 0.20 | 0.43 | 0.89 | 1.71 |
| 0.26 | 0.24 | 0.27 | 0.40 | 0.92 |
| 0.89 | 0.73 | 1.11 | 1.08 | 0.67 |
| 0.25 | 0.43 | 0.80 | 0.92 | 0.73 |

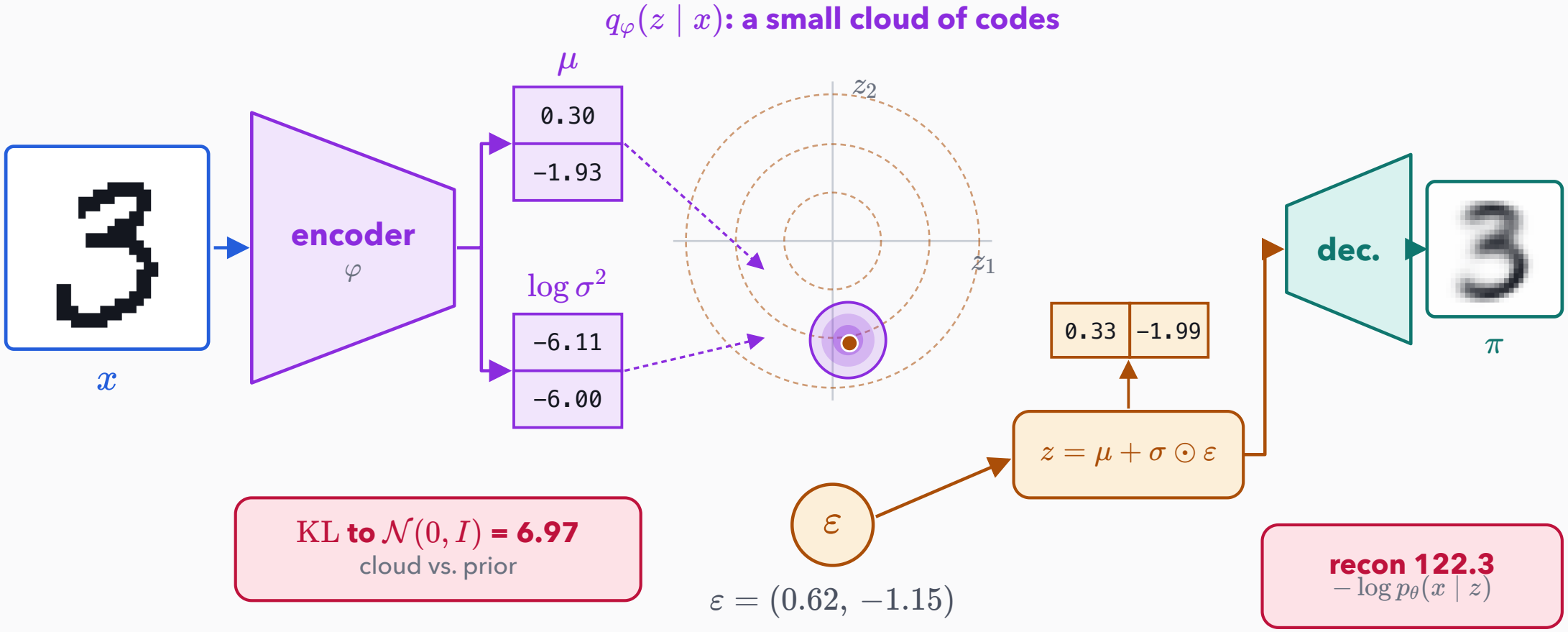
**this 5 × 5 patch: 14.1 nats**

$$-\log p_\theta(x \mid z) = -\sum_{i=1}^{784} \Big[ x_i \log \pi_i + (1 - x_i) \log(1 - \pi_i) \Big] = 122.3 \text{ nats}$$

> **Binary cross-entropy is not a design choice: it is the Bernoulli log-likelihood. Gaussian pixels would give squared error.**

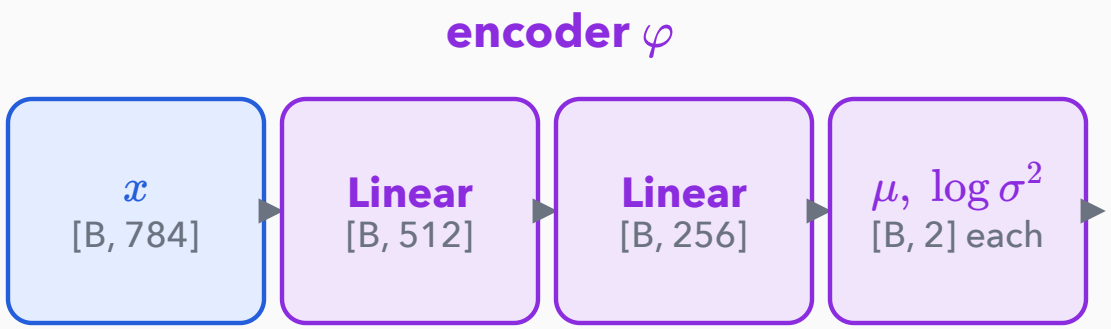
ALL THREE STEPS ON ONE PAGE

# The whole VAE, for our 3



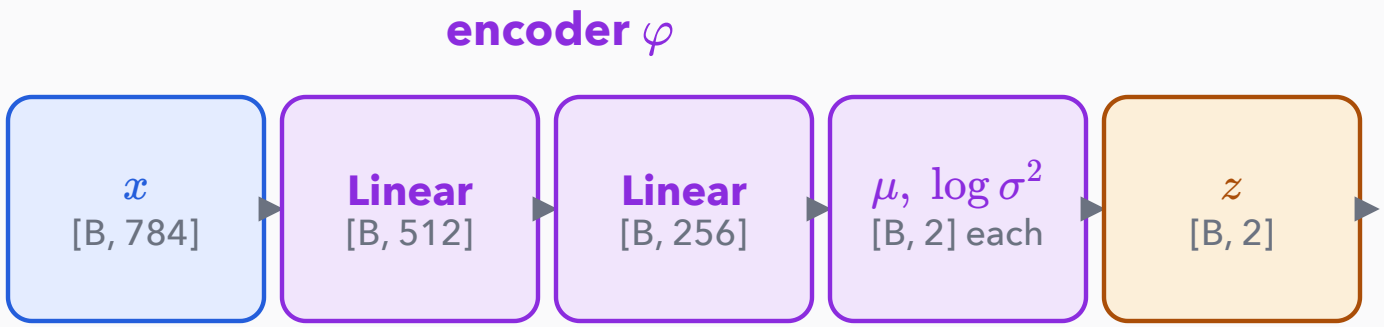
THE SAME NETWORK AS TENSORS

# Shapes, layer by layer



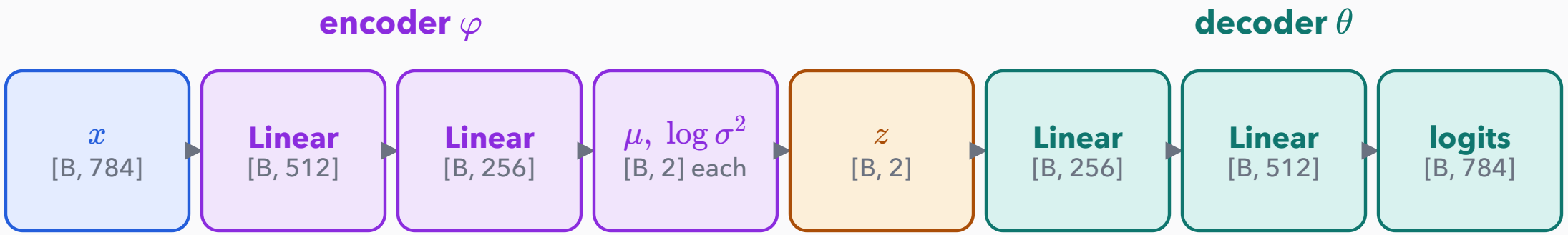
THE SAME NETWORK AS TENSORS

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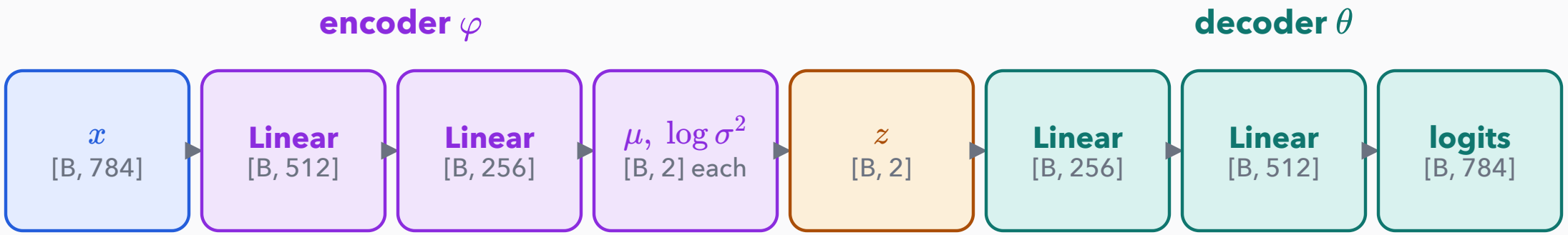
THE SAME NETWORK AS TENSORS

# Shapes, layer by layer



THE SAME NETWORK AS TENSORS

# Shapes, layer by layer




B = batch size. Only the 2 in the middle is the code; $\varepsilon$ has the same shape as $\mu$.

Loss per image: BCE summed over 784 pixels + KL summed over 2 coordinates.
Then average over the B images in the batch.

THE KL TERM FOR OUR 3

# A tight cloud is expensive

$$\mathrm{KL} = \tfrac{1}{2} \sum_{j} \left( \mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2 \right)$$

| coordinate | $\mu_j$ | $\sigma_j$ | $\mu_j^2$ | $\sigma_j^2 - 1$ | $-\log \sigma_j^2$ | $\frac{1}{2}(\text{sum})$ |
|---|---|---|---|---|---|---|
| | | | | | | |
| | | | | | | |

THE KL TERM FOR OUR 3

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|---|---|---|---|---|---|---|
| $z_1$ | 0.3 | 0.047 | 0.09 | $-1.00$ | 6.11 | **2.6** |

THE KL TERM FOR OUR 3

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|---|---|---|---|---|---|---|
| $z_1$ | 0.3 | 0.047 | 0.09 | $-1.00$ | 6.11 | **2.6** |
| $z_2$ | −1.93 | 0.05 | 3.72 | $-1.00$ | 6.00 | **4.37** |

THE KL TERM FOR OUR 3

# A tight cloud is expensive

$$\mathrm{KL} = \tfrac{1}{2}\sum_{j}\left(\mu_j^2 + \sigma_j^2 - 1 - \log\sigma_j^2\right)$$

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|---|---|---|---|---|---|---|
| $z_1$ | 0.3 | 0.047 | 0.09 | $-1.00$ | 6.11 | **2.6** |
| $z_2$ | −1.93 | 0.05 | 3.72 | $-1.00$ | 6.00 | **4.37** |
| **total** | | | | | | **6.97 nats** |

$$-\log\sigma_1^2 = -\log(0.047^2) = -\log 0.0022 = 6.11$$

> **Most of the cost is $-\log\sigma^2$: the price of being precise. Reconstruction wants precision; the KL resists.**

ONE TRAINING STEP

# The whole algorithm fits on a slide

```
mu, logvar = model.encoder(x)              # [B,2], [B,2]
z = mu + torch.exp(0.5*logvar) * torch.randn_like(mu)
logits = model.decoder(z)                  # [B,784]
recon = F.binary_cross_entropy_with_logits(
          logits, x.flatten(1), reduction="none").sum(1)
kl = 0.5 * (mu**2 + logvar.exp() - 1 - logvar).sum(1)
loss = (recon + kl).mean()                 # per image, then batch
opt.zero_grad(); loss.backward(); opt.step()
```

Encode, reparameterise, decode.

ONE TRAINING STEP

# The whole algorithm fits on a slide

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mu, logvar = model.encoder(x)              # [B,2], [B,2]
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loss = (recon + kl).mean()                 # per image, then batch
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```

Sum the BCE over all 784 pixels of each image.

ONE TRAINING STEP

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z = mu + torch.exp(0.5*logvar) * torch.randn_like(mu)
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kl = 0.5 * (mu**2 + logvar.exp() - 1 - logvar).sum(1)
loss = (recon + kl).mean()                 # per image, then batch
opt.zero_grad(); loss.backward(); opt.step()
```

Sum the KL over the 2 code coordinates of each image.

ONE TRAINING STEP

# The whole algorithm fits on a slide

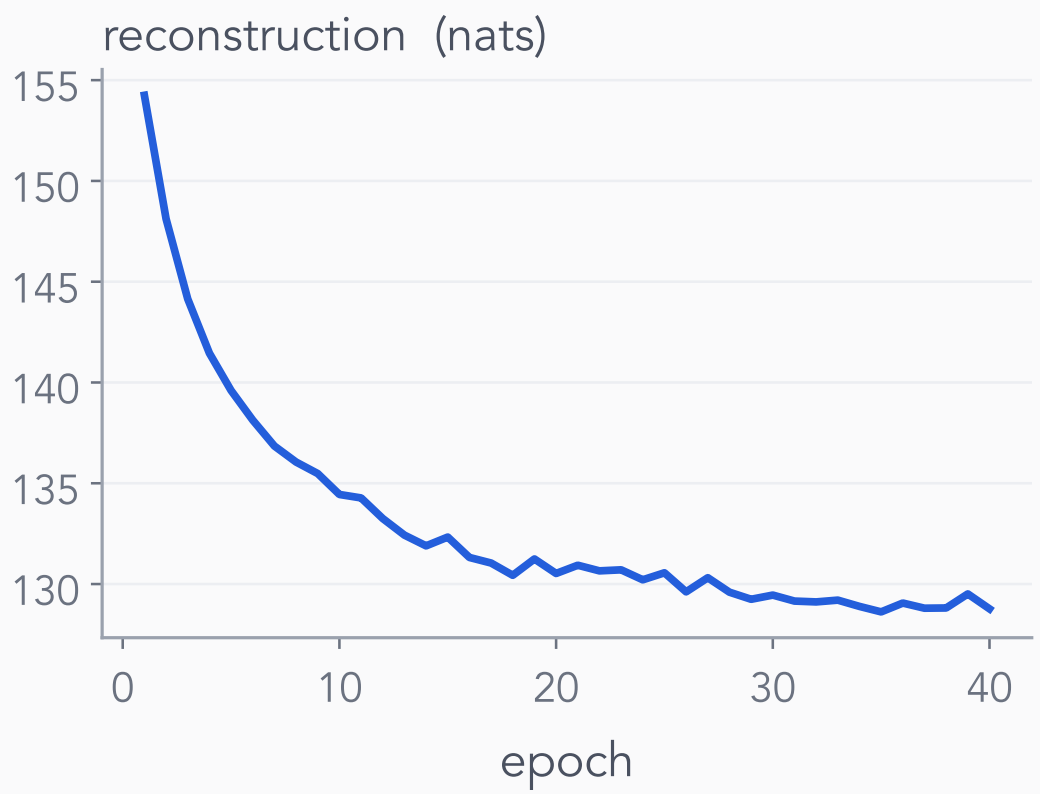
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loss = (recon + kl).mean()                 # per image, then batch
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```

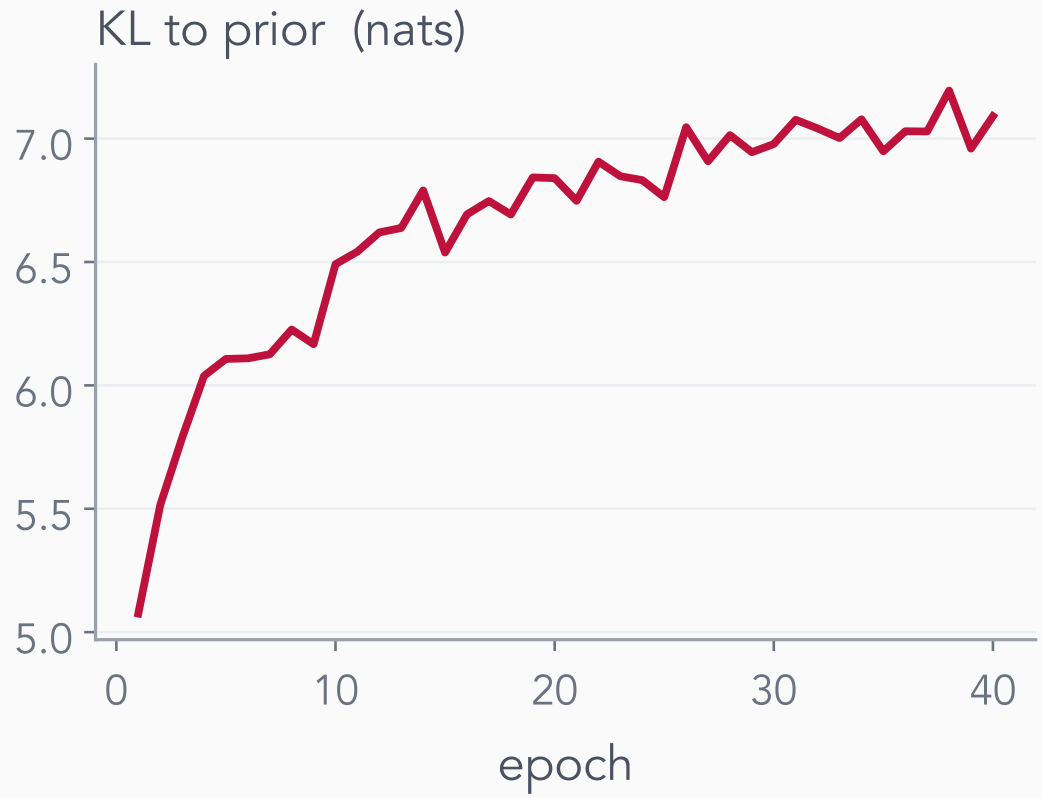
Add per image, average over the batch, step both encoder and decoder.

Averaging BCE over pixels instead of summing silently makes the KL 784× stronger.

TRAINING RUN, TEST SET

# Reconstruction falls, KL rises, then both settle

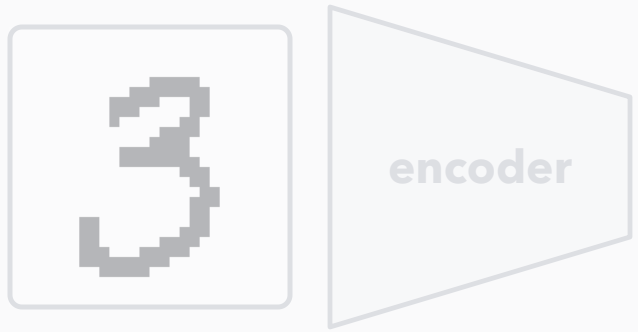



**After 40 epochs: reconstruction 128.8 + KL 7.1 = 135.9 nats per test image.**

THE PAYOFF

# Generate: random code → decoder. No encoder.

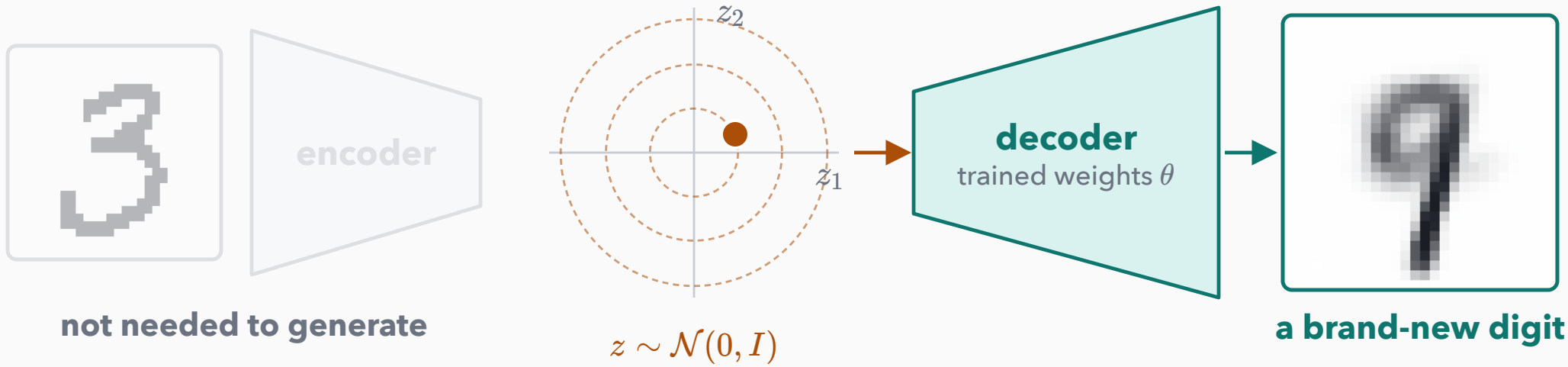



not needed to generate

> This is the whole generator. The encoder was scaffolding for training.

THE PAYOFF

# Generate: random code → decoder. No encoder.




This is the whole generator. The encoder was scaffolding for training.

48 RANDOM CODES EACH

# Samples from two VAEs



2-number code · reconstruction 128.8

48 RANDOM CODES EACH

# Samples from two VAEs



2-number code · reconstruction 128.8

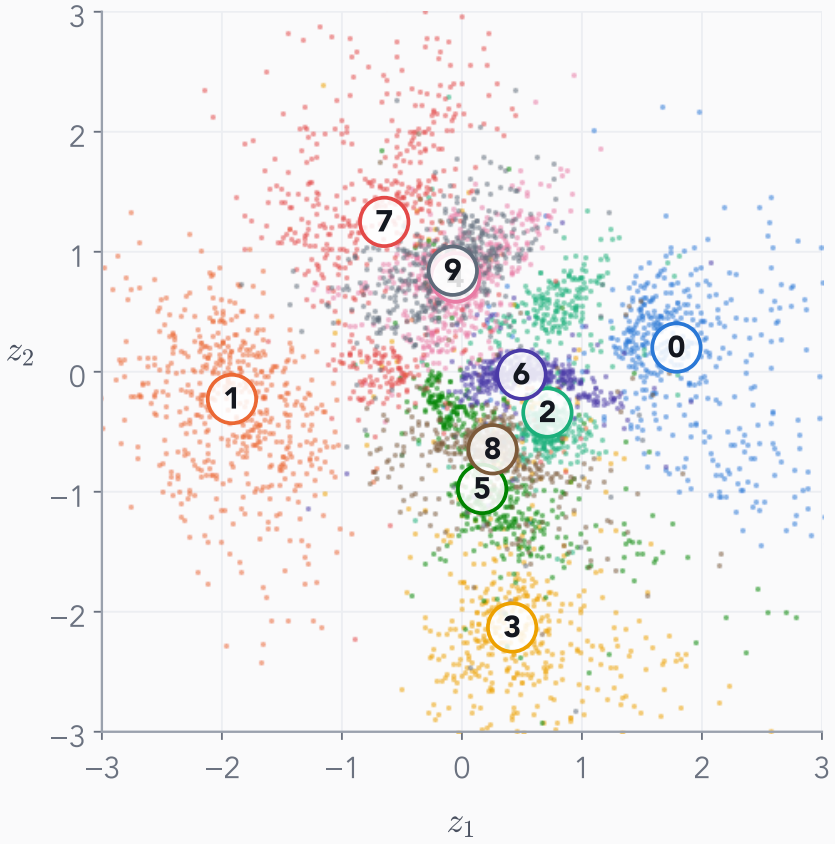


16-number code · reconstruction 57.8

**More code dimensions: sharper and more varied, but some samples are odd. Neither is perfect.**

BECAUSE THE CODE IS 2-D, WE CAN DRAW THE WHOLE MAP

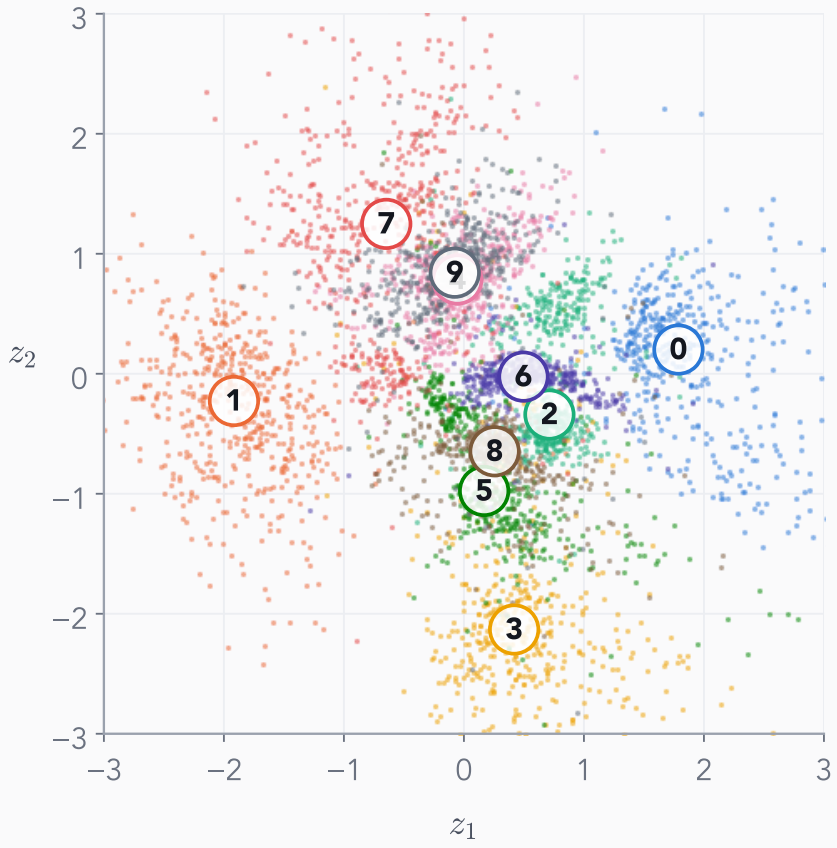
# Every point in the code space is a digit



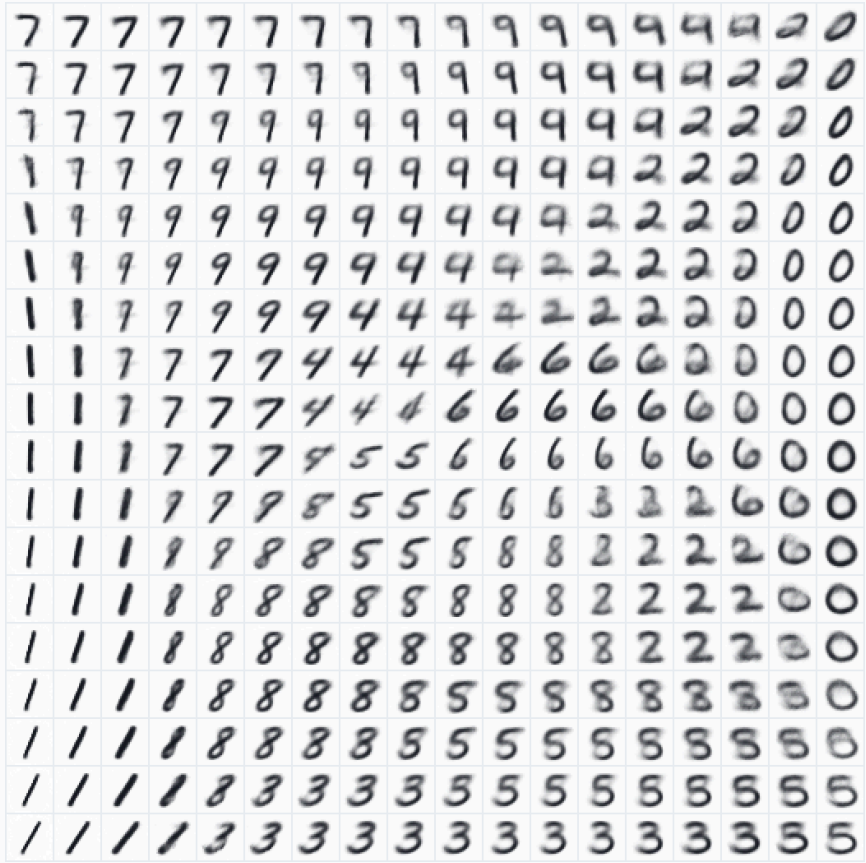

where test digits land

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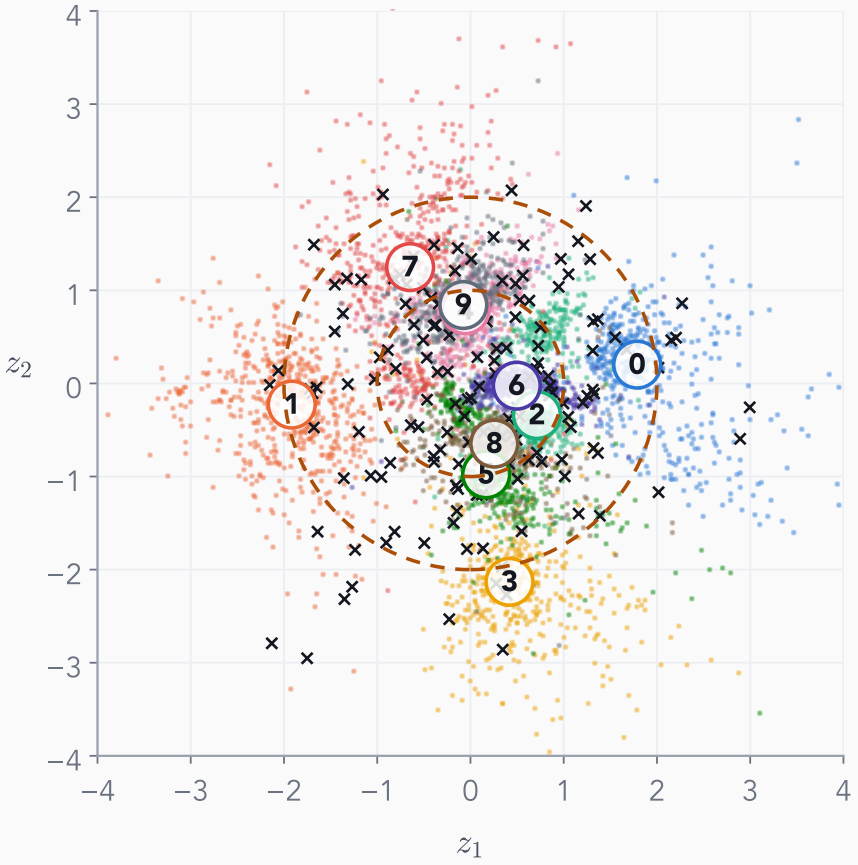


decoded on an 18 × 18 grid, spaced by quantiles of $\mathcal{N}(0, 1)$

**Neighbouring codes give similar digits, and every region decodes to something plausible.**

TRY IT: THE TRAINED VAE, IN YOUR BROWSER

# The same app, now for the VAE




Now random codes from $\mathcal{N}(0, I)$ land among the dots, and almost every one decodes to a digit.

THE COVER, EXPLAINED

# Walk in a straight line from our 3 to a 7



encode both, take 12 evenly spaced points between their $\mu$'s, decode each (16-number VAE)

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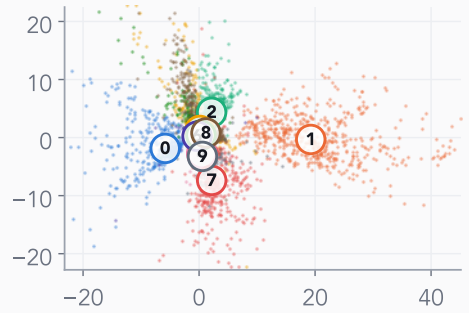


the same walk in the 2-number VAE: the line passes through the 5, 8, 4 and 9 regions on the map

THE TUG OF WAR, MADE VISIBLE

# No KL, normal KL, too much KL

**autoencoder (no KL)**



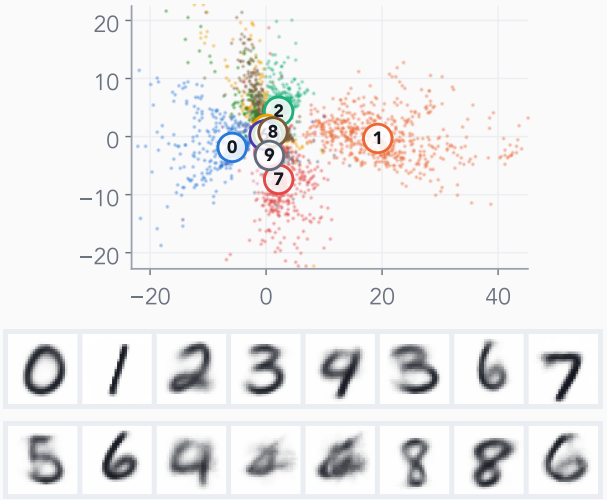



recon 128.1 · no KL term

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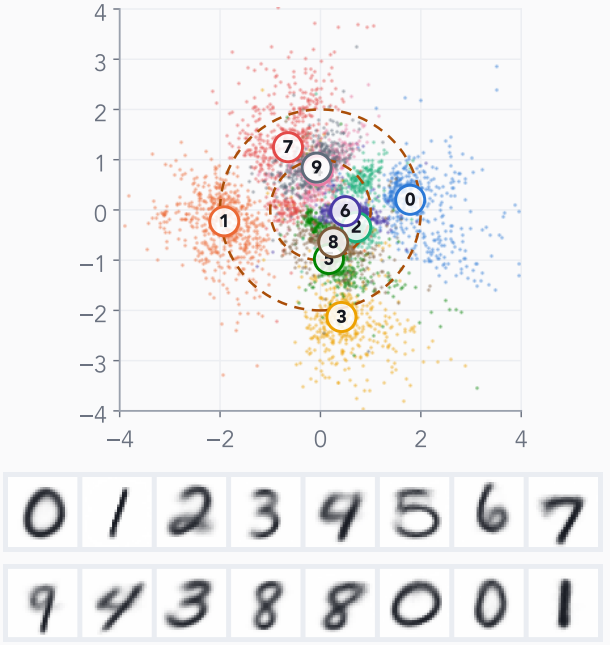
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**VAE, loss = recon + KL**

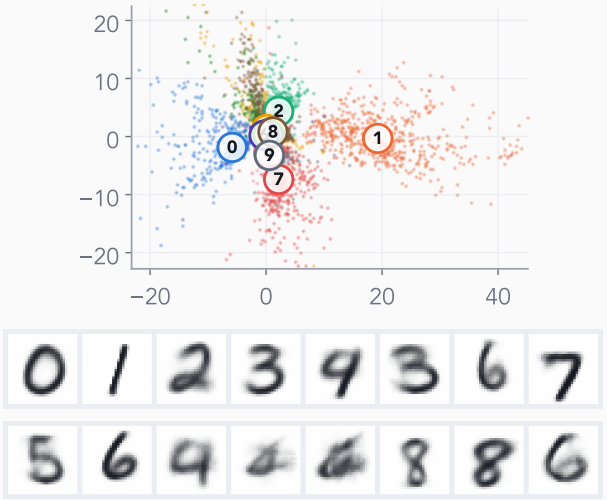



recon 128.8 · KL 7.1

THE TUG OF WAR, MADE VISIBLE

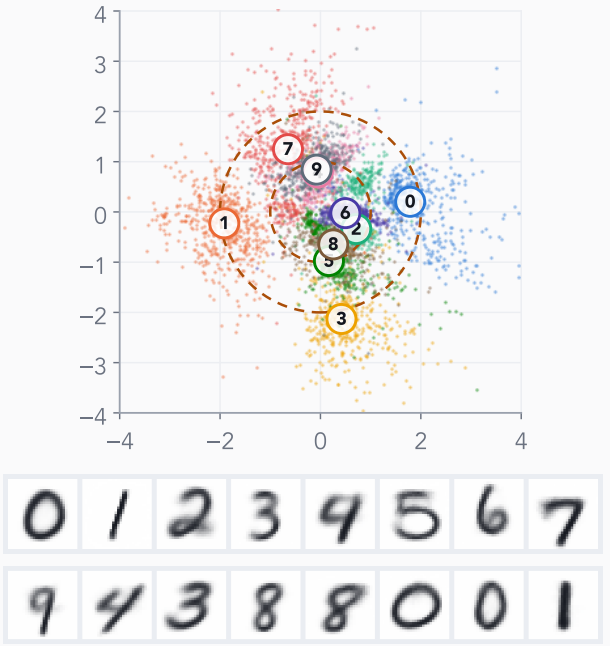
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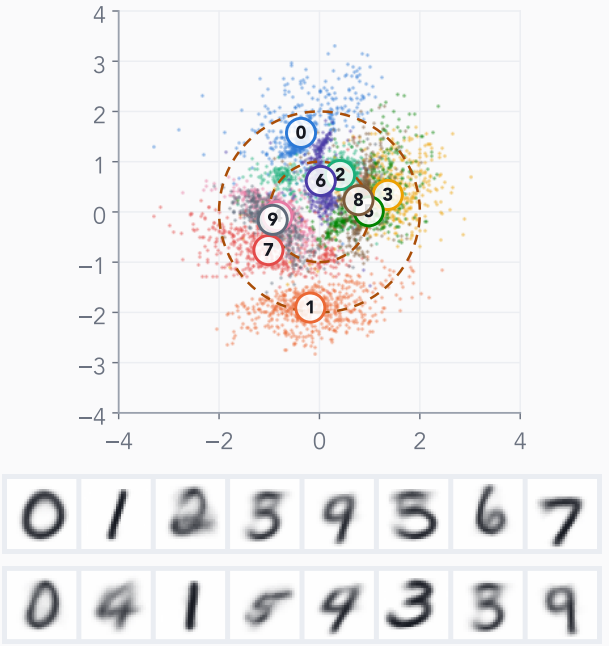
recon 128.1 · no KL term

### VAE, loss = recon + KL



recon 128.8 · KL 7.1

### recon + 8 × KL



recon 140.4 · KL 3.9

> **Rows: code space · reconstructions · random samples. Too little KL: samples fail. Too much: everything blurs together.**

AN HONEST LOOK

# Why are VAE samples blurry?

**1. It fits $\mathrm{KL}(\text{data} \,\|\, \text{model})$.** Like the wide Gaussian on our bumps, it must cover every training digit, so when unsure it spreads probability.



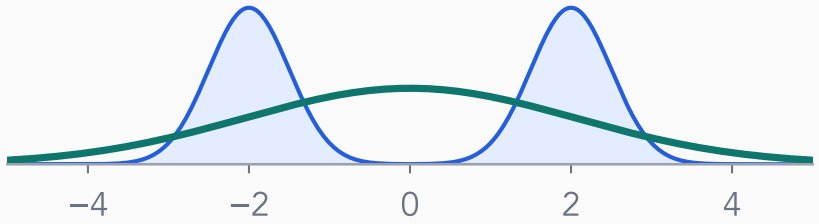
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best $q = \mathcal{N}(0.0,\ 2.06^2)$



AN HONEST LOOK

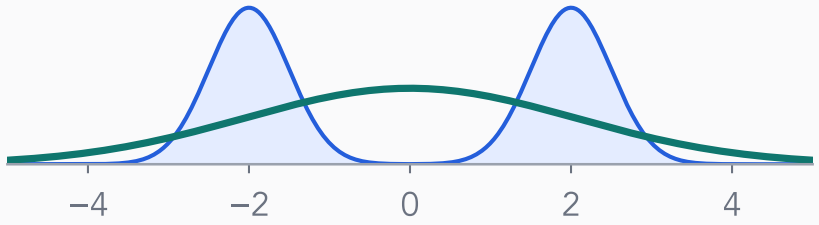
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**3. We show $\pi$, the mean,** not a sample. A sample would be speckled instead of grey.

best $q = \mathcal{N}(0.0,\ 2.06^2)$






$\pi$ (the mean)



a sample

**Not a law of nature: bigger codes, better decoders and better likelihoods all sharpen VAEs.**

END OF PART 7

# What we did, and what comes next

### WHAT WE DID

1. Our 3: encode to a cloud, sample a code, decode, score: loss $= 129.3$.
2. Training: reconstruction falls, KL rises, both settle.
3. Generation needs only the prior and the decoder.
4. The KL weight trades sharp reconstructions against a usable code space; samples blur.

END OF PART 7

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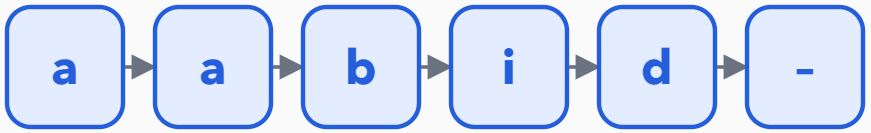
### NEXT LECTURE: A DIFFERENT IDEA

VAEs maximise a bound on $p_\theta(x)$. GANs skip $p_\theta(x)$ entirely: a second network learns to judge real from fake.

THREE WAYS TO GENERATE

# Same goal, three different training signals

**autoregressive**
one token at a time

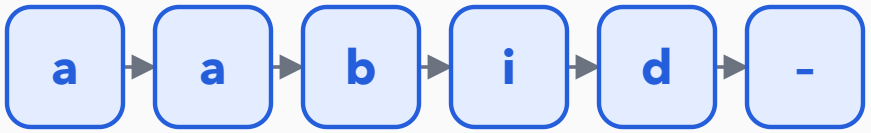



**exact likelihood**
slow sampling

THREE WAYS TO GENERATE

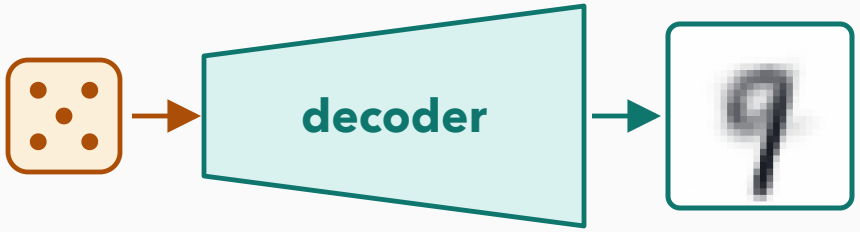
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**autoregressive**
one token at a time




**exact likelihood**
slow sampling

**VAE**
random z, decode once

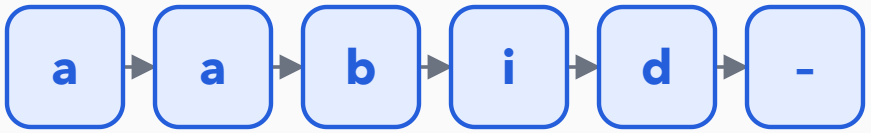



**trained on the ELBO**
tends to blur

THREE WAYS TO GENERATE

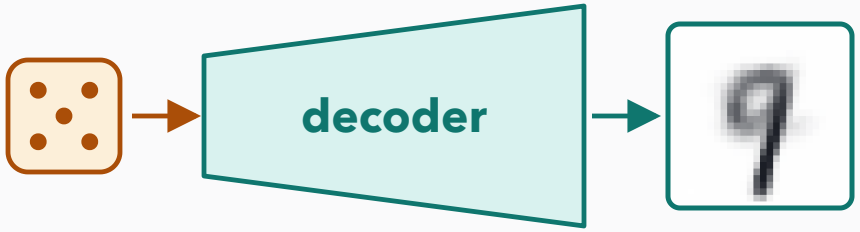
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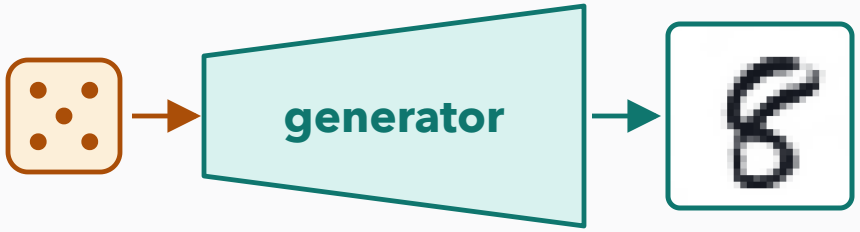

**exact likelihood**
slow sampling

**VAE**
random z, decode once




**trained on the ELBO**
tends to blur

**GAN**
random z, generate once




**next lecture**
trained against a judge

RETURN TO THE OPENING

# What you can now explain



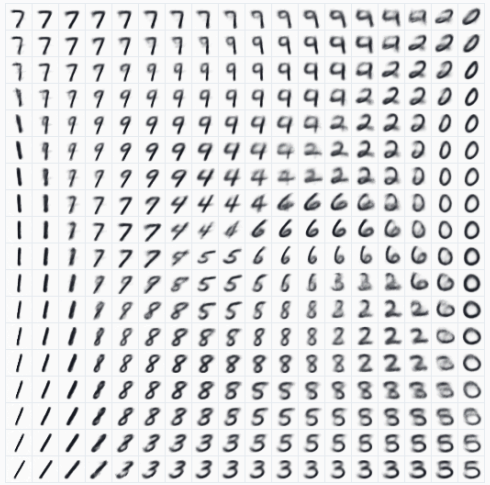
the cover: encode a 3 and a 7, walk a straight line between their codes, decode each point.

RETURN TO THE OPENING

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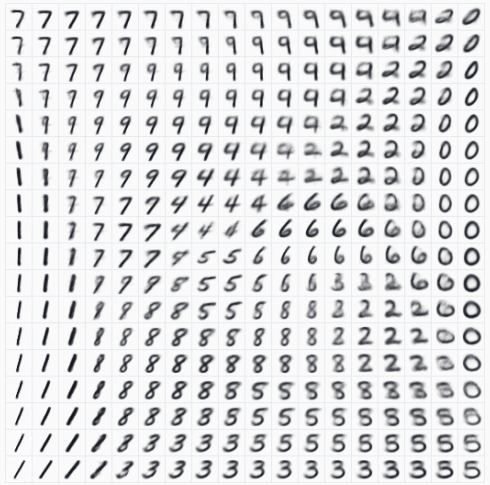
every code in $\mathcal{N}(0, I)$ decodes to a digit: recon + KL, derived from $\mathrm{KL}(q \,\|\, \text{posterior})$.

RETURN TO THE OPENING

# What you can now explain



the cover: encode a 3 and a 7, walk a straight line between their codes, decode each point.



every code in $\mathcal{N}(0, I)$ decodes to a digit: recon + KL, derived from $\mathrm{KL}(q \,\|\, \text{posterior})$.



a conditional VAE: give the label to the encoder and the decoder, and choose which digit to draw.

BEFORE YOU LEAVE

# Three exit questions

1. To generate a new digit with a trained VAE, which parts of the network do you use?

2. Two different KL divergences appear in the VAE story. What does each compare, and which one is in the loss?

3. We never compute $p_\theta(x)$. Why does maximising the ELBO still pull $q$ towards the true posterior?

BEFORE YOU LEAVE

# Three exit questions

1. To generate a new digit with a trained VAE, which parts of the network do you use?

   The prior and the decoder only: $z \sim \mathcal{N}(0, I)$, then decode. The encoder is for training.

2. Two different KL divergences appear in the VAE story. What does each compare, and which one is in the loss?

   $\mathrm{KL}(q \,\|\, \text{prior})$ is in the loss (6.97 for our 3). $\mathrm{KL}(q \,\|\, \text{true posterior})$ is the gap between the ELBO and $\log p_\theta(x)$ (0.55).

3. We never compute $p_\theta(x)$. Why does maximising the ELBO still pull $q$ towards the true posterior?

   $\log p_\theta(x) = \mathrm{ELBO} + \mathrm{KL}(q \,\|\, \text{posterior})$, and the left side does not depend on $q$. Whatever the ELBO gains, the KL loses.

AFTER THE GAN LECTURE

# What if we generate by removing noise, a little at a time?



our 3 → add a little Gaussian noise, again and again → pure noise

> **Adding noise is easy and gives free training pairs. Can a network learn to undo one small step, then walk all the way back from pure noise?**

APPENDIX

# A second route to the ELBO

Part 5 reached the ELBO from $\mathrm{KL}(q \,\|\, \text{posterior})$. Many textbooks reach it from $\log p_\theta(x)$ with Jensen's inequality instead. Same bound, one more tool.

# Start from what we want

$$\log p_\theta(x) = \log \int p_\theta(x, z)\, dz$$

Maximum likelihood wants this number large for our 3, and for every image in the data. The integral over all codes is the part we cannot compute.

**A second route to the bound of Part 5, starting from $\log p_\theta(x)$ instead of the KL.**

# Multiply and divide by the encoder's cloud

$$\log \int p_\theta(x, z)\, dz$$

$$= \log \int q_\varphi(z \mid x)\, \frac{p_\theta(x, z)}{q_\varphi(z \mid x)}\, dz$$

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$$= \log \int q_\varphi(z \mid x)\, \frac{p_\theta(x, z)}{q_\varphi(z \mid x)}\, dz$$

Multiplying and dividing by the same positive number changes nothing. We are free to choose any $q$; we choose the encoder's Gaussian, which is positive everywhere.

**Why bother? Because the next move turns this integral into an average under $q$, and $q$ knows where the patch is.**

# Read the integral as an average under $q$

$$\log \int q_\varphi(z \mid x)\, \frac{p_\theta(x, z)}{q_\varphi(z \mid x)}\, dz$$

$$= \log \mathbb{E}_{z \sim q_\varphi(z|x)} \left[ \frac{p_\theta(x, z)}{q_\varphi(z \mid x)} \right]$$

# Read the integral as an average under $q$

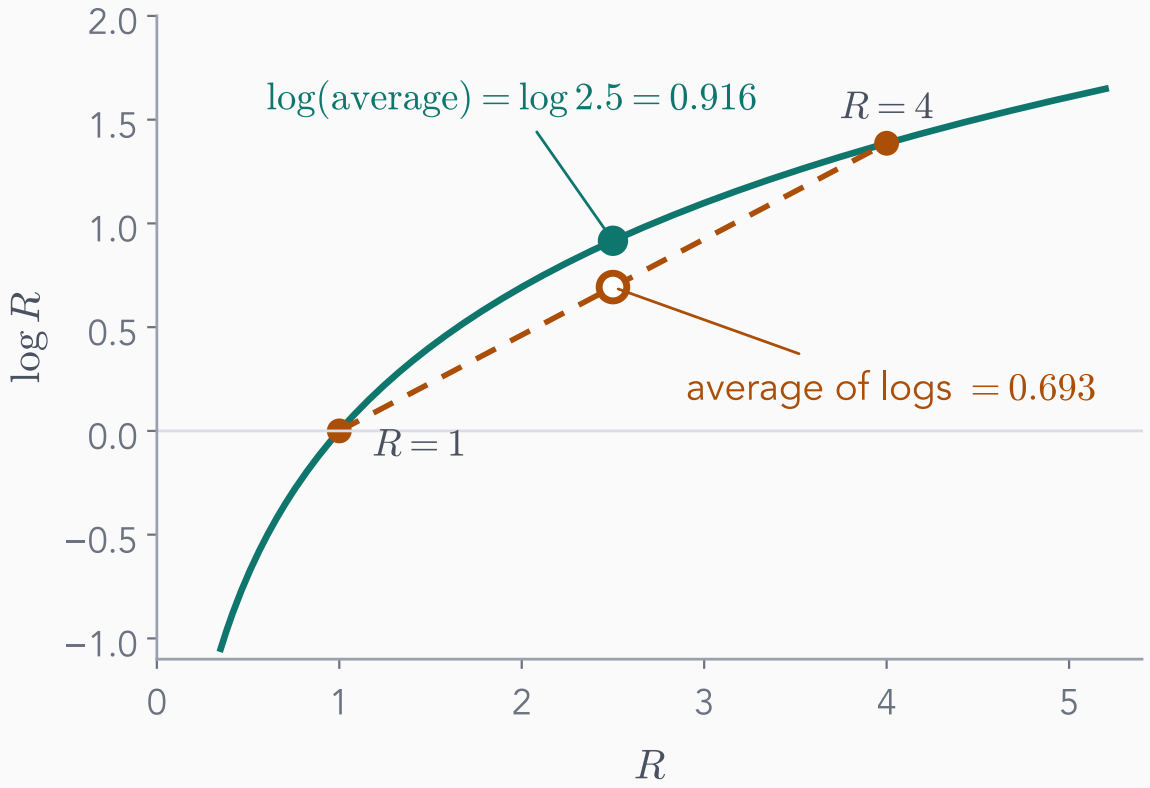
$$\log \int q_\varphi(z \mid x)\, \frac{p_\theta(x, z)}{q_\varphi(z \mid x)}\, dz$$

$$= \log \; \mathbb{E}_{z \sim q_\varphi(z|x)} \left[ \frac{p_\theta(x, z)}{q_\varphi(z \mid x)} \right]$$

From the $\mathbb{E}$ slide: $\int q(z)\, f(z)\, dz = \mathbb{E}_{z \sim q}[f(z)]$. Here $f(z)$ is the ratio $p_\theta(x, z)/q_\varphi(z \mid x)$: one number for each code.

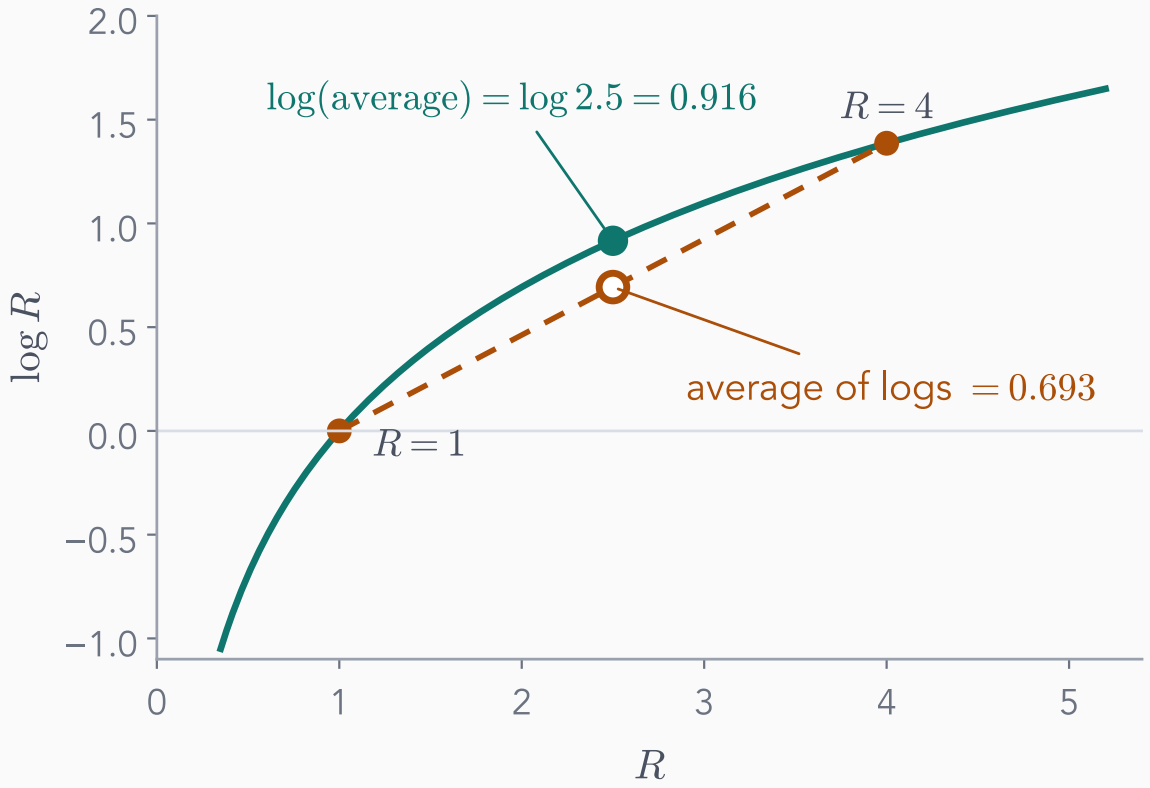
**Still exact. We now average over codes drawn from the encoder's cloud, right where the patch is.**

# Jensen's inequality, in one picture




log curves downwards (it is concave), so every chord lies below the curve.

# Jensen's inequality, in one picture

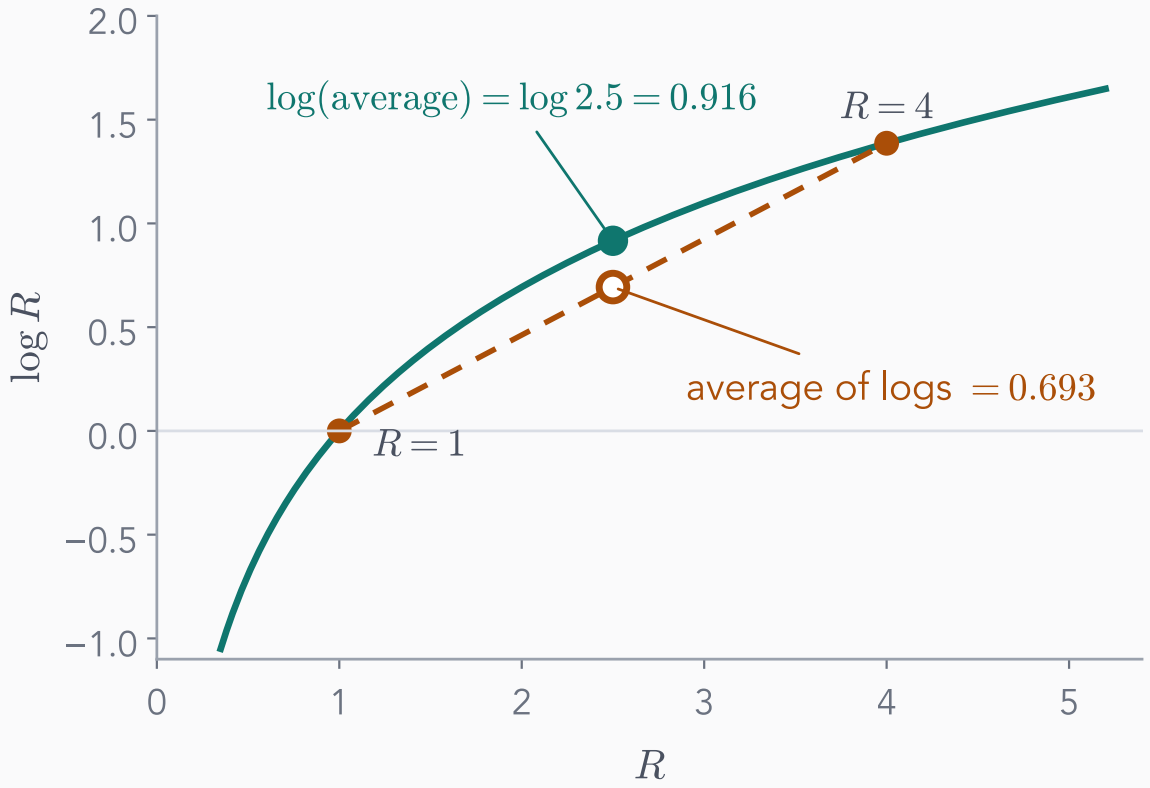



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Two values, $R = 1$ and $R = 4$:

$$\underbrace{\log 2.5}_{\log \text{ of the average}} = 0.916 \geq \underbrace{\tfrac{1}{2}(0 + 1.386)}_{\text{average of the logs}} = 0.693$$

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$$\underbrace{\log 2.5}_{\log \text{ of the average}} = 0.916 \;\geq\; \underbrace{\tfrac{1}{2}(0 + 1.386)}_{\text{average of the logs}} = 0.693$$

$$\log \mathbb{E}[R] \;\geq\; \mathbb{E}[\log R]$$

# Swap the log and the average: the only inequality

$$\log p_\theta(x) = \log \mathbb{E}_q\left[\frac{p_\theta(x,z)}{q_\varphi(z \mid x)}\right]$$

$$\log p_\theta(x) \geq \mathbb{E}_q\left[\log \frac{p_\theta(x,z)}{q_\varphi(z \mid x)}\right] =: \mathrm{ELBO}$$

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Jensen with $R = p_\theta(x,z)/q_\varphi(z \mid x)$. The right side is the **E**vidence **L**ower **BO**und: it can never exceed $\log p_\theta(x)$.

**The same ELBO as Part 5. This route shows it is a bound; Part 5's route also shows the gap is $\mathrm{KL}(q \,\|\, \text{posterior})$.**

FURTHER READING

# Where to go next

- Lilian Weng, *From Autoencoder to Beta-VAE* (lilianweng.github.io): the autoencoder family of Part 2
- Jaan Altosaar, *Tutorial: what is a variational autoencoder?*: the neural-network and probability views side by side
- Kingma & Welling, *Auto-Encoding Variational Bayes*, arXiv 1312.6114
- Blei, Kucukelbir & McAuliffe, *Variational Inference: A Review for Statisticians*, arXiv 1601.00670
- Bishop, *Pattern Recognition and Machine Learning*, §10.1: the ELBO + KL picture of Part 5
- Three YouTube videos on KL divergence that Part 4 draws on (video ids sjgZxuCm_8Q, KHVR587oW8I, tXE23653JrU)
- Stanford CS236 notes on VAEs (deepgenerativemodels.github.io)
- This lecture's code: `scripts/train_models.py` (all models), `scripts/make_figures.py` and `make_figures_extra.py` (every number on the slides)